Receiver POVM (receiver)

Contract, version 0.11.0

This contract turns a declared passive optical receiver into the click-pattern measurement it performs and into lower bounds on its multiclick weight on every photon-number block above a cutoff N; both are recorded in the receiver's artifact, which the flag-state squasher of the reductions module consumes. A receiver may be declared by its port Grams (route (i)), by an instrument matrix (route (ii)) or by a graph of catalogue components (route (iii)); the three routes converge on the port Gram family, which determines the click statistics exactly. A receiver fed by its own probe source as well as by the channel needs a submodule that constructs its setting family and is otherwise refused. The main results of this core module are the click-pattern POVM, which follows by inclusion and exclusion and is block diagonal in the total input photon number (Theorem 4.5); its declared coarse-graining into the protocol's alphabet, under which lower bounds on disjoint raw outcomes add (Theorem 5.6); the weight in [0, 1] emitted for each probe setting, which bounds that setting's multiclick element from below on every block of channel photon number above N (Theorem 8.14); and the range of the weights the artifact carries in its strengthening slot, one per outcome set and setting, which are non-negative and sum to at most one at every setting (Proposition 8.17). The weight comes from an analytic lower bound on the smallest eigenvalue of every multiclick block, made uniform over all blocks above N either by a collapse lemma, which reduces the non-negative part of the infimum over those blocks to that of the value at the first of them, or by the minimum over a finite window of blocks capped by an analytic term for the tail beyond it; an optional exact window uses lower bounds on the smallest eigenvalues of finitely many blocks. Every statement is exact unless it names a receiver tolerance or the eigenvalue slack: where the declaration defines an admissible receiver exactly, the weights are evaluated rigorously, in exact rational arithmetic from enclosures of the model taken on the conservative side, and are lower bounds by theorem with no premise about arithmetic (Theorem 8.13), while elsewhere they are evaluated in binary64 floating point, residuals diagnose the evaluation, and an exact window rests on the assumption that each computed eigenvalue less a declared slack does not exceed the exact one. Separately, for every declaration that defines an admissible receiver exactly, whatever its parameters, the photon-number blocks are enclosed rigorously in exact rational arithmetic, from enclosures of the square roots, cosines and sines the construction takes (Theorem 9.5); where the port Grams are Gaussian-rational the blocks of at most one photon are known exactly (Theorem 9.9), and, read through a port dilation, a block of a receiver without probe inputs can be given besides in enclosed factor form, which fixes its rank where the Gram matrix of its columns has a positive floor, as no entrywise box with positive radii can for a block of lower rank whose support no co-

ordinate vectors span (Theorem 9.7). These enclosures are the reference a snap of the emitted blocks is certified against. The results are conditional on the device assumptions of the Assumptions section, among them a trusted, exactly characterised, passive device whose declared modes carry every path to a detector: the main results rest only on the few of these their proofs use, and the others enter the weight through the corollaries that apply Theorem 8.14 to each setting family. Detector imperfections and the component catalogue are submodules, each with main results of its own.

Grey text, here and throughout, is non-normative: it explains, and nothing in it is assumed or proved.

The design notes (non-normative) explain why the module is built as it is and draw its processing flowchart. The dependencies between the sections are drawn in the section dependency matrix, and the graph index opens every section's dependency graph. The statements digest holds every definition, assumption, result and requirement without the proofs. How to read a contract explains this layout and its links.

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Main results

The module's most important results, those its overview names as what it establishes, with every assumption each rests on, directly or through the claims it uses. Each links to its statement and proof; the statements digest holds them in full.

Theorem 4.5 Click-pattern POVM

The click patterns of a receiver instrument form a POVM whose elements follow from the port Grams by inclusion and exclusion. Every element is block diagonal in the total input photon number, and vacuum gives only the no-click outcome.

Rests on A-PORT, A-VAC

Theorem 5.6 Coarse-graining of the click-pattern POVM

The emitted POVM of a declared coarse-graining is a POVM, block diagonal in the photon number, and lower bounds on disjoint raw outcomes add. A bound on a merged outcome gives nothing about the click patterns it merges.

Rests on A-PORT, A-VAC

Theorem 8.14 Multiclick bounds carried by the emitted weights

Each setting's emitted weight bounds its multiclick element on every photon-number block above the cutoff, and their minimum does so at every setting, under the eigenvalue-slack assumption A-EIG only where computed block eigenvalues enter. Each weight lies in [0, 1] and is finitely computable.

Rests on A-EIG, A-PORT, A-VAC

Proposition 8.17 Range of the strengthening-slot weights

At every setting the strengthening slot's weights (high-photon lower bounds on disjoint sets of raw outcomes) are non-negative and sum to at most one, as are their minima over the settings, and each emitted multiclick weight lies in [0, 1]. That the emitted weights are valid bounds is Theorem 8.14.

Rests on A-EIG, A-PORT, A-VAC

Theorem 16.3 Admissibility of the catalogue entries

Every catalogue entry maps its whole declared parameter domain to contractions on its modes, lossless exactly on the subfamily the catalogue's table lists for it. Only the generic block, an arbitrary matrix on declared modes, needs its contractivity checked at admission.

Rests on no assumption

Theorem 18.1 Detector efficiency in the port Grams

Efficiency η_k uniform on detector port k scales that port's Gram by η_k, keeping its rank if $\eta_k > 0$; varying within a port, it is no scalar and can lower the rank. It enters only as the compiled instrument matrix's last layer, and the loss decomposition Lemma 4.2 checks that loss is counted once.

Rests on A-EXACT, A-MODES, A-NOSEL, A-PASSIVE, A-PORT, A-ROUND, A-TRUST, A-VAC

Theorem 19.2 Dark counts on a photon-number block-diagonal POVM

A dark-count channel keeps the photon-number block-diagonal POVM structure and never moves weight out of the multiclick set. It raises the multiclick bound p_mc to $(1 - r_min)p_mc + r_min$, where r_min is the least multiclick mass over every non-multiclick column, the no-click column included.

Rests on A-EXACT, A-MODES, A-NOSEL, A-PASSIVE, A-PORT, A-ROUND, A-THRESH, A-TRUST, A-VAC

Assumptions at a glance

Every assumption this contract states or its claims assume, with its category and the claims of this contract that assume it directly. An assumption of another contract links into that contract's PDF.

Assumption	Category	Title	Stated as	Assumed directly by
A-EXACT	receiver	Exact declared parameters	Assumption 11.8	Lem. 4.2, Cor. 8.2, Cor. 8.15, Prop. 21.4
A-MODES	receiver	Completeness of the declared modes	Assumption 11.5	Lem. 4.2, Cor. 8.2, Cor. 8.15, Rem. 16.10, Prop. 21.4
A-NOSEL	receiver	Device fixed independently of the adversary	Assumption 11.9	Lem. 4.2, Cor. 8.2, Cor. 8.15, Prop. 21.4
A-PASSIVE	receiver	Passive linear optics	Assumption 11.3	Lem. 4.2, Cor. 8.2, Cor. 8.15, Rem. 16.10, Prop. 21.4
A-PORT	receiver	Mode-insensitive port response	Assumption 11.7	Lem. 2.6, Cor. 8.2, Cor. 8.15, Prop. 21.4
A-ROUND	receiver	Round homogeneity of the instrument, probe and dark counts	Assumption 11.10	Lem. 4.2, Cor. 8.2, Cor. 8.15, Prop. 21.4
A-THRESH	receiver	Threshold detectors without memory	Assumption 11.6	Cor. 8.2, Cor. 8.15, Rem. 16.10, Rem. 19.7, Prop. 21.4
A-TRUST	receiver	Trusted and exactly characterised receiver	Assumption 11.1	Lem. 4.2, Cor. 8.2, Cor. 8.15, Rem. 18.4, Prop. 21.4
A-VAC	receiver	Vacuum on undeclared input modes	Assumption 11.4	Lem. 2.6, Cor. 3.5, Lem. 4.2, Cor. 8.2, Cor. 8.15, Rem. 16.8, Rem. 16.9, Prop. 21.4
A-EIG	numerical trust	Declared slack on the computed block eigenvalues	Assumption 11.11	Rem. 8.11, Thm. 8.14, Cor. 8.15, Thm. 10.10

Notation

The symbols of the repository's notation catalogue (spec/notation.md) that this contract uses, in the catalogue's order, with their meaning.

Symbol	Meaning
$\mathcal{H}(A)$	Hilbert space of the system label A
$\mathcal{L}(A)$	Bounded linear operators on $\mathcal{H}(A)$
$\mathbb{1}$, $\mathbb{1}_C$	Identity operator (contextual / named)
id	Identity channel
P_n, P_n^C	Photon-number projector
$M^{(\leq N)}$	Operator block restriction
$\leq N$, $> N$, $= n$	Block indices for `\blk`
$\mathcal{V}$	The measuring parties' declared joint outcome alphabet — Bob's alone under the default partition
C, P	Channel input; probe input
$\mathcal{Y}$	Bob's coarse-grained detection alphabet, $\lvert\mathcal{Y}\rvert = 6$
$\mathcal{T}_{\text{set}}$	Bob's private setting set, weights q_t (reductions/def:joint-family); in the optical instance the probe settings (register T_{set})
on, off, dc, sq, mc, nc, sc, vac	Mode / outcome abbreviations
$\lambda_{\min}$	Minimum eigenvalue
T	Transpose, X^{T}
$\boldsymbol{v}$	Vector (bold)

Terms

The phrases this contract defines, and the terms of other contracts it uses, each with a one-line gloss and the definition that fixes it. In the text, a linked term opens its definition.

Term	Meaning	Defined in
admissible bound data	Scalars between 0 and 1, a shared γ and c and a β and d per port, with γ at most every β and c at most every d, from which the bound function is built.	Definition 6.1
admissible Gram family	A finite family of positive semidefinite operators whose sum is at most the identity; an exact condition, distinct from exact admissibility of declared data.	Definition 2.2

Term	Meaning	Defined in
angle enclosure	a pair of complex enclosures with rational centres of the cosine and the sine of an angle given as a rational or as a rational multiple of pi, exact at the multiples of pi/2	Definition 9.1
bound data of the setting	The admissible bound data built from a setting's prefactors and the extreme eigenvalues of its operators B.	Definition 8.1
bound function	The function 1 + (K-1)γc^n - Σ_k β_kd$_k{}^n$ of the photon number n, built from admissible bound data.	Definition 6.1
catalogue entries	The ten entry types C1 to C10 of the component catalogue, each with a support, a parameter domain and a matrix.	Definition 16.2
certified	the tier of a lower bound on the minimum of the objective of a declaration, formed with every arithmetic step from that declaration's data exact or rigorously enclosed, so that none of its hypotheses concerns arithmetic	foundations/def: tiers
channel input space	The D-dimensional space whose Fock space a setting family acts on.	Definition 8.1
channel modes	The declared input modes through which the adversary's signal enters.	Definition 2.1
compiled instrument matrix	The matrix D_eta times the ordered product of a well-formed graph's embedded elements.	Definition 3.3
complex enclosure	a Gaussian-rational centre and a non-negative rational radius that bounds the distance of a complex number from the centre; a matrix is enclosed entry by entry	Definition 9.1
component catalogue	A finite set of entry types from which a route-(iii) receiver is assembled.	Definition 3.1
dark-composed family	the click patterns of each setting composed with a dark-count channel, and their sums over the fibres of the declared coarse-graining	Definition 20.1
dark-count channel	A column-stochastic matrix on click patterns that can add clicks and never removes one.	Definition 19.1

Term	Meaning	Defined in
declared coarse-graining	A surjection from the raw click patterns onto the declared emitted alphabet, supplied as data by the protocol declaration.	Definition 5.5
declared dark-count route	The dark-count channel is a matrix R declared as such.	Definition 19.1
declared data	the data an instance states and the data a producer emits as its own declaration, with the sets such data define	foundations/def: declared-data
declared element graph	A finite directed graph whose vertices are catalogue elements, each with an entry type, a parameter tuple and a support.	Definition 3.2
declared input modes	The modes a receiver instrument declares as inputs, the channel modes and the declared source modes; every other mode carries vacuum at the input.	Definition 2.1
declared mode space	A finite, totally ordered set of optical mode labels together with the Hilbert space whose orthonormal basis they index.	Definition 1.1
declared order	a total order on a finite alphabet, fixed together with the alphabet	foundations/def: declared-order
declared source modes	The declared input modes that carry the receiver's own probe state.	Definition 2.1
declared tolerance	a binary64 number, at least zero and below one, that an instance states in its run configuration	foundations/def: tolerance
declared value	a binary64 number stated as a datum, which denotes exactly that dyadic rational and approximates nothing	foundations/def: binary64
detector ports	The pairwise disjoint sets of modes each watched by one threshold detector.	Definition 2.1
eigenvalue slack	The declared number subtracted from each computed block eigenvalue of an exact window.	Definition 8.7
elements	A vertex of a declared element graph: a catalogue component with its parameters and support, not a POVM element.	Definition 3.2

Term	Meaning	Defined in
emitted POVM	The measurement the receiver emits, one element per emitted outcome, each the sum of the click-pattern elements of its fibre.	Definition 5.5
enclosed bound data	Reals that bound the bound data of a setting from the conservative side, c and γ from below and each d and β from above, all within 0 and 1.	Definition 8.12
enclosed weight	The per-setting weight formed on enclosed bound data and, at an exact window, on rigorous window bounds, with no computed eigenvalue and no eigenvalue slack.	Definition 8.12
entry type	A kind of optical component: a support shape, a parameter domain and a rule giving a local element for each parameter tuple.	Definition 3.1
entrywise enclosure	a Gaussian-rational Hermitian centre together with symmetric non-negative rational radii that bound, entry by entry, the modulus of the difference between a matrix and that centre	foundations/def: exact-reference
evaluated off its snap	said of a block of a snap in a run: no number from which the run computes its key length reads the snapped block, save the points at which those numbers are evaluated and the values other than the output length that the run fixes before its first round	foundations/def: evaluated-off-snap
evaluation model	The arithmetic of this contract: its exact claims evaluated in binary64 with diagnostic residuals and no rigorous enclosure of a floating-point number beyond the eigenvalue-slack assumption, beside exact rational enclosures of the blocks and a rigorous evaluation of the weights.	Definition 10.9
exact block bound	The smallest eigenvalue of the n-photon block of the multiclick element, a function of n.	Definition 7.1
exact flag	The flag of a setting, exact or general; it may read exact only where every prefactor is one by construction.	Definition 8.1

Term	Meaning	Defined in
exact island	the computations of the engine, its instances' declarations included, by which a consumer of data or the declarer about to declare them decides whether they satisfy the conditions their declaration states, a declarer computes and decides the reference of a measurement family, a consumer's replacement of an emitted scalar bound is decided to lie on the bound's weakening side, a producer constructs the operator data it emits, the acceptance side of the optimisation is constructed and certified, or a lower bound on a linear minimum over the feasible side of the optimisation is certified	foundations/def: exact-island
exact reference	a reference all of whose radii are zero, so that each block is its Gaussian-rational centre	foundations/def: exact-reference
exact route	the declarations whose declared data determine, exactly, an admissible family of port Grams with Gaussian-rational entries, with a probe of the Gaussian tier	Definition 9.8
exact-window depth	The declared integer up to which the weight of a setting uses, block by block, the larger of the analytic bound and a computed block eigenvalue less the eigenvalue slack.	Definition 8.7
factor enclosure	a matrix box of columns together with an exact positive diagonal core, such that the matrix is the product of a matrix L the box contains, the core and the adjoint of L	foundations/def: factor-enclosure
factored loss	The factor t, between the largest declared efficiency and one, by which a normalized station divides every efficiency.	Definition 21.2
Fock space	The direct sum over n of the symmetric n-photon subspaces of a finite-dimensional mode space.	Definition 1.2
Gaussian tier	The probe declarations whose states are products of single-mode thermal states, vacuum included.	Definition 2.3

Term	Meaning	Defined in
generated dark-count route	The dark-count channel is the independent product form of the declared per-port probabilities.	Definition 19.1
Generic block	Entry C10: an arbitrary matrix on a declared mode subset, admitted after an explicit contractivity check.	Definition 16.2
global embedding	The matrix on the whole mode space that acts as a local element on its support and as the identity elsewhere.	Definition 1.3
handed reference	a reference of a snapped measurement family that its declarer passes, with its centres and radii and unaltered, to every consumer of the snap	foundations/def: evaluated-off-snap
Hermiticity-and-positivity defect	The larger of the spectral norm of X minus its adjoint and the negative part of the smallest eigenvalue of the Hermitian part of X: one number, measured in the spectral norm.	Definition 10.1
independent product form	The dark-count channel built from per-port dark-count probabilities, each port firing independently.	Definition 19.1
instrument matrix	The contraction of a receiver instrument from input modes to output modes.	Definition 2.1
joint no-click elements	For a set J of ports, the sum of the POVM elements of every outcome in which no port of J clicks.	Definition 8.1
lattice value	at a lattice exponent k, a real number that is an integer multiple of two to the power −k and has modulus at most one; a complex number or a matrix is one when the real and the imaginary part of every entry are	foundations/def: binary64
linear extension	An enumeration of the elements of a graph that respects its reachability order.	Definition 3.2
local element	A contraction acting on a chosen subset of the declared modes.	Definition 1.3
loss deficiency	The identity on the declared input modes minus the sum of all port Grams.	Definition 2.2
lossless element	A local element that is an isometry on its support.	Definition 1.3

Term	Meaning	Defined in
matrix box	a matrix of Gaussian-rational centres together with non-negative rational radii that bound, entry by entry, the modulus of the difference between a rectangular complex matrix and that centre	foundations/def: factor-enclosure
measurement isometry	The isometry from the Fock space of the declared input modes that prepares vacuum elsewhere and applies the lifted unitary dilation.	Definition 2.5
multiclick element	The sum of the POVM elements of every outcome in which at least two ports click.	Definition 8.1
multiclick set	The outcomes in which at least two ports click.	Definition 4.4
no-click outcome	The outcome in which no port clicks.	Definition 4.4
normalized station	A receiver instrument with the same channel modes and ports whose port Grams are those of the receiver divided by the factored loss t.	Definition 21.2
occupation-diagonal	Said of a state on the probe Fock space that is diagonal in its occupation basis.	Definition 2.3
optical mode	A label recording a spatial path, a polarization and a reserved temporal or spectral index.	Definition 1.1
outcome	The set of detector ports that click in one round.	Definition 4.4
path	A spatial index carrying the two polarization modes H and V.	Definition 16.2
per-setting weight	The high-photon multiclick weight of one setting: its collapse value, its window and tail, or its exact window.	Definition 8.3
polarization-dependent beamsplitter	A beamsplitter that declares its transmittance and reflection phase separately for the H and the V modes.	Definition 16.2
port dilation	a finite ordered mode set with disjoint ports and an isometry from the declared input modes into it whose Gram on each port is that port's Gram matrix	Definition 9.6

Term	Meaning	Defined in
port Gram matrices	The positive operators on the declared input modes, one per detector port, whose quadratic form is the squared norm of the amplitude a receiver instrument sends to that port.	Definition 2.2
predicate (X)	The condition that a setting's exact flag reads exact; it selects the collapse branch of the per-setting weight.	Definition 8.3
probe declaration	A finite alphabet of probe settings, vacuum among them, with a declared occupation-diagonal state on the declared source modes for each setting.	Definition 2.3
pure-loss channel	The channel on the channel system that sends every channel mode through a beamsplitter of transmittance t with vacuum in its other input and discards the reflected light.	Definition 21.1
raw alphabet	The set of all outcomes, the subsets of the K ports.	Definition 4.4
receiver instrument	A declared mode space with channel and probe inputs, disjoint detector ports and a contractive instrument matrix.	Definition 2.1
receiver tolerances	The eight declared numbers at which the receiver decides its exact conditions on declared data, bounds its residuals and spends its eigenvalue slack; not the instance's declared tolerance.	Definition 10.4
reference	a family of Hermitian positive semidefinite block lists complete at the constant on every block, read together with an entrywise enclosure of every block of every element	foundations/def: exact-reference
residuals	The six consistency checks on the floating-point evaluation of this contract, each exactly zero in exact arithmetic.	Definition 10.1
Retarder	Entry C3: a wave plate with a retardance, a fast-axis angle and a scalar loss, with half-wave and quarter-wave presets.	Definition 16.2

Term	Meaning	Defined in
rigorous window bounds	Reals at most the exact block bound of a setting at each index of its exact window.	Definition 8.12
rigorously enclosed	said of a computation each of whose operations returns a set of numbers, such as an interval or a ball, containing the exact result of the operation at every value its inputs' sets contain	foundations/def: tiers
rotator	Entry C4: the circular retarder, a real rotation of the two polarization modes of one path.	Definition 16.2
round-homogeneous	one fixed list of objects describes every round, and every round behaves as they describe whatever the other rounds do	foundations/def: round-homogeneity
sector map	a block kind together with a non-empty, ordered list of block identifiers, each with a dimension	foundations/def: sector-map
setting family	A finite set of settings, each with a photon-number block-diagonal POVM whose joint no-click elements have the prefactor form, and an exact flag.	Definition 8.1
single-click set	The outcomes in which exactly one port clicks.	Definition 4.4
square-root enclosure	a complex enclosure with a rational centre of the square root of a non-negative rational, exact where that rational is the square of a rational	Definition 9.1
strengthening slot	The list of disjoint raw-outcome sets with per-setting high-photon lower-bound weights that the artifact carries.	Definition 8.16
structural support	of a block of an element: the set of indices whose row or column holds an entry that is not structurally zero	foundations/def: snap
supplier	a module whose own claims decide their conditions on the data it reads or emits in floating point, at slacks an instance declares to it, rather than exactly	foundations/def: declared-data
support	The set of modes a local element acts on.	Definition 1.3

Term	Meaning	Defined in
support comparability	Condition (E3): any two elements whose supports intersect are ordered by the graph.	Definition 3.2
terminal detector-efficiency layer	The diagonal matrix D_eta applying each port's detector efficiency after every element.	Definition 3.3
uniform weight	The minimum of the per-setting weights over all settings of a setting family.	Definition 8.3
uniform weight of an entry	The minimum over the settings of a slot entry's weights; it bounds the entry at every setting.	Definition 8.16
well formed	Said of a declared element graph that satisfies (E1) to (E3): valid parameters, acyclicity and support comparability.	Definition 3.2

Part I
Mathematics

1 Modes, Fock space and the mode algebra

A receiver acts on a finite set of declared optical modes, and its photon statistics live in the Fock space over them. This section lifts a matrix on the modes to every photon-number sector and records the algebra the later proofs run on: embedding, products and adjoints preserve contractions and elements on disjoint modes commute (Proposition 1.4), and the lift respects composition, adjoints and projectors (Lemma 1.6), also for maps between two mode spaces, where the lift of a direct sum is a tensor product (Lemma 1.7). The lift of $1 - A$ is called a lifted no-click operator because, with A the sum of the port Grams of a set of ports, it is the operator for the event that no port of the set clicks, pulled back to the declared inputs (Theorem 4.1). Its spectrum on each photon-number block is computed from the eigenvalues of A (Lemma 1.9), which is where the analytic high-photon bound starts. The normal-ordered form is stated for comparison with the optics literature; no proof uses it.

A declared mode space is a finite, totally ordered set of optical modes, each labelled by a path, a polarization and a reserved temporal or spectral index that takes one declared value. Matrices on the modes follow this order; on the input modes the channel modes come first (Definition 2.1).

The temporal or spectral index of a mode label is reserved: it takes a single value in this contract, so mode labels carry no temporal semantics. The order compares polarizations last, so the H and V modes of a path are adjacent.

Definition 1.1 (Declared mode space)**.** A *declared mode space* is a finite totally ordered set $\mathbb{M}$, each of whose elements is an *optical mode*, together with the Hilbert space $\mathbb{C}^{\mathbb{M}}$ carrying the orthonormal basis $\{|m\rangle\}_{m\in\mathbb{M}}$ indexed in the declared order. A mode is a label $m = (p, \pi, \tau)$ recording a spatial path p, a polarization $\pi \in \{H, V\}$ and a temporal/spectral index τ; the order is lexicographic in (p, τ, π) with H before V. The index τ is *reserved*: it is carried in the label and in the emitted artifact, it may take only a single declared value in this contract, and no catalogue entry (Definition 16.2) reads or writes it. For $\Sigma \subseteq \mathbb{M}$ we write $E_\Sigma = \sum_{m\in\Sigma} |m\rangle\langle m|$ for the coordinate projector, and $\mathbb{M}$ is identified with $\{0, \ldots, |\mathbb{M}| - 1\}$ when matrix indices are needed.

Used by Prop. 1.4, Def. 2.1, Ass. 11.6 · newly derived · accepted

The Fock space over a finite-dimensional space is the direct sum of its symmetric photon-number sectors, and a linear map lifts to it sector by sector. Every photon-number block of the receiver's measurement is assembled from blocks of such lifts.

The lift is written for a map between two spaces as well as for an operator on one; the two-space form is used for coordinate inclusions of mode spaces, among them the inclusion of the declared input modes into the dilated mode space (Definition 2.5).

Definition 1.2 (Fock space and the second-quantization functor)**.** For a finite-dimensional Hilbert space $\mathcal{K}_0$ the *Fock space* is $\mathcal{F}(\mathcal{K}_0) = \bigoplus_{n\geq 0} \mathrm{Sym}^n(\mathcal{K}_0)$, where $\mathrm{Sym}^n(\mathcal{K}_0) \subseteq \mathcal{K}_0^{\otimes n}$ is the subspace invariant under every permutation of the n tensor factors and $\mathrm{Sym}^0(\mathcal{K}_0) = \mathbb{C}\,|\mathrm{vac}\rangle$. For a linear operator X on $\mathcal{K}_0$ put

$$\mathfrak{F}(X) \coloneqq \bigoplus_{n\geq 0} \left(X^{\otimes n}\right)\big|_{\mathrm{Sym}^n}, \qquad \mathfrak{F}(X)\big|_{\mathrm{Sym}^0} = 1. \tag{1.1}$$

This is well defined because $X^{\otimes n}$ commutes with every permutation of the factors and therefore preserves $\mathrm{Sym}^n(\mathcal{K}_0)$ (Lemma 1.6). The same formula defines $\mathfrak{F}(X) : \mathcal{F}(\mathcal{K}_0) \to \mathcal{F}(\mathcal{K}_1)$ for a linear map $X : \mathcal{K}_0 \to \mathcal{K}_1$ between two spaces, because $X^{\otimes n}$ intertwines the permutation actions and therefore carries $\mathrm{Sym}^n(\mathcal{K}_0)$ into $\mathrm{Sym}^n(\mathcal{K}_1)$. Occupation vectors $\boldsymbol{m} \in \mathbb{N}_0^{\dim \mathcal{K}_0}$ with $|\boldsymbol{m}| = n$ index the orthonormal basis $\{|\boldsymbol{m}\rangle\}$ of $\mathrm{Sym}^n(\mathcal{K}_0)$ obtained by symmetrizing and normalizing $|e_1\rangle^{\otimes m_1} \otimes \cdots$ in a fixed orthonormal basis $\{|e_j\rangle\}$ of $\mathcal{K}_0$; a_j and $a_j^\dagger$ denote the annihilation and creation operators of $|e_j\rangle$.

Used by Lem. 1.6, Lem. 1.7, Def. 2.1, Def. 2.3, Def. 2.5, Lem. 2.6, Def. 7.1, Def. 8.1, Lem. 9.3, Lem. 9.4, Def. 9.6, Thm. 9.7, Def. 20.1, Prop. 20.3, Def. 21.1 · newly derived · accepted

A local element is a contraction on a subset of the modes, its support, and its global embedding acts as the identity on every other mode. Embedding is what lets component descriptions compose as matrices on the whole mode space.

Definition 1.3 (Local element and its embedding)**.** A *local element* on a mode subset $\Sigma \subseteq \mathbb{M}$ is a matrix $G_\Sigma \in \mathbb{C}^{\Sigma\times\Sigma}$ with $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}_\Sigma$; Σ is its *support*. Its *global embedding* is

$$\hat{G} \coloneqq G_\Sigma \oplus \mathbb{1}_{\mathbb{M}\setminus\Sigma} \;\in\; \mathbb{C}^{\mathbb{M}\times\mathbb{M}}, \tag{1.2}$$

that is, $\hat{G}\,E_\Sigma = G_\Sigma E_\Sigma$ and $\hat{G}\,(\mathbb{1} - E_\Sigma) = \mathbb{1} - E_\Sigma$. A local element with $G_\Sigma^\dagger G_\Sigma = \mathbb{1}_\Sigma$ is a *lossless element*.

Used by Prop. 1.4, Def. 3.1, Thm. 16.3, Lem. 16.4 · newly derived · accepted

Embedding preserves contractivity, products of contractions are contractions, elements on disjoint supports commute, and embedding respects adjoints and unitarity. Graph compilation rests on these four facts.

Proposition 1.4 (Algebra of embedded local elements)**.** Let $\hat{G}, \hat{G}'$ be global embeddings of local elements with supports Σ, Σ'.

(i) *Embedding preserves contractivity:* $\hat{G}^\dagger \hat{G} \preceq \mathbb{1}_{\mathbb{M}}$, with equality exactly when the element is lossless.

(ii) *Products of contractions are contractions:* if $X^\dagger X \preceq \mathbb{1}$ and $Y^\dagger Y \preceq \mathbb{1}$ then $(XY)^\dagger (XY) \preceq \mathbb{1}$, with equality for all vectors exactly when Y is an isometry and X is isometric on $\operatorname{ran} Y$.

(iii) *Disjoint supports commute:* if $\Sigma \cap \Sigma' = \varnothing$ then $\hat{G}\hat{G}' = \hat{G}'\hat{G}$.

(iv) *Adjoints and unitaries:* $\hat{G}^\dagger$ is the embedding of $G_\Sigma^\dagger$, and $\hat{G}$ is unitary exactly when G_Σ is.

Uses Def. 1.1, Def. 1.3 · *Used by* Thm. 3.4, Thm. 18.1, Def. 21.2 · newly derived · accepted · *Oracles* registry_receiver_povm.py

Proof. (i) In the block decomposition $\mathbb{C}^{\mathbb{M}} = \mathbb{C}^{\Sigma} \oplus \mathbb{C}^{\mathbb{M}\setminus\Sigma}$ one has $\hat{G}^\dagger \hat{G} = (G_\Sigma^\dagger G_\Sigma) \oplus \mathbb{1}_{\mathbb{M}\setminus\Sigma}$, which is $\preceq \mathbb{1}$ precisely when $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}_\Sigma$, and equals $\mathbb{1}$ precisely when $G_\Sigma^\dagger G_\Sigma = \mathbb{1}_\Sigma$.

(ii) For any vector v, $\|XYv\| \le \|Yv\| \le \|v\|$, using $\|Xw\| \le \|w\|$ at $w = Yv$ and then $\|Yv\| \le \|v\|$. Hence $v^\dagger (XY)^\dagger (XY) v \le v^\dagger v$ for all v, which is $(XY)^\dagger (XY) \preceq \mathbb{1}$. Equality for all v forces equality in both steps for all v: the second gives $\|Yv\| = \|v\|$ for all v, i.e. Y isometric, and the first then gives $\|Xw\| = \|w\|$ for every $w \in \operatorname{ran} Y$. Conversely, if Y is an isometry and X is isometric on $\operatorname{ran} Y$, then $\|XYv\| = \|Yv\| = \|v\|$ for every v, which is equality for all vectors.

(iii) Write $\mathbb{C}^{\mathbb{M}} = \mathbb{C}^{\Sigma} \oplus \mathbb{C}^{\Sigma'} \oplus \mathbb{C}^{\mathbb{M}\setminus(\Sigma\cup\Sigma')}$, legitimate because $\Sigma \cap \Sigma' = \varnothing$. In that decomposition $\hat{G} = G_\Sigma \oplus \mathbb{1} \oplus \mathbb{1}$ and $\hat{G}' = \mathbb{1} \oplus G'_{\Sigma'} \oplus \mathbb{1}$; both products equal $G_\Sigma \oplus G'_{\Sigma'} \oplus \mathbb{1}$.

(iv) Immediate from the block form. □

The contract's only commutation result, Proposition 1.4 (iii), needs disjoint mode supports; being unrelated in a declared graph is not enough. Unrelated elements that share a mode can depend on their order, which is why a well-formed graph must order them.

Remark 1.5 (Disjoint mode supports as the commutation hypothesis). Proposition 1.4 (iii) is the *only* commutation this contract has, and it needs disjointness of the *mode supports*, not disjointness in the wiring diagram. Two elements that are unrelated in a declared graph may still touch a common mode, and then their product can depend on the order. That is exactly why Definition 3.2 carries condition (E3) and why Theorem 3.4 is a theorem rather than a formality.

newly derived · accepted

The Fock lift respects composition and adjoints, preserves every photon-number sector, and takes unitaries, contractions and projectors to unitaries, contractions and projectors. The lift of a unitary implements the mode transformation on the annihilation operators.

The later derivations work block by block on the lift of Definition 1.2; this lemma collects the properties of the lift they use.

Lemma 1.6 (Functoriality of the Fock lift)**.** Let $\mathcal{K}_0$ be finite-dimensional and let X, Y be operators on $\mathcal{K}_0$. Then, with $\mathfrak{F}$ as in (1.1):

(i) $X^{\otimes n}$ preserves $\mathrm{Sym}^n(\mathcal{K}_0)$, so $\mathfrak{F}(X)$ is well defined, and it is block diagonal with respect to the photon-number decomposition: $\mathfrak{F}(X)P_n = P_n\mathfrak{F}(X)$ for every n;

(ii) $\mathfrak{F}(XY) = \mathfrak{F}(X)\mathfrak{F}(Y)$, $\mathfrak{F}(\mathbb{1}) = \mathbb{1}$, and $\mathfrak{F}(X)^\dagger = \mathfrak{F}(X^\dagger)$;

(iii) if U is unitary then $\mathfrak{F}(U)$ is unitary; if $X^\dagger X \preceq \mathbb{1}$ then $\|\mathfrak{F}(X)\| \leq 1$, so $\mathfrak{F}(X)$ is a bounded operator on $\mathcal{F}(\mathcal{K}_0)$;

(iv) if P is an orthogonal projector then $\mathfrak{F}(P)$ is the orthogonal projector onto $\mathcal{F}(\operatorname{ran} P) \subseteq \mathcal{F}(\mathcal{K}_0)$ — equivalently, with the occupation vectors formed in an orthonormal basis of $\mathcal{K}_0$ adapted to $\operatorname{ran} P \oplus \ker P$, onto the closed span of those carrying no photon outside $\operatorname{ran} P$;

(v) if U is unitary then $\mathfrak{F}(U)$ implements the mode transformation on the annihilation operators, $\mathfrak{F}(U)^\dagger a_i \mathfrak{F}(U) = \sum_j U_{ij} a_j$.

Uses Def. 1.2 · *Used by* Lem. 1.7, Rem. 1.8, Lem. 1.9, Lem. 1.10, Prop. 1.11, Rem. 1.12, Rem. 1.13, Def. 2.5, Lem. 2.6, Thm. 4.1, Thm. 4.5, Lem. 7.2, Lem. 9.3, Thm. 9.7, Ex. 17.3, Prop. 20.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.fock:fock_block · *Oracles* registry_receiver_povm.py

Proof. (i) For a permutation π of n factors let Π_π be the corresponding unitary on $\mathcal{K}_0^{\otimes n}$. Then $\Pi_\pi X^{\otimes n} = X^{\otimes n}\Pi_\pi$, because both sides send $v_1 \otimes \cdots \otimes v_n$ to the tensor product of the Xv_i permuted by π. Hence $X^{\otimes n}$ commutes with the orthogonal projector $P_{\mathrm{Sym}^n} = \frac{1}{n!}\sum_\pi \Pi_\pi$ and therefore preserves its range. Block diagonality in n is the definition of $\mathfrak{F}$ as a direct sum.

(ii) On Sym^n, $(XY)^{\otimes n} = X^{\otimes n}Y^{\otimes n}$ and both factors preserve Sym^n, so the restriction of the product is the product of the restrictions; $\mathbb{1}^{\otimes n} = \mathbb{1}$; and $(X^{\otimes n})^\dagger = (X^\dagger)^{\otimes n}$, whose restriction to the invariant subspace Sym^n is the adjoint of the restriction. Summing over n gives the three identities.

(iii) With U unitary, (ii) gives $\mathfrak{F}(U)^\dagger\mathfrak{F}(U) = \mathfrak{F}(U^\dagger U) = \mathbb{1}$ and likewise on the other side. With $X^\dagger X \preceq \mathbb{1}$, $\|X\| \leq 1$, so $\|X^{\otimes n}\| = \|X\|^n \leq 1$; restriction to a subspace does not increase the norm, so every block of $\mathfrak{F}(X)$ has norm ≤ 1 and the direct sum is bounded by 1.

(iv) $P^{\otimes n}$ is an orthogonal projector on $\mathcal{K}_0^{\otimes n}$ with range $(\operatorname{ran} P)^{\otimes n}$, and it commutes with P_{Sym^n} by the computation in (i). The product of two

commuting orthogonal projectors is the orthogonal projector onto the intersection of their ranges, so the restriction of $P^{\otimes n}$ to Sym^n is the orthogonal projector onto $\mathrm{Sym}^n(\mathcal{K}_0) \cap (\operatorname{ran} P)^{\otimes n} = \mathrm{Sym}^n(\operatorname{ran} P)$. Choosing an orthonormal basis of $\mathcal{K}_0$ adapted to $\operatorname{ran} P \oplus \ker P$ identifies that subspace with the span of the occupation vectors supported on $\operatorname{ran} P$.

(v) Fix the orthonormal basis $\{|e_j\rangle\}$ used in Definition 1.2. The creation operator satisfies $a_i^\dagger |\psi\rangle = \sqrt{n+1}\, P_{\mathrm{Sym}^{n+1}}(|e_i\rangle \otimes |\psi\rangle)$ for $|\psi\rangle \in \mathrm{Sym}^n$, which is the standard identification of the mode description with the symmetric-tensor description. For unitary U, $\mathfrak{F}(U) P_{\mathrm{Sym}^{n+1}} = P_{\mathrm{Sym}^{n+1}} U^{\otimes(n+1)}$ on $\mathcal{K}_0^{\otimes(n+1)}$ by (i), hence

$$\begin{aligned}\mathfrak{F}(U)\, a_i^\dagger |\psi\rangle &= \sqrt{n+1}\, P_{\mathrm{Sym}^{n+1}}\big(U|e_i\rangle \otimes U^{\otimes n}|\psi\rangle\big) \\ &= \sum_j U_{ji}\sqrt{n+1}\, P_{\mathrm{Sym}^{n+1}}\big(|e_j\rangle \otimes \mathfrak{F}(U)|\psi\rangle\big) = \Big(\sum_j U_{ji} a_j^\dagger\Big)\mathfrak{F}(U)|\psi\rangle\,,\end{aligned} \tag{1.3}$$

using $U|e_i\rangle = \sum_j U_{ji}|e_j\rangle$. Therefore $\mathfrak{F}(U) a_i^\dagger \mathfrak{F}(U)^\dagger = \sum_j U_{ji} a_j^\dagger$; taking adjoints gives $\mathfrak{F}(U) a_i \mathfrak{F}(U)^\dagger = \sum_j \overline{U_{ji}}\, a_j$, and replacing U by $U^\dagger$ gives $\mathfrak{F}(U)^\dagger a_i \mathfrak{F}(U) = \sum_j \overline{(U^\dagger)_{ji}} a_j = \sum_j U_{ij} a_j$. □

The Fock lift respects products and adjoints of maps between two mode spaces and takes contractions to contractions and isometries to isometries. On disjoint mode sets the occupation identification is unitary, and the lift of a direct sum is the tensor product of the lifts.

Lemma 1.6 is stated for operators on one space. A loss in front of a receiver is an isometry into a larger mode space, so the same properties are needed for maps between two spaces, together with the identification of the Fock space over two disjoint mode sets with the tensor product of their Fock spaces.

Lemma 1.7 (The lift of maps between mode spaces and of direct sums). Let $\mathcal{K}_0, \mathcal{K}_1, \mathcal{K}_2$ be finite-dimensional, and let $X\colon \mathcal{K}_0 \to \mathcal{K}_1$ and $Y\colon \mathcal{K}_1 \to \mathcal{K}_2$ be linear maps.

(i) $\mathfrak{F}(YX) = \mathfrak{F}(Y)\mathfrak{F}(X)$ and $\mathfrak{F}(X)^\dagger = \mathfrak{F}(X^\dagger)$; if $X^\dagger X \preceq \mathbb{1}$ then $\|\mathfrak{F}(X)\| \le 1$, and if $X^\dagger X = \mathbb{1}$ then $\mathfrak{F}(X)$ is an isometry.

(ii) Let S and T be disjoint finite mode sets. The linear map $\mathcal{F}(\mathbb{C}^S) \otimes \mathcal{F}(\mathbb{C}^T) \to \mathcal{F}(\mathbb{C}^{S \sqcup T})$ defined on occupation vectors (Definition 1.2) by $|\boldsymbol{m}_1\rangle \otimes |\boldsymbol{m}_2\rangle \mapsto |\boldsymbol{m}_1, \boldsymbol{m}_2\rangle$, the *occupation identification*, is unitary, and on its n-photon level it is $|\varphi\rangle \otimes |\chi\rangle \mapsto \binom{n}{p}^{1/2} P_n(\varphi \otimes \chi)$ for $\varphi \in \mathrm{Sym}^p(\mathbb{C}^S)$, $\chi \in \mathrm{Sym}^q(\mathbb{C}^T)$, $p+q=n$, with P_n the orthogonal projector of $(\mathbb{C}^{S \sqcup T})^{\otimes n}$ onto its symmetric subspace.

(iii) Under that identification, for operators X_1 on $\mathbb{C}^S$ and X_2 on $\mathbb{C}^T$, $\mathfrak{F}(X_1 \oplus X_2) = \mathfrak{F}(X_1) \otimes \mathfrak{F}(X_2)$.

Uses Def. 1.2, Lem. 1.6 · *Used by* Def. 2.5, Lem. 2.6, Def. 21.1, Lem. 21.3 · newly derived · accepted

Proof. (i) $X^{\otimes n}$ maps $\mathrm{Sym}^n(\mathcal{K}_0)$ into $\mathrm{Sym}^n(\mathcal{K}_1)$ (Definition 1.2), and so does $(X^\dagger)^{\otimes n}$ from $\mathrm{Sym}^n(\mathcal{K}_1)$ back, both intertwining the permutation actions. Then $(YX)^{\otimes n} = Y^{\otimes n}X^{\otimes n}$ restricted to $\mathrm{Sym}^n(\mathcal{K}_0)$ is the product of the restrictions, and the restriction of $(X^{\otimes n})^\dagger = (X^\dagger)^{\otimes n}$ to $\mathrm{Sym}^n(\mathcal{K}_1)$ is the adjoint of the restriction of $X^{\otimes n}$, since both restrictions map between the invariant subspaces; summing over n gives the two identities, as in Lemma 1.6 (ii). If $X^\dagger X \preceq \mathbb{1}$ then $\|X\| \le 1$, so every block $X^{\otimes n}$ has norm at most one, as in Lemma 1.6 (iii). If $X^\dagger X = \mathbb{1}$, the two identities give $\mathfrak{F}(X)^\dagger\mathfrak{F}(X) = \mathfrak{F}(X^\dagger X) = \mathfrak{F}(\mathbb{1}) = \mathbb{1}$.

(ii) For an occupation vector $\boldsymbol{m}$ with $|\boldsymbol{m}| = n$ write $e^{\otimes \boldsymbol{m}}$ for the tensor product of m_1 copies of the first basis vector, then m_2 of the second, and so on, and $\boldsymbol{m}! := \prod_j m_j!$. $P_n e^{\otimes \boldsymbol{m}}$ is $\boldsymbol{m}!/n!$ times the sum of the $n!/\boldsymbol{m}!$ distinct reorderings of $e^{\otimes \boldsymbol{m}}$, which are orthonormal, so $\|P_n e^{\otimes \boldsymbol{m}}\|^2 = \boldsymbol{m}!/n!$ and the normalized symmetrization of Definition 1.2 is $|\boldsymbol{m}\rangle = (n!/\boldsymbol{m}!)^{1/2} P_n e^{\otimes \boldsymbol{m}}$. Let $J_{p,q}$ be the displayed map on the level $p+q=n$. Since $P_n(P_p \otimes P_q) = P_n$, the product $P_p \otimes P_q$ averaging over a subgroup of the permutations, and a reordering of the factors is a permutation,

$$J_{p,q}\big(\,|\boldsymbol{m}_1\rangle \otimes |\boldsymbol{m}_2\rangle\,\big) = \left(\tbinom{n}{p}\frac{p!\,q!}{\boldsymbol{m}_1!\,\boldsymbol{m}_2!}\right)^{1/2} P_n\big(e^{\otimes \boldsymbol{m}_1} \otimes e^{\otimes \boldsymbol{m}_2}\big) = |\boldsymbol{m}_1, \boldsymbol{m}_2\rangle\,.$$

So $\bigoplus_{p+q=n} J_{p,q}$ maps an orthonormal basis of the n-photon level of the tensor product onto one of $\mathrm{Sym}^n(\mathbb{C}^{S \sqcup T})$, and the identification, their sum over n, is unitary.

(iii) $(X_1 \oplus X_2)^{\otimes n}$ commutes with P_n (Lemma 1.6 (i)) and acts on $(\mathbb{C}^S)^{\otimes p} \otimes (\mathbb{C}^T)^{\otimes q}$ as $X_1^{\otimes p} \otimes X_2^{\otimes q}$, so $(X_1 \oplus X_2)^{\otimes n} J_{p,q}(\varphi \otimes \chi) = J_{p,q}\big(X_1^{\otimes p}\varphi \otimes X_2^{\otimes q}\chi\big)$ for $\varphi \in \mathrm{Sym}^p(\mathbb{C}^S)$ and $\chi \in \mathrm{Sym}^q(\mathbb{C}^T)$, which is the claim on every level. □

The lift of a unitary acts on n photons as its n-fold tensor power on the symmetric subspace, the standard dictionary between the mode and particle descriptions. A published statement of it corroborates the lemma; no claim depends on the citation.

Remark 1.8 (Mode and particle descriptions in the literature). Lemma 1.6 (v) with (i) is the standard dictionary between the mode (second-quantized) and particle (symmetric-tensor) descriptions of n photons in d modes: a mode transformation $a_i \mapsto \sum_j U_{ij} a_j$ acts on the n-photon sector as $U^{\otimes n}$ restricted to Sym^n. The same correspondence is recorded in [3, Eqs. (9)–(10)], where Eq. (9) is the mode-level transformation of the creation operators and Eq. (10) states that it is "equivalent to the action of the same U on each particle", $|\psi'\rangle_P = U^{\otimes n}|\psi\rangle_P$ in the particle representation. That reference is corroboration only: the lemma is proved in full, and no claim of this contract depends on the reference.

Uses Lem. 1.6 · newly derived · accepted

On the n-photon block, the lift of $1 - A$ has as eigenvalues the products of n factors $1 - \lambda$ over the eigenvalues λ of A, so it lies between the n-th powers of $1 - \lambda max$ and $1 - \lambda min$, both attained. The analytic high-photon bound of the receiver starts from these two extremes.

For A the sum of the port Grams of a set of ports, $\mathfrak{F}(\mathbb{1} - A)$ is the operator for the event that no port of the set clicks, pulled back to the declared input modes (Theorem 4.1 (iii)); hence the name lifted no-click operator.

Lemma 1.9 (Spectrum of a lifted no-click operator on a photon-number block)**.** Let A be Hermitian on $\mathcal{K}_0$ with $0 \preceq A \preceq \mathbb{1}$ and eigenvalues $\lambda_1, \dots, \lambda_d$ ($d = \dim \mathcal{K}_0$). Then $\mathfrak{F}(\mathbb{1} - A)$ is a positive semidefinite contraction, its n-photon block $\mathfrak{F}(\mathbb{1} - A)^{(n)} = \big((\mathbb{1} - A)^{\otimes n}\big)\big|_{\mathrm{Sym}^n}$ has spectrum

$$\Big\{ \textstyle\prod_{j=1}^{d} (1 - \lambda_j)^{m_j} \;:\; \boldsymbol{m} \in \mathbb{N}_0^d,\ |\boldsymbol{m}| = n \Big\}, \tag{1.4}$$

and consequently

$$\big(1 - \lambda_{\max}(A)\big)^n \mathbb{1} \;\preceq\; \mathfrak{F}(\mathbb{1} - A)^{(n)} \;\preceq\; \big(1 - \lambda_{\min}(A)\big)^n \mathbb{1}, \tag{1.5}$$

both bounds being attained as eigenvalues.

Uses Lem. 1.6 · *Used by* Prop. 1.11, Thm. 5.6, Def. 6.1, Thm. 6.2, Lem. 7.2, Rem. 19.5, Prop. 20.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key. receiver.fock:block_spectrum_from_eigenvalues · *Oracles* registry_receiver_povm.py

Proof. Diagonalize $A = U^\dagger \Lambda U$ with U unitary and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$; then $\mathbb{1} - A = U^\dagger(\mathbb{1} - \Lambda)U$ and Lemma 1.6 (ii),(iii) give $\mathfrak{F}(\mathbb{1} - A) = \mathfrak{F}(U)^\dagger \mathfrak{F}(\mathbb{1} - \Lambda)\mathfrak{F}(U)$ with $\mathfrak{F}(U)$ unitary and photon-number preserving, so the two operators have the same spectrum block by block. In the occupation basis built from the eigenbasis of A, the product vector $|e_{j_1}\rangle \otimes \cdots \otimes |e_{j_n}\rangle$ is an eigenvector of $(\mathbb{1} - \Lambda)^{\otimes n}$ with eigenvalue $\prod_r (1 - \lambda_{j_r})$, and that eigenvalue is unchanged by permuting the factors; hence the symmetrized basis vector $|\boldsymbol{m}\rangle$ is an eigenvector of $(\mathbb{1} - \Lambda)^{\otimes n}|_{\mathrm{Sym}^n}$ with eigenvalue $\prod_j (1 - \lambda_j)^{m_j}$. These $|\boldsymbol{m}\rangle$, $|\boldsymbol{m}| = n$, form an orthonormal basis of Sym^n, so (1.4) is the full spectrum. Each factor satisfies $1 - \lambda_j \in [1 - \lambda_{\max}(A), 1 - \lambda_{\min}(A)] \subseteq [0, 1]$, so a product of n of them lies in $[(1 - \lambda_{\max}(A))^n, (1 - \lambda_{\min}(A))^n]$, which is (1.5); the endpoints are attained by the occupation vectors putting all n photons on an eigenvector for $\lambda_{\max}(A)$, respectively for $\lambda_{\min}(A)$. Positivity and contractivity follow from (1.5) and Lemma 1.6 (iii). □

The Fock lift preserves the Loewner order on positive semidefinite operators. A larger detection operator A therefore gives a smaller lifted no-click operator, the lift of $1 - A$.

Lemma 1.10 (Order monotonicity of the Fock lift)**.** If $X \succeq Y \succeq 0$ on $\mathcal{K}_0$ then $\mathfrak{F}(X) \succeq \mathfrak{F}(Y)$. Consequently, if $0 \preceq A \preceq A' \preceq \mathbb{1}$ then $\mathfrak{F}(\mathbb{1} - A) \succeq \mathfrak{F}(\mathbb{1} - A')$.

Uses Lem. 1.6 · *Used by* Rem. 4.6, Lem. 7.2 · newly derived · accepted · *Oracles* registry_receiver_povm.py

Proof. Telescoping,

$$X^{\otimes n} - Y^{\otimes n} = \sum_{j=0}^{n-1} X^{\otimes j} \otimes (X - Y) \otimes Y^{\otimes (n-1-j)}. \tag{1.6}$$

Each summand is a tensor product of positive semidefinite operators and is therefore positive semidefinite, so $X^{\otimes n} \succeq Y^{\otimes n}$ on $\mathcal{K}_0^{\otimes n}$. Both $X^{\otimes n}$ and $Y^{\otimes n}$ preserve Sym^n by Lemma 1.6 (i), so for $|\psi\rangle \in \mathrm{Sym}^n$ the difference of the restrictions is $\langle\psi| (X^{\otimes n} - Y^{\otimes n}) |\psi\rangle \geq 0$: the restriction of a positive semidefinite operator to an invariant subspace is positive semidefinite. Summing over n gives $\mathfrak{F}(X) \succeq \mathfrak{F}(Y)$. The second statement is the first with $X = \mathbb{1} - A$, $Y = \mathbb{1} - A'$, both positive semidefinite because $A, A' \preceq \mathbb{1}$. □

The lift of $1 - A$ equals the normally ordered exponential of minus the quadratic form of A in the mode operators, a series that is a finite sum on each photon-number block. It is the form of the optics literature; no proof of this contract uses it.

Remark 1.13 says why the derivations of this contract do without this form.

Proposition 1.11 (Normal-ordered form of a lifted no-click operator)**.** For $0 \preceq A \preceq \mathbb{1}$ on $\mathcal{K}_0$,

$$\mathfrak{F}(\mathbb{1}-A) = :\exp\bigl(-\boldsymbol{a}^\dagger A \boldsymbol{a}\bigr): \coloneqq \sum_{p\geq 0} \frac{(-1)^p}{p!} \sum_{i_1 \ldots i_p} \sum_{j_1 \ldots j_p} \Bigl(\prod_{r=1}^{p} A_{i_r j_r}\Bigr) a_{i_1}^\dagger \cdots a_{i_p}^\dagger a_{j_1} \cdots a_{j_p}, \tag{1.7}$$

the series acting on each photon-number block as a finite sum.

Uses Lem. 1.6, Lem. 1.9 · *Used by* Rem. 1.12, Rem. 1.13, Rem. 4.7, Rem. 18.5 · newly derived · accepted

Proof. Both sides are unchanged in form under $A \mapsto UAU^\dagger$ combined with conjugation by $\mathfrak{F}(U)$: for the left side by Lemma 1.6 (ii), for the right side because Lemma 1.6 (v) maps creation operators to combinations of creation operators and annihilation operators to combinations of annihilation operators, so no creator crosses an annihilator and the normal-ordered form survives with A replaced by $UAU^\dagger$. It therefore suffices to prove (1.7) for diagonal $A = \Lambda$. For diagonal Λ only the terms with $i_r = j_r$ survive, operators of distinct modes commute, and collecting the $p!/\boldsymbol{p}!$ tuples with a given multiplicity vector $\boldsymbol{p}$ (where $\boldsymbol{p}! \coloneqq \prod_j p_j!$) turns the right side into $\prod_j \sum_{p_j \geq 0} \frac{(-\lambda_j)^{p_j}}{p_j!} a_j^{\dagger p_j} a_j^{p_j}$. On an occupation vector, the standard identity $a_j^{\dagger p} a_j^p |\boldsymbol{m}\rangle = \frac{m_j!}{(m_j - p)!} |\boldsymbol{m}\rangle$ for $p \leq m_j$ and 0 otherwise gives, by the binomial theorem,

$$\sum_{p\geq 0} \frac{(-\lambda_j)^p}{p!} a_j^{\dagger p} a_j^p |\boldsymbol{m}\rangle = \sum_{p=0}^{m_j} \binom{m_j}{p} (-\lambda_j)^p |\boldsymbol{m}\rangle = (1 - \lambda_j)^{m_j} |\boldsymbol{m}\rangle, \tag{1.8}$$

so the right side of (1.7) acts on $|\boldsymbol{m}\rangle$ as $\prod_j (1-\lambda_j)^{m_j}$ — which is exactly the action of $\mathfrak{F}(\mathbb{1}-\Lambda)$ computed in the proof of Lemma 1.9. Both sides agree on an orthonormal basis, hence are equal. □

The single-mode step of the normal-ordered form is the published identity that the normally ordered exponential of z times the number operator is (1 + z) to the number operator, and its λ = 1 case is the perfect on/off detector. Both citations corroborate; neither is used.

Remark 1.12 (The single-mode normal-ordering identity in the literature). Equation (1.8) is the single-mode statement $:e^{z\hat{n}}: = (1+z)^{\hat{n}}$ at $z = -\lambda_j$, where $\hat{n} = a^\dagger a$. That operator identity is [2, Eq. (3.23)], which reads $G_f(z;\hat{n}) = (1+z)^{\hat{n}} = :e^{z\hat{n}}:$ and is proved there without any assumption on the state of the field. Its $\lambda = 1$ case, $:e^{-\hat{n}}: = |\mathrm{vac}\rangle\langle\mathrm{vac}|$, is the no-click element of a perfect on/off detector, recorded as [4, Eq. (6)] in the form $\hat{\pi}_0 = |0\rangle\langle 0| = :e^{-\hat{n}}:$, $\hat{\pi}_1 = :\hat{1} - e^{-\hat{n}}:$. Both are corroboration: (1.8) is proved in Proposition 1.11, and Theorem 4.5 derives the no-click element from the projector picture of Lemma 1.6 (iv) without using normal ordering at all.

Uses Lem. 1.6, Prop. 1.11 · newly derived · accepted

The normal-ordered form is stated for comparison and used in no proof. Every operator identity from the port Grams to the emitted weight follows from the functoriality of the Fock lift and vacuum elimination, block by block in finite dimension, with no convergence question.

Remark 1.13 (Derivations without normal ordering). Proposition 1.11 is stated because it is the form in which these operators appear in the optics literature and in every numerical implementation, and because it makes the efficiency substitution of Theorem 18.1 transparent. It is not used in the proof of any claim. Every operator identity of the sections from the port Grams to the emitted weight is obtained from Lemma 1.6 (i)–(iv) and from the vacuum elimination of Lemma 2.6, which involve only finite-dimensional linear algebra on each photon-number block: no unbounded operator, no operator-valued series and no convergence question arises anywhere in Theorems 4.1, 4.5, 5.1, 5.6, 6.2 and 8.14. A submodule whose closed forms need series or integrals, as conditioning on a probe state can, says so where it uses them.

Uses Lem. 1.6, Prop. 1.11 · newly derived · accepted

2 The receiver instrument and its physical model

A receiver instrument is a declared mode space, a split of the declared inputs into channel modes, through which the adversary's signal enters, and probe modes, disjoint detector ports and a contractive instrument matrix; its port Grams are the compact data the measurement is derived from. The probe modes carry the receiver's own probe light, in a declared state for each probe setting, which the receiver chooses privately in each round. A contraction becomes a physical device by a unitary dilation with vacuum environment modes, and the detectors are commuting threshold measurements on the ports. Vacuum on every undeclared input makes the restriction to the de-

clared input columns exact (Lemma 2.6); a probe mode carries its declared state and is never eliminated.

A receiver instrument is a declared mode space with its inputs split into at least one channel mode and the probe modes, disjoint detector ports and a contractive instrument matrix. Undeclared inputs carry vacuum, and a mode in no port is undetected.

Besides the channel, a receiver may be fed by its own probe source, such as the light source of a detector-attack countermeasure; its light enters through the inputs $\mathbb{I}_P$, whose subscript P is the shared notation's label for the probe input. The definition ends by identifying the shared notation's channel-input system $\mathcal{H}(C)$ with the Fock space over $\mathbb{I}_C$. Its photon-number blocks $\mathcal{H}(C_n)$, grouped at a cutoff N into the blocks below and above it, are the canonical instance of the sectors that the platform's security analysis declares on the receiver's space (`spec/notation.md` §2.4).

Definition 2.1 (Receiver instrument)**.** A *receiver instrument* is a tuple

$$\mathfrak{R} = \Big(\mathbb{M};\ \mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P;\ \{\mathbb{P}_k\}_{k=1}^{K};\ G\Big) \tag{2.1}$$

in which

(i) $\mathbb{M}$ is a declared mode space (Definition 1.1);

(ii) $\mathbb{I} \subseteq \mathbb{M}$ is the set of *declared input modes*, split into the *channel modes* $\mathbb{I}_C \neq \varnothing$ (the adversary's access) and the *declared source modes* $\mathbb{I}_P$ (the probe); every mode of $\mathbb{M} \setminus \mathbb{I}$ carries vacuum at the input (Assumption 11.4);

(iii) $\mathbb{P}_1, \dots, \mathbb{P}_K$ are pairwise disjoint subsets of $\mathbb{M}$, the *detector ports*, with $K \geq 1$; a mode in no port is undetected;

(iv) $G \in \mathbb{C}^{\mathbb{M}\times\mathbb{M}}$ is the *instrument matrix*, indexed rows-by-outputs and columns-by-inputs, with $G^\dagger G \preceq \mathbb{1}$.

The basis $\{|m\rangle\}_{m\in\mathbb{I}}$ of $\mathbb{C}^{\mathbb{I}}$ is ordered with the channel modes $\mathbb{I}_C$ before the declared source modes $\mathbb{I}_P$, each part in the declared order of $\mathbb{M}$; matrices on $\mathbb{C}^{\mathbb{I}}$ and the columns of a matrix with column set $\mathbb{I}$ are indexed in this order, which differs from the declared order of $\mathbb{M}$ exactly when some declared source mode precedes some channel mode in it. We write $G_{\mathbb{I}} \in \mathbb{C}^{\mathbb{M}\times\mathbb{I}}$ for G restricted to the columns $\mathbb{I}$, and $E_J \coloneqq E_{\bigcup_{k\in J}\mathbb{P}_k}$ for $J \subseteq [K]$, where $[K] \coloneqq \{1, \dots, K\}$. The channel-input system of the shared notation is the Fock space over the channel modes: $\mathcal{H}(C) = \bigoplus_n \mathcal{H}(C_n)$ with $\mathcal{H}(C_n) = \mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}_C})$ and photon-number projectors P_n^C.

Uses Def. 1.1, Def. 1.2 · *Used by* Def. 2.2, Def. 2.3, Def. 2.5, Def. 3.3, Cor. 3.5, Lem. 4.2, Def. 5.5, Cor. 8.2, Cor. 8.15, Thm. 9.5, Def. 9.6, Def. 9.8, Def. 10.4, Lem. 16.4, Prop. 20.3, Def. 21.1, Def. 21.2 · newly derived · accepted

The port Gram matrix of port k is the positive operator on the declared inputs whose quadratic form is the squared norm of the amplitude reaching that port, and the loss deficiency is the identity minus their sum. These Grams are the compact data the click measurement is derived from.

The admissibility defined at the end is an exact condition, while a declared family is admitted by a floating-point test, in the evaluation model of Definition 10.9. Theorem 10.10 states what this contract asserts about the exact family and what about its floating-point evaluation.

Definition 2.2 (Port Grams and the loss deficiency)**.** The *port Gram matrices* of $\mathfrak{R}$ are the operators on $\mathbb{C}^{\mathbb{I}}$

$$A_k \coloneqq G_{\mathbb{I}}^{\dagger} E_{\{k\}} G_{\mathbb{I}}, \qquad k = 1, \ldots, K, \tag{2.2}$$

and for $J \subseteq [K]$ we write $A_J \coloneqq \sum_{k \in J} A_k$ and $A_\Sigma \coloneqq A_{[K]}$. The *loss deficiency* is

$$\mathbb{L} \coloneqq \mathbb{1}_{\mathbb{I}} - A_\Sigma. \tag{2.3}$$

A family $\{A_k\}_{k=1}^K$ of operators on a finite-dimensional space is an *admissible Gram family* when $A_k \succeq 0$ for every k and $\sum_k A_k \preceq \mathbb{1}$. This is an exact condition; on a *declared* family it is *decided* in floating point at the receiver tolerance ε_2, by Definition 10.4 (i).

Uses Def. 2.1 · *Used by* Thm. 4.1, Lem. 4.2, Thm. 5.1, Cor. 8.2, Thm. 9.5, Def. 9.6, Thm. 9.7, Def. 9.8, Def. 10.4, Rem. 10.5, Req. 14.6, Lem. 16.4, Def. 21.2, Prop. 21.4 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.grams:GramFamily

A probe declaration is a finite alphabet of probe settings, vacuum among them, with a declared probe state per setting that is diagonal in the occupation basis. In the Gaussian tier each state is a product of thermal states given by their mean photon numbers.

A probe setting is classical data of known bias that the receiver chooses privately in each round; for a detector-attack countermeasure it fixes whether, and how brightly, the probe light shines. The bias, the weights q_t of the shared notation, belongs to the protocol declaration, and this contract uses only the probe state of each setting. The word tier names a class of probe states here and is unrelated to the tiers a reported key rate carries.

Definition 2.3 (Probe declaration)**.** A *probe declaration* is a pair $\left(\mathcal{T}_{\text{set}}, \{\rho^{(t)}\}_{t \in \mathcal{T}_{\text{set}}}\right)$ in which $\mathcal{T}_{\text{set}}$ is a finite non-empty alphabet of probe settings (including the setting off, whose state is vacuum) and, for each t, $\rho^{(t)}$ is a declared state on $\mathcal{F}(\mathbb{C}^{\mathbb{I}_P})$ that is *occupation-diagonal*: diagonal in the occupation basis of $\mathcal{F}(\mathbb{C}^{\mathbb{I}_P})$ (diagonal in the Fock basis). In the *Gaussian tier* each $\rho^{(t)}$ is a zero-mean Gaussian state with diagonal normally-ordered second-moment matrix $\Gamma^{(t)} = \operatorname{diag}(\bar{n}_m^{(t)})_{m \in \mathbb{I}_P} \succeq 0$, that is, a product of single-mode thermal states with mean photon numbers $\bar{n}_m^{(t)} \geq 0$; $\Gamma^{(t)} = 0$ is vacuum. A state of the Gaussian tier is occupation-diagonal by construction, a thermal state

being diagonal in the occupation basis. When $\mathbb{I}_P = \varnothing$ the declaration is $\mathcal{T}_{\text{set}} = \{\text{off}\}$ and every statement about conditioning is vacuous.

Uses Def. 1.2, Def. 2.1 · *Used by* Rem. 2.7, Def. 9.8, Req. 12.2, Rem. 16.8, Ex. 17.2 · newly derived · accepted

Every contractive instrument matrix is the input-to-output block of a unitary on the modes together with a copy of them as environment modes. The amplitude the matrix removes leaves through those extra outputs, which makes the matrix a passive physical device.

An instrument matrix is a contraction, and a contraction is not by itself the mode transformation of a physical device: the amplitude it removes has to leave through some output, and the input ports it does not use have to be filled with a state. The dilation of this lemma makes it physical, and Lemma 2.6 then shows that the dilation changes nothing: restricting to the declared input columns is *exact*, not an approximation.

Lemma 2.4 (Unitary dilation of a contraction)**.** Let G be a contraction on $\mathbb{C}^{\mathbb{M}}$ and let $\mathbb{E}$ be a disjoint copy of $\mathbb{M}$. With the defect operators $D_{\text{out}} \coloneqq (\mathbb{1} - GG^\dagger)^{1/2}$ and $D_{\text{in}} \coloneqq (\mathbb{1} - G^\dagger G)^{1/2}$, the matrix

$$U_G = \begin{pmatrix} G & D_{\text{out}} \\ D_{\text{in}} & -G^\dagger \end{pmatrix} \quad \text{on} \quad \mathbb{C}^{\mathbb{M}} \oplus \mathbb{C}^{\mathbb{E}} \tag{2.4}$$

is unitary, and its block from the inputs $\mathbb{M}$ to the outputs $\mathbb{M}$ is G.

Used by Def. 2.5, Thm. 4.1 · newly derived · accepted

Proof. Both defect operators exist: $G^\dagger G \preceq \mathbb{1}$ by hypothesis, and $GG^\dagger$ has the same nonzero eigenvalues as $G^\dagger G$, so $GG^\dagger \preceq \mathbb{1}$ as well; the positive semidefinite square root of a positive semidefinite operator is unique. Write a singular value decomposition $G = \sum_j s_j\, y_j x_j^\dagger$ with $s_j \in [0,1]$ and orthonormal $\{x_j\}, \{y_j\}$. Then $G^\dagger G = \sum_j s_j^2 x_j x_j^\dagger$ and $GG^\dagger = \sum_j s_j^2 y_j y_j^\dagger$, so $D_{\text{in}} x_j = \sqrt{1 - s_j^2}\, x_j$ and $D_{\text{out}} y_j = \sqrt{1 - s_j^2}\, y_j$, both acting as the identity on the orthogonal complements. Hence

$$D_{\text{out}} G = \sum_j s_j \sqrt{1 - s_j^2}\, y_j x_j^\dagger = G D_{\text{in}}. \tag{2.5}$$

Multiplying (2.4) by its adjoint blockwise and using $D_{\text{out}}^\dagger = D_{\text{out}}$, $D_{\text{in}}^\dagger = D_{\text{in}}$,

$$U_G U_G^\dagger = \begin{pmatrix} GG^\dagger + D_{\text{out}}^2 & GD_{\text{in}} - D_{\text{out}} G \\ D_{\text{in}} G^\dagger - G^\dagger D_{\text{out}} & D_{\text{in}}^2 + G^\dagger G \end{pmatrix} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix}, \tag{2.6}$$

the diagonal blocks by the definitions of the defect operators and the off-diagonal blocks by (2.5) and its adjoint. A square matrix with $UU^\dagger = \mathbb{1}$ is unitary. □

The receiver's measurement is a unitary dilation of its instrument matrix applied to the declared inputs with vacuum everywhere else, followed by commuting threshold measurements on the detector ports. The input-side click POVM is derived from this model.

Definition 2.5 (The physical measurement model)**.** Fix a receiver instrument $\mathfrak{R}$ and a unitary extension U of G on $\mathbb{C}^{\mathbb{M}\sqcup\mathbb{E}}$ (Lemma 2.4 supplies one; any other may be used, and by Theorem 4.1 (v) the choice does not matter). Let $\iota : \mathbb{C}^{\mathbb{I}} \hookrightarrow \mathbb{C}^{\mathbb{M}\sqcup\mathbb{E}}$ be the inclusion of the declared input modes and $\mathfrak{F}(\iota) : \mathcal{F}(\mathbb{C}^{\mathbb{I}}) \to \mathcal{F}(\mathbb{C}^{\mathbb{M}\sqcup\mathbb{E}})$ the induced map, which under the factorization $\mathcal{F}(\mathbb{C}^{\mathbb{M}\sqcup\mathbb{E}}) \cong \mathcal{F}(\mathbb{C}^{\mathbb{I}}) \otimes \mathcal{F}(\mathbb{C}^{(\mathbb{M}\sqcup\mathbb{E})\setminus\mathbb{I}})$, the occupation identification of Lemma 1.7 (ii), is $|\psi\rangle \mapsto |\psi\rangle \otimes |\mathrm{vac}\rangle$: it prepares vacuum on every undeclared input mode and on every dilation mode (Assumption 11.4). The *measurement isometry* of the receiver is

$$V \coloneqq \mathfrak{F}(U)\,\mathfrak{F}(\iota) \;:\; \mathcal{F}(\mathbb{C}^{\mathbb{I}}) \longrightarrow \mathcal{F}(\mathbb{C}^{\mathbb{M}\sqcup\mathbb{E}}). \tag{2.7}$$

The detectors measure, on the output side, the K commuting binary projective measurements $\{\Pi_k^0, \mathbb{1}-\Pi_k^0\}$ where $\Pi_k^0 \coloneqq \mathfrak{F}(\mathbb{1}-E_{\{k\}})$, with the projector $E_{\{k\}}$ of $\mathbb{C}^{\mathbb{M}}$ extended by zero to $\mathbb{C}^{\mathbb{M}\sqcup\mathbb{E}}$, is the projector onto "no photon in the modes of $\mathbb{P}_k$" (Lemma 1.6 (iv), Assumption 11.7). For $J \subseteq [K]$ put $\Pi_J^0 \coloneqq \prod_{k\in J} \Pi_k^0$.

Uses Def. 1.2, Lem. 1.6, Lem. 1.7, Def. 2.1, Lem. 2.4 · *Used by* Lem. 2.6, Thm. 4.5, Rem. 4.6, Prop. 20.3, Prop. 21.4 · newly derived · accepted

Compressing the lift of any operator by vacuum on the modes outside a coordinate subspace gives exactly the lift of its block on that subspace, and the measurement isometry is an isometry. Dropping the undeclared input columns is therefore exact under the vacuum assumption.

Assumption 11.4 is what licenses restricting G to the columns $\mathbb{I}$; the lemma shows that the restriction is exact.

Lemma 2.6 (Exactness of vacuum elimination)**.** Let $S \subseteq T$ be finite sets of modes and $\iota : \mathbb{C}^S \hookrightarrow \mathbb{C}^T$ the coordinate inclusion. Under the factorization $\mathcal{F}(\mathbb{C}^T) \cong \mathcal{F}(\mathbb{C}^S) \otimes \mathcal{F}(\mathbb{C}^{T\setminus S})$ of occupation bases, the occupation identification of Lemma 1.7 (ii), $\mathfrak{F}(\iota)$ is $|\psi\rangle \mapsto |\psi\rangle \otimes |\mathrm{vac}\rangle$; and for any operator X on $\mathbb{C}^T$,

$$\mathfrak{F}(\iota)^\dagger\,\mathfrak{F}(X)\,\mathfrak{F}(\iota) \;=\; \mathfrak{F}\big(\iota^\dagger X \iota\big), \tag{2.8}$$

where $\iota^\dagger X \iota$ is the $S \times S$ block of X. In particular $\mathfrak{F}(\iota)$ is an isometry. The inclusion ι of Definition 2.5, with $S = \mathbb{I}$ and $T = \mathbb{M}\sqcup\mathbb{E}$, is one instance, and for it V of (2.7) is an isometry as well.

Uses Def. 1.2, Lem. 1.6, Lem. 1.7, Def. 2.5 · *Assumes* A-VAC, A-PORT · *Used by* Rem. 2.7, Thm. 4.1, Thm. 4.5, Ex. 17.1, Ex. 17.2 · newly derived · accepted

Proof. ι maps each basis vector $|m\rangle$, $m \in S$, to the basis vector $|m\rangle$ of $\mathbb{C}^T$, so $\iota^{\otimes n}$ maps the symmetrized and normalized $|e_{m_1}\rangle \otimes \cdots \otimes |e_{m_n}\rangle$ of

Definition 1.2 to the same symmetrized and normalized vector over T: the occupation vector $|\boldsymbol{m}\rangle$ of S goes to the occupation vector of T that agrees with $\boldsymbol{m}$ on S and is 0 on $T \setminus S$, which the factorization identifies with $|\boldsymbol{m}\rangle \otimes |\mathrm{vac}\rangle$, and by linearity $\mathfrak{F}(\iota)\,|\psi\rangle = |\psi\rangle \otimes |\mathrm{vac}\rangle$ for every $|\psi\rangle$. Next, $\mathfrak{F}(\iota)$ maps $\mathrm{Sym}^n(\mathbb{C}^S)$ into $\mathrm{Sym}^n(\mathbb{C}^T)$ by $\iota^{\otimes n}$, and distinct n land in orthogonal summands, so both sides of (2.8) are block diagonal in n and it suffices to compare blocks. For $|\psi\rangle, |\phi\rangle \in \mathrm{Sym}^n(\mathbb{C}^S)$,

$$\langle\psi|\,\mathfrak{F}(\iota)^\dagger \mathfrak{F}(X)\mathfrak{F}(\iota)\,|\phi\rangle = \left\langle \iota^{\otimes n}\psi \right| X^{\otimes n} \left|\iota^{\otimes n}\phi\right\rangle = \langle\psi|\,(\iota^\dagger X \iota)^{\otimes n}\,|\phi\rangle\,, \tag{2.9}$$

using $(\iota^\dagger)^{\otimes n} X^{\otimes n} \iota^{\otimes n} = (\iota^\dagger X \iota)^{\otimes n}$, which holds factor by factor. Taking $X = \mathbb{1}$ gives $\mathfrak{F}(\iota)^\dagger \mathfrak{F}(\iota) = \mathfrak{F}(\mathbb{1}_S) = \mathbb{1}$, so $\mathfrak{F}(\iota)$ is an isometry; for the inclusion of Definition 2.5, $\mathfrak{F}(U)$ is unitary by Lemma 1.6 (iii), so V is an isometry. □

Vacuum elimination removes an input mode only because it carries vacuum; a declared source mode carries the probe state and stays a column of the instrument matrix. Eliminating it would replace the probe by vacuum and describe a different device, a security-relevant mismatch.

Remark 2.7 (Loss ancillas and probe inputs). Lemma 2.6 eliminates an input mode precisely because that mode carries *vacuum*. A declared source mode does not: it carries $\rho^{(t)}$, it stays as a column of G, and its state is supplied (Definition 2.3). The difference between "a loss ancilla" and "a probe input" is exactly the difference between an undeclared vacuum input and a declared-state input, and the two must not be conflated: eliminating a probe mode would silently replace the probe by vacuum, so the derived POVM would describe a different device than the one deployed; since the probe is a countermeasure, that mismatch is security-relevant.

Uses Def. 2.3, Lem. 2.6 · newly derived · accepted

3 Component graphs and compilation

A receiver may be declared as a directed graph of catalogue components. The graph determines an instrument matrix when it is well formed (parameters in their domains, acyclic, and any two elements that share a mode ordered): every topological order then compiles to the same contraction (Theorem 3.4). A well-formed graph, with its declared input modes, ports and efficiencies, then gives a receiver instrument (Corollary 3.5). Without the ordering condition two sorts of the same graph can give different devices, which is why it is a well-formedness condition rather than a modelling remark.

A component catalogue is a finite set of entry types, each fixing a support shape, a parameter domain and a rule that yields a local element. The catalogue submodule declares the entries and proves that each maps its whole domain into local elements.

Route (iii) declares a receiver as a graph of standard components rather than as a matrix. This section shows when such a graph determines the instrument matrix: it does, under a condition that is stated rather than assumed.

Definition 3.1 (Component catalogue)**.** A *component catalogue* $\mathfrak{C}$ is a finite set of entry types. An *entry type* fixes a support shape (which modes of $\mathbb{M}$ it acts on), a parameter domain, and a rule assigning to every parameter tuple in that domain a local element in the sense of Definition 1.3. The catalogue of this contract is the one of Definition 16.2 (Table 4) in the component-catalogue submodule, and Theorem 16.3 there states that every entry type maps its whole declared domain into local elements.

Uses Def. 1.3 · *Used by* Def. 3.2 · newly derived · accepted

A declared element graph is well formed when every element's parameters lie in their domain, the graph is acyclic, and any two elements sharing a mode are ordered by it. Support comparability keeps an arbitrary sort from supplying a physical ordering the declaration left open.

Definition 3.2 (Declared element graph)**.** A *declared element graph* is a finite directed graph $\mathfrak{E} = (V, E)$ whose vertices $e \in V$ are *elements*, each carrying a catalogue entry type, a parameter tuple, and a support $\Sigma_e \subseteq \mathbb{M}$. Write $\preceq_{\mathfrak{E}}$ for the reachability relation: $e \preceq_{\mathfrak{E}} f$ when $e = f$ or there is a directed path from e to f. The graph is *well formed* when

(E1) every element's parameter tuple lies in the declared domain of its entry type, so that, by Definition 3.1, G_{Σ_e} is a local element: defined, with $G^{\dagger}_{\Sigma_e} G_{\Sigma_e} \preceq \mathbb{1}$;

(E2) $\mathfrak{E}$ is acyclic, so that $\preceq_{\mathfrak{E}}$ is a partial order;

(E3) (*support comparability*) any two elements whose supports intersect are comparable: $\Sigma_e \cap \Sigma_f \neq \varnothing \implies e \preceq_{\mathfrak{E}} f$ or $f \preceq_{\mathfrak{E}} e$.

A *linear extension* of $\mathfrak{E}$ is an enumeration $e_1, \dots, e_L$ of V with $e_i \preceq_{\mathfrak{E}} e_j \implies i \leq j$.

Uses Def. 3.1 · *Used by* Def. 3.3, Thm. 3.4, Cor. 3.5, Rem. 3.8, Def. 10.4, Req. 12.1, Thm. 16.3, Lem. 16.4, Rem. 16.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.graph:ElementGraph

The compiled instrument matrix is the product of the embedded elements along a linear extension, later elements multiplying from the left, followed by the terminal detector-efficiency layer. This fixes the matrix convention of the whole receiver contract.

The product below follows the matrix convention the interface records (`spec/modules/receiver/interface.yaml`): a factor acts on column vectors indexed by the modes, so the element that light meets later stands further to the left.

Definition 3.3 (Compilation of a declared element graph)**.** Let $\mathfrak{E}$ be a well-formed graph; it is in particular acyclic (condition (E2) of Definition 3.2), so that a linear extension exists. For a linear extension $\sigma = (e_1, \dots, e_L)$ of $\mathfrak{E}$ put

$$G^{\sigma}_{\mathfrak{E}} \coloneqq \hat{G}_{e_L}\, \hat{G}_{e_{L-1}} \cdots \hat{G}_{e_1} \;\in\; \mathbb{C}^{\mathbb{M}\times\mathbb{M}}, \tag{3.1}$$

later elements multiplying from the left. The *compiled instrument matrix* of the receiver is $G \coloneqq D_\eta\, G^{\sigma}_{\mathfrak{E}}$, where $D_\eta = \sum_k \sqrt{\eta_k}\, E_{\{k\}} + E_{\mathbb{M}\setminus\bigcup_k \mathbb{P}_k}$ is the *terminal detector-efficiency layer*, applied after every element, and $\eta_k \in [0,1]$ is the declared efficiency of port k (the effect of that layer on the port Grams is Theorem 18.1 of the detector-imperfections submodule). The detector ports $\mathbb{P}_k$ and the port-embedding abbreviation $E_{\{k\}}$ written here are those of a receiver instrument tuple: clause (iii) of Definition 2.1 and its closing notation.

Uses Def. 2.1, Def. 3.2 · *Used by* Thm. 3.4, Cor. 3.5, Thm. 9.5, Req. 14.6, Thm. 16.3, Lem. 16.4, Rem. 16.8, Ex. 17.1, Ex. 17.2, Ex. 17.3, Thm. 18.1, Def. 21.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.graph:compile_element_graph

Every linear extension of a well-formed graph compiles to the same contraction before the efficiency layer, which is unitary exactly when every element is lossless, and well-formedness is decidable by finite inspection. A well-formed route-(iii) graph therefore determines its instrument matrix.

Two elements that share a mode but are not ordered by the graph can make the compiled product depend on the sort (Remark 3.6). Support comparability rules this out, and item (i) shows that it is enough for every sort to give the same matrix.

Theorem 3.4 (Well-definedness of compilation)**.** In items (i)–(iii), let $\mathfrak{E}$ be a well-formed declared element graph (Definition 3.2); in particular $\mathfrak{E}$ is acyclic (condition (E2) of Definition 3.2), so that $\preceq_{\mathfrak{E}}$ is a partial order and linear extensions of $\mathfrak{E}$ and of each of its induced subgraphs exist. Then:

(i) *Independence of the sort.* $G^{\sigma}_{\mathfrak{E}}$ is the same matrix for every linear extension σ of $\mathfrak{E}$; write $G_{\mathfrak{E}}$ for it. In particular, if V is partitioned into *layers* $L_1, \dots, L_r$ such that elements of one layer have pairwise disjoint supports and $e \preceq_{\mathfrak{E}} f$ implies $\mathrm{layer}(e) \le \mathrm{layer}(f)$, then $G_{\mathfrak{E}} = \big(\prod_{e\in L_r} \hat{G}_e\big) \cdots \big(\prod_{e\in L_1} \hat{G}_e\big)$ with each intra-layer product order-independent.

(ii) *Admissibility by construction.* $G_{\mathfrak{E}}^{\dagger} G_{\mathfrak{E}} \preceq \mathbb{1}$, and hence also $G^{\dagger} G \preceq \mathbb{1}$ for the compiled instrument matrix of Definition 3.3.

(iii) *Losslessness.* $G_{\mathfrak{E}}$ is unitary if and only if every element of $\mathfrak{E}$ is lossless on its support. If in addition every $\eta_k = 1$ then G is unitary.

(iv) *Decidability.* For an arbitrary declared element graph, not necessarily well formed, conditions (E1)–(E3) are decidable by finite inspection: (E1) by evaluating the declared domain predicate of each element, (E2) by a topological sort, (E3) by testing $\Sigma_e \cap \Sigma_f \neq \varnothing$ against the transitive closure of E for each of the $\binom{|V|}{2}$ pairs.

Uses Prop. 1.4, Def. 3.2, Def. 3.3 · *Used by* Cor. 3.5, Req. 12.1, Req. 14.6, Thm. 16.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key. receiver_graph:compile_element_graph · *Oracles* registry_receiver_povm.py

Proof. (i) Induction on $L = |V|$. For $L \leq 1$ there is nothing to prove. Let σ and τ be linear extensions and let $e := \sigma(1)$. Since σ is a linear extension and e is first, no element other than e satisfies $f \preceq_{\mathfrak{E}} e$; that is, e is minimal for $\preceq_{\mathfrak{E}}$. Let j be the position of e in τ. Every element $f = \tau(i)$ with $i < j$ satisfies $e \not\preceq_{\mathfrak{E}} f$ (else f would have to follow e in τ) and $f \not\preceq_{\mathfrak{E}} e$ (minimality of e), so f and e are incomparable. By (E3), $\Sigma_f \cap \Sigma_e = \varnothing$, and by Proposition 1.4 (iii) the embeddings $\hat{G}_f$ and $\hat{G}_e$ commute. In the product $G_{\mathfrak{E}}^{\tau} = \hat{G}_{\tau(L)} \cdots \hat{G}_{\tau(1)}$ the factor $\hat{G}_e$ sits at position j from the right and may therefore be moved past the $j-1$ factors to its right one at a time, giving $G_{\mathfrak{E}}^{\tau} = \big(\hat{G}_{\tau(L)} \cdots \widehat{\hat{G}_e} \cdots \hat{G}_{\tau(1)}\big) \hat{G}_e$, where the hat denotes omission and the remaining order is τ restricted to $V \setminus \{e\}$. The induced subgraph on $V \setminus \{e\}$ is itself well formed: (E1) and (E2) pass to any induced subgraph, and since e is minimal no directed path between two other elements passes through e, so reachability among them, and with it (E3), is inherited. That restriction of τ is therefore a linear extension of the induced subgraph; so is σ restricted to $V \setminus \{e\}$, and both compile to the same matrix by the induction hypothesis. Multiplying on the right by $\hat{G}_e$ gives $G_{\mathfrak{E}}^{\tau} = G_{\mathfrak{E}}^{\sigma}$.

For the layered form, note that a layering as described yields a linear extension by listing L_1 in an order compatible with $\preceq_{\mathfrak{E}}$ (one exists, the restriction of $\preceq_{\mathfrak{E}}$ to a layer being a partial order), then L_2 likewise, and so on: an element preceding another in $\preceq_{\mathfrak{E}}$ lies in an earlier layer or earlier within the same one. The intra-layer products are order-independent by Proposition 1.4 (iii), since the elements of one layer have pairwise disjoint supports, so the layered product equals the product along that linear extension, which is $G_{\mathfrak{E}}$ by the first part.

(ii) Each $\hat{G}_e$ is a contraction by (E1) and Proposition 1.4 (i), so the product is a contraction by repeated application of Proposition 1.4 (ii). D_η is diagonal with entries in $[0, 1]$, hence a contraction, and one more application of (ii) gives $G^{\dagger} G \preceq \mathbb{1}$.

(iii) If every element is lossless then every $\hat{G}_e$ is unitary (Proposition 1.4 (i) and (iv), a square isometry being unitary), and so is their product. Conversely suppose $G_{\mathfrak{E}}$ is unitary, and write $X_i \coloneqq \hat{G}_{e_i} \cdots \hat{G}_{e_1}$. For every vector v, $\|v\| = \|X_L v\| \le \|X_{L-1} v\| \le \cdots \le \|X_1 v\| \le \|v\|$, so every inequality is an equality for every v. From $\|X_1 v\| = \|v\|$ for all v, $\hat{G}_{e_1}$ is a square isometry, hence unitary, hence $\operatorname{ran} X_1 = \mathbb{C}^{\mathbb{M}}$; then $\|X_2 v\| = \|X_1 v\|$ for all v says $\hat{G}_{e_2}$ is isometric on all of $\mathbb{C}^{\mathbb{M}}$, hence unitary, and so on. Each $\hat{G}_{e_i}$ unitary forces $G_{\Sigma_{e_i}}$ unitary, i.e. lossless. The statement about D_η is the same argument applied to the one extra factor, D_η being unitary exactly when every $\eta_k = 1$.

(iv) Each test is a finite computation on finite data, as described. □

A well-formed graph with a declared mode space, input split, port map and efficiencies yields a receiver instrument whose matrix does not depend on the sort. This joins the graph route to the object the measurement theorems are stated of.

The three input routes converge on a receiver instrument. This corollary is the step for route (iii). Once it holds, a result this contract proves for every receiver instrument applies to a route-(iii) declaration by instantiation; a result proved of a receiver instrument carrying further hypotheses applies only where the declaration meets those too.

Corollary 3.5 (Receiver instrument from a well-formed graph declaration)**.** Let a route-(iii) declaration supply a declared mode space $\mathbb{M}$; a well-formed declared element graph $\mathfrak{E}$ on $\mathbb{M}$ (Definition 3.2), in particular acyclic (condition (E2) of Definition 3.2); a set $\mathbb{I} \subseteq \mathbb{M}$ of declared input modes with a split $\mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P$, $\mathbb{I}_C \neq \varnothing$, every mode of $\mathbb{M} \setminus \mathbb{I}$ carrying vacuum at the input (Assumption 11.4); and a port map $\mathbb{P}_1, \ldots, \mathbb{P}_K$, pairwise disjoint subsets of $\mathbb{M}$ with $K \ge 1$, carrying detector efficiencies η_k in the domain $[0, 1]$ declared for them in Definition 3.3. Let σ be a linear extension of $\mathfrak{E}$ and let $G \coloneqq D_\eta\, G^{\sigma}_{\mathfrak{E}}$ be the compiled instrument matrix of Definition 3.3. Then

(i) G does not depend on σ, so it is a function of the declaration alone; and

(ii) the tuple $\mathfrak{R} = \big(\mathbb{M};\ \mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P;\ \{\mathbb{P}_k\}_{k=1}^{K};\ G\big)$ is a receiver instrument in the sense of Definition 2.1: it has the shape (2.1) and satisfies its four clauses.

Uses Def. 2.1, Def. 3.2, Def. 3.3, Thm. 3.4 · *Assumes* A-VAC · *Used by* Ex. 17.1 · newly derived · accepted

Proof. Clauses (i) to (iii) of Definition 2.1. Clause (i) is the declared mode space and clause (iii) is the declared port map; that the ports are pairwise disjoint and that $K \ge 1$ are conditions on the declaration, decided by finite inspection of it, and nothing here derives them. Clause (ii) is the declared split $\mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P$ with $\mathbb{I}_C \neq \varnothing$, again a condition on the declaration, together with the condition that every mode of $\mathbb{M} \setminus \mathbb{I}$ carries vacuum at the input,

which is Assumption 11.4: it is a physical condition on the declaration that the tuple's data does not encode and nothing here derives, so this corollary declares it in its own right, as Definition 2.1 (ii) does. What remains is clause (iv) and item (i), and both are read off Theorem 3.4.

Item (i). Because $\mathfrak{E}$ is well formed it is acyclic (condition (E2) of Definition 3.2), so $\preceq_{\mathfrak{E}}$ is a partial order and a linear extension σ of $\mathfrak{E}$ exists at all. Hence the product (3.1) is defined and $G^{\sigma}_{\mathfrak{E}} \in \mathbb{C}^{\mathbb{M}\times\mathbb{M}}$. By Theorem 3.4 (i) it is the same matrix for every linear extension of $\mathfrak{E}$; write $G_{\mathfrak{E}}$ for that common value. The terminal layer D_η is built from the port map and the efficiencies alone and contains no element of $\mathfrak{E}$ and no ordering, so $G = D_\eta\, G^{\sigma}_{\mathfrak{E}} = D_\eta\, G_{\mathfrak{E}}$ is independent of σ as well and depends on $\mathfrak{E}$ and on the declared pairs $\{(\mathbb{P}_k, \eta_k)\}_{k=1}^{K}$ alone.

Clause (iv). The terminal layer $D_\eta = \sum_k \sqrt{\eta_k}\, E_{\{k\}} + E_{\mathbb{M}\setminus\bigcup_k \mathbb{P}_k}$ of Definition 3.3 is diagonal, with entry $\sqrt{\eta_k}$ on the modes of $\mathbb{P}_k$ (well defined as a diagonal because the ports are pairwise disjoint, so that no mode is given two entries) and with entries in $[0,1]$ because $\eta_k \in [0,1]$; it is therefore a contraction. The indexing of G is that of (3.1): each factor $\hat{G}_e$ acts on column vectors indexed by $\mathbb{M}$, so the product, and with it $G \in \mathbb{C}^{\mathbb{M}\times\mathbb{M}}$, is indexed rows by outputs and columns by inputs, which is what clause (iv) of Definition 2.1 requires of an instrument matrix. Finally Theorem 3.4 (ii) states $G^{\dagger}G \preceq \mathbb{1}$ of exactly this compiled instrument matrix. So clause (iv) holds and $\mathfrak{R}$ is a receiver instrument. □

An acyclic graph of a rotation and a polarizer on the same two modes, with no edge between them, compiles at generic angles to two different contractions under its two sorts. Acyclicity alone therefore does not make a declared graph determine an instrument.

Remark 3.6 (Order dependence without support comparability). Take $\mathbb{M} = \{m\}$ and two elements e, f on $\Sigma_e = \Sigma_f = \{m\}$ with no edge between them, $G_{\Sigma_e} = (\frac{1}{2})$ and $G_{\Sigma_f} = (\frac{1}{2})$: here the two orders agree, and the example looks harmless. Now take $\mathbb{M} = \{m_1, m_2\}$ with $\Sigma_e = \Sigma_f = \mathbb{M}$, $G_{\Sigma_e} = R(\theta)$ a rotation and $G_{\Sigma_f} = \mathrm{diag}(1,0)$ a polarizer. The graph $\mathfrak{E} = (\{e, f\}, \varnothing)$ is acyclic and every element is admissible, yet $G_{\Sigma_f} G_{\Sigma_e} \neq G_{\Sigma_e} G_{\Sigma_f}$ for $\theta \notin \frac{\pi}{2}\mathbb{Z}$: rotating and then projecting is not projecting and then rotating. Two topological sorts of the *same* declared receiver would compile to two different instruments, and, because both are contractions with the same declared port map, both would pass every other validation in this contract. Without (E3), "the declared receiver" does not denote.

newly derived · accepted

An edge fixes the order in which two elements act; moving light between modes is done by the elements' matrices. Components drawn on different arms carry no edge and have disjoint supports, so support comparability constrains only an ambiguous drawing.

Remark 3.7 (Meaning of an edge). An edge $e \to f$ fixes the order in which e and f act: e first. Moving light from one mode to another is done by the elements' matrices, for instance by a mode-permutation element (entry C9 of Table 4), not

by the edge; between elements with disjoint supports the order the edge fixes does not by itself change the compiled matrix, since they commute (Proposition 1.4 (iii)); such an edge matters only through the order it induces, by reachability, between elements whose supports intersect. A user who draws two components in series draws the edge; a user who draws them on different arms of a beamsplitter draws none, and (E3) then requires their mode supports to be disjoint, which, on different arms, they are. The condition therefore constrains exactly the drawings that were ambiguous.

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A cavity, a ring resonator or any other feedback path is a cycle, which (E2) excludes. Compiling one need a fixed-point solve whose contractivity must itself be certified, so such a receiver needs a new claim rather than a larger computation.

Remark 3.8 (Feedback paths outside the acyclic scope). A cycle in $\mathfrak{E}$, such as a cavity, a ring resonator or any feedback path, is excluded by condition (E2) of Definition 3.2. Compiling one would require solving a fixed-point equation on the mode graph, and the resulting matrix is a contraction only under a spectral condition that would itself have to be certified. This contract does neither: a declared cycle leaves no finite product to compile, and it is refused, not solved (Requirement 12.1). A receiver that needs a feedback path is a new claim, not a bigger computation.

Uses Def. 3.2 · newly derived · accepted

The claims that read only the Gram substitution of the efficiency theorem involve the terminal efficiency layer alone, a function of the port map and efficiencies, and no element graph. The theorem's items on the compiled product and its record hold only for a well-formed, hence acyclic, graph.

Remark 3.9 (Where the element graph enters the efficiency theorem). The two remarks that use Theorem 18.1 (Remarks 18.4 and 18.5) and Requirement 22.3, which rests on the first of them, read only the substitution of item (i) of that theorem: they involve the terminal detector-efficiency layer D_η of Definition 3.3, which is a function of the port map and the per-port efficiencies alone, and no graph enters their statements or proofs. The same holds of items (i), (ii) and (v) of the theorem. Items (iii) and (iv) are statements about the compiled product of Definition 3.3 and its record, which exist only for a well-formed, hence acyclic, graph (condition (E2) of Definition 3.2); so are Requirements 22.1 and 22.2, which state their engine side.

newly derived · accepted

4 Port Grams and the click-pattern POVM

The joint no-click element of any set of ports pulls back to the lift of the identity minus the sum of the Grams of the ports in that set, whatever dilation is chosen (Theorem 4.1), and the loss deficiency splits exactly into loss through the dilation and light reaching unmonitored declared modes. An outcome is the set of ports that click. The main result, Theorem 4.5,

is the click-pattern POVM: it follows by inclusion and exclusion, is positive by construction rather than by its alternating form, and is block diagonal in the total input photon number.

The port Grams form an admissible family, and every joint no-click element pulls back to the lift of the identity minus the sum of the Grams of the ports in its set. The measurement depends only on the declared input columns and on no choice of dilation.

The port Grams are the compact data the whole measurement is derived from. This theorem shows that they are admissible, that the joint no-click elements are functions of them alone, and that the unitary dilation used to define the physical model leaves no trace in the result.

Theorem 4.1 (Port Grams and dilation independence)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}$ of Definition 2.2.

(i) *Admissibility.* $A_k \succeq 0$ for every k; $\sum_k A_k = G_{\mathbb{I}}^\dagger E_{[K]} G_{\mathbb{I}} \preceq G_{\mathbb{I}}^\dagger G_{\mathbb{I}} \preceq \mathbb{1}_{\mathbb{I}}$, so $\{A_k\}$ is admissible and the loss deficiency $\mathbb{L} = \mathbb{1}_{\mathbb{I}} - A_\Sigma$ is positive semidefinite.

(ii) *Rank.* $\operatorname{rank} A_k \le \min\big(|\mathbb{I}|, |\mathbb{P}_k|\big)$.

(iii) *Pullback.* For every $J \subseteq [K]$,

$$V^\dagger \Pi_J^0 V = \mathfrak{F}\big(\mathbb{1}_{\mathbb{I}} - A_J\big). \tag{4.1}$$

(iv) *Vacuum elimination is exact.* The right-hand side of (4.1) depends on the instrument only through $G_{\mathbb{I}}$, the columns of G indexed by the declared input modes. Dropping the other columns is therefore an identity, not an approximation.

(v) *Dilation independence.* (4.1) holds for *every* unitary U on $\mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}'}$, with any finite mode set $\mathbb{E}'$, whose block from the inputs $\mathbb{I}$ to the outputs $\bigcup_k \mathbb{P}_k$ equals the corresponding block of G. In particular the sub-unitary description (a contraction G on the declared modes) and the unitary-dilation description (a unitary on system plus environment, with vacuum environment and the environment discarded) induce the *same* measurement on $\mathcal{F}(\mathbb{C}^{\mathbb{I}})$.

Uses Lem. 1.6, Def. 2.2, Lem. 2.4, Lem. 2.6 · *Used by* Thm. 4.5, Thm. 5.1, Cor. 8.2, Rem. 10.5, Req. 14.6, Ex. 17.1, Prop. 20.3, Prop. 21.4 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.grams: check_gram_family_admissible · *Oracles* registry_receiver_povm.py

Proof. (i) $A_k = (E_{\{k\}} G_{\mathbb{I}})^\dagger (E_{\{k\}} G_{\mathbb{I}}) \succeq 0$ because $E_{\{k\}}$ is an orthogonal projector, hence idempotent and self-adjoint. The ports are pairwise disjoint, so $\sum_k E_{\{k\}} = E_{[K]} \preceq \mathbb{1}_{\mathbb{M}}$, and conjugating that inequality by $G_{\mathbb{I}}$ gives

$\sum_k A_k \preceq G_{\mathbb{I}}^\dagger G_{\mathbb{I}}$, which is $\preceq \mathbb{1}_{\mathbb{I}}$ because $G^\dagger G \preceq \mathbb{1}_{\mathbb{M}}$ restricts to the principal submatrix on $\mathbb{I}$.

(ii) A_k is the Gram matrix of the rows of $G_{\mathbb{I}}$ indexed by $\mathbb{P}_k$, so $\operatorname{rank} A_k = \operatorname{rank}(E_{\{k\}} G_{\mathbb{I}})$, and that matrix has at most $|\mathbb{P}_k|$ nonzero rows and $|\mathbb{I}|$ columns.

(iii) By Lemma 1.6 (iv), Π_k^0 is the orthogonal projector onto the Fock space over the modes outside $\mathbb{P}_k$, and since the $\mathbb{P}_k$ are pairwise disjoint the Π_k^0 are commuting projectors with $\Pi_J^0 = \mathfrak{F}(\mathbb{1} - E_J)$: indeed $\prod_{k\in J} \mathfrak{F}(\mathbb{1} - E_{\{k\}}) = \mathfrak{F}\big(\prod_{k\in J}(\mathbb{1} - E_{\{k\}})\big) = \mathfrak{F}(\mathbb{1} - E_J)$ by Lemma 1.6 (ii) and disjointness. Then, using Lemma 1.6 (ii) and Lemma 2.6,

$$V^\dagger \Pi_J^0 V = \mathfrak{F}(\iota)^\dagger \mathfrak{F}\big(U^\dagger(\mathbb{1} - E_J)U\big)\mathfrak{F}(\iota) = \mathfrak{F}\Big(\iota^\dagger U^\dagger (\mathbb{1} - E_J) U \iota\Big). \tag{4.2}$$

Now $\iota^\dagger U^\dagger(\mathbb{1} - E_J)U\iota = \mathbb{1}_{\mathbb{I}} - (E_J U\iota)^\dagger (E_J U\iota)$, and $E_J U\iota$ is the submatrix of U with rows in $\bigcup_{k\in J} \mathbb{P}_k \subseteq \mathbb{M}$ and columns in $\mathbb{I} \subseteq \mathbb{M}$, which is $E_J G_{\mathbb{I}}$ because the $\mathbb{M}\times\mathbb{M}$ block of U is G. Hence $\iota^\dagger U^\dagger(\mathbb{1} - E_J)U\iota = \mathbb{1}_{\mathbb{I}} - G_{\mathbb{I}}^\dagger E_J G_{\mathbb{I}} = \mathbb{1}_{\mathbb{I}} - A_J$.

(iv) Immediate from the closed form just obtained: A_J is built from $G_{\mathbb{I}}$ alone.

(v) The computation in (iii) used only two facts about U: that it is unitary, and that $E_J U\iota = E_J G_{\mathbb{I}}$ for every J, i.e. that its block from $\mathbb{I}$ to the port modes agrees with that of G. Any dilation with that property therefore yields the same $V^\dagger \Pi_J^0 V$ for every J. It yields the same measurement too: the Π_k^0 commute, so expanding $\prod_{k\in S}(\mathbb{1} - \Pi_k^0) = \sum_{T\subseteq S} (-1)^{|T|} \prod_{k\in T} \Pi_k^0$ gives $V^\dagger\big[\prod_{k\in S}(\mathbb{1} - \Pi_k^0)\prod_{k\notin S}\Pi_k^0\big]V = \sum_{T\subseteq S}(-1)^{|T|} V^\dagger \Pi^0_{([K]\setminus S)\cup T} V$, a combination of those operators, for every click pattern S (the elements of the POVM of Theorem 4.5). Conversely a contraction G always admits such a dilation (Lemma 2.4), so the two descriptions are interchangeable. □

The undetected fraction splits exactly into light lost through the dilation and light reaching declared modes that no port monitors, both positive semidefinite. It is the check that loss is counted exactly once.

The identity below holds for every receiver instrument and is the correct check that loss was accounted for exactly once.

Lemma 4.2 (Decomposition of the loss deficiency)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}$ and let E_{un} be the projector onto the declared modes lying in no port. Then

$$\mathbb{L} \;=\; \mathbb{1}_{\mathbb{I}} - A_\Sigma \;=\; \underbrace{\big(\mathbb{1}_{\mathbb{I}} - G_{\mathbb{I}}^\dagger G_{\mathbb{I}}\big)}_{\text{left through the dilation}} \;+\; \underbrace{G_{\mathbb{I}}^\dagger\, E_{\mathrm{un}}\, G_{\mathbb{I}}}_{\text{reached an unmonitored declared mode}}, \tag{4.3}$$

and both summands are positive semidefinite.

Uses Def. 2.1, Def. 2.2 · *Assumes* A-TRUST, A-PASSIVE, A-VAC, A-MODES, A-EXACT, A-NOSEL, A-ROUND · *Used by* Def. 10.1, Thm. 18.1, Rem. 18.3 · newly derived · accepted · *Oracles* registry_receiver_povm.py

Proof. The ports and the unmonitored declared modes partition $\mathbb{M}$, so $E_{[K]} + E_{\mathrm{un}} = \mathbb{1}_{\mathbb{M}}$ and hence $G_{\mathbb{I}}^{\dagger} G_{\mathbb{I}} = A_{\Sigma} + G_{\mathbb{I}}^{\dagger} E_{\mathrm{un}} G_{\mathbb{I}}$. Rearranging gives (4.3). The first summand is positive semidefinite because G is a contraction and the restriction of $\mathbb{1} - G^{\dagger} G \succeq 0$ to the principal submatrix on $\mathbb{I}$ is $\mathbb{1}_{\mathbb{I}} - G_{\mathbb{I}}^{\dagger} G_{\mathbb{I}}$; the second is a conjugated projector. □

Deleting the row of an output mode that no port monitors changes no port Gram and hence no POVM element; the deleted amplitude is carried by the dilation. Which modes lie in a port is a declaration with no observable content beyond the Grams it produces.

Remark 4.3 (Unmonitored output modes). A mode that lies in no port is undetected: the amplitude reaching it is loss. Nothing is lost by that convention. If a model retains an output mode that no port monitors, deleting the corresponding row of G changes no A_J (the row does not appear in any $E_J G_{\mathbb{I}}$), hence changes no POVM element; the deleted amplitude is carried by the dilation instead. The partition into "in some port" and "in no port" is therefore a *declaration* with no observable content beyond the A_k it produces.

Used by Req. 12.3 · newly derived · accepted

A raw outcome is the set of detector ports that click, and the raw alphabet splits into the no-click outcome, the single-click set and the multiclick set. Any merging into a protocol's alphabet comes later.

At $K = 4$ the sets defined here are the raw, single-click and multiclick sets of the optical-profile notation (`spec/notation.md#37-optical-profile-symbols`).

Definition 4.4 (Click-pattern alphabet)**.** An *outcome* is the set $S \subseteq [K]$ of ports that click; the *raw alphabet* is $2^{[K]}$, in bijection with $\{0,1\}^K$ by indicator vectors. We write

$$\mathsf{SC} \coloneqq \big\{\{k\} : k \in [K]\big\}, \qquad \mathsf{MC} \coloneqq \big\{S \subseteq [K] : |S| \geq 2\big\} \tag{4.4}$$

for the *single-click set* and the *multiclick set*, so that $2^{[K]} = \{\varnothing\} \sqcup \mathsf{SC} \sqcup \mathsf{MC}$, and $\varnothing$ is the *no-click outcome*.

Used by Def. 8.1 · newly derived · accepted

The click patterns of a receiver instrument form a POVM whose elements follow from the port Grams by inclusion and exclusion. Every element is block diagonal in the total input photon number, and vacuum gives only the no-click outcome.

The element of an outcome S is the pull-back of "every port in S clicks and no other does". Positivity and completeness come from that projective form, and item (ii) then gives the closed form in the port Grams.

Theorem 4.5 (Click-pattern POVM). Let $\mathfrak{R}$ be a receiver instrument with measurement isometry V (Definition 2.5) and port Grams $\{A_k\}$. Define

$$M_S \coloneqq V^{\dagger}\Big[\prod_{k \in S} (\mathbb{1} - \Pi_k^0) \prod_{k \notin S} \Pi_k^0\Big] V, \qquad S \subseteq [K]. \tag{4.5}$$

Then:

(i) $\{M_S\}$ *is a POVM.* $M_S \succeq 0$ for every S and $\sum_{S\subseteq[K]} M_S = \mathbb{1}$. Moreover $\Pi_{\mathrm{vac}}(J) \coloneqq V^\dagger \Pi^0_J V = \sum_{S\subseteq[K]\setminus J} M_S$ for every $J \subseteq [K]$.

(ii) *Closed form.* $\Pi_{\mathrm{vac}}(J) = \mathfrak{F}(\mathbb{1}_{\mathbb{I}} - A_J)$ and

$$M_S = \sum_{T\subseteq S} (-1)^{|T|}\, \mathfrak{F}\big(\mathbb{1}_{\mathbb{I}} - A_{([K]\setminus S)\cup T}\big). \tag{4.6}$$

(iii) *Photon-number block structure.* Every M_S commutes with the total photon number of the declared inputs, so $M_S = \bigoplus_{n\geq 0} M_S^{(=n)}$ with $M_S^{(=n)}$ acting on $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$, and

$$M_S^{(=n)} = \sum_{T\subseteq S} (-1)^{|T|} \Big(\big(\mathbb{1} - A_{([K]\setminus S)\cup T}\big)^{\otimes n} \Big)\Big|_{\mathrm{Sym}^n}. \tag{4.7}$$

In particular $M_S^{((=0))} = \delta_{S,\varnothing}$: no photon, no click.

(iv) *The coarse elements.* With $A_\Sigma = A_{[K]}$,

$$M_{\mathrm{nc}} \coloneqq M_\varnothing = \mathfrak{F}(\mathbb{1} - A_\Sigma), \tag{4.8}$$

$$M_{\mathrm{sc}_k} \coloneqq M_{\{k\}} = \mathfrak{F}\big(\mathbb{1} - A_{[K]\setminus\{k\}}\big) - \mathfrak{F}(\mathbb{1} - A_\Sigma), \tag{4.9}$$

$$M_{\mathrm{mc}} \coloneqq \sum_{S\in\mathsf{MC}} M_S = \mathbb{1} + (K-1)\,\mathfrak{F}(\mathbb{1} - A_\Sigma) - \sum_{k=1}^{K} \mathfrak{F}\big(\mathbb{1} - A_{[K]\setminus\{k\}}\big). \tag{4.10}$$

Uses Lem. 1.6, Def. 2.5, Lem. 2.6, Thm. 4.1 · *Used by* Rem. 4.6, Rem. 4.7, Rem. 4.8, Rem. 4.9, Thm. 5.1, Def. 5.5, Thm. 5.6, Rem. 6.3, Def. 7.1, Cor. 8.2, Rem. 8.18, Thm. 9.5, Thm. 9.7, Thm. 9.9, Thm. 19.2, Prop. 19.4, Rem. 19.6, Prop. 20.3, Prop. 21.4, Rem. 21.5, reductions/req:recv-row-provenance, reductions/wpfss/cor:normalized-flat, reductions/wpfss/prop:route, reductions/wpfss/rem:receiver-imports · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.povm: unconditioned_family · *Oracles* registry_receiver_povm.py

Proof. (i) The Π^0_k are orthogonal projectors (Lemma 1.6 (iv)) acting on the Fock spaces over pairwise disjoint mode sets, hence they commute: $\Pi^0_k\Pi^0_j = \mathfrak{F}\big((\mathbb{1} - E_{\{k\}})(\mathbb{1} - E_{\{j\}})\big) = \Pi^0_j\Pi^0_k$ by Lemma 1.6 (ii) and commutativity of the coordinate projectors. A product of commuting orthogonal projectors is an orthogonal projector, so the bracket in (4.5) is a product of commuting orthogonal projectors and is therefore positive semidefinite; conjugating by V preserves that. Completeness: expanding the commuting product $\prod_k \big(\Pi^0_k + (\mathbb{1} - \Pi^0_k)\big) = \mathbb{1}$ enumerates exactly the brackets of (4.5) over $S \subseteq [K]$, so their sum is $\mathbb{1}$, and $V^\dagger \mathbb{1} V = \mathbb{1}$ because V is an isometry (Lemma 2.6). Summing (4.5) over $S \subseteq [K]\setminus J$ leaves the terms in which every $k \in J$ contributes the factor Π^0_k, and the same expansion of the remaining product gives $\prod_{k\in J} \Pi^0_k = \Pi^0_J$; conjugating by V gives the last

identity.

(ii) $\Pi_{\mathrm{vac}}(J) = \mathfrak{F}(\mathbb{1} - A_J)$ is Theorem 4.1 (iii). Expanding the commuting product, $\prod_{k\in S}(\mathbb{1} - \Pi_k^0) = \sum_{T\subseteq S}(-1)^{|T|}\prod_{k\in T}\Pi_k^0$, and multiplying by $\prod_{k\notin S}\Pi_k^0$ gives $\sum_{T\subseteq S}(-1)^{|T|}\Pi^0_{([K]\setminus S)\cup T}$, since $T \subseteq S$ is disjoint from $[K]\setminus S$. Applying $V^\dagger(\cdot)V$ termwise and using $\Pi_{\mathrm{vac}}(J) = \mathfrak{F}(\mathbb{1} - A_J)$ gives (4.6).

(iii) Each $\mathfrak{F}(\cdot)$ is photon-number block diagonal by Lemma 1.6 (i), and (4.7) is (4.6) restricted to $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$. At $n = 0$ every $\mathfrak{F}(\cdot)^{(0)} = 1$, so $M_S^{((=0))} = \sum_{T\subseteq S}(-1)^{|T|} = \sum_{j=0}^{|S|}\binom{|S|}{j}(-1)^j = (1-1)^{|S|}$, which is 1 for $S = \varnothing$ and 0 otherwise.

(iv) (4.8) is (4.6) at $S = \varnothing$ and (4.9) at $S = \{k\}$. For (4.10), completeness gives $M_{\mathrm{mc}} = \mathbb{1} - M_\varnothing - \sum_k M_{\{k\}}$, and substituting the previous two lines contributes $-\mathfrak{F}(\mathbb{1} - A_\Sigma)$ once from $M_\varnothing$ and $+\mathfrak{F}(\mathbb{1} - A_\Sigma)$ once from each of the K single-click terms, for a net coefficient $K - 1$. □

The inclusion-exclusion formula is an alternating sum and nothing in its form makes it positive. Positivity comes first, from commuting projective measurements pulled back through an isometry, and the single-click case also follows from monotonicity of the lift.

Remark 4.6 (Source of positivity of the click-pattern elements). Formula (4.6) is an alternating sum of positive operators and nothing about its form makes it positive; the element of a non-empty outcome S has $2^{|S|}$ terms, half of them with a minus sign. Positivity is established *before* the expansion, in part (i) of Theorem 4.5, from the physical construction: K commuting binary projective measurements on disjoint output mode sets, pulled back through an isometry. Definition 2.5 is set up for that order of argument. A derivation that started from (4.6) would have to prove positivity of an alternating sum of Fock functors, which is both harder and unnecessary. For a small independent check, $(4.9) \succeq 0$ follows separately from Lemma 1.10, since $A_{[K]\setminus\{k\}} \preceq A_\Sigma$ gives $\mathfrak{F}(\mathbb{1} - A_{[K]\setminus\{k\}}) \succeq \mathfrak{F}(\mathbb{1} - A_\Sigma)$.

Uses Lem. 1.10, Def. 2.5, Thm. 4.5 · newly derived · accepted

Each joint no-click element is the normal-ordered exponential of the summed Gram A_J of its ports, the form used in the optics literature. For one port and one input mode with Gram A_1 = 1 this is the perfect on/off detector pair of the cited literature.

Remark 4.7 (Click-pattern elements in normal-ordered form). By Proposition 1.11, $\Pi_{\mathrm{vac}}(J) = {:}\exp\!\big(-\boldsymbol{a}^\dagger A_J \boldsymbol{a}\big){:}$, which is the form these operators take in the optics literature; (4.6) then reads $M_S = \sum_{T\subseteq S}(-1)^{|T|}{:}\exp\!\big(-\boldsymbol{a}^\dagger A_{([K]\setminus S)\cup T}\boldsymbol{a}\big){:}$. At $K = 1$ and a single input mode with $A_1 = \lambda$ this is $M_\varnothing = {:}e^{-\lambda\hat{n}}{:}$ and $M_{\{1\}} = {:}\mathbb{1} - e^{-\lambda\hat{n}}{:}$, which at $\lambda = 1$ is the perfect on/off detector pair $\hat{\pi}_0 = |0\rangle\langle 0| = {:}e^{-\hat{n}}{:}$, $\hat{\pi}_1 = {:}\hat{1} - e^{-\hat{n}}{:}$ of [4, Eq. (6)]. The comparison is corroboration only: the identity is Proposition 1.11 and the POVM property is Theorem 4.5 (i).

Uses Prop. 1.11, Thm. 4.5 · newly derived · accepted

With one port the multiclick element vanishes, as the closed form confirms; with two ports each single-click element is the other port dark minus both dark. These cases check the inclusion-exclusion formula by hand.

Remark 4.8 (Small numbers of ports). $K = 1$: $M_{\varnothing} = \mathfrak{F}(\mathbb{1} - A_1)$, $M_{\{1\}} = \mathbb{1} - \mathfrak{F}(\mathbb{1} - A_1)$, and $\mathsf{MC} = \varnothing$ so $M_{\mathrm{mc}} = 0$, consistent with (4.10), whose right-hand side is $\mathbb{1} + 0 - \mathfrak{F}(\mathbb{1} - A_{\varnothing}) = \mathbb{1} - \mathfrak{F}(\mathbb{1}) = 0$. $K = 2$, $S = \{1\}$: $M_{\{1\}} = \mathfrak{F}(\mathbb{1} - A_2) - \mathfrak{F}(\mathbb{1} - A_1 - A_2)$, i.e. "port 2 dark" minus "both dark". $K = 2$, $\mathsf{MC} = \{\{1, 2\}\}$: $M_{\{1,2\}} = \mathbb{1} + \mathfrak{F}(\mathbb{1} - A_{\Sigma}) - \mathfrak{F}(\mathbb{1} - A_2) - \mathfrak{F}(\mathbb{1} - A_1)$.

Uses Thm. 4.5 · newly derived · accepted

The block structure is in the total input photon number, which equals the channel photon number only without a probe; an occupation-diagonal probe state suffices to recover the channel grading. The emitted low blocks decompose the POVM and are not compressions or a squashing argument.

The difference between an operator block and a compression is the third caveat of the sector convention (`spec/notation.md#24-sector-decompositions-and-block-restri`

Remark 4.9 (Scope of the photon-number block structure). The grading in Theorem 4.5 (iii) is by the *total* photon number of the declared inputs $\mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P$. When a probe is declared, that is not the same as the channel photon number, and the downstream modules need the *channel* grading; recovering it is what the occupation-diagonality of the probe states (Definition 2.3) is required for, and the submodule that conditions on the probe states proves that it suffices. When no probe is declared ($\mathbb{I}_P = \varnothing$) the two gradings coincide. The emitted low-photon blocks $M_S^{(\leq N)}$ are operators on $\mathcal{H}(C_{\leq N})$ and are *not* the compressions $P_{\leq N} M_S P_{\leq N}$; emitting them is a decomposition of the physical POVM, never a squashing argument.

Uses Thm. 4.5 · newly derived · accepted

5 Representation independence and coarse-graining

Two receivers induce the same click POVM exactly when their port Grams agree, every admissible Gram family is realised by some instrument, and a one-mode-per-port description exists exactly when every Gram has rank at most one (Theorem 5.1). Phases inside a single port after all interference are invisible; a phase on an earlier element whose output paths are later recombined is not covered and can change the Grams. The main result, Theorem 5.6, is that under a declared coarse-graining, which merges raw outcomes into the protocol's alphabet, lower bounds on disjoint raw outcomes add, and a bound on a merged outcome gives nothing about its parts.

Two receivers have the same click-pattern POVM exactly when their port Grams agree, every admissible Gram family is realised by some instrument, and a one-mode-per-port description exists exactly when every Gram has rank at most one.

The port Grams carry exactly the information the click statistics can reveal, no more and no less. The theorem makes this precise in both directions and says when a description with one optical mode per detector port exists. Under the mode-insensitive port response of Assumption 11.7, item (iii) is what makes A_k, rather than the individual rows feeding port k, the right object.

Theorem 5.1 (Representation independence of the click-pattern POVM)**.** Let $\mathfrak{R}, \mathfrak{R}'$ be receiver instruments on the same declared input space $\mathbb{C}^{\mathbb{I}}$ with the same number of ports K.

(i) *Determination, in both directions.* $\mathfrak{R}$ and $\mathfrak{R}'$ induce the same click-pattern POVM if and only if $A_k = A'_k$ for every k. In particular the POVM depends on the instrument matrix only through the port Grams.

(ii) *Realizability.* Every admissible family $\{A_k\}_{k=1}^{K}$ on $\mathbb{C}^{\mathbb{I}}$ (Definition 2.2) is the port-Gram family of some receiver instrument, in which port k collects at most rank $A_k \le |\mathbb{I}|$ modes.

(iii) *Port-local unitaries are invisible.* If $G' = \big(\bigoplus_k V_k \oplus V_{\mathrm{rest}}\big)G$ with each V_k unitary on $\mathbb{C}^{\mathbb{P}_k}$ and V_{rest} unitary on the remaining modes, then $A'_k = A_k$ for every k.

(iv) *One mode per port.* There is a receiver instrument with $|\mathbb{P}_k| = 1$ for every k inducing the same POVM as $\mathfrak{R}$ if and only if $\operatorname{rank} A_k \le 1$ for every k. In that case, writing $A_k = \lambda_k u_k u_k^\dagger$ with $\lambda_k \ge 0$ and u_k a unit vector, the single row of port k is $g_k = e^{i\varphi_k}\sqrt{\lambda_k}\, u_k^\dagger$, unique up to the phase φ_k.

Uses Def. 2.2, Thm. 4.1, Thm. 4.5 · *Used by* Cor. 5.2, Rem. 5.4, Def. 9.8, Def. 10.1, Req. 12.4, Ex. 17.1, Thm. 18.1, Rem. 18.2, Def. 21.2, reductions/req:recv-row-provenance · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.grams: realizing_instrument · *Oracles* registry_receiver_povm.py

Proof. Idea. The joint no-click elements are lifts of $\mathbb{1} - A_J$, and the click-pattern POVM and these elements determine each other; the one-photon block of a lift is the matrix itself, which recovers the Grams. Realisation takes square roots of the Grams as rows. For (iv), by (i) a one-mode-per-port instrument with the same POVM has the same Grams, each of rank at most one by Theorem 4.1 (ii); conversely, Grams of rank at most one give single rows that realise the family.

(i) If $A_k = A'_k$ for all k then $A_J = A'_J$ for all J, so by Theorem 4.1 (iii) the two receivers have the same joint no-click elements $\Pi_{\mathrm{vac}}(J) = \mathfrak{F}(\mathbb{1} - A_J)$,

and by (4.6) of Theorem 4.5 the same M_S for every S. Conversely, suppose the POVMs agree. "No port in J clicks" is the union of the disjoint outcomes whose click set avoids J, so

$$\Pi_{\mathrm{vac}}(J) = \sum_{S \subseteq [K] \setminus J} M_S \tag{5.1}$$

(Theorem 4.5 (i)), and hence $\mathfrak{F}(\mathbb{1} - A_J) = \mathfrak{F}(\mathbb{1} - A'_J)$. Reading off the one-photon block, whose value is $\mathfrak{F}(\mathbb{1} - A_J)^{(1)} = \mathbb{1} - A_J$, gives $A_J = A'_J$ for all J, and in particular $A_{\{k\}} = A_k = A'_k$.

(ii) Write $r_k \coloneqq \operatorname{rank} A_k \leq |\mathbb{I}|$ and diagonalize $A_k = U_k \Lambda_k U_k^\dagger$ over its non-zero eigenvalues, with Λ_k the positive diagonal $r_k \times r_k$ matrix of those eigenvalues and $U_k \in \mathbb{C}^{\mathbb{I} \times r_k}$ with orthonormal columns. Put $C_k \coloneqq \Lambda_k^{1/2} U_k^\dagger \in \mathbb{C}^{r_k \times \mathbb{I}}$, so that $C_k^\dagger C_k = U_k \Lambda_k U_k^\dagger = A_k$ with r_k rows. Let $\mathbb{M}'$ consist of the $|\mathbb{I}|$ input modes, with the split of $\mathbb{I}$ that $\mathfrak{R}$ declares, together with K blocks of $r_1, \ldots, r_K$ output modes, let $\mathbb{P}'_k$ be the k-th output block, and let G' have the rows of C_k in the block $\mathbb{P}'_k$ and zeros elsewhere. Then $E_{\{k\}} G'_{\mathbb{I}} = C_k$, so $A'_k = C_k^\dagger C_k = A_k$, and $G'^\dagger_{\mathbb{I}} G'_{\mathbb{I}} = \sum_k A_k \preceq \mathbb{1}$, so G' is a contraction and extends to a contraction on all of $\mathbb{C}^{\mathbb{M}'}$ by padding with zeros. Port k collects $r_k = \operatorname{rank} A_k$ modes.

(iii) The ports are pairwise disjoint, so $E_{\{k\}} \big(\bigoplus_j V_j \oplus V_{\mathrm{rest}} \big) = V_k E_{\{k\}}$ as maps into $\mathbb{C}^{\mathbb{P}_k}$. Hence $A'_k = G_{\mathbb{I}}^\dagger E_{\{k\}} V_k^\dagger V_k E_{\{k\}} G_{\mathbb{I}} = G_{\mathbb{I}}^\dagger E_{\{k\}} G_{\mathbb{I}} = A_k$.

(iv) If a one-mode-per-port instrument induces the same POVM then by (i) its Grams are the A_k, and by Theorem 4.1 (ii) each has rank at most $|\mathbb{P}_k| = 1$. Conversely, if $\operatorname{rank} A_k \leq 1$, write $A_k = \lambda_k u_k u_k^\dagger$ (possible for a rank-≤ 1 positive semidefinite matrix, with $\lambda_k = 0$ allowed) and set $g_k \coloneqq \sqrt{\lambda_k}\, u_k^\dagger$. Stacking the g_k as K single rows gives a matrix with $\sum_k g_k^\dagger g_k = \sum_k A_k \preceq \mathbb{1}$, hence a contraction, whose port Grams are the A_k; by (i) it induces the same POVM. For uniqueness, let g be a row vector with $g^\dagger g = \lambda u u^\dagger$. Taking traces, $\|g\|^2 = \lambda$. The range of $g^\dagger g$ is spanned by $g^\dagger$ when $g \neq 0$, and equals $\operatorname{span}\{u\}$, so $g^\dagger = cu$ for some $c \in \mathbb{C}$ with $|c|^2 = \lambda$, i.e. $g = \bar{c}\, u^\dagger$ with $\bar{c} = e^{i\varphi}\sqrt{\lambda}$. If $\lambda = 0$ then $g = 0$ and the statement is vacuous. □

A unitary acting inside a single detector port after all interference changes no click statistic. A phase inserted earlier, on an element whose output paths are later recombined, is not covered by this invariance.

"Global phases do not matter" holds here only with a stated hypothesis: the unitary acts inside one port after all interference. That is why every entry of the component catalogue carries its phases, rather than dropping those that look global on the element they belong to (`spec/modules/receiver/design.md`).

Corollary 5.2 (Unobservability of per-port phases)**.** (i) Multiplying the output row of a port by a phase, or more generally applying any

unitary *inside* a single port's mode set, changes no click statistic (Theorem 5.1 (iii)).

(ii) *The hypothesis matters*: the unitary must act inside one port's modes and after all interference has taken place. A phase that is global on an element in the *middle* of the receiver can become relative once that element's output paths are recombined with other paths, and it can then change the Grams.

Uses Thm. 5.1 · *Used by* Ex. 17.1 · newly derived · accepted

Proof. Item (i) is Theorem 5.1 (iii) applied to the unitary in question, which is port-local by hypothesis. For item (ii), a unitary that acts on an element in the middle of the receiver is not of the form $\bigoplus_k V_k \oplus V_{\text{rest}}$ on the output modes once its paths are recombined, so part (iii) does not apply to it; and it can change the Grams. Take two modes a, b with $\mathbb{I} = \mathbb{I}_C = \{a\}$, b in vacuum, and ports $\{a\}$ and $\{b\}$; apply the balanced beamsplitter $H = 2^{-1/2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ to (a, b), then the phase $e^{i\theta}$ on a, a global phase on that one-mode element, then H again. The instrument matrix is $G = H\,\mathrm{diag}(e^{i\theta}, 1)H$, whose input column is $\frac{1}{2}\big(e^{i\theta}+1,\ e^{i\theta}-1\big)^{\mathsf{T}}$, so $A_1 = \big|\frac{1}{2}(e^{i\theta}+1)\big|^2 = \cos^2(\theta/2)$ depends on θ. □

At two input modes a nonzero port Gram has rank one exactly when the port extinguishes some polarization completely. At more input modes one null direction bounds the rank only by the number of input modes minus one.

The rank criterion of the one-mode-per-port description has a laboratory reading when there are two input modes, such as two polarizations.

Corollary 5.3 (Extinction criterion for rank one at two input modes)**.** Suppose $|\mathbb{I}| = 2$. Then $\operatorname{rank} A_k \le 1$ holds if and only if either $A_k = 0$ or there is a unit vector $v \in \mathbb{C}^2$ with $v^\dagger A_k v = 0$, that is, a polarization the port extinguishes completely. For $|\mathbb{I}| > 2$ the two conditions are *not* equivalent: the existence of one null direction of a positive semidefinite A_k says only $\operatorname{rank} A_k \le |\mathbb{I}| - 1$.

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Proof. For $A_k \succeq 0$ and v a unit vector, $v^\dagger A_k v = 0$ iff $A_k^{1/2} v = 0$ iff $A_k v = 0$, so a null direction is a kernel vector. On $\mathbb{C}^2$ a nonzero positive semidefinite matrix with a kernel vector has rank exactly 1; conversely a rank-1 matrix on $\mathbb{C}^2$ has a one-dimensional kernel. On $\mathbb{C}^d$ a kernel vector only gives $\operatorname{rank} A_k \le d - 1$. □

No result of this contract assumes one optical mode per detector port: a port collecting several modes can have a Gram of rank above one. The one-mode-per-port description holds exactly under the rank criterion of Theorem 5.1 (iv).

Remark 5.4 (Ports of rank greater than one). Example 17.1 ends with four rank-one Grams, which is why the textbook 4×2 BB84 instrument is legitimate *there*. Nothing in this contract assumes one mode per port anywhere else: the click-pattern POVM, the theorems built on it and the decomposition of the loss deficiency, among them Theorems 4.5, 5.6, 6.2, 8.14 and 10.10 and Lemma 4.2 of the core and Theorems 18.1 and 19.2 of its submodules, are proved for arbitrary $\{A_k\}$, and a port that collects several modes, such as a multimode fiber onto one active area or two spatial modes onto one detector, can have rank $A_k > 1$, and every result above holds when it does. The criterion for a one-mode-per-port description is Theorem 5.1 (iv); it is decidable from the emitted Grams, and Remark 18.2 exhibits the one operation in this contract, apart from declaring a zero efficiency, that can change its answer. Requirement 12.4 states the rule the engine follows.

Uses Thm. 5.1 · *Used by* Rem. 18.2 · newly derived · accepted

A declared surjection from raw click patterns onto the protocol's outcome alphabet defines the emitted measurement. Each emitted element is the sum of the click-pattern elements in its fibre.

The protocol reads an outcome alphabet that is usually coarser than the raw click pattern, for instance one that merges every pattern with two or more clicks into a single multiclick outcome.

Definition 5.5 (Declared coarse-graining and the emitted POVM)**.** A *declared coarse-graining* is a surjection $\varpi : 2^{[K]} \to \mathcal{V}$ onto the declared emitted alphabet, supplied as data by the protocol declaration, $[K]$ indexing the detector ports of a receiver instrument (Definition 2.1). With M_S the click-pattern elements of Theorem 4.5 ((4.5)), the *emitted POVM* is

$$F_v := \sum_{S \in \varpi^{-1}(v)} M_S, \qquad v \in \mathcal{V}. \tag{5.2}$$

Uses Def. 2.1, Thm. 4.5 · *Used by* Thm. 9.5, Thm. 9.7, Def. 20.1, Prop. 20.3, Prop. 21.4 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.coarse:CoarseGraining

The emitted POVM of a declared coarse-graining is a POVM, block diagonal in the photon number, and lower bounds on disjoint raw outcomes add. A bound on a merged outcome gives nothing about the click patterns it merges.

Merging outcomes keeps everything the POVM structure needs and loses information in one direction only. Part (iii) is why the strengthening slot of Definition 8.16 is indexed by outcome sets rather than by emitted letters: by (ii) a bound on a set of raw outcomes passes to every emitted outcome whose fibre contains the set, while by (iii) a bound on an emitted outcome

cannot be split back.

Theorem 5.6 (Coarse-graining of the click-pattern POVM). Let ϖ be a declared coarse-graining.

(i) $\{F_v\}_{v\in\mathcal{V}}$ is a POVM on $\mathcal{F}(\mathbb{C}^{\mathbb{I}})$, and every F_v is block diagonal in the photon number with $F_v^{(=n)} = \sum_{S\in\varpi^{-1}(v)} M_S^{(=n)}$.

(ii) *Bounds add.* If $\mathcal{O} \subseteq 2^{[K]}$ and $M_S^{(>N)} \succeq p_S\,\mathbb{1}^{(>N)}$ for every $S \in \mathcal{O}$ with $p_S \geq 0$, then $\sum_{S\in\mathcal{O}} M_S^{(>N)} \succeq \big(\sum_{S\in\mathcal{O}} p_S\big)\mathbb{1}^{(>N)}$. In particular, if $\varpi^{-1}(v) \supseteq \mathcal{O}$ then $F_v^{(>N)} \succeq \big(\sum_{S\in\mathcal{O}} p_S\big)\mathbb{1}^{(>N)}$.

(iii) *Coarsening is lossy in one direction only.* A bound on a coarse outcome does not descend to its fibres: $F_v \succeq p\,\mathbb{1}$ implies nothing about the individual M_S with $\varpi(S) = v$.

Uses Lem. 1.9, Thm. 4.5 · *Used by* Thm. 9.5, Thm. 9.9, Def. 10.1, Prop. 10.2, reductions/req:recv-row-provenance, reductions/wpfss/cor:normalized-flat, reductions/wpfss/prop:route, reductions/wpfss/rem:receiver-imports · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.coarse: coarse_block · *Oracles* registry_receiver_povm.py

Proof. (i) Each F_v is a finite sum of positive operators, hence positive; the fibres $\varpi^{-1}(v)$ partition $2^{[K]}$ because ϖ is a map, so $\sum_v F_v = \sum_S M_S = \mathbb{1}$ by Theorem 4.5 (i). Block diagonality is inherited from each summand (Theorem 4.5 (iii)), and the block of a finite sum is the sum of the blocks.

(ii) The Löwner order is preserved by sums: $\sum_{S\in\mathcal{O}} M_S^{(>N)} - \big(\sum_S p_S\big)\mathbb{1}^{(>N)} = \sum_{S\in\mathcal{O}} \big(M_S^{(>N)} - p_S\mathbb{1}^{(>N)}\big) \succeq 0$. For the second sentence add the remaining, positive, elements of the fibre.

(iii) Take $K = 2$ and $\mathcal{V} = \{v\}$ with ϖ constant, so $F_v = \mathbb{1}$ and the bound $F_v \succeq 1 \cdot \mathbb{1}$ is available. Choose port Grams with $\lambda_{\min}(A_\Sigma) = 0$, i.e. A_Σ singular: any lossy receiver with an input direction no port sees. By Lemma 1.9, $\lambda_{\max}\big(M_\varnothing^{(=n)}\big) = (1 - \lambda_{\min}(A_\Sigma))^n = 1$, so on every block there is a unit vector $|\psi\rangle$ with $\langle\psi|\, M_\varnothing^{(=n)}\, |\psi\rangle = 1$; completeness and positivity then force $\langle\psi|\, M_S^{(=n)}\, |\psi\rangle = 0$ for every $S \neq \varnothing$, and in particular $\lambda_{\min}\big(M_{\{1,2\}}^{(=n)}\big) = 0$. So a bound on the coarse outcome gives nothing about its fibres. □

6 The analytic high-photon bound

A squashing argument needs a lower bound on the multiclick element valid on every photon-number block above the cutoff. Extreme eigenvalues of the no-click operators give a scalar bound function on each block (Theorem 6.2); it is not monotone in general. Where its value at zero photons is at most zero, the bound cannot decrease from any photon number n ≥ 1 at which it is non-negative, so the non-negative part of the infimum over the tail is that of

its value at the first photon number above the cutoff (Lemma 6.5); otherwise a finite window with an analytic tail gives a valid bound that converges to the infimum when c < 1, with a stated residual gap (Lemma 6.8).

Ordered scalars between 0 and 1, one pair shared and one pair per port, define an exponential bound function of the photon number for the multiclick element. Admissibility is exact, and distinct from the tolerance rules that admit a floating-point evaluation.

What a squashing argument consumes is, for every declared probe setting, one number: a lower bound on the smallest eigenvalue of that setting's multiclick element, valid on *every* photon-number block above the declared cutoff. The bound itself is three lines from Lemma 1.9, which gives the spectrum of a lifted no-click operator on each block. Making the quantifier "every block above the cutoff" into a finite computation is the work of this section. The definition closes with how its one computed condition is decided in the floating-point evaluation model of Definition 10.9.

Definition 6.1 (Admissible bound data and the bound function)**.** A tuple $(\gamma, \{\beta_k\}, c, \{d_k\})$ with $K \geq 1$ entries is *admissible bound data* when

$$0 \leq \gamma \leq \beta_k \leq 1, \qquad 0 \leq c \leq d_k \leq 1 \qquad (k = 1, \ldots, K), \tag{6.1}$$

and the associated *bound function* is

$$\beta(n) \coloneqq 1 + (K-1)\,\gamma\, c^{\,n} - \sum_{k=1}^{K} \beta_k\, d_k^{\,n}, \qquad n \geq 0, \tag{6.2}$$

with the convention $0^0 = 1$. Admissibility (6.1) is exact. The one part of it that this contract decides on computed data is $0 \leq c$; the other parts are not separately tested (Definition 10.4 (iv)). In floating point that part is decided by the widened test $c \geq -\varepsilon_2$ of Definition 10.4 (iv), which decides it on *every* input route and is not displaced by the route-(ii) admission of Definition 10.4 (v); the computed c enters (6.2) unclamped.

Uses Lem. 1.9 · *Used by* Thm. 6.2, Lem. 6.5, Rem. 6.7, Lem. 6.8, Def. 8.1, Def. 8.3, Rem. 8.6, Rem. 8.11, Def. 8.12, Thm. 8.13, Def. 10.4, Rem. 10.5, Rem. 10.8, Thm. 10.10, Rem. 10.12, Rem. 19.6 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:BoundData

For no-click operators of prefactor form, every n-photon block of the multiclick combination is bounded below by the bound function at n, built from extreme eigenvalues of the mode operators B and B_k. The bound is at most one.

The multiclick element is the identity plus a multiple of the all-dark element, the joint no-click element of all K ports, minus the joint no-click elements that leave out one port. Bounding each lifted operator by an extreme eigenvalue of its mode-space operator, from below where it enters

with a non-negative coefficient and from above where it is subtracted, gives a scalar bound on each photon-number block.

Theorem 6.2 (Lower bound on every multiclick block)**.** Let $B, B_1, \dots, B_K$ be Hermitian operators on a finite-dimensional space with

$$0 \preceq B_k \preceq B \preceq \mathbb{1} \qquad (k = 1, \dots, K), \tag{6.3}$$

let $\gamma, \beta_1, \dots, \beta_K \in (0, 1]$ with $\gamma \le \beta_k$ for every k, let $\Pi_{\mathrm{vac}}([K])$ and $\Pi_{\mathrm{vac}}([K] \setminus \{k\})$ $(k = 1, \dots, K)$ be operators on the Fock space of that space of the form

$$\Pi_{\mathrm{vac}}([K]) = \gamma\, \mathfrak{F}(\mathbb{1} - B), \qquad \Pi_{\mathrm{vac}}([K] \setminus \{k\}) = \beta_k\, \mathfrak{F}\big(\mathbb{1} - B_k\big), \tag{6.4}$$

and put $M_{\mathrm{mc}} \coloneqq \mathbb{1} + (K-1)\Pi_{\mathrm{vac}}([K]) - \sum_k \Pi_{\mathrm{vac}}([K] \setminus \{k\})$. Put

$$c \coloneqq 1 - \lambda_{\max}(B), \qquad d_k \coloneqq 1 - \lambda_{\min}\big(B_k\big). \tag{6.5}$$

Then $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ is admissible bound data, and for every $n \ge 0$

$$\lambda_{\min}\big(M_{\mathrm{mc}}^{(=n)}\big) \;\ge\; \beta(n), \qquad \text{equivalently} \qquad M_{\mathrm{mc}}^{(=n)} \;\succeq\; \beta(n)\, \mathbb{1}^{(=n)}. \tag{6.6}$$

Moreover $\beta(n) \le 1$ for every n.

Uses Lem. 1.9, Def. 6.1 · *Used by* Rem. 6.4, Def. 7.1, Lem. 7.2, Rem. 7.3, Lem. 7.4, Def. 8.1, Cor. 8.2, Rem. 8.11, Thm. 8.13, Thm. 8.14, Def. 10.4, Rem. 10.5, Rem. 10.6, Thm. 10.10, Rem. 10.12, Req. 13.5, Ex. 17.3, Prop. 20.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound: setting_bound · *Oracles* registry_receiver_povm.py

Proof. Admissibility of the data. $0 \preceq B \preceq \mathbb{1}$ gives $\lambda_{\max}(B) \in [0, 1]$, hence $c \in [0, 1]$; likewise $0 \preceq B_k \preceq \mathbb{1}$ gives $d_k \in [0, 1]$. For $c \le d_k$: from $B_k \preceq B$ we get $\lambda_{\min}(B_k) \le \lambda_{\min}(B) \le \lambda_{\max}(B)$, and subtracting from 1 reverses the inequality, so $d_k = 1 - \lambda_{\min}(B_k) \ge 1 - \lambda_{\max}(B) = c$. The relation $\gamma \le \beta_k$ is a hypothesis.

The bound. By Lemma 1.9 applied to B and to each B_k — legitimate because (6.3) puts all of them in $[0, \mathbb{1}]$ —

$$\mathfrak{F}(\mathbb{1} - B)^{(n)} \succeq c^n \mathbb{1}^{(=n)}, \qquad \mathfrak{F}\big(\mathbb{1} - B_k\big)^{(n)} \preceq d_k^n \mathbb{1}^{(=n)}. \tag{6.7}$$

The Löwner order is preserved by sums and by multiplication with non-negative scalars, and $K - 1 \ge 0$, $\gamma > 0$, $\beta_k > 0$. Taking the n-photon block of $M_{\mathrm{mc}} = \mathbb{1} + (K-1)\Pi_{\mathrm{vac}}([K]) - \sum_k \Pi_{\mathrm{vac}}([K] \setminus \{k\})$, bounding the term with the non-negative coefficient from below and each subtracted term from above, gives $M_{\mathrm{mc}}^{(=n)} \succeq \big(1 + (K-1)\gamma c^n - \sum_k \beta_k d_k^n\big)\mathbb{1}^{(=n)}$, and $X \succeq t\,\mathbb{1}$ says exactly $\lambda_{\min}(X) \ge t$.

$\beta \le 1$. Since $d_k \ge c$ and $\beta_k \ge \gamma$, every term of $\sum_k \beta_k d_k^n$ is at least γc^n, so the sum is at least $K\gamma c^n \ge (K-1)\gamma c^n$, whence $\beta(n) \le 1$. □

The all-dark element (no port clicks) enters the multiclick element with coefficient $K-1$: each of the K single-click elements contributes one, and the no-click element, which is this same operator, takes one away. An error by one would show only where $c > 0$.

Remark 6.3 (The coefficient $K-1$ of the all-dark element). The multiclick element is the complement of the no-click and the K single-click elements, and each single-click element already contains one subtraction of the all-dark element (4.9). The K subtractions and the one from $M_{\varnothing}$ combine into the coefficient $K-1$ (Theorem 4.5 (iv)). An error by one in this coefficient changes the bound by γc^n, a term that vanishes on every block $n \geq 1$ when $c = 0$ and survives when $c > 0$: the error would be invisible in the first case and material in the second.

Uses Thm. 4.5 · newly derived · accepted

A published construction bounds a photon-number subspace weight from an observed multiclick rate, for one row per detector. Theorem 6.2 is instead an operator inequality for every input state and ports of any rank; no step here depends on the citation.

Remark 6.4 (Relation to a published multiclick weight bound). A published route to a related quantity is [1, Thm. 1, Eq. (58)], which writes a multiclick POVM element as $M^{\mathrm{mult}}(G) = \mathbb{1} - |0\rangle\langle 0| - \sum_i F_{\mathrm{sic}}$ for a passive linear-optical setup with mode transformation G, and then bounds the weight of a photon-number subspace using an *observed* multiclick rate together with constants obtained by partitioning the rows of G. That is a different object from (6.6): it is a statement about a measured statistic and a subspace weight, whereas Theorem 6.2 is an operator inequality on each photon-number block that holds for every input state and is used as a device fact rather than as an observation. The construction is also less general in the direction that matters here: it is written for a one-row-per-detector transformation, whereas Theorem 6.2 is proved for arbitrary port Grams, hence for ports of arbitrary rank. The citation is for orientation, and no step of this contract depends on it.

Uses Thm. 6.2 · newly derived · accepted

If the bound function is at most zero at zero photons, it cannot decrease from any $n \geq 1$ at which it is non-negative. The non-negative part of its infimum above a cutoff is then that of its value at the next photon number.

A squashing argument consumes not $\beta(N+1)$ but the infimum $\inf_{n>N} \lambda_{\min}\big(M_{\mathrm{mc}}^{(=n)}\big)$, and a squashing request whose bounds are not uniform over *all* photon numbers above the cutoff fails (`spec/architecture.md#54-failure-policy`). The bound function is not monotone in general: it is the sum of a non-increasing term $(K-1)\gamma c^n$ and a non-decreasing term $-\sum_k \beta_k d_k^n$. The lemma shows that the non-monotone regime is confined to where the bound

is useless anyway.

Lemma 6.5 (Collapse of the tail infimum to the first photon number)**.** Let $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ be admissible bound data satisfying

$$\sum_{k=1}^{K} \beta_k \;\geq\; 1 + (K-1)\gamma \qquad \text{(equivalently, } \beta(0) \leq 0\text{)}. \tag{6.8}$$

Then for every $n \geq 1$,

$$\beta(n) \geq 0 \implies \beta(n+1) \;\geq\; \beta(n). \tag{6.9}$$

Consequently, for every cutoff $N \geq 0$,

$$\max\Big\{0,\; \inf_{n>N} \beta(n)\Big\} \;=\; \max\big\{0,\; \beta(N+1)\big\}, \tag{6.10}$$

and if $\beta(N+1) \geq 0$ then β is non-decreasing on $\{N+1, N+2, \dots\}$ and $\inf_{n>N} \beta(n) = \beta(N+1)$. Hypothesis (6.8) holds automatically when $\gamma = \beta_k = 1$, where it reads $K \geq K$.

Uses Def. 6.1 · *Used by* Rem. 6.6, Rem. 6.7, Lem. 6.8, Lem. 7.4, Rem. 7.5, Def. 8.1, Cor. 8.2, Def. 8.3, Rem. 8.4, Rem. 8.6, Def. 8.7, Rem. 8.8, Prop. 8.9, Rem. 8.10, Thm. 8.13, Thm. 8.14, Rem. 8.18, Thm. 10.10, Rem. 10.12, Ex. 17.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:collapse_slack

Proof. Idea. The increment $\beta(n+1) - \beta(n)$ is non-negative exactly when a weighted sum over the ports dominates the all-dark term. After the substitution $u_k = \beta_k d_k^n$ each port's contribution is a concave function of u_k, so it lies above its chord; the constraint $\beta(n) \geq 0$ bounds $\sum_k u_k$, and the hypothesis at zero photons closes the gap. An induction then turns the step into the statement about the infimum.

Step 1 (reduction to one inequality). Fix $n \geq 1$ and assume $\beta(n) \geq 0$, that is

$$\sum_{k=1}^{K} \beta_k d_k^n \;\leq\; T := 1 + (K-1)\gamma c^n. \tag{6.11}$$

Since $\beta(n+1) - \beta(n) = \sum_k \beta_k d_k^n (1-d_k) - (K-1)\gamma c^n (1-c)$, the implication (6.9) is exactly the statement

$$\sum_{k=1}^{K} \beta_k d_k^n (1-d_k) \;\geq\; (K-1)\gamma\, c^n (1-c) \tag{6.12}$$

under the constraint (6.11) and $c \leq d_k \leq 1$.

Step 2 (the case $c = 1$). If $c = 1$ then $d_k = 1$ for every k and both sides of (6.12) vanish. Assume therefore $c < 1$, so that $c^n < 1$ for $n \geq 1$.

Step 3 (the chord bound). Substitute $u_k \coloneqq \beta_k d_k^n$, a strictly increasing bijection from $d_k \in [c,1]$ onto $u_k \in [\beta_k c^n, \beta_k]$ (for $\beta_k > 0$; a $\beta_k = 0$ contributes 0 to both sides of the chord bound below and may be dropped). Under it

$$\beta_k d_k^n (1-d_k) = \psi_k(u_k), \qquad \psi_k(u) \coloneqq u - \beta_k^{-1/n}\, u^{1+1/n}, \tag{6.13}$$

because $\beta_k^{-1/n} u^{1+1/n} = \beta_k^{-1/n} (\beta_k d_k^n)^{1+1/n} = \beta_k d_k^{n+1}$. Two values and one convexity fact are all that is needed:

$$\psi_k(\beta_k) = 0, \qquad \psi_k(\beta_k c^n) = \beta_k c^n (1-c), \qquad \psi_k''(u) = -\beta_k^{-1/n}\Big(1+\tfrac{1}{n}\Big)\tfrac{1}{n}\, u^{1/n-1} \le 0, \tag{6.14}$$

so ψ_k is concave on $[0,\beta_k]$ and therefore lies *above* the chord joining its values at the two endpoints of $[\beta_k c^n, \beta_k]$. Writing a point of that interval as $u = \lambda \beta_k c^n + (1-\lambda)\beta_k$ with $\lambda = (\beta_k - u)/\big(\beta_k(1-c^n)\big) \in [0,1]$, the chord bound is

$$\psi_k(u) \;\ge\; \lambda\, \psi_k(\beta_k c^n) + (1-\lambda)\, \psi_k(\beta_k) \;=\; \frac{c^n(1-c)}{1-c^n}\,\big(\beta_k - u\big). \tag{6.15}$$

Summing (6.15) over k and using $\sum_k u_k \le T$ from (6.11) together with $\frac{c^n(1-c)}{1-c^n} \ge 0$,

$$\sum_{k=1}^{K} \beta_k d_k^n (1-d_k) = \sum_{k=1}^{K} \psi_k(u_k) \;\ge\; \frac{c^n(1-c)}{1-c^n}\Big(\sum_k \beta_k - \sum_k u_k\Big) \;\ge\; \frac{c^n(1-c)}{1-c^n}\Big(\sum_k \beta_k - T\Big). \tag{6.16}$$

Step 4 (the hypothesis at zero photons). Hypothesis (6.8) gives

$$\sum_k \beta_k - T \;\ge\; \big(1+(K-1)\gamma\big) - 1 - (K-1)\gamma c^n \;=\; (K-1)\gamma\big(1-c^n\big) \;\ge\; 0, \tag{6.17}$$

and substituting (6.17) into (6.16) cancels the factor $1-c^n > 0$ and yields exactly (6.12).

Step 5 (from (6.9) to (6.10)). The induction runs over $n \ge N+1 \ge 1$, which is where (6.9) was proved. If $\beta(N+1) \ge 0$ then, inductively, $\beta(n+1) \ge \beta(n) \ge \beta(N+1) \ge 0$ for every $n \ge N+1$, so β is non-decreasing on $\{N+1, N+2, \dots\}$ and $\inf_{n>N} \beta(n) = \beta(N+1)$, which gives (6.10). If $\beta(N+1) < 0$ then $\inf_{n>N} \beta(n) \le \beta(N+1) < 0$ and both sides of (6.10) equal 0. □

The monotonicity step of Lemma 6.5 fails at $n = 0$, with an explicit two-port receiver as witness. Nothing is lost, because the infimum runs over photon numbers above the cutoff, all at least one.

Remark 6.6 (The monotonicity step at zero photons). Lemma 6.5 is stated from $n \geq 1$ and cannot be extended to $n = 0$. Take $K \geq 2$, $\gamma = \beta_k = 1$ and $d_k = 1$ for every k, and $c < 1$: then $\beta(0) = 1 + (K-1) - K = 0 \geq 0$, while $\beta(1) = 1 + (K-1)c - K = (K-1)(c-1) < 0$, so (6.9) fails at $n = 0$. The data are realised, for instance, by $K = 2$ on two input modes with $A_1 = a\,|0\rangle\langle 0|$, $A_2 = a\,|1\rangle\langle 1|$, $a \in (0,1)$: there $c = 1 - a$ and $d_1 = d_2 = 1$. The substitution $u_k = \beta_k d_k^n$ degenerates at $n = 0$, which is precisely where the proof breaks. Nothing is lost: the infimum in (6.10) runs over $n > N \geq 0$, so the induction never starts below $n = 1$.

Uses Lem. 6.5 · newly derived · accepted

At the data $\gamma = \beta_k = 1$, $d_1 = 1$ and $d_k = c$ for $k \geq 2$, the bound function is identically zero, so the monotonicity step of Lemma 6.5 cannot be strengthened to strict increase.

Remark 6.7 (Sharpness of the collapse lemma). Equality holds in (6.12) at the data $\gamma = \beta_k = 1$, $d_1 = 1$, $d_2 = \cdots = d_K = c$: there $\beta(n) = 1 + (K-1)c^n - 1 - (K-1)c^n = 0$ for every n, so β is constant and the implication (6.9) cannot be strengthened to strict increase. More generally, at $\gamma = \beta_k = 1$ every $d_1 = 1$ gives $\beta(n) \leq 0$ for every n, because $d_k \geq c$ by (6.1) makes $\beta(n) \leq 1 + (K-1)c^n - 1 - (K-1)c^n$. Physically, $d_1 = 1$ is a singular B_1, and at a receiver without a probe, where $B_1 = A_{[K]\setminus\{1\}}$ (Corollary 8.2), that is a receiver in which some input polarization reaches no port other than port 1: no multiclick weight can be guaranteed at any photon number, and the bound correctly returns 0. The data above attain equality; at a receiver without a probe they force ports 2 to K to receive nothing, since $d_k = c$ means $\lambda_{\min}(B_k) = \lambda_{\max}(B)$, which with $B_k \preceq B$ forces $B_k = B$, that is $A_k = 0$.

Uses Def. 6.1, Lem. 6.5, Cor. 8.2 · *Used by* Rem. 10.12 · newly derived · accepted

The minimum of the bound function over a finite window, capped by an analytic tail term, is a valid uniform bound above the cutoff. It improves as the window widens, converges to the infimum when $c < 1$, and a residual gap bounds what remains.

Where the hypothesis of Lemma 6.5 fails, the infimum of the bound function over all photon numbers above the cutoff is still bounded from below by a finite computation, which converges to it when $c < 1$: evaluate β on a finite window and bound the rest by a term that drops the non-negative all-dark contribution.

Lemma 6.8 (Window-and-tail bound on the tail infimum)**.** Let $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ be admissible bound data and $N \geq 0$. Define, for an integer $m > N$,

$$\underline{\beta}_{N,m} \coloneqq \min\big\{\, W_{N,m},\, T_m \,\big\}, \qquad W_{N,m} \coloneqq \min_{N<n\leq m} \beta(n), \qquad T_m \coloneqq 1 - \sum_{k=1}^{K} \beta_k d_k^m. \tag{6.18}$$

The window $\{n : N < n \leq m\}$ is non-empty for every $m > N$; at $m = N+1$ it is the single index $N+1$. Then:

(i) $\underline{\beta}_{N,m} \le \inf_{n>N} \beta(n)$ for every $m > N$: the quantity is a valid uniform bound;

(ii) $\underline{\beta}_{N,m}$ is non-decreasing in m, and $\lim_{m\to\infty} \underline{\beta}_{N,m} = \inf_{n>N} \beta(n)$ whenever $c < 1$; moreover the *residual gap*

$$g_{N,m} \coloneqq \max\{0,\ W_{N,m} - T_m\} \tag{6.19}$$

is a finite non-negative real for every $m > N$ — at $m = N+1$ it equals $(K-1)\gamma c^{N+1}$ — and satisfies $\inf_{n>N} \beta(n) - \underline{\beta}_{N,m} \le g_{N,m}$, so the evaluation is convergent and $g_{N,m}$ bounds what widening the window could still recover;

(iii) if $c = 0$ then $\beta(n) = 1 - \sum_k \beta_k d_k^n$ for $n \ge 1$ is non-decreasing, $\underline{\beta}_{N,N+1} = \beta(N+1)$ is exact, and $g_{N,N+1} = 0$;

(iv) the two lemmas are consistent: under the hypothesis of Lemma 6.5 with $\beta(N+1) \ge 0$ *and* $c < 1$, (i) and (ii) give $\underline{\beta}_{N,m} \le \beta(N+1)$ for every m with equality in the limit $m \to \infty$, so the first branch of Definition 8.3 is the exact value the window-and-tail evaluation converges to, and using it loses nothing. The hypothesis $c < 1$ is needed: at $c = 1$ admissibility forces $d_k = 1$ for every k, β is constant, and under Lemma 6.5's hypothesis with $\beta(N+1) \ge 0$ that constant is 0 while $T_m = 1 - \sum_k \beta_k = -(K-1)\gamma$ for every m, so the limit clause fails whenever $\gamma > 0$ and $K \ge 2$.

Uses Def. 6.1, Lem. 6.5 · *Used by* Def. 7.1, Lem. 7.4, Rem. 7.5, Def. 8.3, Rem. 8.4, Rem. 8.6, Def. 8.7, Rem. 8.8, Prop. 8.9, Rem. 8.11, Def. 8.12, Thm. 8.13, Thm. 8.14, Rem. 8.18, Def. 10.4, Thm. 10.10, Rem. 10.12, Req. 14.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:window_floor

Proof. (i) For $N < n \le m$ the value $\beta(n)$ appears in $W_{N,m}$. For $n > m$, $d_k \le 1$ gives $d_k^n \le d_k^m$ and $\gamma c^n \ge 0$, so $\beta(n) \ge 1 - \sum_k \beta_k d_k^n \ge T_m$. Hence every $\beta(n)$ with $n > N$ dominates $\min\{W_{N,m}, T_m\}$.

(ii) *Monotonicity.* $W_{N,m+1} = \min\{W_{N,m}, \beta(m+1)\}$ and $T_{m+1} \ge T_m$ because $d_k^{m+1} \le d_k^m$; and $\beta(m+1) \ge T_m$ by the computation in (i) at $n = m+1 > m$. Each of $W_{N,m}$, $\beta(m+1)$ and T_{m+1} is therefore at least $\min\{W_{N,m}, T_m\} = \underline{\beta}_{N,m}$, and $\underline{\beta}_{N,m+1}$ is the minimum of those three.

Limit. $W_{N,m}$ is non-increasing in m and the windows exhaust $\{n : n > N\}$, so $W_{N,m} \downarrow \inf_{n>N} \beta(n)$. Put $D \coloneqq \{k : d_k = 1\}$; then $T_m \uparrow 1 - \sum_{k\in D} \beta_k$, and since $c < 1$ also $\beta(n) \to 1 - \sum_{k\in D} \beta_k$, so $\lim_m T_m \ge \inf_{n>N} \beta(n)$. Both sequences converge, hence $\lim_m \min\{W_{N,m}, T_m\} = \min\{\inf_{n>N} \beta(n), \lim_m T_m\} = \inf_{n>N} \beta(n)$.

The residual gap. Both $W_{N,m}$ and T_m are minima of finitely many reals, so $g_{N,m}$ is a finite non-negative real; at $m = N+1$ the window is $\{N+1\}$, so $W_{N,N+1} = \beta(N+1) = T_{N+1} + (K-1)\gamma c^{N+1}$ and $g_{N,N+1} = (K-1)\gamma c^{N+1}$.

For the bound, write $W = W_{N,m}$, $T = T_m$. If $T \le W$ then $\underline{\beta}_{N,m} = T$ and $\inf_{n>N} \beta(n) \le W$ (the window values are among those infimized), so the difference is at most $W - T = g_{N,m}$. If $W < T$ then $\underline{\beta}_{N,m} = W = \beta(n_0)$ for some $N < n_0 \le m$, so $\inf_{n>N} \beta(n) \le \underline{\beta}_{N,m}$ and the difference is at most $0 = g_{N,m}$.

(iii) $c = 0$ makes $\gamma c^n = 0$ for $n \ge 1$, so $\beta(n) = 1 - \sum_k \beta_k d_k^n$ there, and each d_k^n is non-increasing in n; in particular $W_{N,N+1} = \beta(N+1) = T_{N+1}$, so $\underline{\beta}_{N,N+1} = \beta(N+1)$ and $g_{N,N+1} = 0$.

(iv) Under the hypothesis of Lemma 6.5 with $\beta(N+1) \ge 0$ and $c < 1$, that lemma gives $\inf_{n>N} \beta(n) = \beta(N+1)$; then (i) reads $\underline{\beta}_{N,m} \le \beta(N+1)$ and (ii) gives $\underline{\beta}_{N,m} \to \beta(N+1)$. □

7 Exact block eigenvalues

The analytic bound bounds three operators separately and loses slack that no wider window recovers. Computing the smallest eigenvalue of a block itself recovers it: that exact block bound dominates the analytic bound, is at most one, and is non-negative wherever the multiclick element is positive (Lemma 7.2). Lower estimates on finitely many exact blocks combine with an admissible analytic remainder into a uniform bound over the whole tail (Lemma 7.4).

The exact block bound $\beta^*(n)$ is the smallest eigenvalue of the n-photon block of the multiclick element. Computing it on finitely many blocks recovers the slack the analytic bound loses by bounding three operators separately.

Both evaluations of the uniform bound so far, the collapse of Lemma 6.5 and the window and tail of Lemma 6.8, bound $\lambda_{\min}(M_{\mathrm{mc}}^{(=n)})$ by the analytic $\beta(n)$, and (6.6) is where all of the slack enters: Theorem 6.2 bounds three operators separately. Widening the analytic window does not recover it: at $c = 0$, for instance at a receiver with no probe whose instrument sends every input entirely into detector ports ($A_\Sigma = \mathbb{1}$), Lemma 6.8 (iii) makes the residual gap exactly zero. What recovers it is computing the block itself, on finitely many indices. Lemma 7.2 and Lemma 7.4 say what may be computed, what must still be argued, and how the two compose.

Definition 7.1 (The exact block bound function)**.** Under the hypotheses of Theorem 6.2, the *exact block bound* is

$$\beta^*(n) \;:=\; \lambda_{\min}\big(M_{\mathrm{mc}}^{(=n)}\big) \;=\; \lambda_{\min}\Big(\mathbb{1}^{(=n)} + (K-1)\gamma\,\mathfrak{F}(\mathbb{1}-B)^{(n)} - \sum_{k=1}^{K} \beta_k\,\mathfrak{F}\big(\mathbb{1}-B_k\big)^{(n)}\Big), \tag{7.1}$$

for $n \ge 0$, the second equality being (6.4) restricted to the n-photon block. The operator inside is a Hermitian matrix of dimension $\binom{D+n-1}{n}$, where D

is the dimension of the space B acts on: the channel input space of the setting family (Definition 8.1), which is $|\mathbb{I}_C|$ at both instances this module constructs. No new object is introduced: each $\mathfrak{F}(\cdot)^{(n)} = \big((\cdot)^{\otimes n}\big)\big|_{\mathrm{Sym}^n}$ is the construction (4.7) already uses for the blocks the artifact emits at $n \leq N$, in the occupation basis and order that Definition 1.2 and the interface conventions of this module fix, evaluated at an index above the cutoff instead of below it.

Uses Def. 1.2, Thm. 4.5, Thm. 6.2, Lem. 6.8 · *Used by* Lem. 7.2, Def. 8.7, Def. 8.12, Thm. 8.13, Thm. 8.14, Rem. 8.18, Ass. 11.11, Req. 13.5, Ex. 17.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:exact_block_bound

The exact block bound lies between the analytic bound and one on every photon-number block, and it is non-negative when the multiclick element is positive. That positivity holds at every setting of a setting family but does not follow from the multiclick bound's hypotheses.

The three bounds of the lemma need different hypotheses. The comparison with the analytic bound and the upper bound hold under the hypotheses of Theorem 6.2 alone. Non-negativity needs the multiclick element itself to be positive, (7.3), which holds at every setting of a setting family (Definition 8.1 (a)), because there M_{mc} is a sum of elements of a POVM; those are the only instances to which this module applies Theorem 6.2. It does not follow from (6.3) and (6.4), and Remark 7.3 is the instance that shows so.

Lemma 7.2 (Range of the exact block bound)**.** Let the hypotheses of Theorem 6.2 hold. Then, for every $n \geq 0$,

$$\beta(n) \;\leq\; \beta^*(n) \;\leq\; 1. \tag{7.2}$$

If in addition

$$M_{\mathrm{mc}} \;\succeq\; 0, \tag{7.3}$$

then $\beta^*(n) \geq 0$ as well, so that $\beta^*(n) \in [0, 1]$.

Uses Lem. 1.6, Lem. 1.9, Lem. 1.10, Thm. 6.2, Def. 7.1 · *Used by* Rem. 7.3, Lem. 7.4, Rem. 7.5, Def. 8.1, Rem. 8.11, Thm. 8.13, Thm. 8.14, Def. 10.1, Rem. 10.6, Thm. 10.10, Rem. 10.12 · newly derived · accepted

Proof. Lower comparison. $\beta(n) \leq \beta^*(n)$ is (6.6).

Upper bound. For $\beta^*(n) \leq 1$ it suffices that $M_{\mathrm{mc}} \preceq \mathbb{1}$, which is the operator form of the argument that closes the proof of Theorem 6.2. From $B_k \preceq B$, Lemma 1.10 gives $\mathfrak{F}(\mathbb{1} - B_k) \succeq \mathfrak{F}(\mathbb{1} - B)$; multiplying by $\beta_k > 0$, summing, and then using $\beta_k \geq \gamma$ together with $\mathfrak{F}(\mathbb{1} - B) \succeq 0$ (Lemma 1.9),

$$\sum_{k=1}^{K} \beta_k \mathfrak{F}\big(\mathbb{1}-B_k\big) \;\succeq\; \Big(\sum_{k=1}^{K} \beta_k\Big)\mathfrak{F}(\mathbb{1}-B) \;\succeq\; K\gamma\,\mathfrak{F}(\mathbb{1}-B) \;\succeq\; (K-1)\gamma\,\mathfrak{F}(\mathbb{1}-B), \tag{7.4}$$

so (6.4) gives $M_{\mathrm{mc}} = \mathbb{1} + (K-1)\gamma\mathfrak{F}(\mathbb{1}-B) - \sum_k \beta_k \mathfrak{F}(\mathbb{1}-B_k) \preceq \mathbb{1}$. Every summand is photon-number block diagonal (Lemma 1.6 (i)), so the n-photon block of M_{mc} is its compression by Π_n and inherits the bound: $M_{\mathrm{mc}}^{(=n)} \preceq \mathbb{1}^{(=n)}$, whence $\beta^*(n) \le 1$.

Non-negativity. Under (7.3) the same compression inherits positivity, since $\langle\psi|\,\Pi_n M_{\mathrm{mc}} \Pi_n\,|\psi\rangle = \langle\Pi_n\psi|\,M_{\mathrm{mc}}\,|\Pi_n\psi\rangle \ge 0$ for every $|\psi\rangle$; so $M_{\mathrm{mc}}^{(=n)} \succeq 0$ and $\beta^*(n) \ge 0$. □

Data satisfying every hypothesis of the multiclick bound can display a multiclick element with exact block bound −3, so non-negativity of the exact block bound needs positivity as its own hypothesis. No receiver of this contract produces such an operator.

Remark 7.3 (Positivity of the multiclick element as a separate hypothesis). Hypothesis (7.3) does not follow from (6.3) and (6.4), and the failure is not delicate. Take $K = 4$, $\gamma = \beta_k = 1$, $B = \mathbb{1}$ and $B_k = 0$: then $0 \preceq B_k \preceq B \preceq \mathbb{1}$ and $\gamma \le \beta_k \le 1$, so every hypothesis of Theorem 6.2 holds, while the operator M_{mc} that the theorem forms from the operators (6.4) is $M_{\mathrm{mc}} = \mathbb{1} + 3\,\mathfrak{F}(0) - 4\,\mathfrak{F}(\mathbb{1})$, which on every block $n \ge 1$ is $-3\,\mathbb{1}^{(=n)}$ because $\mathfrak{F}(0)^{(n)} = 0$ there. So $\beta^*(n) = -3 < 0$.

Theorem 6.2 is untouched by the instance: the same data give $c = 1-\lambda_{\max}(B) = 0$ and $d_k = 1 - \lambda_{\min}(B_k) = 1$, hence $\beta(n) = 1 + 0 - 4 = -3$, and (6.6) holds with equality. A *lower* bound on $\lambda_{\min}$ needs no positivity; the statement $\beta^* \ge 0$ is about the operator itself and does. What the instance excludes is the operator, not the numbers: no receiver of this contract produces it, because there the left-hand sides of (6.4) are the joint no-click elements of a POVM and M_{mc} is a sum of that POVM's elements. Stating (7.3) explicitly, rather than reading it off the notation (6.4) uses, is what keeps Lemma 7.2 and Lemma 7.4 true as printed; Theorem 8.14 discharges it at each of the two instances, and no emitted value depends on the difference.

Uses Thm. 6.2, Lem. 7.2 · newly derived · accepted

Lower estimates of the exact block bound on finitely many blocks above the cutoff, combined with an admissible remainder for the blocks beyond them, give a weight in [0,1] that bounds the multiclick element on every block above the cutoff. Three admissible choices are named.

The lemma turns information about finitely many blocks into a bound uniform over all blocks above the cutoff: a lower estimate on each exact block in a window, and a remainder that covers every block beyond it. The three choices listed at its end are all this contract uses.

Lemma 7.4 (Exact window with an analytic remainder)**.** Let the hypotheses of Theorem 6.2 hold together with (7.3), let $N \ge 0$, and let $m^* > N$ be an integer. Let $\ell(n)$, $N < n \le m^*$, be reals with

$$\ell(n) \;\le\; \beta^*(n) \qquad \text{for } N < n \le m^*, \tag{7.5}$$

and let τ be a real with

$$\tau \le 0 \qquad \text{or} \qquad \tau \le \inf_{n > m^*} \beta^*(n). \tag{7.6}$$

Then

$$P \;:=\; \max\Big\{0,\; \min\Big\{\; \min_{N<n\le m^*} \ell(n),\;\; \tau \;\Big\}\Big\} \tag{7.7}$$

satisfies $0 \le P \le 1$ and

$$M_{\mathrm{mc}}^{(>N)} \;\succeq\; P\,\mathbb{1}^{(>N)}. \tag{7.8}$$

Three choices are admissible:

(i) $\tau = T_{m^*}$ of (6.18), formed from the bound data $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ that Theorem 6.2 forms by (6.5), satisfies (7.6) with no further hypothesis;

(ii) if the bound data satisfy (6.8), then $\tau = \beta(m^*+1)$ satisfies (7.6);

(iii) $\ell(n) = \beta(n)$ satisfies (7.5), and the pointwise maximum of finitely many admissible choices of ℓ is admissible.

Uses Thm. 6.2, Lem. 6.5, Lem. 6.8, Lem. 7.2 · *Used by* Rem. 7.5, Def. 8.7, Rem. 8.8, Rem. 8.11, Thm. 8.13, Thm. 8.14, Rem. 8.18, Thm. 10.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:exact_composite

Proof. Step 1 (range of P). $0 \le P$ is the outer maximum. For $P \le 1$: either $P = 0$, or $P \le \ell(N+1) \le \beta^*(N+1) \le 1$ by (7.5) and (7.2), which needs no more than the hypotheses of Theorem 6.2, the window being non-empty because $m^* > N$.

Step 2 (the bound (7.8)). If $P = 0$ the assertion is $M_{\mathrm{mc}}^{(>N)} \succeq 0$, and this is the one step that uses (7.3): that hypothesis gives $M_{\mathrm{mc}}^{(=n)} \succeq 0$ on every block (Lemma 7.2), hence on their direct sum. So let $P > 0$. Then $P = \min\{\min_{N<n\le m^*} \ell(n),\; \tau\}$, and in particular $\tau \ge P > 0$, so the first disjunct of (7.6) fails and the second holds. Take any $n > N$. If $n \le m^*$ then $\beta^*(n) \ge \ell(n) \ge P$ by (7.5); if $n > m^*$ then $\beta^*(n) \ge \inf_{j>m^*} \beta^*(j) \ge \tau \ge P$. Either way $\lambda_{\min}(M_{\mathrm{mc}}^{(=n)}) \ge P$, that is $M_{\mathrm{mc}}^{(=n)} \succeq P\,\mathbb{1}^{(=n)}$; the blocks with $n > N$ are orthogonal direct summands, and a lower bound holding on each holds on the direct sum.

Step 3 (choice (i)). For $n > m^*$, $d_k \le 1$ gives $d_k^n \le d_k^{m^*}$ and $\gamma c^n \ge 0$, so $\beta(n) \ge 1 - \sum_k \beta_k d_k^n \ge 1 - \sum_k \beta_k d_k^{m^*} = T_{m^*}$, the tail step of Lemma 6.8 (i); and $\beta^*(n) \ge \beta(n)$ by Lemma 7.2. Hence $\inf_{n>m^*} \beta^*(n) \ge T_{m^*}$.

Step 4 (choice (ii)). If $\beta(m^*+1) \le 0$, the first disjunct of (7.6) holds. Otherwise $\beta(m^*+1) > 0$ and Lemma 6.5, applied with cutoff m^* in place of N, gives $\inf_{n>m^*} \beta(n) = \beta(m^*+1)$; with $\beta^* \ge \beta$ pointwise this is $\inf_{n>m^*} \beta^*(n) \ge \beta(m^*+1)$, the second disjunct.

Step 5 (choice (iii)). The first statement is Lemma 7.2. For the second, a maximum of finitely many numbers each at most $\beta^*(n)$ is at most $\beta^*(n)$. □

The exact-window lemma makes finitely many blocks yield a bound uniform above the cutoff, from remainders already proved, and uses positivity at one step only. It is stated on the exact block bound and makes no computed eigenvalue available by itself.

A squashing request fails when a bound does not hold at every photon number above the cutoff (`spec/architecture.md` §5.4), so a bound on finitely many blocks is of no use there by itself.

Remark 7.5 (Scope of the exact-window lemma). *What the lemma buys.* Lemma 7.4 converts information about *finitely many* blocks into a bound uniform over the infinitely many above the cutoff, which is what a squashing request requires and what Remark 8.18 names as the obstacle to using block eigenvalues at all. The conversion needs no new analysis: both admissible remainders are results already proved, (i) being the tail step inside Lemma 6.8 (i) and (ii) being Lemma 6.5 read at a larger cutoff. Some hypotheses are *not* used: (i) is free of (6.8), neither part needs predicate (X), which enters only in choosing between them, and (7.3) is used at exactly one step, the $P = 0$ branch that the outer $\max\{0, \cdot\}$ of (7.7) creates.

What it does not do. The lemma does not make an exact eigenvalue available. It is stated on β^*, an exact quantity, and (7.5) is a hypothesis on whatever numbers are supplied for it. Definition 8.7 is where those numbers come from, and Remark 8.11 is where the price of computing them is stated.

Uses Lem. 6.5, Lem. 6.8, Lem. 7.2, Lem. 7.4 · newly derived · accepted

8 Setting families and the emitted weight

A setting family packages, for each declared setting, a photon-number block-diagonal POVM whose joint no-click elements have the prefactor form of the multiclick bound; a probe-free receiver is the one-setting family, and a submodule that constructs another family proves there that it is one. Where a depth is declared, a setting's weight is an exact window; elsewhere it is the collapse value where the declaration marks the setting exact and a declared window and tail otherwise; their minimum is the uniform weight. Formed on bounds that lie on the conservative side of the exact data, the same formulas give the enclosed weight, a bound with no assumption (Theorem 8.13). The first main result, Theorem 8.14, is that each per-setting weight bounds its setting's multiclick element on every block above the cutoff, and the uniform weight bounds it at every setting, under the eigenvalue-slack assumption only where a computed block eigenvalue enters. The second, Proposition 8.17, is that the strengthening slot in which the artifact carries the weights has weights that are non-negative and sum to at most one at every setting, as a squashing request requires.

A setting family gives, per setting, a photon-number block-diagonal POVM whose joint no-click elements have the prefactor form of the multiclick bound, and a flag that reads exact only where every prefactor is one by construction. The high-photon weights are stated on it.

The weights built on the multiclick bound (Theorem 6.2 onwards) and the residuals of Definition 10.1 are stated per *setting*: a receiver without a probe has one setting, a receiver with a probe has one per declared probe setting, and in both cases the object they are stated on is the same. This definition names that object once, so that the emitted value and its guarantee are proved for it and each instance is established separately. The core constructs the unconditioned receiver as an instance (Corollary 8.2, one setting); a submodule that constructs another, such as a receiver conditioned on its own probe with one setting per declared probe setting, proves there that it is one.

Definition 8.1 (Setting family)**.** Let $D \geq 1$ and write $\mathcal{F}_D \coloneqq \mathcal{F}(\mathbb{C}^D)$ for the Fock space of Definition 1.2 over a D-dimensional *channel input space*, with photon-number projectors Π_n onto $\mathrm{Sym}^n(\mathbb{C}^D)$. A *setting family* of $K \geq 1$ ports on $\mathcal{F}_D$ is a finite non-empty set $\mathcal{T}_{\mathrm{set}}$ of *settings* together with, for every $t \in \mathcal{T}_{\mathrm{set}}$:

(a) a POVM $\{M_{S|t}\}_{S\subseteq[K]}$ on $\mathcal{F}_D$ every element of which commutes with the photon number, so that $M_{S|t} = \bigoplus_{n\geq 0} M_{S|t}^{(=n)}$ with $M_{S|t}^{(=n)}$ acting on $\mathrm{Sym}^n(\mathbb{C}^D)$. Its *joint no-click elements* and its *multiclick element* are

$$\Pi_{\mathrm{vac}}(J)^{(t)} \coloneqq \sum_{S\subseteq[K]\setminus J} M_{S|t} \quad (J \subseteq [K]), \qquad M_{\mathrm{mc}|t} \coloneqq \sum_{S\in\mathsf{MC}} M_{S|t}, \tag{8.1}$$

with MC the multiclick set of Definition 4.4;

(b) Hermitian operators $B^{(t)}, B_1^{(t)}, \ldots, B_K^{(t)}$ on $\mathbb{C}^D$ and scalars $\gamma^{(t)}, \beta_1^{(t)}, \ldots, \beta_K^{(t)} \in (0,1]$ satisfying the hypotheses of Theorem 6.2, namely (6.3) and $\gamma^{(t)} \leq \beta_k^{(t)}$ for every k, such that the joint no-click elements have the prefactor form (6.4),

$$\Pi_{\mathrm{vac}}([K])^{(t)} = \gamma^{(t)}\,\mathfrak{F}\big(\mathbb{1}-B^{(t)}\big), \qquad \Pi_{\mathrm{vac}}([K]\setminus\{k\})^{(t)} = \beta_k^{(t)}\,\mathfrak{F}\big(\mathbb{1}-B_k^{(t)}\big); \tag{8.2}$$

(c) a flag $\chi^{(t)} \in \{\mathrm{exact}, \mathrm{general}\}$, the setting's *exact flag*, which may be exact only if $\gamma^{(t)} = \beta_k^{(t)} = 1$ for every k holds *exactly, by the construction that produced the setting*, read off the declaration with no arithmetic performed, so that at an exact setting (6.8) reads $K \geq K$ and holds with equality by construction.

The *bound data of the setting* are $\big(\gamma^{(t)}, \{\beta_k^{(t)}\}, c^{(t)}, \{d_k^{(t)}\}\big)$ with $c^{(t)} \coloneqq 1 - \lambda_{\max}(B^{(t)})$ and $d_k^{(t)} \coloneqq 1 - \lambda_{\min}(B_k^{(t)})$ as in (6.5); they are admissible bound

data (Definition 6.1) by Theorem 6.2, and $\beta^{(t)}$ denotes their bound function (6.2).

Two consequences are used throughout. First, the multiclick element has the inclusion–exclusion form

$$M_{\mathrm{mc}|t} = \mathbb{1} + (K-1)\,\Pi_{\mathrm{vac}}([K])^{(t)} - \sum_{k=1}^{K} \Pi_{\mathrm{vac}}([K]\setminus\{k\})^{(t)}, \tag{8.3}$$

because completeness gives $M_{\mathrm{mc}|t} = \mathbb{1} - M_{\varnothing|t} - \sum_k M_{\{k\}|t}$ while (8.1) gives $M_{\varnothing|t} = \Pi_{\mathrm{vac}}([K])^{(t)}$ and $M_{\{k\}|t} = \Pi_{\mathrm{vac}}([K]\setminus\{k\})^{(t)} - \Pi_{\mathrm{vac}}([K])^{(t)}$; so Theorem 6.2 applies to every setting, with $\lambda_{\min}(M_{\mathrm{mc}|t}^{(=n)}) \geq \beta^{(t)}(n)$ for every $n \geq 0$. Second, $M_{\mathrm{mc}|t}$ is a sum of elements of a POVM, so hypothesis (7.3) of Lemma 7.2, $M_{\mathrm{mc}|t} \succeq 0$, holds at every setting.

Uses Def. 1.2, Def. 4.4, Def. 6.1, Thm. 6.2, Lem. 6.5, Lem. 7.2 · *Used by* Cor. 8.2, Def. 8.3, Rem. 8.5, Rem. 8.6, Def. 8.7, Def. 8.12, Thm. 8.13, Thm. 8.14, Def. 8.16, Prop. 8.17, Def. 10.1, Prop. 10.2, Def. 10.4, Def. 10.9, Thm. 10.10, Req. 13.5, Thm. 19.2, Prop. 19.4, Prop. 20.3, reductions/req:recv-row-provenance, reductions/wpfss/prop:route, reductions/wpfss/rem:receiver-imports · newly derived · accepted

A receiver without probe modes is a one-setting family whose prefactors are all exactly one, with bound data read off its port Grams. Its flag is exact by construction, so the collapse branch applies to it.

Corollary 8.2 (The unconditioned receiver is a setting family)**.** Let $\mathfrak{R}$ be a receiver instrument (Definition 2.1) with $\mathbb{I}_P = \varnothing$, so that $\mathbb{I} = \mathbb{I}_C$, and let $\{A_k\}$ be its port Grams and $\{M_S\}$ its click-pattern POVM (Theorem 4.5). Then $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$ with

$$M_{S|\mathrm{off}} \coloneqq M_S, \qquad B^{(\mathrm{off})} \coloneqq A_\Sigma, \qquad B_k^{(\mathrm{off})} \coloneqq A_{[K]\setminus\{k\}}, \qquad \gamma^{(\mathrm{off})} = \beta_k^{(\mathrm{off})} \coloneqq 1, \qquad \chi^{(\mathrm{off})} \coloneqq \mathrm{exact}, \tag{8.4}$$

is a setting family (Definition 8.1) on $\mathcal{F}(\mathbb{C}^{\mathbb{I}})$, with $D = |\mathbb{I}|$. Its bound data are $c = 1 - \lambda_{\max}(A_\Sigma)$ and $d_k = 1 - \lambda_{\min}(A_{[K]\setminus\{k\}})$, and (6.8) holds with equality.

Uses Def. 2.1, Def. 2.2, Thm. 4.1, Thm. 4.5, Thm. 6.2, Lem. 6.5, Def. 8.1 · *Assumes* A-TRUST, A-PASSIVE, A-VAC, A-MODES, A-THRESH, A-PORT, A-EXACT, A-NOSEL, A-ROUND · *Used by* Rem. 6.7, Rem. 8.5, Cor. 8.15, Def. 10.1, Rem. 10.5, Thm. 10.10, Req. 13.5, reductions/req:recv-row-provenance, reductions/wpfss/prop:route, reductions/wpfss/rem:receiver-imports · newly derived · accepted

Proof. The dimension is $D = |\mathbb{I}| = |\mathbb{I}_C| \geq 1$, since $\mathbb{I}_C \neq \varnothing$ by Definition 2.1 (ii), as Definition 8.1 requires. We check the three items of Definition 8.1.

(a) $\{M_S\}$ is a POVM by Theorem 4.5 (i), block diagonal in the photon number of $\mathbb{I} = \mathbb{I}_C$ by Theorem 4.5 (iii), and by Theorem 4.5 (i) its joint no-click elements in the sense of (8.1) are $\Pi_{\mathrm{vac}}(J) = \sum_{S\subseteq[K]\setminus J} M_S$.

(b) Theorem 4.5 (ii) gives $\Pi_{\mathrm{vac}}(J) = \mathfrak{F}(\mathbb{1} - A_J)$ for every J, which is (8.2) with $\gamma = \beta_k = 1$, $B = A_\Sigma$ and $B_k = A_{[K]\setminus\{k\}}$. Hypothesis (6.3) is $0 \preceq A_{[K]\setminus\{k\}} \preceq A_\Sigma \preceq \mathbb{1}$: the first two inequalities because every $A_k \succeq 0$ and $A_\Sigma - A_{[K]\setminus\{k\}} = A_k$ (Theorem 4.1 (i), Definition 2.2), the last by Theorem 4.1 (i). $\gamma \le \beta_k$ is $1 \le 1$.

(c) $\gamma = \beta_k = 1$ are identities of Theorem 4.5 (ii), read off the single declared fact $\mathbb{I}_P = \varnothing$ with no arithmetic performed, so the flag exact is admissible; at $\gamma = \beta_k = 1$ hypothesis (6.8) of Lemma 6.5 reads $K \ge K$, which is equality. The bound data are (6.5) at these operators. □

At a setting without an exact-window depth, the per-setting weight is the collapse value where the flag is exact and a declared window with its tail otherwise. The uniform weight is the minimum of the per-setting weights over all settings.

Each setting without an exact-window depth gets its own weight, by the collapse of Lemma 6.5 where the setting's flag guarantees its hypothesis and by the window and tail of Lemma 6.8 otherwise. Remark 8.5 says where the flag is set at the two instances this module constructs.

Definition 8.3 (The uniform high-photon multiclick weight)**.** Fix a cutoff N and a setting family (Definition 8.1); for each setting $t \in \mathcal{T}_{\mathrm{set}}$ let $\beta^{(t)}$ be the bound function (6.2) of the bound data of the setting. At every setting t whose declaration carries *no* exact-window depth (Definition 8.7), the *per-setting weight* is

$$p_{\mathrm{mc}}^{(t)} := \begin{cases} \max\{0,\ \beta^{(t)}(N+1)\} & \text{if (X) holds at } t, \\ \max\{0,\ \underline{\beta}^{(t)}_{N,m_t}\} & \text{otherwise, for the declared window } m_t > N, \end{cases} \tag{8.5}$$

and the *uniform weight* is $p_{\mathrm{mc}} := \min_{t\in\mathcal{T}_{\mathrm{set}}} p_{\mathrm{mc}}^{(t)}$, where the branch-1 *predicate (X)* is

(X) ***exact by construction, decidable from the declaration:*** the setting's exact flag is $\chi^{(t)} = \mathrm{exact}$ (Definition 8.1 (c)). Then $\gamma^{(t)} = \beta_k^{(t)} = 1$ exactly, so (6.8) reads $K \ge K$ and holds with *equality*, no arithmetic having been performed.

Where the exact window replaces this. At a setting whose declaration carries an exact-window depth m_t^*, (8.5) does not apply and $p_{\mathrm{mc}}^{(t)}$ is (8.7) instead (Definition 8.7); $p_{\mathrm{mc}} = \min_{t\in\mathcal{T}_{\mathrm{set}}} p_{\mathrm{mc}}^{(t)}$ is unchanged, and the two kinds of setting may be mixed in one declaration. A declaration carrying no depth at any setting is governed by (8.5) alone: no β^* is formed, no eigenvalue of any block above the cutoff is computed, and Assumption 11.11 is not invoked.

Uses Def. 6.1, Lem. 6.5, Lem. 6.8, Def. 8.1 · *Used by* Rem. 8.4, Rem. 8.5, Rem. 8.6, Def. 8.7, Rem. 8.8, Prop. 8.9, Rem. 8.10, Def. 8.12, Thm. 8.13, Thm. 8.14, Cor. 8.15, Def. 10.4, Thm. 10.10, Req. 13.1, Req. 13.2, Req. 13.5, Req. 14.2, Ex. 17.2, Ex. 17.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:SettingBound

The collapse branch is taken only where predicate (X) holds, because it can overstate the bound where the collapse hypothesis fails, while the window branch is valid always. A computed margin test would never fire, since the collapse slack is never positive.

Remark 8.4 (Choice of the collapse predicate). *Why the predicate is exactly (X).* Branch 2 of (8.5) is valid for *every* admissible bound data with no hypothesis (Lemma 6.8 (i)), whereas branch 1 needs (6.8) and can, where that fails, return a value strictly above $\inf_{n>N} \beta^{(t)}(n)$: an over-large lower bound, which is the unsafe direction. An instance on admissible bound data is $K = 4$, $\gamma^{(t)} = \beta_k^{(t)} = \frac{1}{100}$ and $d_k^{(t)} = 1$ for every k, and $0 < c^{(t)} < 1$, for which $\beta^{(t)}(n) = \frac{96}{100} + \frac{3}{100}\,(c^{(t)})^n$ is positive and strictly decreasing in n, so that branch 1 returns $\beta^{(t)}(N+1) > \frac{96}{100} = \inf_{n>N} \beta^{(t)}(n)$. Predicate (X) is decided on the declaration, before any arithmetic, and admits the equality case, which is the generic one: (6.8) is satisfied with slack exactly zero by every probe-free receiver and at every probe-off setting, which is precisely the configuration Lemma 6.5 was proved for. (X) is *sufficient, not necessary*: a receiver whose slack happens to be zero for some other reason, such as a probe that is on but decoupled from every port, where $\Gamma^{1/2} A_{PP} \Gamma^{1/2} = 0$ makes $\gamma^{(t)} = \beta_k^{(t)} = 1$ as well, fails (X) and is served by branch 2. That is conservative, and Lemma 6.8 (i) makes it sound.

Why there is no computed companion to (X). A test of the form "take branch 1 when the evaluated slack $\sum_k \beta_k^{(t)} - 1 - (K-1)\gamma^{(t)}$ clears a declared margin" would never fire, and Remark 8.6 proves it: that slack equals $-\beta^{(t)}(0)$, which is ≤ 0 for every receiver, at every setting, at every parameter value. A margin test on it could therefore be met only by floating-point noise, which is the one case a margin exists to exclude. There is no such predicate here, and there is no declared margin among the inputs.

Uses Lem. 6.5, Lem. 6.8, Def. 8.3, Rem. 8.6 · newly derived · accepted

The construction of each setting sets its exact flag; at the one setting of a probe-free receiver it is exact. Every setting whose flag is general takes the general branch of whichever weight definition governs it, read off the declaration alone.

Remark 8.5 (Where the exact flag is set, and what a general flag selects). Predicate (X) of Definition 8.3 is the setting's exact flag (Definition 8.1 (c)), and the flag is set by the construction that produced the setting. At the instance the core constructs, the one setting of a probe-free receiver, it is set (Corollary 8.2); a submodule that constructs a setting family states at which of its settings it is set. At every setting whose flag is general the general branch of whichever definition governs the setting is taken: branch 2 of (8.5) where the declaration carries no exact-window depth, and the second branch of (8.7) where it carries one (Definition 8.7). Which of the two flag values applies is read off the declaration alone, before any arithmetic.

Uses Def. 8.1, Cor. 8.2, Def. 8.3, Def. 8.7 · newly derived · accepted

At every setting of a setting family the collapse slack is $-\beta(0)$, where $\beta(0)$ equals the vacuum block of the multiclick element, so it is never positive and the collapse hypothesis holds only with equality. A computed margin test for branch 1 could therefore never fire.

Remark 8.6 (Sign of the collapse slack). *The slack is the negative vacuum block.* Fix a setting t and write $\sigma \coloneqq \sum_k \beta_k^{(t)} - 1 - (K-1)\gamma^{(t)}$ for the slack in (6.8). Evaluating (6.2) at $n = 0$ under the convention $0^0 = 1$ gives $\beta^{(t)}(0) = 1 + (K-1)\gamma^{(t)} - \sum_k \beta_k^{(t)} = -\sigma$, so the two are negatives of each other. For a setting of a setting family (Definition 8.1) the vacuum block of the multiclick element is $M_{\mathrm{mc}|t}^{((=0))} = \beta^{(t)}(0)$ exactly: that block is one dimensional and every $\mathfrak{F}(\cdot)^{(0)} = 1$, so (8.2) reads $\Pi_{\mathrm{vac}}([K])^{((=0))} = \gamma^{(t)}$ and $\Pi_{\mathrm{vac}}([K] \setminus \{k\})^{((=0))} = \beta_k^{(t)}$, and the $n = 0$ block of (8.3) is $1 + (K-1)\gamma^{(t)} - \sum_k \beta_k^{(t)} = \beta^{(t)}(0)$. That block is a diagonal block of a positive operator (Definition 8.1 (a)), hence a non-negative real. So $\beta^{(t)}(0) \geq 0$ and $\sigma \leq 0$ always, and (6.8) holds *if and only if* $\beta^{(t)}(0) = 0$, that is $\sigma = 0$ exactly.

No computed margin test can select branch 1. A test demanding $\sigma \geq \varepsilon > 0$ is therefore unsatisfiable, and a test demanding $\sigma \geq 0$ is satisfied only in the exact-equality case, which is what (X) already guarantees from the declaration without computing anything. So branch 1 of (8.5) is taken only through (X), and it is by design that no arithmetic decides it. The declared $\varepsilon_{\mathrm{eig}}$ of Definition 8.7 is not a margin of this kind and decides no predicate: it is a slack subtracted from a computed eigenvalue, and the exact window is selected by the presence of a declared depth, which is again a property of the declaration.

Nothing is lost. Branch 2 is unconditionally valid (Lemma 6.8 (i)). At a setting where (6.8) holds but (X) does not, it recovers branch 1's value in the limit of a widening window whenever $c^{(t)} < 1$: by Lemma 6.8 (ii) its value $\max\{0, \underline{\beta}_{N,m}^{(t)}\}$ tends to $\max\{0, \inf_{n>N} \beta^{(t)}(n)\}$, which is $\max\{0, \beta^{(t)}(N+1)\}$ by (6.10).

Uses Def. 6.1, Lem. 6.5, Lem. 6.8, Def. 8.1, Def. 8.3 · *Used by* Rem. 8.4 · newly derived · accepted

At a setting with a declared depth, the weight uses on each block above the cutoff up to the depth the larger of the analytic bound and the computed block eigenvalue less the eigenvalue slack, and keeps the setting's analytic remainder above. A depth changes only its own setting's weight.

The analytic weight of Definition 8.3 bounds three operators separately and so loses slack (Remark 8.18). Where a declaration asks for it, the exact window uses, on finitely many blocks above the cutoff, the larger of the analytic value and a computed block eigenvalue less a declared slack, and keeps the setting's analytic remainder above those blocks; Lemma 7.4 makes the combination a bound under Assumption 11.11, and Remark 8.8 explains the remainder each branch keeps. What a declaration must supply, and which depths are refused, is Requirement 13.2.

Definition 8.7 (Exact-window weight of a setting with a declared depth)**.** Let $t \in \mathcal{T}_{\mathrm{set}}$ be a setting whose declaration carries an *exact-window depth* m_t^*: an integer with $m_t^* > N$ and, where predicate (X) fails at t, also $m_t^* \leq m_t$. Write $\widehat{\beta}^{*(t)}(n)$ for the value an implementation computes for $\beta^{*(t)}(n)$ of (7.1)

at that setting, let $\varepsilon_{\mathrm{eig}}$ be the declared *eigenvalue slack*, and put

$$\ell^{(t)}(n) \;:=\; \max\big\{\; \beta^{(t)}(n),\;\; \widehat{\beta}^{*(t)}(n) - \varepsilon_{\mathrm{eig}} \;\big\}, \qquad N < n \le m_t^*. \tag{8.6}$$

Then

$$p_{\mathrm{mc}}^{(t)} := \begin{cases} \max\Big\{0,\; \min\big\{\; \min\limits_{N<n\le m_t^*} \ell^{(t)}(n),\;\; \beta^{(t)}\big(m_t^*+1\big) \;\big\}\Big\} & \text{if (X) holds at } t, \\ \max\Big\{0,\; \min\big\{\; \min\limits_{N<n\le m_t^*} \ell^{(t)}(n),\;\; \min\limits_{m_t^*<n\le m_t} \beta^{(t)}(n),\;\; T_{m_t}^{(t)} \;\big\}\Big\} & \text{otherwise,} \end{cases} \tag{8.7}$$

the middle minimum in the second case running over the empty index set — and so equal to $+\infty$ and inactive — when $m_t^* = m_t$.

Per setting. The depth is optional and per setting, and each per-setting weight $p_{\mathrm{mc}}^{(t)}$ is its own setting's: a depth changes the weight of the setting that declares it and of no other. A declaration that carries a depth at some settings and not at others therefore obtains the exact-window value as the weight of each setting with one, while the uniform weight $p_{\mathrm{mc}} = \min_{t\in\mathcal{T}_{\mathrm{set}}} p_{\mathrm{mc}}^{(t)}$ is bounded by the settings without one.

Uses Lem. 6.5, Lem. 6.8, Def. 7.1, Lem. 7.4, Def. 8.1, Def. 8.3 · *Used by* Rem. 8.5, Rem. 8.8, Prop. 8.9, Rem. 8.11, Def. 8.12, Thm. 8.13, Thm. 8.14, Cor. 8.15, Rem. 8.18, Def. 10.1, Def. 10.4, Rem. 10.6, Thm. 10.10, Rem. 10.12, Ass. 11.11, Req. 13.2, Req. 13.5, Ex. 17.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:ExactWindowRecord

The exact window covers finitely many blocks, so each branch keeps an admissible remainder above it: branch 1 the bound function one past the depth, branch 2 its own analytic window and tail. Neither remainder is new or weakened.

Remark 8.8 (The analytic remainder of each exact-window branch). The exact window of Definition 8.7 covers $N < n \le m_t^*$ and nothing above it, so a remainder admissible in the sense of (7.6) is still required, and each branch of (8.7) already has one. Branch 1 has (6.8) exactly, by predicate (X) of Definition 8.3, and takes $\beta^{(t)}(m_t^*+1)$ by Lemma 7.4 (ii). Branch 2 has no hypothesis available and keeps its own analytic window and tail (6.18) above the exact one; this is why $m_t^* \le m_t$ is required there, so that the two windows compose into the single window $N < n \le m_t$ to which Lemma 7.4 is applied. Neither remainder is new and neither is weakened.

Uses Lem. 6.5, Lem. 6.8, Lem. 7.4, Def. 8.3, Def. 8.7 · newly derived · accepted

At every setting with a declared depth, the exact-window weight is at least the analytic weight on the same bound data. The comparison is between exact expressions; Remark 8.10 says which of its steps survive an evaluation.

Proposition 8.9 (Dominance of the exact-window weight over the analytic weight)**.** At every setting carrying a declared exact-window depth, the value (8.7) is at least the value (8.5) gives at that setting on the same bound data.

Uses Lem. 6.5, Lem. 6.8, Def. 8.3, Def. 8.7 · *Used by* Rem. 8.10, Rem. 8.11 · newly derived · accepted

Proof. Write P^* for (8.7) and P for (8.5) and drop the superscript (t). From the maximum in (8.6), $\ell(n) \geq \beta(n)$ for every n in the exact window.

Branch 2. By (6.18), $P = \max\{0, \min\{\min_{N<n\leq m_t} \beta(n),\, T_{m_t}\}\}$. Splitting the window at m_t^*, $\min_{N<n\leq m_t} \beta(n) = \min\{\min_{N<n\leq m_t^*} \beta(n),\, \min_{m_t^*<n\leq m_t} \beta(n)\}$, and (8.7) is that same expression with the first of the two replaced by $\min_{N<n\leq m_t^*} \ell(n)$, which is at least it. A minimum is non-decreasing in each argument and $\max\{0, \cdot\}$ is non-decreasing, so $P^* \geq P$.

Branch 1. $P = \max\{0, \beta(N+1)\}$. If $\beta(N+1) < 0$ then $P = 0 \leq P^*$. If $\beta(N+1) \geq 0$ then, predicate (X) making (6.8) hold with equality, Lemma 6.5 makes β non-decreasing on $\{N+1, N+2, \dots\}$; hence $\ell(n) \geq \beta(n) \geq \beta(N+1)$ for $N < n \leq m_t^*$, and $\beta(m_t^*+1) \geq \beta(N+1)$. Both arguments of the inner minimum in (8.7) are therefore at least $\beta(N+1)$, so $P^* \geq \max\{0, \beta(N+1)\} = P$. □

Two steps of Proposition 8.9 compare evaluated numbers and survive an evaluation; branch 1's use of the collapse lemma does not. There the exact-window weight can sit below the analytic one by evaluation errors of the bound function between the first block above the cutoff and one past the depth.

Remark 8.10 (The exact-window comparison under floating-point evaluation). Proposition 8.9 compares two *expressions* in exact arithmetic. Two of its three steps survive an evaluation term for term, because they compare evaluated numbers with each other: $\ell(n) \geq \beta(n)$ is an evaluated maximum, and branch 2 minimises over exactly the evaluated $\beta(n)$ that (8.5) minimises over. The step that does not is branch 1's use of Lemma 6.5, $\beta(n) \geq \beta(N+1)$ for $N+1 < n \leq m_t^*+1$: a statement about exact values, and those further $\beta(n)$ are quantities (8.5) never evaluates, whose evaluated values can fall below the evaluated $\beta(N+1)$. So on branch 1 the emitted value can sit below the one (8.5) would produce by the evaluation errors of those further values together with that of $\beta(N+1)$, and this contract claims no bound on either (Theorem 10.10 (ii)). Unaffected either way are the result that underwrites the emitted number (Theorem 8.14) and the fact that a declaration carrying no depth performs none of this arithmetic.

Uses Lem. 6.5, Def. 8.3, Prop. 8.9 · newly derived · accepted

In the exact-window maximum the analytic entry is a proven lower bound, while the computed eigenvalue less the slack is one only under A-EIG; the rigorous evaluation replaces it by a proven bound. No test catches an overshoot, r_6 detects only an undershoot, and each weight stays in [0, 1].

The exact window trades the analytic bound for a sharper value on finitely many blocks, and in the floating-point evaluation the price is one assumption on the computed eigenvalues. The rigorous evaluation pays no such price: it puts a rigorous lower bound on the block eigenvalue in the computed value's place (Definition 8.12, Requirement 13.5), and the weight is then a bound by Theorem 8.13.

Remark 8.11 (The eigenvalue-slack assumption behind the exact window). (8.6) is a maximum of two lower bounds on $\beta^{*(t)}(n)$, and they are of two different *kinds*. The first, $\beta^{(t)}(n) \leq \beta^{*(t)}(n)$, is Theorem 6.2: an inequality between exact quantities,

holding with no hypothesis about any computation. The second, $\widehat{\beta}^{*(t)}(n) - \varepsilon_{\text{eig}} \leq \beta^{*(t)}(n)$, is *not a theorem of this contract*: the evaluation model of Definition 10.9 asserts no bound on the difference between a quantity computed in floating point and the exact one, and $\widehat{\beta}^{*}$ is the output of an extreme-eigenvalue computation on a matrix of dimension $\binom{|\mathbb{I}_C|+n-1}{n}$. It is Assumption 11.11, and that assumption is invoked *only* at the settings and indices where a depth is declared and the weight is that of Definition 8.7. What is a theorem is the same maximum with a rigorous window bound in the second entry: in the enclosed weight of Definition 8.12 the entry is a real proved to lie below $\beta^{*(t)}(n)$, not an evaluation of it, no slack is subtracted, and Theorem 8.13 makes the weight a bound with no assumption. Everything below concerns the floating-point evaluation.

The direction. A computed eigenvalue landing *below* the exact one costs tightness and nothing else, and (8.6) floors the value at $\beta^{(t)}(n)$, which is sound with no assumption at all; so that direction cannot take the emitted weight below what the analytic evaluation emits (Proposition 8.9). A computed eigenvalue landing *above* the exact one by more than ε_{eig} can make the emitted weight $p_{\text{mc}}^{(t)}$ of that setting exceed $\inf_{n>N} \lambda_{\min}(M_{\text{mc}|t}^{(=n)})$: an over-large lower bound, the unsafe direction, and one that reaches a key length through Definition 8.16. No range check catches it, whatever the size of the overshoot. Every branch of (8.7) is capped by an analytic remainder that is at most 1 and does not depend on $\widehat{\beta}^{*(t)}$, so $p_{\text{mc}}^{(t)}$ stays in $[0, 1]$ (the *Range* paragraph of this remark), and neither the "$p_y < 0$" nor the "$\sum_y p_y > 1$" failure condition that Proposition 8.17 names can fire on it. An overshoot of any size is caught by nothing. That asymmetry is why Assumption 11.11 is stated one-sidedly, on the side that matters, and why ε_{eig} is *subtracted* rather than used as a comparison threshold: it decides nothing, it lowers a number.

A slack, not a repair. The computed $\widehat{\beta}^{*(t)}(n)$ enters (8.6) unmodified, and ε_{eig} enters only as a subtracted declared slack, so that (8.6) can be recomputed from the recorded values (Requirement 13.3). This is the distinction Remark 19.6 draws for $\bar{r}$ and Definition 10.4 (iv) draws for c: a coefficient or a slack inside a stated inequality is not a repair of a datum.

What r_6 tests, and the direction it is blind to. Since $\beta(n) \leq \beta^{*}(n)$ is a theorem (Lemma 7.2), the window consistency defect r_6 of (10.6) has exact value 0, and it is a residual in exactly the sense of the other five: what is emitted is its floating-point evaluation, and a value above ε_6 refuses. It catches an evaluation that contradicts an inequality this contract proves — a block formed at the wrong index, a wrongly ordered occupation basis, a conditioned family taken at the wrong setting, an iteration that has not converged — which is the failure mode most likely to be gross. It does *not* verify Assumption 11.11: r_6 fires when $\widehat{\beta}^{*}$ lands too *low*, and Assumption 11.11 constrains it from landing too *high*. No test in this contract bounds the unsafe direction; the assumption carries it.

Range. $0 \leq p_{\text{mc}}^{(t)} \leq 1$ holds exactly, whatever the computed values $\widehat{\beta}^{*(t)}(n)$ and whether or not Assumption 11.11 holds: the outer maximum of (8.7) gives $p_{\text{mc}}^{(t)} \geq 0$, and the inner minimum of each branch contains an analytic remainder at most 1 — $\beta^{(t)}(m_t^{*} + 1) \leq 1$ on branch 1 by Theorem 6.2, and the tail value $T_{m_t}^{(t)} = 1 - \sum_k \beta_k^{(t)} \big(d_k^{(t)}\big)^{m_t} \leq 1$ of (6.18) on branch 2 because $\beta_k^{(t)}, d_k^{(t)} \geq 0$ by (6.1). Lemma 7.4 gives the same range only through (7.5), which at a setting carrying a depth is Assumption 11.11. So Theorem 8.14's range statement and Proposition 8.17 hold unchanged; their floating-point evaluation is governed by

Theorem 10.10 (ii) here as everywhere else, and no computed value is clamped into $[0, 1]$ to make them look true.

Uses Def. 6.1, Thm. 6.2, Lem. 6.8, Lem. 7.2, Lem. 7.4, Def. 8.7, Prop. 8.9 · *Assumes* A-EIG · *Used by* Rem. 10.6, Req. 13.3 · newly derived · accepted

Enclosed bound data take c and γ from below and each d and β from above, within $[0, 1]$, and rigorous window bounds lie below the exact block eigenvalues. The enclosed weight is the per-setting weight formed on them, with no computed eigenvalue and no eigenvalue slack.

The weights of Definitions 8.3 and 8.7 are formulas in the bound data of a setting and, where a depth is declared, in computed block eigenvalues. The same formulas, evaluated on numbers known to lie on the conservative side of the exact data, give a weight that rests on no premise about how it was computed (Theorem 8.13). The four directions are those Remark 10.12 reads off the bound function, and in the window a rigorous lower bound on the exact block eigenvalue takes the place of the computed eigenvalue less the slack, which is why the slack does not enter (Remark 8.11).

Definition 8.12 (The weight evaluated on enclosed bound data)**.** Fix a cutoff N, a setting family (Definition 8.1) and a setting $t \in \mathcal{T}_{\mathrm{set}}$, and write $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$ for the bound data of the setting, the index t omitted.

(i) Reals $\big(\check\gamma, \{\check\beta_k\}, \check c, \{\check d_k\}\big)$ are *enclosed bound data* of t when

$$0 \le \check\gamma \le \gamma, \qquad \beta_k \le \check\beta_k \le 1, \qquad 0 \le \check c \le c, \qquad d_k \le \check d_k \le 1 \qquad (k = 1, \dots, K). \tag{8.8}$$

$\check\beta$ denotes the bound function (6.2) of this tuple, and $\check T_m$ and $\underline{\check\beta}_{N,m}$ the quantities T_m and $\underline{\beta}_{N,m}$ of (6.18) formed from it.

(ii) Where t carries an exact-window depth m_t^* (Definition 8.7), reals $w(n)$, $N < n \le m_t^*$, are *rigorous window bounds* of t when $w(n) \le \beta^{*(t)}(n)$ for each such n, $\beta^{*(t)}$ being the exact block bound (7.1) at t.

(iii) The *enclosed weight* $\check p_{\mathrm{mc}}^{(t)}$ of t formed from them is (8.5) where t carries no exact-window depth and (8.7) where it carries one, its branch chosen by predicate (X) at t as there, each read with $\check\beta$, $\underline{\check\beta}_{N,m_t}$ and $\check T_{m_t}$ in place of $\beta^{(t)}$, $\underline{\beta}^{(t)}_{N,m_t}$ and $T^{(t)}_{m_t}$, and with

$$\check\ell(n) \;:=\; \max\big\{\check\beta(n),\; w(n)\big\}, \qquad N < n \le m_t^*, \tag{8.9}$$

in place of $\ell^{(t)}(n)$ of (8.6). No computed eigenvalue of a block and no eigenvalue slack enters it.

Uses Def. 6.1, Lem. 6.8, Def. 7.1, Def. 8.1, Def. 8.3, Def. 8.7 · *Used by* Thm. 8.13, Thm. 8.14, Cor. 8.15, Def. 10.9, Thm. 10.10, Req. 13.5 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.rigorous:enclosed_weight

Bounds on the four inputs and on the window blocks, taken on the conservative side and clipped to [0, 1], give enclosed bound data, and the enclosed weight bounds its setting's multiclick element on every block above the cutoff. No premise about arithmetic enters, so no eigenvalue slack.

Each formula of the weight is non-decreasing in c and in γ and non-increasing in each d_k and β_k on $[0,1]$, and each minimum and maximum in it is non-decreasing in its arguments, so the weight formed on enclosed bound data lies below the weight formed on the exact data, entry by entry; this is the argument of Remark 10.12. In the exact window the lemma that combines the window with its remainder admits any lower bounds on the block eigenvalues and any lower remainder, so a rigorous lower bound serves where a computed eigenvalue needed an assumption. The theorem is the rigorous counterpart of Theorem 8.14 at one setting, and it has no assumption at all.

Theorem 8.13 (The weight on enclosed bound data bounds the multiclick element)**.** Let a setting family (Definition 8.1) be given, let $N \geq 0$, and let $t \in \mathcal{T}_{\mathrm{set}}$, with operators $B = B^{(t)}$ and $B_k = B_k^{(t)}$, bound data of the setting $\big(\gamma, \{\beta_k\}, c, \{d_k\}\big)$, bound function $\beta = \beta^{(t)}$ and T_m of (6.18) formed from those data. Write $\mathrm{clip}(x) \coloneqq \min\{1, \max\{0, x\}\}$ for real x.

(i) *Construction.* Let u, $\ell_1, \dots, \ell_K$, g and $b_1, \dots, b_K$ be reals with $u \geq \lambda_{\max}(B)$, $\ell_k \leq \lambda_{\min}(B_k)$, $g \leq \gamma$ and $b_k \geq \beta_k$. Then

$$\big(\,\mathrm{clip}(g),\ \{\mathrm{clip}(b_k)\},\ \mathrm{clip}(1-u),\ \{\mathrm{clip}(1-\ell_k)\}\big) \tag{8.10}$$

are enclosed bound data of t. Where t carries an exact-window depth m_t^* and reals λ_n satisfy $\lambda_n \leq \lambda_{\min}\big(M_{\mathrm{mc}|t}^{(=n)}\big)$ for $N < n \leq m_t^*$, the reals $\max\{0, \lambda_n\}$ are rigorous window bounds of t.

(ii) *Comparison.* Enclosed bound data $\big(\check{\gamma}, \{\check{\beta}_k\}, \check{c}, \{\check{d}_k\}\big)$ of t are admissible bound data (Definition 6.1), and their bound function $\check{\beta}$ and $\check{T}_m$ satisfy

$$\check{\beta}(n) \ \leq\ \beta(n) \quad (n \geq 0), \qquad \check{T}_m \ \leq\ T_m \quad (m \geq 0). \tag{8.11}$$

(iii) *The bound.* For enclosed bound data of t and, where t carries an exact-window depth, rigorous window bounds of t, the enclosed weight $\check{p}_{\mathrm{mc}}^{(t)}$ formed from them (Definition 8.12) satisfies $0 \leq \check{p}_{\mathrm{mc}}^{(t)} \leq 1$ and

$$M_{\mathrm{mc}|t}^{(>N)} \ \succeq\ \check{p}_{\mathrm{mc}}^{(t)}\, \mathbb{1}^{(>N)}, \tag{8.12}$$

and every real p with $0 \leq p \leq \check{p}_{\mathrm{mc}}^{(t)}$ satisfies $0 \leq p \leq 1$ and $M_{\mathrm{mc}|t}^{(>N)} \succeq p\, \mathbb{1}^{(>N)}$.

Reading: Where the receiver's model is known exactly, its high-photon multiclick weight can be computed from numbers that are proven to lie on the safe side of the exact ones, and the weight so computed, or any smaller number, is a guaranteed lower bound on

the multiclick probability of every block above the cutoff. It assumes nothing about the accuracy of floating-point arithmetic or of an eigenvalue routine, and it needs no eigenvalue slack. It still rests on the receiver's device assumptions, which say that the declared model is your receiver.

Uses Def. 6.1, Thm. 6.2, Lem. 6.5, Lem. 6.8, Def. 7.1, Lem. 7.2, Lem. 7.4, Def. 8.1, Def. 8.3, Def. 8.7, Def. 8.12 · *Used by* Thm. 8.14, Thm. 10.10, Req. 13.5, reductions/req:recv-row-provenance · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.rigorous:rigorous_weights

Proof. Two facts about the setting are used throughout. By Definition 8.1 (b) the hypotheses of Theorem 6.2 hold at t, so its bound data are admissible bound data, $\gamma, \beta_k \in (0,1]$, and by (8.3)

$$M_{\mathrm{mc}|t}^{(=n)} \succeq \beta(n)\, \mathbb{1}^{(=n)} \qquad \text{for every } n \geq 0 \tag{F1}$$

(Theorem 6.2, (6.6)). And $M_{\mathrm{mc}|t} \succeq 0$, the consequence (7.3) recorded in Definition 8.1, so Lemma 7.2 gives $\beta^{*(t)}(n) \in [0,1]$ for every n, that is,

$$M_{\mathrm{mc}|t}^{(=n)} \succeq 0 \qquad \text{for every } n \geq 0. \tag{F2}$$

The blocks $n > N$ are orthogonal direct summands of $M_{\mathrm{mc}|t}^{(>N)}$, so a lower bound $s\,\mathbb{1}^{(=n)}$ holding on each of them holds as $s\,\mathbb{1}^{(>N)}$ on their sum.

(i) The map clip is non-decreasing, takes values in $[0,1]$, and is the identity on $[0,1]$. Since $\gamma \in [0,1]$ and $g \leq \gamma$, $0 \leq \mathrm{clip}(g) \leq \mathrm{clip}(\gamma) = \gamma$; since $\beta_k \in [0,1]$ and $b_k \geq \beta_k$, $\beta_k = \mathrm{clip}(\beta_k) \leq \mathrm{clip}(b_k) \leq 1$. By (6.5), $c = 1 - \lambda_{\max}(B) \in [0,1]$ and $u \geq \lambda_{\max}(B)$ give $1 - u \leq c$, so $0 \leq \mathrm{clip}(1-u) \leq c$; and $d_k = 1 - \lambda_{\min}(B_k) \in [0,1]$ and $\ell_k \leq \lambda_{\min}(B_k)$ give $1 - \ell_k \geq d_k$, so $d_k \leq \mathrm{clip}(1-\ell_k) \leq 1$. These are the four conditions of (8.8). For the window, $\lambda_{\min}(M_{\mathrm{mc}|t}^{(=n)})$ is $\beta^{*(t)}(n)$ by (7.1), which is at least 0 by (F2), so $\max\{0, \lambda_n\} \leq \beta^{*(t)}(n)$.

(ii) By (8.8) and the admissibility of the exact data, $0 \leq \check{\gamma} \leq \gamma \leq \beta_k \leq \check{\beta}_k \leq 1$ and $0 \leq \check{c} \leq c \leq d_k \leq \check{d}_k \leq 1$ for every k, which is (6.1) for the enclosed tuple. Fix $n \geq 0$. With the convention $0^0 = 1$ the map $x \mapsto x^n$ is non-decreasing on $[0,\infty)$ (at $n = 0$ it is the constant 1). From $0 \leq \check{c} \leq c$ follows $0 \leq \check{c}^n \leq c^n$, and with $0 \leq \check{\gamma} \leq \gamma$, $\check{\gamma}\,\check{c}^n \leq \gamma\,\check{c}^n \leq \gamma\, c^n$; as $K - 1 \geq 0$, $(K-1)\check{\gamma}\check{c}^n \leq (K-1)\gamma c^n$. From $0 \leq d_k \leq \check{d}_k$ follows $d_k^n \leq \check{d}_k^n$, and with $0 \leq \beta_k \leq \check{\beta}_k$, $\beta_k d_k^n \leq \beta_k \check{d}_k^n \leq \check{\beta}_k \check{d}_k^n$. Adding, $\check{\beta}(n) = 1 + (K-1)\check{\gamma}\check{c}^n - \sum_k \check{\beta}_k \check{d}_k^n \leq 1 + (K-1)\gamma c^n - \sum_k \beta_k d_k^n = \beta(n)$. The second of these comparisons, with m in place of n, gives $\check{T}_m = 1 - \sum_k \check{\beta}_k \check{d}_k^m \leq 1 - \sum_k \beta_k d_k^m = T_m$.

(iii) Write $\check{p} := \check{p}_{\mathrm{mc}}^{(t)}$. Minima and maxima of finitely many reals are non-decreasing in each argument, which is all that is used of them below.

Case 1: t carries no exact-window depth. If predicate (X) holds at t, $\check{p} = \max\{0, \check{\beta}(N+1)\} \leq \max\{0, \beta(N+1)\}$ by (ii); the flag gives $\gamma = \beta_k = 1$ (Definition 8.1 (c)), so (6.8) holds and (6.10) of Lemma 6.5 gives $\max\{0, \beta(N+1)\} = \max\{0, \inf_{n>N} \beta(n)\}$. Otherwise $\check{p} = \max\{0, \underline{\check{\beta}}_{N,m_t}\}$,

and $\underline{\check{\beta}}_{N,m_t} = \min\{\min_{N<n\le m_t}\check{\beta}(n), \check{T}_{m_t}\} \le \min\{\min_{N<n\le m_t}\beta(n), T_{m_t}\} = \underline{\beta}_{N,m_t} \le \inf_{n>N}\beta(n)$ by (ii) and Lemma 6.8 (i). In both cases $\check{p} \le q \coloneqq \max\{0, \inf_{n>N}\beta(n)\}$. For every $n > N$, (F1) gives $M^{(=n)}_{\mathrm{mc}|t} \succeq \beta(n)\,\mathbb{1}^{(=n)} \succeq \big(\inf_{j>N}\beta(j)\big)\mathbb{1}^{(=n)}$ and (F2) gives $M^{(=n)}_{\mathrm{mc}|t} \succeq 0$, so $M^{(=n)}_{\mathrm{mc}|t} \succeq q\,\mathbb{1}^{(=n)} \succeq \check{p}\,\mathbb{1}^{(=n)}$, and (8.12) follows on the direct sum. $\check{p} \ge 0$ by the outer maximum, and $\check{p} \le q \le \max\{0, \beta(N+1)\} \le 1$, since $\beta \le 1$ (Theorem 6.2).

Case 2: a depth m^*_t, and predicate (X) holds at t. Apply Lemma 7.4 at t, whose hypotheses are those of Theorem 6.2 and (7.3), with $m^* = m^*_t$, $\ell = \check{\ell}$ of (8.9) and $\tau = \check{\beta}(m^*_t + 1)$. Hypothesis (7.5): for $N < n \le m^*_t$, $\check{\beta}(n) \le \beta(n) \le \beta^{*(t)}(n)$ by (ii) and (7.2), and $w(n) \le \beta^{*(t)}(n)$ by Definition 8.12 (ii), so their maximum $\check{\ell}(n)$ is at most $\beta^{*(t)}(n)$. Hypothesis (7.6): the flag makes (6.8) hold, as in Case 1, so $\tau_0 \coloneqq \beta(m^*_t + 1)$ satisfies (7.6) by Lemma 7.4 (ii), and $\tau \le \tau_0$ by (ii). A real below one that satisfies (7.6) satisfies it too: $\tau_0 \le 0$ gives $\tau \le 0$, and $\tau_0 \le \inf_{n>m^*_t}\beta^{*(t)}(n)$ gives $\tau \le \inf_{n>m^*_t}\beta^{*(t)}(n)$. With these choices (7.7) is the first branch of (8.7) read as Definition 8.12 (iii) reads it, that is $P = \check{p}$, and the lemma gives $0 \le \check{p} \le 1$ and (7.8), which is (8.12).

Case 3: a depth m^*_t, and predicate (X) fails at t. Then $m^*_t \le m_t$ (Definition 8.7). Apply Lemma 7.4 at t with $m^* = m_t$, with $\ell(n) = \check{\ell}(n)$ for $N < n \le m^*_t$ and $\ell(n) = \check{\beta}(n)$ for $m^*_t < n \le m_t$, and with $\tau = \check{T}_{m_t}$. Hypothesis (7.5) holds on the first range as in Case 2 and on the second by $\check{\beta}(n) \le \beta(n) \le \beta^{*(t)}(n)$. Hypothesis (7.6): T_{m_t} satisfies it by Lemma 7.4 (i), and $\check{T}_{m_t} \le T_{m_t}$ by (ii), lowered as in Case 2. With these choices (7.7) reads $\max\{0, \min\{\min_{N<n\le m^*_t}\check{\ell}(n), \min_{m^*_t<n\le m_t}\check{\beta}(n), \check{T}_{m_t}\}\}$, the middle minimum being over the empty set and inactive when $m^*_t = m_t$; this is the second branch of (8.7) read as Definition 8.12 (iii) reads it, so $P = \check{p}$, and the lemma concludes as in Case 2.

Any smaller weight. For $0 \le p \le \check{p}$, $M^{(>N)}_{\mathrm{mc}|t} \succeq \check{p}\,\mathbb{1}^{(>N)} \succeq p\,\mathbb{1}^{(>N)}$, since $(\check{p} - p)\mathbb{1}^{(>N)} \succeq 0$, and $0 \le p \le \check{p} \le 1$. □

Each setting's emitted weight bounds its multiclick element on every photon-number block above the cutoff, and their minimum does so at every setting, under the eigenvalue-slack assumption A-EIG only where computed block eigenvalues enter. Each weight lies in [0, 1] and is finitely computable.

This is the theorem the emitted weights rest on. Each setting's weight is the analytic weight of Definition 8.3 or, where a depth is declared, the exact-window weight of Definition 8.7; where the engine evaluates the weight rigorously (Requirement 13.5) it is the enclosed weight of Definition 8.12, formed by the same formulas from bounds on the conservative side. The theorem says that each bounds the multiclick element of its own setting on every photon-number block above the cutoff, and that the minimum over the settings bounds it at every setting, under Assumption 11.11 only where a computed block eigenvalue enters, at a setting governed by Definition 8.7.

Remark 8.11 explains why that entry rests on the assumption rather than on a theorem, and Theorem 8.13 is the theorem that replaces it.

Theorem 8.14 (Multiclick bounds carried by the emitted weights). Let a setting family (Definition 8.1) be given. For every setting $t \in \mathcal{T}_{\mathrm{set}}$ let $p_{\mathrm{mc}}^{(t)}$ be as in Definition 8.3 if t carries no exact-window depth and as in Definition 8.7 if it carries one, or, at any setting, let it be the enclosed weight $\check{p}_{\mathrm{mc}}^{(t)}$ of Definition 8.12 formed from enclosed bound data of t and, where t carries a depth, from rigorous window bounds of t, or a real p with $0 \le p \le \check{p}_{\mathrm{mc}}^{(t)}$; and let $p_{\mathrm{mc}} = \min_{t \in \mathcal{T}_{\mathrm{set}}} p_{\mathrm{mc}}^{(t)}$ as in Definition 8.3. Then

$$M_{\mathrm{mc}|t}^{(>N)} \succeq p_{\mathrm{mc}}^{(t)} \mathbb{1}^{(>N)} \qquad \text{for every } t \in \mathcal{T}_{\mathrm{set}}, \tag{8.13}$$

where $M_{\mathrm{mc}|t}^{(>N)}$ is the restriction of the multiclick element $M_{\mathrm{mc}|t}$ of (8.1) at setting t to the photon-number blocks $n > N$; and consequently, with the one weight p_{mc} at every setting,

$$M_{\mathrm{mc}|t}^{(>N)} \succeq p_{\mathrm{mc}} \mathbb{1}^{(>N)} \qquad \text{for every } t \in \mathcal{T}_{\mathrm{set}}. \tag{8.14}$$

At a setting governed by Definition 8.3 or by Definition 8.12 both inequalities hold at that setting with no hypothesis beyond this contract's ledger; at a setting governed by Definition 8.7 they hold there *under Assumption 11.11*, which is what makes the second entry of (8.6) a lower bound. At a one-setting family, $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$, one has $p_{\mathrm{mc}} = p_{\mathrm{mc}}^{(\mathrm{off})}$ and the two inequalities are one. Moreover $0 \le p_{\mathrm{mc}}^{(t)} \le 1$ at every setting, hence $0 \le p_{\mathrm{mc}} \le 1$, and each $p_{\mathrm{mc}}^{(t)}$ of Definition 8.3 or 8.7 is computable by finitely many arithmetic operations on $\{\lambda_{\max}(B^{(t)}), \lambda_{\min}(B_k^{(t)}), \gamma^{(t)}, \beta_k^{(t)}\}$ and the declared N, m_t and m_t^* — where $B^{(t)}, B_k^{(t)}$ are the operators of the setting (Definition 8.1 (b)) — together with, where the setting carries a declared depth, the declared $\varepsilon_{\mathrm{eig}}$ and the $m_t^* - N$ extreme eigenvalues $\widehat{\beta}^{*(t)}(n)$ of the Hermitian matrices (7.1), each of dimension $\binom{D+n-1}{n}$; each enclosed weight by finitely many on the enclosed bound data, the rigorous window bounds and the declared N, m_t and m_t^*; and p_{mc} by finitely many comparisons among them.

Uses Thm. 6.2, Lem. 6.5, Lem. 6.8, Def. 7.1, Lem. 7.2, Lem. 7.4, Def. 8.1, Def. 8.3, Def. 8.7, Def. 8.12, Thm. 8.13 · *Assumes* A-EIG · *Used by* Cor. 8.15, Prop. 8.17, Rem. 10.6, Thm. 10.10, reductions/wpfss/prop:route, reductions/wpfss/rem:minweight, reductions/wpfss/rem:receiver-imports · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound: emitted_weights · *Oracles* registry_receiver_povm.py

Proof. Idea. At a fixed setting, Theorem 6.2 bounds the multiclick element block by block. Lemma 6.5 or Lemma 6.8 turns the block bounds into the analytic weight, and Lemma 7.4 turns them into the exact-window weight; Theorem 8.13 does both for the enclosed weight. The uniform statement follows because a minimum lies below each of its terms.

Step 1 (the analytic weight at a fixed setting). Fix t. By Theorem 6.2, applicable at every setting by (8.3), $M_{\mathrm{mc}|t}^{(=n)} \succeq \beta^{(t)}(n)\mathbb{1}$ for every n, so $M_{\mathrm{mc}|t}^{(>N)} \succeq \big(\inf_{n>N} \beta^{(t)}(n)\big)\mathbb{1}^{(>N)}$ — the blocks are orthogonal summands and a lower bound holding on each holds on the direct sum. Since $M_{\mathrm{mc}|t}^{(>N)} \succeq 0$ as well ($M_{\mathrm{mc}|t}$ is a sum of POVM elements, Definition 8.1 (a)), the bound may be replaced by its positive part. In the first branch of (8.5) — where the flag is exact, so that (6.8) holds with equality by construction (Definition 8.1 (c)) — Lemma 6.5 identifies that positive part with $\max\{0, \beta^{(t)}(N+1)\}$; in the second, Lemma 6.8 (i) gives $\underline{\beta}_{N,m_t}^{(t)} \leq \inf_{n>N} \beta^{(t)}(n)$, so its positive part is a valid lower bound as well.

Step 2 (the exact-window weight at a fixed setting). At a setting carrying a declared depth, (7.8) of Lemma 7.4 gives the conclusion directly for (8.7); its hypothesis (7.3) holds at every setting, because $M_{\mathrm{mc}|t} = \sum_{S \in \mathsf{MC}} M_{S|t}$ is a sum of elements of a POVM (Definition 8.1 (a)), which is the discharge Remark 7.3 says is needed. On branch 1, apply it with $m^* = m_t^*$, $\ell = \ell^{(t)}$ and $\tau = \beta^{(t)}(m_t^* + 1)$, the remainder being admissible by part (ii) because predicate (X) makes (6.8) hold. On branch 2, apply it with $m^* = m_t$, $\tau = T_{m_t}^{(t)}$ admissible by part (i), and the window function $\ell(n) = \ell^{(t)}(n)$ for $N < n \leq m_t^*$, $\ell(n) = \beta^{(t)}(n)$ for $m_t^* < n \leq m_t$ — the two ranges together being $N < n \leq m_t$ because $m_t^* \leq m_t$ is required there. In both cases hypothesis (7.5) holds entry by entry: for $\beta^{(t)}(n)$ by Lemma 7.2, and for $\widehat{\beta}^{*(t)}(n) - \varepsilon_{\mathrm{eig}}$ by Assumption 11.11; a maximum of the two is admissible by part (iii).

Step 2a (the enclosed weight at a fixed setting). At a setting whose weight is the enclosed weight $\check{p}_{\mathrm{mc}}^{(t)}$ or a real p with $0 \leq p \leq \check{p}_{\mathrm{mc}}^{(t)}$, Theorem 8.13 (iii) gives (8.13) at t and $0 \leq p_{\mathrm{mc}}^{(t)} \leq 1$, for the enclosed weight and for every real between 0 and it, with no hypothesis beyond Definition 8.1, whether or not t carries a depth.

Step 3 (the uniform weight). So (8.13) holds at t, under Assumption 11.11 exactly where t is governed by Definition 8.7. Since p_{mc} is the minimum of the $p_{\mathrm{mc}}^{(s)}$ over the finite set $\mathcal{T}_{\mathrm{set}}$, $p_{\mathrm{mc}} \leq p_{\mathrm{mc}}^{(t)}$, and $p_{\mathrm{mc}}^{(t)}\mathbb{1}^{(>N)} \succeq p_{\mathrm{mc}}\,\mathbb{1}^{(>N)}$ turns (8.13) at t into (8.14) at t, with no further hypothesis; at $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$ the minimum is over one number.

Step 4 (range and computability). The range $0 \leq p_{\mathrm{mc}}^{(t)} \leq 1$ follows at every setting from the positive part and from an entry at most 1 in the inner minimum. At a setting governed by Definition 8.3 every entry is a value of $\beta^{(t)}$ or the tail value $T_{m_t}^{(t)}$, and $\beta \leq 1$ (Theorem 6.2). At a setting governed by Definition 8.7 the window entries $\ell^{(t)}(n)$ may exceed 1, but the inner minimum of each branch contains an analytic remainder at most 1: $\beta^{(t)}(m_t^*+1) \leq 1$ on branch 1, and $T_{m_t}^{(t)} = 1 - \sum_k \beta_k^{(t)} \big(d_k^{(t)}\big)^{m_t} \leq 1$ on branch 2 because $\beta_k^{(t)}, d_k^{(t)} \geq 0$ (the *Range* paragraph of Remark 8.11). At a setting governed by Definition 8.12 the range is Step 2a. And p_{mc}, a minimum of

finitely many numbers in $[0,1]$, lies there with them. Computability of each $p_{\mathrm{mc}}^{(t)}$ is the finiteness of its window, and of p_{mc} the finiteness of $\mathcal{T}_{\mathrm{set}}$. □

For a receiver without probe modes the one emitted weight bounds the multiclick element on every block above the cutoff and lies in [0, 1], under A-EIG only where a computed block eigenvalue enters it. The slot entry of the multiclick set meets the guarantee of Definition 8.16.

Corollary 8.15 (Emitted multiclick weight of a probe-free receiver)**.** Let $\mathfrak{R}$ be a receiver instrument with $\mathbb{I}_P = \varnothing$ (Definition 2.1), let N be the declared cutoff, and let p_{mc} be computed by Definition 8.3, or by Definition 8.7 where an exact-window depth is declared, from the one-setting family of Corollary 8.2, or let it be a real with $0 \le p_{\mathrm{mc}} \le \check{p}_{\mathrm{mc}}$, where $\check{p}_{\mathrm{mc}}$ is an enclosed weight of that family's one setting (Definition 8.12), $\check{p}_{\mathrm{mc}}$ itself included. Then

$$M_{\mathrm{mc}}^{(>N)} \succeq p_{\mathrm{mc}}\, \mathbb{1}^{(>N)}, \qquad 0 \le p_{\mathrm{mc}} \le 1, \tag{8.15}$$

with no hypothesis beyond the ledger where p_{mc} is given by Definition 8.3 or 8.12 and under Assumption 11.11 where it is given by Definition 8.7; and the strengthening slot $\{(\mathsf{MC}, p_{\mathrm{mc}})\}$ satisfies the guarantee of Definition 8.16.

Uses Def. 2.1, Cor. 8.2, Def. 8.3, Def. 8.7, Def. 8.12, Thm. 8.14, Def. 8.16 · *Assumes* A-TRUST, A-PASSIVE, A-VAC, A-MODES, A-THRESH, A-PORT, A-EXACT, A-NOSEL, A-ROUND, A-EIG · *Used by* Rem. 21.5 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:emitted_weight

Proof. By Corollary 8.2 the receiver is a setting family with $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$ and $M_{\mathrm{mc|off}} = M_{\mathrm{mc}}$, so Theorem 8.14 applies and (8.14) at the one setting is (8.15); the range and the hypothesis clause are the theorem's. The slot guarantee is (8.15) read as $\sum_{S \in \mathsf{MC}} M_S^{(>N)} \succeq p_{\mathrm{mc}}\, \mathbb{1}^{(>N)}$ at the one setting. □

The artifact carries high-photon lower bounds on disjoint sets of raw outcomes, one weight per set and setting; a set's uniform weight is its minimum over the settings. The receiver carries one entry, the multiclick set with the per-setting multiclick weights.

The slot is indexed by sets of raw outcomes rather than by emitted letters: lower bounds on disjoint raw outcomes add, while a bound on a merged outcome implies nothing about its parts (Theorem 5.6).

Definition 8.16 (The strengthening slot)**.** The *strengthening slot* is the finite list of pairs in which the artifact carries the high-photon evidence,

$$\Big\{\big(\mathcal{O}_i,\ (p_i^{(t)})_{t \in \mathcal{T}_{\mathrm{set}}}\big)\Big\}_{i=1}^{r}, \qquad \mathcal{O}_i \subseteq 2^{[K]} \text{ pairwise disjoint}, \quad p_i^{(t)} \ge 0, \tag{8.16}$$

one weight per entry and per setting $t \in \mathcal{T}_{\mathrm{set}}$ of the setting family (Definition 8.1), with the guarantee $\sum_{S \in \mathcal{O}_i} M_{S|t}^{(>N)} \succeq p_i^{(t)}\, \mathbb{1}^{(>N)}$ for every setting

$t \in \mathcal{T}_{\mathrm{set}}$ and every i. The *uniform weight of an entry* i is $p_i := \min_{t \in \mathcal{T}_{\mathrm{set}}} p_i^{(t)}$; since $0 \le p_i \le p_i^{(t)}$, it carries the guarantee $\sum_{S \in \mathcal{O}_i} M_{S|t}^{(>N)} \succeq p_i \, \mathbb{1}^{(>N)}$ for every setting $t \in \mathcal{T}_{\mathrm{set}}$ and every i, one weight at every setting. At a one-setting family, $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$, an entry has the single weight $p_i^{(\mathrm{off})} = p_i$ and is written $(\mathcal{O}_i, p_i)$. The artifact carries $r = 1$ with $(\mathcal{O}_1, (p_1^{(t)})_{t \in \mathcal{T}_{\mathrm{set}}}) = (\mathsf{MC}, (p_{\mathrm{mc}}^{(t)})_{t \in \mathcal{T}_{\mathrm{set}}})$, the per-setting weights of Theorem 8.14, each given at its setting by Definition 8.3 or 8.7 or as a non-negative real at most the enclosed weight of Definition 8.12, whose uniform weight is p_{mc}; at a one-setting family that entry is $(\mathsf{MC}, p_{\mathrm{mc}})$.

Uses Def. 8.1 · *Used by* Cor. 8.15, Prop. 8.17, Ass. 11.11, Req. 13.5, reductions/req:recv-row-provenance, reductions/wpfss/prop:route, reductions/wpfss/rem:absorbneeded, reductions/wpfss/rem:minweight, reductions/wpfss/rem:receiver-imports · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.bound:StrengtheningSlot

At every setting the strengthening slot's weights (high-photon lower bounds on disjoint sets of raw outcomes) are non-negative and sum to at most one, as are their minima over the settings, and each emitted multiclick weight lies in [0, 1]. That the emitted weights are valid bounds is Theorem 8.14.

Among the conditions under which a squashing request fails are a negative weight, weights summing to more than one, and a bound that does not hold where it is stated (`spec/architecture.md#54-failure-policy`). The proposition rules out the first two for every slot and names the result that rules out the third for the slot this module emits, under Assumption 11.11 only where a computed block eigenvalue enters a weight.

Proposition 8.17 (Range of the strengthening-slot weights). For any list (8.16) satisfying the stated guarantee with pairwise disjoint $\mathcal{O}_i$, one has at every setting $t \in \mathcal{T}_{\mathrm{set}}$ that $p_i^{(t)} \ge 0$ for every i and $\sum_i p_i^{(t)} \le 1$; hence the uniform weights satisfy $p_i \ge 0$ for every i and $\sum_i p_i \le 1$. In particular the list this module emits — by Corollary 8.15 and, for a setting family a submodule constructs, by that submodule's instance of Theorem 8.14 — satisfies $p_{\mathrm{mc}}^{(t)} \in [0, 1]$ at every setting, so the failure conditions "$p_y < 0$" and "$\sum_y p_y > 1$" of a squashing request cannot be triggered by this module's output; and what the further failure condition of a squashing request, "a bound does not hold at every declared setting it is stated for and every photon number above the cutoff", requires to hold — each $p_{\mathrm{mc}}^{(t)}$ at its own setting t and p_{mc} at every setting, for every photon number above the cutoff — is Theorem 8.14.

Uses Def. 8.1, Thm. 8.14, Def. 8.16 · *Used by* reductions/wpfss/prop:route, reductions/wpfss/rem:receiver-imports · newly derived · accepted

Proof. Non-negativity is part of (8.16). For the sum, fix any t and any unit vector $|\psi\rangle$ in a block $n > N$. Since the $\mathcal{O}_i$ are pairwise disjoint subsets of $2^{[K]}$ and $\{M_{S|t}\}$ is a POVM (Definition 8.1 (a)), $\sum_i \sum_{S \in \mathcal{O}_i} M_{S|t}^{(=n)} \preceq$

$\sum_{S\subseteq[K]} M^{(=n)}_{S|t} = \mathbb{1}^{(=n)}$, so by the guarantee at t, $\sum_i p_i^{(t)} \le \sum_i \langle\psi| \sum_{S\in\mathcal{O}_i} M^{(=n)}_{S|t} |\psi\rangle \le \langle\psi| \mathbb{1} |\psi\rangle = 1$. The uniform weights satisfy $0 \le p_i \le p_i^{(t)}$ (Definition 8.16), so $p_i \ge 0$ and $\sum_i p_i \le \sum_i p_i^{(t)} \le 1$ at any t. □

The analytic bound bounds three operators separately and need not be tight, while the exact block eigenvalue covers one block at a time. The exact window combines them: computed block eigenvalues less the slack, floored at the analytic bound, on finitely many blocks, an analytic remainder above.

Remark 8.18 (Slack of the analytic bound against the block eigenvalue). $\beta(n)$ bounds three operators separately and so need not be tight (Example 17.3 exhibits a strict gap at $n = 2$): on a given block, $\lambda_{\min}(M^{(=n)}_{\mathrm{mc}})$ can be computed exactly from (4.7) by diagonalizing a matrix of dimension $\binom{|\mathbb{I}_C|+n-1}{n}$, which is what Definition 7.1 names $\beta^*(n)$. The analytic bound is what the module emits where no depth is declared, because it is uniform in n — by Lemma 6.5 where predicate (X) of Definition 8.3 holds and by Lemma 6.8 (i) where it does not — while a per-block computation covers finitely many blocks. The exact window does not answer that by computing more blocks: on the finitely many blocks up to a declared depth it uses the computed block eigenvalues less the eigenvalue slack, floored at the analytic bound, and it keeps an analytic remainder above them (Lemma 7.4, Definition 8.7). Which of the two evaluations enters the artifact, and what it records beside each weight, is Requirement 13.4.

Uses Thm. 4.5, Lem. 6.5, Lem. 6.8, Def. 7.1, Lem. 7.4, Def. 8.7 · *Used by* Req. 13.4 · newly derived · accepted

9 The reference, enclosed and exact

The reference a run hands beside the snap of the emitted blocks, and evaluates its first-order bound on, must be known through rational boxes, and the binary64 evaluation is not one. Complex enclosures, a Gaussian-rational centre with a rational radius, are carried through sums, products and reciprocals by explicit rules (Lemma 9.2). An entry of a lifted operator in the occupation basis is a permanent divided by square roots of factorials (Lemma 9.3), so enclosures of the port Grams, themselves obtained from enclosures of square roots, cosines and sines, enclose every block, and the emitted blocks up to the cutoff are a reference for any declared parameters (Theorem 9.5). Where the port Grams are Gaussian-rational, the exact route (Definition 9.8), every block of at most one photon is known exactly, which a snap needs in order to keep a block unmoved (Theorem 9.9). Read through a port dilation, an isometry from the inputs that reproduces the port Grams (Definition 9.6), every block is also a weighted sum of rank-one terms whose columns are enclosed the same way, a factor enclosure, which fixes the block's rank where those columns are certified independent (Theorem 9.7).

Imports foundations/def:binary64, foundations/def:exact-reference, foundations/def:factor-enclosure, foundations/def:sector-map

A complex enclosure of a number is a Gaussian-rational centre with a rational radius that bounds its distance from the number; a matrix is enclosed entry by entry. Square roots, cosines and sines are enclosed by any such pair with a rational centre, exactly where the value is plainly rational.

A reference of a measurement family is read through rational boxes around its entries (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)). The receiver's photon-number blocks are built from the declared parameters by finitely many arithmetic operations, square roots, cosines and sines, so a box around each block follows from boxes around those square roots and trigonometric values, carried through the arithmetic (Theorem 9.5). The definition fixes what such a box must satisfy and leaves open how it is computed; a series with a bounded remainder, or bisection on the squares for a square root, are ways to obtain one.

Definition 9.1 (Complex enclosures, and enclosures of square roots and angles)**.** Write $\mathbb{Q}(i) = \{x + iy : x, y \in \mathbb{Q}\}$ and, for $z \in \mathbb{Q}(i)$, $|z|_1 := |\operatorname{Re} z| + |\operatorname{Im} z|$, a rational with $|z| \le |z|_1$. A *complex enclosure* of a complex number x is a pair (x_c, r) of $x_c \in \mathbb{Q}(i)$, its *centre*, and a rational $r \ge 0$, its *radius*, with $|x - x_c| \le r$. It is *exact* when $r = 0$, and then $x = x_c$. A complex enclosure of a complex matrix X is a pair (X_c, r) of a matrix X_c of the same shape with entries in $\mathbb{Q}(i)$ and a matrix r of non-negative rationals such that $\big((X_c)_{il}, r_{il}\big)$ is a complex enclosure of X_{il} for all i, l.

A *square-root enclosure* of a rational $p \ge 0$ is a complex enclosure of $\sqrt{p}$ with a rational centre, exact where p is the square of a rational. An *angle enclosure* of a real number θ that is given either as a rational, such as a declared value (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)), or as $c\pi$ with c a rational, is a pair of complex enclosures with rational centres of $\cos\theta$ and of $\sin\theta$, both exact where $\theta = c\pi$ with $2c$ an integer.

Uses foundations/def:binary64 · *Used by* Lem. 9.2, Thm. 9.5, Thm. 9.7, Thm. 9.9, Def. 10.9, Thm. 10.10, Req. 13.5, Req. 14.6, Cor. 20.2, Rem. 20.4, Req. 22.9, Req. 22.10, reductions/cor:reference-companion · newly derived · implemented · *Code* certiqs_key.receiver.enclosure:Ball

Sums, products, conjugates and reciprocals of enclosed numbers are enclosed by explicit rational rules, so any finite expression in enclosed numbers is enclosed and stays exact where its inputs are. Averaging a Hermitian matrix's enclosure with its adjoint makes the centre Hermitian.

This is interval arithmetic on complex discs with Gaussian-rational centres, written so that every rule is computed in exact rational arithmetic. A rule never shrinks a radius, and an enclosure obtained by it can be far wider than the true uncertainty when the same quantity enters an expression twice;

soundness is all the lemma asserts.

Lemma 9.2 (Arithmetic on complex enclosures). Let (x_c, r_x) and (y_c, r_y) be complex enclosures of complex numbers x and y (Definition 9.1).

(i) *Sums and conjugates.* $(x_c + y_c,\ r_x + r_y)$, $(-x_c,\ r_x)$ and $(\overline{x_c},\ r_x)$ are complex enclosures of $x + y$, $-x$ and $\overline{x}$.

(ii) *Products.* $\big(x_c y_c,\ |x_c|_1\, r_y + r_x\, |y_c|_1 + r_x r_y\big)$ is a complex enclosure of xy.

(iii) *Reciprocals.* If ℓ is a rational with $\ell > 0$, $\ell^2 \le |x_c|^2$ and $r_x < \ell$, then $x \neq 0$, $x_c \neq 0$, $1/x_c = \overline{x_c}/|x_c|^2$ lies in $\mathbb{Q}(i)$, and $\big(1/x_c,\ r_x/(\ell(\ell - r_x))\big)$ is a complex enclosure of $1/x$.

(iv) *Expressions.* Let E be a finite expression whose leaves are complex numbers and whose nodes are sums, differences, products, conjugates and reciprocals, and let a complex enclosure of every leaf be given. Assign to every node, in an order in which its operands come first, the enclosure that (i)–(iii) give from the enclosures of its operands, a reciprocal node receiving one only where some rational ℓ meets the condition of (iii). If every reciprocal node receives one, the enclosure of every node is a complex enclosure of its value, and a node whose operands have exact enclosures has an exact one. This applies entry by entry to sums and products of matrices, to permanents and determinants, which are sums of products of entries, and to the inverse of an invertible matrix, whose entries are signed minors divided by its determinant.

(v) *Hermitian centres.* If X is a Hermitian $d \times d$ matrix and (X_c, r) a complex enclosure of X, then $\big(\tfrac{1}{2}(X_c + X_c^{\dagger}),\ \tfrac{1}{2}(r + r^{\mathsf{T}})\big)$ is a complex enclosure of X whose centre is Hermitian with entries in $\mathbb{Q}(i)$ and whose radii are symmetric; it is therefore an entrywise enclosure of X (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)), and its entry (i, l) is exact where $r_{il} = r_{li} = 0$.

Uses Def. 9.1, foundations/def:exact-reference · *Used by* Thm. 9.5, Thm. 9.7, Thm. 9.9, Thm. 10.10, Req. 13.5, Req. 14.6, Cor. 20.2, Req. 22.9, Req. 22.10, reductions/cor:reference-companion · newly derived · implemented · *Code* certiqs_key.receiver.enclosure:Ball

Proof. (i) $|(x + y) - (x_c + y_c)| \le |x - x_c| + |y - y_c| \le r_x + r_y$; negation and conjugation preserve moduli, and $\mathbb{Q}(i)$ is closed under both.

(ii) $xy - x_c y_c = x_c(y - y_c) + (x - x_c)y_c + (x - x_c)(y - y_c)$, so $|xy - x_c y_c| \le |x_c| r_y + r_x |y_c| + r_x r_y$, and $|z| \le |z|_1$ for $z \in \mathbb{Q}(i)$. The centre lies in $\mathbb{Q}(i)$ and the radius is a non-negative rational.

(iii) $|x_c| \geq \ell > 0$, so $x_c \neq 0$ and $1/x_c = \overline{x_c}/|x_c|^2$, with $|x_c|^2$ a positive rational. Further $|x| \geq |x_c| - r_x \geq \ell - r_x > 0$, so $x \neq 0$ and

$$\left|\frac{1}{x} - \frac{1}{x_c}\right| = \frac{|x_c - x|}{|x|\,|x_c|} \leq \frac{r_x}{(\ell - r_x)\,\ell},$$

a non-negative rational.

(iv) By induction along the order of assignment: a leaf has a complex enclosure by hypothesis, and a node whose operands have complex enclosures receives one by (i)–(iii). If every operand radius is zero, the radii of (i) and (ii) are zero, and so is that of (iii), its numerator being $r_x = 0$. A sum or product of matrices has entries that are sums of products of entries; a permanent or determinant is a sum of products of entries, with signs for the determinant; and Cramer's rule writes every entry of an inverse as a signed minor, a determinant, times the reciprocal of the determinant of the matrix.

(v) The centre $Z \coloneqq \frac{1}{2}(X_c + X_c^\dagger)$ is Hermitian with entries in $\mathbb{Q}(i)$. Since $X_{il} = \overline{X_{li}}$,

$$|X_{il} - Z_{il}| = \tfrac{1}{2}\left|(X_{il} - (X_c)_{il}) + \overline{(X_{li} - (X_c)_{li})}\right| \leq \tfrac{1}{2}\,(r_{il} + r_{li}),$$

and $\frac{1}{2}(r + r^{\mathsf{T}})$ is a symmetric matrix of non-negative rationals. These are the data of an entrywise enclosure of the Hermitian matrix X, and the radius of entry (i, l) is zero where $r_{il} = r_{li} = 0$. □

An entry of the n-photon block of a lifted operator is a permanent of a submatrix with repeated rows and columns, divided by the square roots of two products of factorials. For a Gaussian-rational operator the entry is Gaussian-rational on the diagonal and on every block of at most one photon.

Lemma 9.3 (Entries of a lifted operator in the occupation basis). Let I be a finite mode set with orthonormal basis $\{|e_i\rangle\}_{i \in I}$, fix an order on I, and let X be an operator on $\mathbb{C}^I$ and $n \geq 0$. For an occupation vector $\boldsymbol{m} \in \mathbb{N}_0^I$ with $|\boldsymbol{m}| = n$ let $w_{\boldsymbol{m}} = (w_{\boldsymbol{m}}(1), \ldots, w_{\boldsymbol{m}}(n))$ be the word listing each $i \in I$ exactly m_i times in the order of I, write $\boldsymbol{m}! \coloneqq \prod_{i \in I} m_i!$, and for two such vectors let $X[\boldsymbol{m}|\boldsymbol{m}']$ be the $n \times n$ matrix with entries $X_{w_{\boldsymbol{m}}(r),\, w_{\boldsymbol{m}'}(s)}$. Write $\operatorname{per} Y \coloneqq \sum_\pi \prod_{r=1}^n Y_{r,\pi(r)}$, the sum over the permutations of $\{1, \ldots, n\}$, with $\operatorname{per} Y = 1$ for the empty matrix.

(i) In the occupation basis of Definition 1.2,

$$\langle \boldsymbol{m}|\, \mathfrak{F}(X)^{(n)}\, |\boldsymbol{m}'\rangle = \frac{\operatorname{per} X[\boldsymbol{m}|\boldsymbol{m}']}{\sqrt{\boldsymbol{m}!\,\boldsymbol{m}'!}}. \tag{9.1}$$

(ii) If X has entries in $\mathbb{Q}(i) = \{x + iy : x, y \in \mathbb{Q}\}$, then $\operatorname{per} X[\boldsymbol{m}|\boldsymbol{m}']$ lies in $\mathbb{Q}(i)$, and the entry (9.1) lies in $\mathbb{Q}(i)$ wherever $\boldsymbol{m}!\,\boldsymbol{m}'!$ is the square of an integer; in particular on the diagonal, $\boldsymbol{m} = \boldsymbol{m}'$, and for every $\boldsymbol{m}, \boldsymbol{m}'$ when $n \leq 1$.

Uses Def. 1.2, Lem. 1.6 · *Used by* Lem. 9.4, Thm. 9.5, Thm. 9.9 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.generic:fock_block

Proof. (i) Write $P_{\mathrm{Sym}^n} = \frac{1}{n!}\sum_\pi \Pi_\pi$ for the orthogonal projector onto $\mathrm{Sym}^n(\mathbb{C}^I)$, Π_π permuting the tensor factors, and $|w_{\boldsymbol{m}}\rangle := \left|e_{w_{\boldsymbol{m}}(1)}\right\rangle \otimes \cdots \otimes \left|e_{w_{\boldsymbol{m}}(n)}\right\rangle$. The permutations with $\Pi_\pi |w_{\boldsymbol{m}}\rangle = |w_{\boldsymbol{m}}\rangle$ are those that permute equal letters among themselves, of which there are $\boldsymbol{m}!$; every other $\Pi_\pi |w_{\boldsymbol{m}}\rangle$ is a product basis vector orthogonal to $|w_{\boldsymbol{m}}\rangle$. Hence $\|P_{\mathrm{Sym}^n}|w_{\boldsymbol{m}}\rangle\|^2 = \langle w_{\boldsymbol{m}}| P_{\mathrm{Sym}^n} |w_{\boldsymbol{m}}\rangle = \boldsymbol{m}!/n!$, and the symmetrized and normalized vector of Definition 1.2 is $|\boldsymbol{m}\rangle = \sqrt{n!/\boldsymbol{m}!}\, P_{\mathrm{Sym}^n}|w_{\boldsymbol{m}}\rangle$. The block $\mathfrak{F}(X)^{(n)}$ is the restriction of $X^{\otimes n}$ to $\mathrm{Sym}^n(\mathbb{C}^I)$, and $X^{\otimes n}$ commutes with P_{Sym^n} (proof of Lemma 1.6 (i)), so

$$\langle \boldsymbol{m}| X^{\otimes n} |\boldsymbol{m}'\rangle = \frac{n!}{\sqrt{\boldsymbol{m}!\,\boldsymbol{m}'!}} \langle w_{\boldsymbol{m}}| X^{\otimes n} P_{\mathrm{Sym}^n} |w_{\boldsymbol{m}'}\rangle = \frac{1}{\sqrt{\boldsymbol{m}!\,\boldsymbol{m}'!}} \sum_\pi \langle w_{\boldsymbol{m}}| X^{\otimes n} \Pi_\pi |w_{\boldsymbol{m}'}\rangle ,$$

using $P_{\mathrm{Sym}^n}^\dagger = P_{\mathrm{Sym}^n} = P_{\mathrm{Sym}^n}^2$. The vector $\Pi_\pi |w_{\boldsymbol{m}'}\rangle$ is the product vector whose r-th factor is $\left|e_{w_{\boldsymbol{m}'}(\sigma(r))}\right\rangle$ for a permutation σ determined by π, and $\pi \mapsto \sigma$ is a bijection of the permutation group. The summand is therefore $\prod_r \left\langle e_{w_{\boldsymbol{m}}(r)}\right| X \left|e_{w_{\boldsymbol{m}'}(\sigma(r))}\right\rangle = \prod_r X_{w_{\boldsymbol{m}}(r),\, w_{\boldsymbol{m}'}(\sigma(r))}$, and the sum over σ is $\operatorname{per} X[\boldsymbol{m}|\boldsymbol{m}']$. At $n = 0$ both sides are 1.

(ii) The permanent is a sum of products of entries of X with coefficient 1, so it lies in $\mathbb{Q}(i)$, which is closed under sums and products. If $\boldsymbol{m}!\,\boldsymbol{m}'! = s^2$ with s a positive integer, the entry (9.1) is $\operatorname{per} X[\boldsymbol{m}|\boldsymbol{m}']/s$, which lies in $\mathbb{Q}(i)$. On the diagonal $\boldsymbol{m}!\,\boldsymbol{m}'! = (\boldsymbol{m}!)^2$, and for $n \le 1$ every factorial in $\boldsymbol{m}!$ and $\boldsymbol{m}'!$ is 0! or 1!, so $\boldsymbol{m}!\,\boldsymbol{m}'! = 1$. □

The Fock lift of a linear map between two mode sets respects composition and adjoints and keeps an isometry an isometry, and its entries in the occupation bases are permanents divided by square roots of factorials. Reading a block through a port dilation needs both.

Lemma 1.6 and Lemma 9.3 are stated for an operator on one mode space. Theorem 9.7 lifts an isometry from the declared inputs into a different set of modes, and needs the same two facts for it: that the lift composes and takes adjoints, and that its entries are permanents.

Lemma 9.4 (The lift of a map between two mode sets). Let I, O and O' be finite mode sets, each with an orthonormal basis listed in a fixed order, let $X : \mathbb{C}^I \to \mathbb{C}^O$ and $X' : \mathbb{C}^O \to \mathbb{C}^{O'}$ be linear maps, and let $\mathfrak{F}(X)$ be the lift of Definition 1.2.

(i) $\mathfrak{F}(X)$ maps $\mathrm{Sym}^n(\mathbb{C}^I)$ into $\mathrm{Sym}^n(\mathbb{C}^O)$ for every $n \ge 0$; $\mathfrak{F}(X'X) = \mathfrak{F}(X')\mathfrak{F}(X)$ and $\mathfrak{F}(X)^\dagger = \mathfrak{F}(X^\dagger)$; and if $X^\dagger X = \mathbb{1}_I$, then $\mathfrak{F}(X)^\dagger\mathfrak{F}(X)$ is the identity of $\mathcal{F}(\mathbb{C}^I)$, so that $\mathfrak{F}(X)$ is an isometry.

(ii) Let $\boldsymbol{m} \in \mathbb{N}_0^O$ and $\boldsymbol{m}' \in \mathbb{N}_0^I$ be occupation vectors with $|\boldsymbol{m}| = |\boldsymbol{m}'| = n$, let $w_{\boldsymbol{m}}$, $w_{\boldsymbol{m}'}$, $\boldsymbol{m}!$ and $\boldsymbol{m}'!$ be the words and the factorial products of Lemma 9.3, each formed in its own mode set, and let $X[\boldsymbol{m}|\boldsymbol{m}']$ be

the $n \times n$ matrix with entries $X_{w_{\boldsymbol{m}}(r),\, w_{\boldsymbol{m}'}(s)}$. Then

$$\langle \boldsymbol{m}| \, \mathfrak{F}(X)^{(n)} \, |\boldsymbol{m}'\rangle = \frac{\operatorname{per} X[\boldsymbol{m}|\boldsymbol{m}']}{\sqrt{\boldsymbol{m}!\, \boldsymbol{m}'!}}, \tag{9.2}$$

with per as in Lemma 9.3; at $n = 0$ both sides are 1.

Uses Def. 1.2, Lem. 9.3 · *Used by* Thm. 9.7 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.factor:_column

Proof. (i) On $\mathrm{Sym}^n(\mathbb{C}^I)$ the lift $\mathfrak{F}(X)$ is the restriction of $X^{\otimes n}$, which carries $\mathrm{Sym}^n(\mathbb{C}^I)$ into $\mathrm{Sym}^n(\mathbb{C}^O)$ (Definition 1.2); so $\mathfrak{F}(X)$ maps each photon-number summand into the summand of the same n. Factor by factor, $(X'X)^{\otimes n} = X'^{\otimes n} X^{\otimes n}$, and since $X^{\otimes n}$ lands in $\mathrm{Sym}^n(\mathbb{C}^O)$, the restriction of the product to $\mathrm{Sym}^n(\mathbb{C}^I)$ is the product of the restrictions; summing over n gives $\mathfrak{F}(X'X) = \mathfrak{F}(X')\mathfrak{F}(X)$. On the full tensor powers $(X^{\otimes n})^\dagger = (X^\dagger)^{\otimes n}$, and $(X^\dagger)^{\otimes n}$ carries $\mathrm{Sym}^n(\mathbb{C}^O)$ into $\mathrm{Sym}^n(\mathbb{C}^I)$ (Definition 1.2 for the map $X^\dagger$). So for $\psi \in \mathrm{Sym}^n(\mathbb{C}^O)$ and $\phi \in \mathrm{Sym}^n(\mathbb{C}^I)$, $\langle \psi, X^{\otimes n}\phi\rangle = \langle (X^\dagger)^{\otimes n}\psi, \phi\rangle$ with both vectors on the right in their symmetric subspaces, and the restriction of $(X^\dagger)^{\otimes n}$ to $\mathrm{Sym}^n(\mathbb{C}^O)$ is the adjoint of the restriction of $X^{\otimes n}$ to $\mathrm{Sym}^n(\mathbb{C}^I)$. Distinct n give orthogonal summands, so $\mathfrak{F}(X)^\dagger = \mathfrak{F}(X^\dagger)$. If $X^\dagger X = \mathbb{1}_I$, these give $\mathfrak{F}(X)^\dagger \mathfrak{F}(X) = \mathfrak{F}(X^\dagger X) = \mathfrak{F}(\mathbb{1}_I)$, and $\mathbb{1}^{\otimes n}$ restricts to the identity of every $\mathrm{Sym}^n(\mathbb{C}^I)$.

(ii) Let $P_I = \frac{1}{n!}\sum_\pi \Pi_\pi^I$ and $P_O = \frac{1}{n!}\sum_\pi \Pi_\pi^O$ be the orthogonal projectors onto $\mathrm{Sym}^n(\mathbb{C}^I)$ and $\mathrm{Sym}^n(\mathbb{C}^O)$, Π_π^I and Π_π^O permuting the tensor factors, and let $|w_{\boldsymbol{m}}\rangle$ and $|w_{\boldsymbol{m}'}\rangle$ be the product basis vectors whose r-th factors are the basis vectors of the letters $w_{\boldsymbol{m}}(r)$ and $w_{\boldsymbol{m}'}(r)$. The permutations that fix $|w_{\boldsymbol{m}}\rangle$ are those that permute equal letters among themselves, $\boldsymbol{m}!$ of them, and every other Π_π^O sends it to an orthogonal product basis vector; so $\|P_O\,|w_{\boldsymbol{m}}\rangle\|^2 = \langle w_{\boldsymbol{m}}|\, P_O\, |w_{\boldsymbol{m}}\rangle = \boldsymbol{m}!/n!$, and the symmetrized and normalized vector of Definition 1.2 is $|\boldsymbol{m}\rangle = \sqrt{n!/\boldsymbol{m}!}\, P_O\, |w_{\boldsymbol{m}}\rangle$, and likewise $|\boldsymbol{m}'\rangle = \sqrt{n!/\boldsymbol{m}'!}\, P_I\, |w_{\boldsymbol{m}'}\rangle$. For every permutation π, $X^{\otimes n}\Pi_\pi^I = \Pi_\pi^O X^{\otimes n}$, both sending $v_1 \otimes \cdots \otimes v_n$ to the tensor product of the Xv_r permuted by π; so $X^{\otimes n} P_I = P_O X^{\otimes n}$. With $P_O^\dagger = P_O$ and $P_I^2 = P_I$,

$$\langle \boldsymbol{m}|\, X^{\otimes n}\, |\boldsymbol{m}'\rangle = \frac{n!}{\sqrt{\boldsymbol{m}!\, \boldsymbol{m}'!}}\, \langle w_{\boldsymbol{m}}|\, P_O X^{\otimes n} P_I\, |w_{\boldsymbol{m}'}\rangle = \frac{n!}{\sqrt{\boldsymbol{m}!\, \boldsymbol{m}'!}}\, \langle w_{\boldsymbol{m}}|\, X^{\otimes n} P_I\, |w_{\boldsymbol{m}'}\rangle = \frac{1}{\sqrt{\boldsymbol{m}!\, \boldsymbol{m}'!}} \sum_\pi \langle w_{\boldsymbol{m}}$$

The vector $\Pi_\pi^I\, |w_{\boldsymbol{m}'}\rangle$ is the product vector whose r-th factor is the basis vector of $w_{\boldsymbol{m}'}(\sigma(r))$ for a permutation σ determined by π, and $\pi \mapsto \sigma$ is a bijection of the permutation group. The summand is therefore $\prod_r X_{w_{\boldsymbol{m}}(r),\, w_{\boldsymbol{m}'}(\sigma(r))}$, and the sum over σ is $\operatorname{per} X[\boldsymbol{m}|\boldsymbol{m}']$. The left-hand side is the entry of $\mathfrak{F}(X)^{(n)}$, $|\boldsymbol{m}'\rangle$ lying in $\mathrm{Sym}^n(\mathbb{C}^I)$. At $n = 0$ the lift is 1 on the vacuum, the permanent of the empty matrix is 1 and both factorial products are 1. □

Enclosures of the port Grams, obtained from enclosures of square roots, cosines and sines, carry through permanents, inclusion and exclusion and the coarse-graining to enclosures of every block. With Hermitian centres they make the emitted blocks up to any cutoff a reference.

Every block of the receiver's measurement is a finite expression in the entries of the port Grams: a permanent divided by square roots of factorials (Lemma 9.3), combined by inclusion and exclusion and summed over the fibres of the coarse-graining. A compiled receiver's port Grams are in turn finite expressions in the entries of its elements, which are square roots, cosines and sines of the declared parameters. Enclosures of these, carried through the expressions by Lemma 9.2, therefore enclose the blocks of the model at the declared parameters, whatever the parameters are, with no floating-point step. The enclosures depend on the order in which the products are formed; the value they enclose does not (Theorem 3.4 (i)). Where the enclosures are exact, the reference is exact (Theorem 9.9).

Theorem 9.5 (Rigorous enclosures of the photon-number blocks)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}_{k=1}^{K}$ on $\mathbb{C}^{\mathbb{I}}$ (Definitions 2.1 and 2.2), $\{M_S\}_{S\subseteq[K]}$ its click-pattern POVM (Theorem 4.5), ϖ a declared coarse-graining onto $\mathcal{V}$ and $\{F_v\}_{v\in\mathcal{V}}$ the emitted POVM (Definition 5.5). Fix $N \geq 0$, and read every block $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$, $n \leq N$, in its occupation basis listed in an order fixed for that block. Enclosures are complex enclosures (Definition 9.1), and "the rules" are those of Lemma 9.2 (i)–(iv).

(i) *Port Grams of a compiled receiver.* Suppose that the instrument matrix of $\mathfrak{R}$ is the compiled instrument matrix $G = D_\eta G^\sigma_{\mathfrak{E}}$ of a well-formed graph for a linear extension σ (Definition 3.3), and that enclosures are given of the matrix $\hat{G}_e$ of every element e and of $\sqrt{\eta_k}$ for every port k. Enclose every other entry of D_η, and every entry of each $E_{\{k\}}$, exactly by itself. Then the rules, applied to the products (3.1) in the order of σ, to the restriction of G to the columns $\mathbb{I}$, and to (2.2), give enclosures of the A_k.

(ii) *Blocks.* Let (C_k, r_k) be enclosures of the A_k, and for every $n \leq N$ and occupation vectors $\boldsymbol{m}, \boldsymbol{m}'$ of weight n let $(a_{\boldsymbol{m}\boldsymbol{m}'}, e_{\boldsymbol{m}\boldsymbol{m}'})$ be a square-root enclosure of $1/(\boldsymbol{m}!\,\boldsymbol{m}'!)$. For $J \subseteq [K]$ enclose $\mathbb{1} - A_J$ by the rules from the (C_k, r_k), the identity being exact; enclose the entry $(\boldsymbol{m}, \boldsymbol{m}')$ of $\Pi_{\mathrm{vac}}(J)^{(n)}$ by the rules applied to the product of $\mathrm{per}\big((\mathbb{1} - A_J)[\boldsymbol{m}|\boldsymbol{m}']\big)$ with $\sqrt{1/(\boldsymbol{m}!\,\boldsymbol{m}'!)}$; enclose $M_S^{(=n)}$ by the rules applied to (4.7), the signs being exact; and enclose $F_v^{(=n)}$ by the rules applied to $\sum_{S\in\varpi^{-1}(v)} M_S^{(=n)}$. Each enclosure so obtained is a complex enclosure of the block it names.

(iii) *The reference.* Replace each enclosure of a block $F_v^{(=n)}$ of (ii) by the one with a Hermitian centre of Lemma 9.2 (v). Let $\mathfrak{m}_N$ be a sector

map (Definition "Sector maps and block-list operators" (foundations/def:sector-map)) whose blocks are the photon numbers $n = 0, \dots, N$ of the declared inputs, in increasing order, of dimensions $\binom{|\mathbb{I}|+n-1}{n}$, with any block kind and block identifiers, each block read in its fixed order. Then the family $\{F_v\}$ restricted to the blocks $n \le N$, read with these enclosures, is a reference with constant 1 against $\mathfrak{m}_N$ over $\mathcal{V}$ (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)); the same holds for $\{M_S\}$ over the click patterns.

(iv) *Exactness.* An entry of an enclosure of (i)–(iii) is exact where every enclosure it is computed from is exact; the reference of (iii) is an exact reference when every one of its radii is zero.

Uses Def. 2.1, Def. 2.2, Def. 3.3, Thm. 4.5, Def. 5.5, Thm. 5.6, Def. 9.1, Lem. 9.2, Lem. 9.3, foundations/def:exact-reference, foundations/def:sector-map · *Used by* Thm. 9.9, Thm. 10.10, Req. 13.5, Req. 14.6, Cor. 20.2, Req. 22.9, Req. 22.10, reductions/req:recv-row-provenance · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.generic:element_blocks

Proof. (i) Every entry of a product or a sum of matrices is a sum of products of entries, the adjoint conjugates and transposes entries, and restricting to the columns $\mathbb{I}$ selects entries. The matrix D_η is diagonal with the entry $\sqrt{\eta_k}$ on the modes of port k and 1 elsewhere (Definition 3.3), and $E_{\{k\}}$ is diagonal with entries 0 and 1, so all their entries are enclosed. Each entry of $A_k = G_{\mathbb{I}}^{\dagger} E_{\{k\}} G_{\mathbb{I}}$ is thus a finite expression in enclosed numbers, and Lemma 9.2 (iv), which needs no reciprocal here, encloses it.

(ii) By Theorem 4.5 (ii) and its block structure (iii), $\Pi_{\mathrm{vac}}(J) = \mathfrak{F}(\mathbb{1} - A_J)$ is block diagonal, and by (9.1) of Lemma 9.3 (i) the entry $(\boldsymbol{m}, \boldsymbol{m}')$ of its block n is per $\big((\mathbb{1} - A_J)[\boldsymbol{m}|\boldsymbol{m}']\big) \cdot \sqrt{1/(\boldsymbol{m}!\,\boldsymbol{m}'!)}$. The permanent is a sum of products of entries of $\mathbb{1} - A_J = \mathbb{1} - \sum_{k \in J} A_k$, each of which is enclosed by Lemma 9.2 (i); the square root is enclosed by hypothesis; so Lemma 9.2 (iv) encloses the entry. Equation (4.7) writes $M_S^{(=n)}$ as $\sum_{T \subseteq S} (-1)^{|T|} \Pi_{\mathrm{vac}}(([K] \setminus S) \cup T)^{(n)}$, an integer combination taken entry by entry, and Theorem 5.6 (i) gives $F_v^{(=n)} = \sum_{S \in \varpi^{-1}(v)} M_S^{(=n)}$; both are finite expressions in the enclosed entries, enclosed by the same lemma.

(iii) Each $F_v^{(=n)}$ is Hermitian, being a block of the block-diagonal POVM element F_v, so Lemma 9.2 (v) gives an entrywise enclosure with a Hermitian Gaussian-rational centre and rational radii. The blocks are positive semidefinite, since $M_S \succeq 0$ (Theorem 4.5 (i)) and a block of a block-diagonal positive semidefinite operator is positive semidefinite, and F_v is a sum of such. Completeness on every block holds because $\sum_S M_S = \mathbb{1}$ restricts to $\sum_S M_S^{(=n)} = \mathbb{1}$ on $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$ and the fibres of ϖ partition $2^{[K]}$, so $\sum_v F_v^{(=n)} = \mathbb{1}$ there. The dimension of $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$ is the number of occupation vectors of weight n on $|\mathbb{I}|$ modes, $\binom{|\mathbb{I}|+n-1}{n}$, which the sector map

declares, and the constant 1 is a positive declared value. These are the conditions of a reference. Nothing depends on the block kind or the identifiers; the enclosures were formed in each block's fixed order, which is the order in which the sector map reads the block. The statement for $\{M_S\}$ is the same argument without the fibre sums.

(iv) Lemma 9.2 (iv) gives an exact enclosure at every node whose operands are exactly enclosed, and (v) of the same lemma keeps an entry exact where it and its transposed entry are. A reference all of whose radii are zero is an exact reference by definition. □

A port dilation is an isometry from the declared inputs into a finite mode set with disjoint ports that reproduces every port Gram. Any such map carries the click statistics as the unitary dilation of the physical model does, and it may have far fewer modes.

The measurement isometry of Definition 2.5 applies a unitary on the modes and an environment to the declared inputs, and Theorem 4.1 (iii) and (v) show that the click statistics read of it only the amplitudes it sends from the inputs to each port. The definition names the least that a map from the inputs must satisfy to carry the same statistics, so that a smaller isometry, into the port modes and as many loss modes as the instrument's loss needs, can stand in for the unitary dilation: Theorem 9.7 reads every photon-number block through such a map. The restriction $U\iota$ of a unitary dilation U to the declared inputs, with the ports of the instrument, is one, since $(U\iota)^\dagger U\iota = \mathbb{1}_{\mathbb{I}}$ and the rows of $U\iota$ on a port are those of $G_{\mathbb{I}}$.

Definition 9.6 (Port dilation of a receiver instrument)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}_{k=1}^{K}$ on $\mathbb{C}^{\mathbb{I}}$ (Definitions 2.1 and 2.2). A *port dilation* of $\mathfrak{R}$ is a triple $(\mathbb{O}; \mathbb{O}_1, \dots, \mathbb{O}_K; \Omega)$ of a finite mode set $\mathbb{O}$ with a fixed order, pairwise disjoint subsets $\mathbb{O}_1, \dots, \mathbb{O}_K$ of $\mathbb{O}$, its *ports*, and a linear map $\Omega : \mathbb{C}^{\mathbb{I}} \to \mathbb{C}^{\mathbb{O}}$ with

$$\Omega^\dagger \Omega = \mathbb{1}_{\mathbb{I}} \qquad \text{and} \qquad \Omega^\dagger E_{\mathbb{O}_k} \Omega = A_k \quad (k = 1, \dots, K), \tag{9.3}$$

where $E_{\mathbb{O}_k}$ is the coordinate projector of $\mathbb{C}^{\mathbb{O}}$ onto the modes of $\mathbb{O}_k$. The modes of $\mathbb{O}$ in no port are its *loss modes*, and the *row* of a mode $i \in \mathbb{O}$ is the row of Ω that i indexes. The *click pattern* of an occupation vector $\boldsymbol{m} \in \mathbb{N}_0^{\mathbb{O}}$ (Definition 1.2) is

$$S_{\boldsymbol{m}} := \big\{ k \in [K] : \ m_i \geq 1 \text{ for some } i \in \mathbb{O}_k \big\}. \tag{9.4}$$

Uses Def. 1.2, Def. 2.1, Def. 2.2 · *Used by* Thm. 9.7, Lem. 16.4, Req. 22.9 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.factor:PortDilation

Read through a port dilation, a non-negative combination of click-pattern blocks is a weighted sum of rank-one terms whose columns are permanents of the dilation, enclosed by the same rules as the blocks. With rational weights, columns and weights are a factor enclosure of the block.

Theorem 9.5 encloses every block entry by entry. A block whose rank is below its dimension and whose support no coordinate vectors span is then known only through a box that also contains matrices of higher rank, and no lower bound on a least eigenvalue over that box fixes its rank. The physical model gives such a block in another form. A click pattern is an event on the output modes that is diagonal in their occupation basis, so pulling it back through the measurement writes its block as a sum of rank-one terms, one for each output occupation vector with that pattern, whose columns are the pulled-back occupation vectors. A port dilation (Definition 9.6) serves in place of the unitary dilation, and the columns are permanents of its entries (Lemma 9.4), enclosed by the rules that enclose the blocks. Read through its columns, the block keeps the rank they fix (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure)). A coarse-graining, and a classical channel of dark counts on the recorded patterns, combine click patterns with non-negative weights, which keeps the form.

Theorem 9.7 (Photon-number blocks in enclosed factor form)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}_{k=1}^{K}$ on $\mathbb{C}^{\mathbb{I}}$ and click-pattern POVM $\{M_S\}_{S\subseteq[K]}$ (Theorem 4.5), and let $(\mathbb{O};\mathbb{O}_1,\dots,\mathbb{O}_K;\Omega)$ be a port dilation of $\mathfrak{R}$ (Definition 9.6), with $S_{\boldsymbol{m}}$ the click pattern of an occupation vector $\boldsymbol{m}$ of $\mathbb{O}$. Read every block $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$ in its occupation basis listed in an order fixed for that block, and for an occupation vector $\boldsymbol{m}\in\mathbb{N}_0^{\mathbb{O}}$ of weight n put

$$c_{\boldsymbol{m}} \coloneqq \mathfrak{F}(\Omega)^{\dagger}\,|\boldsymbol{m}\rangle \;\in\; \mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}}). \tag{9.5}$$

Enclosures are complex enclosures (Definition 9.1), and "the rules" are those of Lemma 9.2 (i)–(iv).

(i) *The factor form.* For every $n\ge 0$ and all reals $w_S\ge 0$, $S\subseteq[K]$,

$$\sum_{S\subseteq[K]} w_S\, M_S^{(=n)} \;=\; \sum_{\boldsymbol{m}:\,|\boldsymbol{m}|=n} w_{S_{\boldsymbol{m}}}\, c_{\boldsymbol{m}} c_{\boldsymbol{m}}^{\dagger}, \tag{9.6}$$

the sum on the right over the occupation vectors of $\mathbb{O}$ of weight n. In particular $M_S^{(=n)}$ is the sum of the $c_{\boldsymbol{m}}c_{\boldsymbol{m}}^{\dagger}$ with $S_{\boldsymbol{m}}=S$, and the block $F_v^{(=n)}$ of the emitted POVM of a declared coarse-graining ϖ (Definition 5.5) is (9.6) with $w_S=1$ for $S\in\varpi^{-1}(v)$ and $w_S=0$ otherwise. The entry of $c_{\boldsymbol{m}}$ at an occupation vector $\boldsymbol{m}'$ of $\mathbb{I}$ is $\operatorname{per}\Omega[\boldsymbol{m}|\boldsymbol{m}']/\sqrt{\boldsymbol{m}!\,\boldsymbol{m}'!}$, with $\Omega[\boldsymbol{m}|\boldsymbol{m}']$ as in Lemma 9.4 (ii).

(ii) *Enclosed columns.* Let enclosures of the entries of Ω be given, and, for all occupation vectors $\boldsymbol{m}$ of $\mathbb{O}$ and $\boldsymbol{m}'$ of $\mathbb{I}$ of weight n, a square-root enclosure of $1/(\boldsymbol{m}!\,\boldsymbol{m}'!)$. The rules, applied to the conjugate of

$\operatorname{per}\Omega[\boldsymbol{m}|\boldsymbol{m}']$ times that square root, enclose every entry of every $c_{\boldsymbol{m}}$. An entry so enclosed is exact where every enclosure it is computed from is exact, and it is enclosed exactly as 0 where every term of the permanent, a product of entries of Ω, has a factor enclosed exactly as 0; in particular every entry of $c_{\boldsymbol{m}}$ is, where $\boldsymbol{m}$ has a photon in a mode whose row is enclosed exactly as zero.

(iii) *The enclosure of the factor form.* Let $n \geq 0$, let the weights $w_S \geq 0$ be rationals, let enclosures of the columns $c_{\boldsymbol{m}}$ be given, such as those of (ii), and let $\mathcal{C}$ be the set of the occupation vectors $\boldsymbol{m}$ of $\mathbb{O}$ of weight n with $w_{S_{\boldsymbol{m}}} > 0$ whose column is not enclosed exactly as zero, listed in a fixed order. Let (L_c, ϱ) be the matrix box of complex $\dim \operatorname{Sym}^n(\mathbb{C}^{\mathbb{I}}) \times |\mathcal{C}|$ matrices whose column $\boldsymbol{m}$ has the centres and the radii of the enclosure of $c_{\boldsymbol{m}}$, and $W \coloneqq \operatorname{diag}(w_{S_{\boldsymbol{m}}})_{\boldsymbol{m}\in\mathcal{C}}$. Then (L_c, ϱ, W) is a factor enclosure (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) of the block (9.6), with $|\mathcal{C}|$ columns; where $\mathcal{C}$ is empty, that block is zero.

(iv) *Port dilations from factors.* Let $\mathbb{O}_0, \mathbb{O}_1, \ldots, \mathbb{O}_K$ be pairwise disjoint finite mode sets and $Y_k : \mathbb{C}^{\mathbb{I}} \to \mathbb{C}^{\mathbb{O}_k}$ linear maps with $Y_k^\dagger Y_k = A_k$ for $k = 1, \ldots, K$ and $Y_0^\dagger Y_0 = \mathbb{L}$, the loss deficiency (Definition 2.2). Then the map Ω into $\mathbb{O} \coloneqq \mathbb{O}_0 \sqcup \mathbb{O}_1 \sqcup \cdots \sqcup \mathbb{O}_K$, in any fixed order, whose rows on $\mathbb{O}_k$ are those of Y_k for $k = 0, \ldots, K$, is with the ports $\mathbb{O}_1, \ldots, \mathbb{O}_K$ a port dilation of $\mathfrak{R}$. If a matrix X is $E^\dagger D E$ with D diagonal with positive rational diagonal entries d_i and E with entries in $\mathbb{Q}(i)$, then $Y \coloneqq D^{1/2}E$ satisfies $Y^\dagger Y = X$, and the rules applied to the products of square-root enclosures of the d_i with the entries of E enclose Y, exactly in every row i in which d_i is the square of a rational.

Uses Def. 1.2, Lem. 1.6, Def. 2.2, Thm. 4.5, Def. 5.5, Def. 9.1, Lem. 9.2, Lem. 9.4, Def. 9.6, foundations/def:factor-enclosure · *Used by* Req. 22.9 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.factor:factor_enclosures

Proof. (i). By Lemma 9.4 (i), $\mathfrak{F}(\Omega)$ maps $\operatorname{Sym}^n(\mathbb{C}^{\mathbb{I}})$ into $\operatorname{Sym}^n(\mathbb{C}^{\mathbb{O}})$, and its adjoint $\mathfrak{F}(\Omega^\dagger)$ maps back, so $c_{\boldsymbol{m}}$ lies in $\operatorname{Sym}^n(\mathbb{C}^{\mathbb{I}})$. For $J \subseteq [K]$ put $E_J \coloneqq \sum_{k\in J} E_{\mathbb{O}_k}$, a coordinate projector of $\mathbb{C}^{\mathbb{O}}$ since the ports are pairwise disjoint, and $\Pi_J \coloneqq \mathfrak{F}(\mathbb{1}_{\mathbb{O}} - E_J)$. By the same lemma and (9.3),

$$\mathfrak{F}(\Omega)^\dagger \, \Pi_J \, \mathfrak{F}(\Omega) = \mathfrak{F}\big(\Omega^\dagger(\mathbb{1}_{\mathbb{O}} - E_J)\Omega\big) = \mathfrak{F}\Big(\mathbb{1}_{\mathbb{I}} - \sum_{k\in J} A_k\Big) = \mathfrak{F}(\mathbb{1}_{\mathbb{I}} - A_J).$$

By Lemma 1.6 (iv) on $\mathbb{C}^{\mathbb{O}}$, whose coordinate basis is adapted to the range and kernel of the coordinate projector $\mathbb{1}_{\mathbb{O}} - E_J$, Π_J is the orthogonal projector onto the span of the occupation vectors with no photon in a mode of

$\bigcup_{k\in J}\mathbb{O}_k$; the occupation vectors being an orthonormal basis, $\Pi_J\,|\boldsymbol{m}\rangle = |\boldsymbol{m}\rangle$ where $S_{\boldsymbol{m}} \cap J = \varnothing$ and $\Pi_J\,|\boldsymbol{m}\rangle = 0$ otherwise. Fix $S \subseteq [K]$. By Theorem 4.5 (ii) and the identity above,

$$M_S = \sum_{T\subseteq S} (-1)^{|T|}\,\mathfrak{F}\big(\mathbb{1}_{\mathbb{I}} - A_{([K]\setminus S)\cup T}\big) = \mathfrak{F}(\Omega)^\dagger P_S\,\mathfrak{F}(\Omega), \qquad P_S \coloneqq \sum_{T\subseteq S} (-1)^{|T|}\,\Pi_{([K]\setminus S)\cup T}.$$

Let $\boldsymbol{m}$ be an occupation vector of $\mathbb{O}$. If $S_{\boldsymbol{m}}$ meets $[K]\setminus S$, every term of P_S annihilates $|\boldsymbol{m}\rangle$. Otherwise $S_{\boldsymbol{m}} \subseteq S$, and the term of T keeps $|\boldsymbol{m}\rangle$ exactly when $T \cap S_{\boldsymbol{m}} = \varnothing$, so $P_S\,|\boldsymbol{m}\rangle = \sum_{T\subseteq S\setminus S_{\boldsymbol{m}}} (-1)^{|T|}\,|\boldsymbol{m}\rangle = (1-1)^{|S\setminus S_{\boldsymbol{m}}|}\,|\boldsymbol{m}\rangle$, which is $|\boldsymbol{m}\rangle$ where $S_{\boldsymbol{m}} = S$ and 0 otherwise. So P_S restricted to $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{O}})$ is $\sum_{|\boldsymbol{m}|=n,\,S_{\boldsymbol{m}}=S} |\boldsymbol{m}\rangle\langle\boldsymbol{m}|$, and since $\mathfrak{F}(\Omega)$ maps $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$ into $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{O}})$, the block of M_S on $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$ is

$$M_S^{(=n)} = \sum_{|\boldsymbol{m}|=n,\,S_{\boldsymbol{m}}=S} \mathfrak{F}(\Omega)^\dagger\,|\boldsymbol{m}\rangle\langle\boldsymbol{m}|\,\mathfrak{F}(\Omega) = \sum_{|\boldsymbol{m}|=n,\,S_{\boldsymbol{m}}=S} c_{\boldsymbol{m}} c_{\boldsymbol{m}}^\dagger.$$

Every occupation vector has exactly one click pattern, so the sum of the $w_S M_S^{(=n)}$ is (9.6). The emitted POVM is $F_v = \sum_{S\in\varpi^{-1}(v)} M_S$ (Definition 5.5), and every M_S is block diagonal in the photon number (Theorem 4.5 (iii)), so the block of F_v is the sum of the blocks of its M_S. Finally, $\langle\boldsymbol{m}'|\,c_{\boldsymbol{m}} = \langle\boldsymbol{m}'|\,\mathfrak{F}(\Omega)^\dagger\,|\boldsymbol{m}\rangle = \overline{\langle\boldsymbol{m}|\,\mathfrak{F}(\Omega)\,|\boldsymbol{m}'\rangle}$, which Lemma 9.4 (ii) evaluates, the square root being a positive real.

(ii). The permanent of $\Omega[\boldsymbol{m}|\boldsymbol{m}']$ is a sum of products of entries of Ω, each of them enclosed, and the entry of $c_{\boldsymbol{m}}$ is its conjugate times the square root; Lemma 9.2 (iv) encloses the value of this expression and keeps it exact where its inputs are. A product one of whose operands is enclosed exactly as 0, with centre and radius 0, receives by part (ii) of that lemma the centre 0 and the radius $0\cdot r_y + 0\cdot|y_c|_1 + 0\cdot r_y = 0$, so a term of the permanent with such a factor is enclosed exactly as 0, whatever order its factors are multiplied in; a sum of such terms is enclosed exactly as 0 by part (i), and so are its conjugate and its product with the square root. Where $\boldsymbol{m}$ has a photon in a mode i, the letter i occurs in the word $w_{\boldsymbol{m}}$, so every term of the permanent has a factor from the row of i; if that row is enclosed exactly as zero, every term has a factor enclosed exactly as 0.

(iii). Let L be the matrix whose columns are the $c_{\boldsymbol{m}}$, $\boldsymbol{m}\in\mathcal{C}$, in the fixed order. Each of its entries lies within the radius of the centre of its enclosure, so the box (L_c, ϱ), whose centres have entries in $\mathbb{Q}(i)$ and whose radii are non-negative rationals (Definition 9.1), contains L; and W is diagonal with positive rational diagonal entries. A column enclosed exactly as zero is zero, an exact enclosure being equal to its centre. In (9.6) the terms with $w_{S_{\boldsymbol{m}}} = 0$ and the terms whose column is zero vanish, and the remaining terms are those of $\mathcal{C}$, so the block is $\sum_{\boldsymbol{m}\in\mathcal{C}} w_{S_{\boldsymbol{m}}} c_{\boldsymbol{m}} c_{\boldsymbol{m}}^\dagger = LWL^\dagger$ with L in the box, which is what a factor enclosure asks of the matrix it encloses (Definition

"Factor enclosure of a block" (foundations/def:factor-enclosure)). Where $\mathcal{C}$ is empty the sum is empty and the block is zero.

(iv). The rows of Ω on $\mathbb{O}_k$ are those of Y_k, so $\Omega^\dagger E_{\mathbb{O}_k}\Omega = Y_k^\dagger Y_k = A_k$ for $k = 1, \dots, K$, and, the $\mathbb{O}_k$ partitioning $\mathbb{O}$, $\Omega^\dagger\Omega = \sum_{k=0}^{K} Y_k^\dagger Y_k = \mathbb{L} + \sum_{k=1}^{K} A_k = \mathbb{1}_{\mathbb{I}}$ by (2.3). The ports $\mathbb{O}_1, \dots, \mathbb{O}_K$ are pairwise disjoint, so this is (9.3). For the second sentence, $D^{1/2}$ is the real diagonal matrix of the positive square roots $\sqrt{d_i}$, so $Y^\dagger Y = E^\dagger D^{1/2} D^{1/2} E = E^\dagger D E = X$. The entry (i, l) of Y is $\sqrt{d_i}\, E_{il}$, the product of $\sqrt{d_i}$, which a square-root enclosure of d_i encloses, and of E_{il}, enclosed exactly by itself; part (ii) of Lemma 9.2 encloses it, exactly where the square-root enclosure is exact, which Definition 9.1 makes it where d_i is the square of a rational. □

A declaration is on the exact route when the port Grams its declared data determine, computed exactly, have Gaussian-rational entries and are an admissible Gram family exactly, and its probe is of the Gaussian tier. A declaration of Grams or of an instrument matrix admissible exactly is one.

The receiver's blocks are enclosed for every declaration (Theorem 9.5). Where the port Grams are Gaussian-rational, many entries of the blocks, every entry of a block of at most one photon among them, are Gaussian rationals that can be known exactly (Theorem 9.9), and a snap can keep such a block at weight zero only if it is known exactly. The route is decided on the Grams and not on the instrument matrix: a half-wave plate at $\pi/8$ has entries $\pm 1/\sqrt{2}$, yet behind an ideal polarizing beamsplitter the port Grams it produces are rational.

Definition 9.8 (Gaussian-rational port Grams and the exact route)**.** Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}_{k=1}^{K}$ on $\mathbb{C}^{\mathbb{I}}$ (Definitions 2.1 and 2.2), and write $\mathbb{Q}(i) = \{x + iy : x, y \in \mathbb{Q}\}$. A declaration of a receiver is on the *exact route* when (i) the port Grams $\{A_k\}$ that its declared data determine, computed in exact arithmetic, have entries in $\mathbb{Q}(i)$, are Hermitian, and are an admissible Gram family exactly (Definition 2.2), that is,

$$A_k \succeq 0 \quad (k = 1, \dots, K) \qquad \text{and} \qquad \mathbb{1} - A_{[K]} \succeq 0, \tag{9.7}$$

conditions on Gaussian-rational matrices; and (ii) its probe declaration, where it has one, is of the Gaussian tier (Definition 2.3), whose mean photon numbers $\bar{n}_m^{(t)}$ are declared values and so rationals (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)). Under (i) the port Grams are those of a receiver instrument (Theorem 5.1 (ii)), and every receiver instrument with these port Grams induces the same click-pattern POVM (Theorem 5.1 (i)); where the instrument matrix G, declared or compiled, satisfies $G^\dagger G \preceq \mathbb{1}$ exactly, the declared instrument is one of them. A declared family admitted only at the receiver tolerances (Definition 10.4) and not admissible exactly is not on the exact route. A port Gram family or an instrument matrix declared directly has entries that are

dyadic rationals (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)); its port Grams then have entries in $\mathbb{Q}(i)$, by (2.2) for a declared instrument matrix, and such a declaration satisfies (i) exactly when its port Grams are Hermitian, as they always are for a declared instrument matrix, and admissible exactly.

Uses Def. 2.1, Def. 2.2, Def. 2.3, Thm. 5.1, Def. 10.4, foundations/def:binary64 · *Used by* Thm. 9.9, Def. 10.9, Thm. 10.10, Req. 14.6, Rem. 20.4, Req. 22.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.reference:ExactRouteDecision

With Gaussian-rational port Grams, every block entry whose two occupation vectors have factorial products forming a square, every entry of a block of at most one photon among them, is Gaussian-rational and enclosed exactly. Up to one photon the reference is then exact.

A snap keeps a block only where its reference is exact on the block and every element is already a lattice value (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (ii)). The theorem says where the enclosures of Theorem 9.5 are exact: on every entry that the factorials of the occupation basis do not make irrational. Above one photon the off-diagonal entries generally carry a factor $1/\sqrt{2}$ or similar, so those blocks have non-zero radii even at Gaussian-rational Grams.

Theorem 9.9 (Exact entries of the blocks on the exact route)**.** Let $\mathfrak{R}$ be a receiver instrument whose port Grams $\{A_k\}$ have entries in $\mathbb{Q}(i)$, as on the exact route (Definition 9.8), with click-pattern POVM $\{M_S\}$, a declared coarse-graining ϖ and emitted POVM $\{F_v\}$. Let $n \geq 0$ and let $\boldsymbol{m}, \boldsymbol{m}'$ be occupation vectors of weight n such that $\boldsymbol{m}!\,\boldsymbol{m}'!$ is the square of an integer; this holds on the diagonal, $\boldsymbol{m} = \boldsymbol{m}'$, and for every $\boldsymbol{m}, \boldsymbol{m}'$ when $n \leq 1$.

(i) The entry $(\boldsymbol{m}, \boldsymbol{m}')$ of every block $\Pi_{\mathrm{vac}}(J)^{(n)}$, $M_S^{(=n)}$ and $F_v^{(=n)}$ lies in $\mathbb{Q}(i)$.

(ii) If the enclosures of the A_k in Theorem 9.5 (ii) are exact, that is, each A_k is enclosed by itself with radius zero, then the enclosure of the entry $(\boldsymbol{m}, \boldsymbol{m}')$ of each of these blocks in Theorem 9.5 (ii), and of the blocks $F_v^{(=n)}$ in Theorem 9.5 (iii), is exact, its centre being the entry.

(iii) Under the hypothesis of (ii), every block $n \leq 1$ of the reference of Theorem 9.5 (iii) has radius zero everywhere, and for a cutoff $N \leq 1$ that reference is an exact reference (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)).

Uses Thm. 4.5, Thm. 5.6, Def. 9.1, Lem. 9.2, Lem. 9.3, Thm. 9.5, Def. 9.8, foundations/def:exact-reference · *Used by* Thm. 10.10, Req. 14.6, Rem. 20.4, Req. 22.9, Req. 22.10 · newly derived · accepted

Proof. (i) Every entry of $\mathbb{1} - A_J$ lies in $\mathbb{Q}(i)$, so Lemma 9.3 (ii) puts the entry $(\boldsymbol{m}, \boldsymbol{m}')$ of $\Pi_{\mathrm{vac}}(J)^{(n)} = \mathfrak{F}(\mathbb{1} - A_J)^{(n)}$ (Theorem 4.5 (ii)) in $\mathbb{Q}(i)$. By (4.7) and Theorem 5.6 (i), the entry $(\boldsymbol{m}, \boldsymbol{m}')$ of $M_S^{(=n)}$ and of $F_v^{(=n)}$ is an integer combination of the entries $(\boldsymbol{m}, \boldsymbol{m}')$ of blocks $\Pi_{\mathrm{vac}}(J)^{(n)}$, and $\mathbb{Q}(i)$ is closed under such combinations.

(ii) With exact enclosures of the A_k, every entry of $\mathbb{1} - A_J$ and every permanent formed from them has an exact enclosure (Lemma 9.2 (iv)). If $\boldsymbol{m}!\,\boldsymbol{m}'! = s^2$ with s a positive integer, then $1/(\boldsymbol{m}!\,\boldsymbol{m}'!) = (1/s)^2$ is the square of a rational, so its square-root enclosure is exact (Definition 9.1), and the product of two exactly enclosed numbers is exactly enclosed. The entries $(\boldsymbol{m}, \boldsymbol{m}')$ of $M_S^{(=n)}$ and $F_v^{(=n)}$ are enclosed by integer combinations of these exact enclosures, hence exactly. The condition on $\boldsymbol{m}!\,\boldsymbol{m}'!$ is symmetric in $\boldsymbol{m}$ and $\boldsymbol{m}'$, so the entry $(\boldsymbol{m}', \boldsymbol{m})$ is exact as well, and the Hermitian centre of Lemma 9.2 (v) keeps the entry exact. An exact enclosure has as its centre the number it encloses.

(iii) For $n \leq 1$ every pair $\boldsymbol{m}, \boldsymbol{m}'$ satisfies the condition, so by (ii) every radius of the block is zero. For $N \leq 1$ every block of the reference is of this kind, and a reference all of whose radii are zero is exact. □

10 Residuals, tolerances and floating-point evaluation

Every statement of this contract is exact unless it names a receiver tolerance or the eigenvalue-slack assumption; its evaluation is binary64 floating point, beside a rigorous evaluation of the weights in exact rational arithmetic wherever the port Grams are enclosed (Definition 10.9). Six residuals check the floating-point evaluation's consistency, each with an exact value of zero, and the receiver tolerances, numbers an instance declares to the receiver as a supplier, decide each exact condition on declared data (Definition 10.4), with the direction of the unclamped bound-data constant stated on both sides. None of this encloses a floating-point number: the one relation between a floating-point and an exact quantity is the opt-in eigenvalue-slack assumption (Theorem 10.10), and the floating-point blocks are evaluations of the model's blocks, which the rational enclosures of the previous section contain, while a rigorously evaluated weight is a bound by theorem. The last remark records in which direction each input of the bound is enclosed for the rigorous evaluation.

Imports foundations/cor:least-eigenvalue-over-enclosure, foundations/def:binary64, foundations/def:exact-reference, foundations/lem:enclosure-psd, foundations/lem:least-eigenvalue-bound

Six residuals, each exactly zero in exact arithmetic, check contractivity, Gram positivity, the deficiency identity, POVM completeness, block Hermiticity and positivity, and window consistency. Which exist depends on the input route and on whether an exact window is requested.

Route (iii) changes the nature of the arithmetic. Where the Grams are declared data, the only numerical step is an eigenvalue computation. Where they are *compiled*, G is a product of many matrices, and A_k, $\lambda_{\min}$ and $\lambda_{\max}$ are derived quantities whose floating-point error compounds through the product. The residuals below check the consistency of that evaluation; which of them the artifact carries, and what a residual above its tolerance causes, are Requirements 14.1 and 14.2. Remark 8.11 states which direction of failure the window residual r_6 is blind to.

Definition 10.1 (The six residuals)**.** The *residuals* of an evaluation of this contract are those of the six quantities below that exist on the input route taken and for the evaluation requested (which exist is stated after (10.7)):

$$r_1 \coloneqq \left\| \left(G^\dagger G - \mathbb{1}\right)_+ \right\| \qquad \text{(contractivity defect: the positive p} \tag{10.1}$$

$$r_2 \coloneqq \max_k \ \max\left\{0,\ -\lambda_{\min}(A_k)\right\} \qquad \text{(Gram positivity),} \tag{10.2}$$

$$r_3 \coloneqq \left\| \mathbb{1}_{\mathbb{I}} - A_\Sigma - G_{\mathbb{I}}^\dagger E_{\mathrm{un}} G_{\mathbb{I}} - \left(\mathbb{1}_{\mathbb{I}} - G_{\mathbb{I}}^\dagger G_{\mathbb{I}}\right) \right\| \qquad \text{(deficiency identity (4.3)),} \tag{10.3}$$

$$r_4 \coloneqq \left\| \mathbb{1} - \sum_{S \subseteq [K]} M_{S|t}^{(=n)} \right\| \ \text{ for each emitted } n \text{ and each setting } t \qquad \text{(POVM completeness, blockwise),} \tag{10.4}$$

$$r_5 \coloneqq \max_{v,\,n,\,t} \ \varrho\big(M_{v|t}^{(=n)}\big) \qquad \text{(block Hermiticity and positivity),} \tag{10.5}$$

$$r_6 \coloneqq \max_{t,\ N<n\le m_t^*} \left(\beta^{(t)}(n) - \widehat{\beta}^{*(t)}(n)\right)_+ \qquad \text{(window consistency, Lem. 7.2),} \tag{10.6}$$

in which the *Hermiticity-and-positivity defect* of a square matrix X is

$$\varrho(X) \ \coloneqq\ \max\Big\{ \left\|X - X^\dagger\right\|,\ \ \left(-\lambda_{\min}\big(\tfrac{1}{2}(X + X^\dagger)\big)\right)_+ \Big\}, \tag{10.7}$$

where E_{un} projects onto the declared modes in no port, $(\cdot)_+$ is the positive part, and $\|\cdot\|$ is the spectral norm throughout (10.1)–(10.7). The maximum in (10.5) runs over exactly the blocks $M_{v|t}^{(=n)}$ of the coarse-grained elements, before any dark-count channel, that the artifact emits: every emitted outcome letter $v \in \mathcal{V}$, every photon number $0 \le n \le N$, and every setting

$t \in \mathcal{T}_{\mathrm{set}}$ of the setting family (Definition 8.1; one setting when $\mathbb{I}_P = \varnothing$, Corollary 8.2); the emitted bound of the one-setting family is Corollary 8.15, and that of a setting family a submodule constructs is that submodule's instance of Theorem 8.14. Here $M_{v|t}^{(=n)}$ is the emitted block: the n-photon block of the coarse-grained conditional element $F_{v|t}$, written $F_v^{(=n)}$ in Theorem 5.6 (i) at a single setting.

Which residuals exist on which route. r_2, r_4 and r_5 are functions of the port Grams and of the emitted blocks, which every route produces, so all three exist on all three routes. r_1 (10.1) and r_3 (10.3) are written on G and $G_{\mathbb{I}}$, and route (i) supplies no instrument matrix: there the Grams are the declared datum and no G exists, so r_1 and r_3 are *not defined*. Synthesizing a G from Theorem 5.1 (ii) to evaluate them on would test the synthesis rather than the declaration. This agrees with Theorem 10.10 (iv), where route (i) is the case in which "there is no compilation". On routes (ii) and (iii) all five of $r_1, \ldots, r_5$ exist.

r_6 depends on the request, not on the route. It compares the two evaluations of a single proved inequality, $\beta^{(t)}(n) \le \beta^{*(t)}(n)$ (Lemma 7.2), so its exact value is 0 like the other five, and it exists on all three routes but *only at the settings whose declaration carries an exact-window depth* (Definition 8.7), over exactly the indices $N < n \le m_t^*$ that depth names. A declaration that requests no exact window computes no $\widehat{\beta}^*$ and has nothing to compare, so r_6 is not defined there, as r_1 and r_3 are not on route (i).

No residual bounds the distance between a computed quantity and the exact one (Theorem 10.10 (ii)).

Uses Lem. 4.2, Thm. 5.1, Thm. 5.6, Lem. 7.2, Def. 8.1, Cor. 8.2, Def. 8.7 · *Used by* Prop. 10.2, Rem. 10.3, Def. 10.4, Rem. 10.6, Thm. 10.10, Req. 14.1, Req. 14.2, Rem. 18.3, Def. 19.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.residuals:evaluate_residuals

The defect of a matrix vanishes exactly when it is Hermitian and positive semidefinite, and bounds how far its expectation values stray below zero or off the real axis. Every emitted block before the dark counts has defect zero exactly, so the fifth residual measures the evaluation alone.

Proposition 10.2 (Properties of the Hermiticity-and-positivity defect)**.** Let X be a complex $D \times D$ matrix and $\varrho(X)$ its Hermiticity-and-positivity defect (10.7).

(i) $\varrho(X) = 0$ if and only if $X = X^\dagger$ and $X \succeq 0$.

(ii) If $X = X^\dagger$ then $\varrho(X) = \big(-\lambda_{\min}(X)\big)_+$, which is the form (10.2) takes on a Gram.

(iii) For every unit vector $|\psi\rangle$, $\mathrm{Re}\,\langle\psi|\,X\,|\psi\rangle \ge -\varrho(X)$ and $\big|\,\mathrm{Im}\,\langle\psi|\,X\,|\psi\rangle\,\big| \le \frac{1}{2}\varrho(X)$. Hence $\varrho(X) \le \varepsilon$ bounds every diagonal expectation of X away from the negative reals by ε and off the real axis by $\varepsilon/2$.

(iv) Every block $M^{(=n)}_{v|t}$ of the coarse-grained elements, before any dark-count channel, that the artifact emits satisfies $\varrho = 0$ *exactly*: for each v, n and t, $M^{(=n)}_{v|t}$ is a diagonal block of the positive operator $F_{v|t}$ (Definition 8.1 (a) and Theorem 5.6), and a diagonal block — the compression $\Pi X \Pi$ of a positive operator X by an orthogonal projector Π, read on the range of Π — is Hermitian and positive semidefinite. So r_5 is a residual in exactly the sense of the other five: the exact value is 0, and what is emitted is the floating-point evaluation.

Uses Thm. 5.6, Def. 8.1, Def. 10.1 · *Used by* Req. 14.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.numerics:hermiticity_positivity_defect

Proof. Write $H \coloneqq \frac{1}{2}(X + X^\dagger)$ and $S \coloneqq X - X^\dagger$, so that $X = H + \frac{1}{2}S$ with H Hermitian and S skew-Hermitian.

(i) If $\varrho(X) = 0$ then $\|S\| = 0$, so $X = X^\dagger$ and $H = X$; and $(-\lambda_{\min}(X))_+ = 0$ gives $\lambda_{\min}(X) \geq 0$, i.e. $X \succeq 0$. Conversely if $X = X^\dagger \succeq 0$ then $S = 0$ and $\lambda_{\min}(X) \geq 0$, so both entries of the maximum vanish.

(ii) $S = 0$ gives $\|S\| = 0$ and $H = X$, so the maximum is its second entry.

(iii) $\langle\psi| X |\psi\rangle = \langle\psi| H |\psi\rangle + \frac{1}{2}\langle\psi| S |\psi\rangle$. The first term is real, with $\langle\psi| H |\psi\rangle \geq \lambda_{\min}(H) \geq -\varrho(X)$ by the definition of ϱ; the second is purely imaginary, because iS is Hermitian and therefore $\langle\psi| S |\psi\rangle \in i\mathbb{R}$, with modulus at most $\frac{1}{2}\|S\| \leq \frac{1}{2}\varrho(X)$. Splitting into real and imaginary parts gives both inequalities.

(iv) A POVM element is positive semidefinite, and so is any sum of them, so $F_{v|t} = \sum_{S \in \varpi^{-1}(v)} M_{S|t} \succeq 0$. Each $F_{v|t}$ is block diagonal in the photon number (Definition 8.1 (a) and Theorem 5.6), so $M^{(=n)}_{v|t}$ is the compression $\Pi_n F_{v|t} \Pi_n$ read on the range of Π_n. For any $Y \succeq 0$ and any orthogonal projector Π, $\Pi Y \Pi$ is Hermitian and $\langle\psi| \Pi Y \Pi |\psi\rangle = \langle\Pi\psi| Y |\Pi\psi\rangle \geq 0$, so the compression is Hermitian positive semidefinite. Now apply (i). □

The fifth residual is the only one that looks at the emitted photon-number blocks, which can fail Hermiticity or positivity at the last step while the other residuals pass. It is neither a repair nor an enclosure.

Remark 10.3 (What the fifth residual checks that the others do not). Four of the other five residuals of Definition 10.1 check the *optics*: that the compiled matrix is a contraction, that the Grams are positive, that the loss identity closes, that the POVM sums to the identity. None of them looks at the object the artifact actually emits, the photon-number block, and a block can fail Hermiticity or positivity at the last floating-point step even when all four pass, because r_4 constrains only the *sum* over S and r_2 lives on the Grams, one compilation stage earlier. r_5 makes that defect visible where the receiver is in hand, with the port, the setting and the photon number that produced it.

Two things r_5 is not. It is not a repair: a block that fails is refused and never repaired (Requirement 14.2), because a repaired block is a different operator and

no claim of this contract holds for it. And it is not an enclosure: $r_5 = 0$ says that the *computed* block is Hermitian and positive, not that it is close to the exact one, which is Theorem 10.10 (ii) exactly as for $r_1, \dots, r_4$.

Uses Def. 10.1 · *Used by* Rem. 19.6 · newly derived · accepted

Each exact condition on declared data that the receiver decides in floating point has one receiver tolerance and one test, and every other validation is a domain comparison, a finite inspection or a residual test. Admitting an instrument matrix does not admit the bound data derived from it.

The receiver is a supplier in the sense of Definition "Declared data and the parties to a module boundary" (foundations/def:declared-data) (ii): it decides its own conditions on the data it reads and emits in floating point, at slacks an instance declares to it. Those slacks are declared numbers of their own, never the instance's declared tolerance (Requirement "Scope of the declared tolerance" (foundations/req:tolerance-scope) (iii)); this definition names them and says which condition each one decides. The floating-point evaluation model in which they decide is Definition 10.9. Among the engine's obligations behind the validations of kinds (a) to (c) are the reserved tolerance field (Requirement 15.2), the exclusive dark-count routes (Requirement 22.7), the exact-window declaration (Requirement 13.2) and the residual refusal (Requirement 14.2); Remark 8.11 says why ε_{eig} is no test.

Definition 10.4 (Receiver tolerances and the exact conditions they decide)**.** Several conditions this contract states are *exact* statements about declared data that this contract nevertheless decides in floating point. Each is decided by a declared number, fixed here, and by no other threshold. The *receiver tolerances* are the declared $\varepsilon_1, \dots, \varepsilon_6$, ε_{gap} and ε_{eig}, each in the declared domain $[0, 1)$; every quantitative statement below uses that upper bound and says where. Items (i) to (v) spend ε_1, ε_2 and ε_4; ε_3, ε_5, ε_6 and ε_{gap} decide no exact condition on declared data and bound computed quantities only (kind (c) below); and ε_{eig} decides nothing at all. It is subtracted from a computed eigenvalue in (8.6), so it is the one declared number this contract spends on a *hypothesis* (Assumption 11.11) rather than on a test.

(i) *Route-(i) Gram admissibility*, $A_k \succeq 0$ and $\sum_k A_k \preceq \mathbb{1}$ (Definition 2.2), is decided at ε_2: the family is admitted when $\lambda_{\min}(A_k) \geq -\varepsilon_2$ for every k and $\lambda_{\max}(A_\Sigma) \leq 1 + \varepsilon_2$. ε_2 is the tolerance the contract already attaches to Gram positivity through (10.2), so the same declared number bounds the residual and decides the admission, in the same units.

(ii) *The route-(i) rank consistency check* $\operatorname{rank} A_k \leq |\mathbb{P}_k|$ is decided by the ε_2-rank

$$\operatorname{rank}_{\varepsilon_2}(X) \;:=\; \#\big\{\, i : \lambda_i\big(\tfrac{1}{2}(X + X^\dagger)\big) > \varepsilon_2 \,\big\} \qquad \text{(eigenvalues with multiplicity)}, \tag{10.8}$$

the check being $\operatorname{rank}_{\varepsilon_2}(A_k) \le |\mathbb{P}_k|$ for every k. The threshold is *absolute* and it is ε_2; no separate tolerance is added.

(iii) *Dark-channel admissibility* for a *declared* R is decided in the detector-imperfections submodule, Definition 19.3, which extends this list; the tolerances that definition spends are the ε_4 and ε_2 of this list.

(iv) *Bound-data admissibility*, $0 \le c$ in (6.1), is decided at ε_2: the data are admitted when $c = 1-\lambda_{\max}(B) \ge -\varepsilon_2$. *Nothing is clamped*: the computed c enters (6.2) exactly as it stands. The remaining inequalities of (6.1), $\gamma \le \beta_k \le 1$ and $c \le d_k \le 1$, are *not* separately tested and no threshold decides them: they are redundant by theorem, discharged by the admissibility paragraph of Theorem 6.2 from (6.3) and from $\gamma \le \beta_k$, both hypotheses of the setting family (Definition 8.1 (b)), and their floating-point evaluation is governed by Theorem 10.10 (ii) like every other computed quantity.

Item (iv) decides this admission alone, on every input route, and item (v) does not decide it. Whenever $\varepsilon_1 > \varepsilon_2$ the band

$$c \in \big[\, -\varepsilon_1,\ -\varepsilon_2\big) \tag{10.9}$$

is non-empty, admitted by item (v)'s test and refused by item (iv)'s, and item (iv) wins on it: the band is refused.

(v) *Route-(ii) contractivity, and (E1) at catalogue entry C10.* $G^\dagger G \preceq \mathbb{1}$ (Definition 2.1) is decided at ε_1: the matrix is admitted when $\lambda_{\max}(G^\dagger G) \le 1 + \varepsilon_1$. The same test decides the (E1) predicate of the catalogue's generic block entry C10 (Definition 16.5 of the component-catalogue submodule).

The tolerance that bounds the residual r_1 also decides this admission. $G^\dagger G$ is Hermitian positive semidefinite, so $\big(G^\dagger G - \mathbb{1}\big)_+$ has eigenvalues $\big(\lambda_i(G^\dagger G) - 1\big)_+$ and therefore

$$r_1 = \big\|\big(G^\dagger G - \mathbb{1}\big)_+\big\| = \big(\lambda_{\max}(G^\dagger G) - 1\big)_+, \tag{10.10}$$

so "$r_1 \le \varepsilon_1$" and "$\lambda_{\max}(G^\dagger G) \le 1 + \varepsilon_1$" are the *same test on the same number*, and no separate tolerance is added.

The five items, together with the two extensions the submodules declare in the same form, Definition 19.3, which is item (iii)'s content, and Definition 16.5, are the complete set of exact conditions about declared *data that this module and its submodules decide under a threshold.* Every other validation they state is one of three kinds, and none of them adds a threshold this definition has not named:

(a) a comparison of declared data against a declared domain: every parameter domain of Table 4 except C10's, the detector-layer ranges, $m_t > N$, $m_t^* > N$, $m_t^* \le m_t$, $N \ge 0$, the domain of the receiver tolerances itself, the test of the reserved tolerance field of Assumption 11.8, and the diagonal non-negativity of $\Gamma^{(t)}$;

(b) a combinatorial predicate on the declaration decided by finite inspection: (E2) and (E3) of Definition 3.2, route exclusivity, the port map's disjointness, totality and surjectivity of ϖ, the single-valued τ of Definition 1.1, the exclusivity of the two dark-count routes of Definition 19.1, predicate (X) of Definition 8.3, and the conditional presence of $\varepsilon_{\mathrm{eig}}$ and ε_6 at a declaration carrying an exact-window depth;

(c) the residual tests themselves: r_3, r_4, r_5, r_6 against ε_3, ε_4, ε_5, ε_6, and the residual gap g_{N,m_t} against $\varepsilon_{\mathrm{gap}}$. These are threshold decisions, but on *computed* quantities rather than on declared data, and each is a declared-threshold decision, for the residuals of Definition 10.1 and for the residual gap (6.19), so nothing undeclared decides them either. The tests $r_1 \le \varepsilon_1$ and $r_2 \le \varepsilon_2$ are of this kind too, except where they coincide with an item: on route (ii), where G is declared, the test $r_1 \le \varepsilon_1$ *is* item (v) by (10.10), and on route (i), where the Grams are declared, $r_2 \le \varepsilon_2$ is the positivity half of item (i) written as a residual; on route (iii) r_1 tests the compiled G, and on routes (ii) and (iii) r_2 tests computed Grams. Neither is $\varepsilon_{\mathrm{eig}}$, which is not a test of any kind: it is subtracted from $\widehat{\beta}^*$ in (8.6) and changes an emitted value, not an admission.

Uses Def. 2.1, Def. 2.2, Def. 3.2, Def. 6.1, Thm. 6.2, Lem. 6.8, Def. 8.1, Def. 8.3, Def. 8.7, Def. 10.1 · *Used by* Def. 9.8, Rem. 10.5, Rem. 10.6, Rem. 10.7, Rem. 10.8, Thm. 10.10, Rem. 10.12, Req. 14.3, Req. 14.6, Def. 16.5, Def. 19.1, Def. 19.3, Prop. 19.4, Rem. 19.5, Rem. 19.6, Rem. 19.7, Req. 22.9 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.numerics:tolerances_from_declaration

Where the contractivity test at ε_1 admits and the bound-data test at ε_2 refuses, the bound-data test decides: an exact implication between the two conditions transports no admission between computed numbers. Admitting the band would make a stated error term false on admitted values.

Items (iv) and (v) of Definition 10.4 both widen an exact condition by a receiver tolerance, and in exact arithmetic the condition of item (v) implies that of item (iv). This remark says why the implication nevertheless admits nothing, and what each widened test does and does not guard against.

Remark 10.5 (The bound-data test and the contractivity test on the band). *Why item (iv) is widened at all.* Admissible receivers with $\lambda_{\max}(B) = 1$ exactly are ordinary: at a receiver without a probe $B = A_\Sigma$ (Corollary 8.2), and the channel-only receiver of Example 17.1 has $A_\Sigma = \mathbb{1}_2$. There a computed $\lambda_{\max}(B)$ one rounding unit above 1 makes c negative on a receiver that is admissible, and refusing it would refuse the simplest legitimate device.

The two items decide different objects. A declaration can satisfy one and fail the other: item (v) admits a route-(ii) G at ε_1, item (iv) decides the derived c at ε_2, and whenever $\varepsilon_1 > \varepsilon_2$, the ordinary declaration, the band (10.9) is non-empty, admitted by item (v)'s test and refused by item (iv)'s. On route (i) the band has no analogue, item (i) and item (iv) carrying the same ε_2; but item (iv) is a test there too and not a formality, for the reason step (2) below gives. Item (iv) wins on the band, in four steps.

(1) *In exact arithmetic item (v)'s condition implies item (iv)'s, so item (iv) is redundant, exactly.* $G^\dagger G \preceq \mathbb{1}$ gives $A_\Sigma \preceq G_\mathbb{I}^\dagger G_\mathbb{I} \preceq \mathbb{1}$ by Theorem 4.1 (i), hence $\lambda_{\max}(A_\Sigma) \leq 1$ and $c \geq 0$; and at every setting $B^{(t)} \preceq \mathbb{1}$ by (6.3), hence $c \geq 0$ again.

(2) *But item (v) never establishes its exact condition.* It admits under a widened test on a *computed* number, and Theorem 10.10 (ii) bounds the distance from that number to the exact one by nothing at all. So what step (1) transports is an implication between exact quantities and *no implication whatever between the two computed quantities*, which are separated by at least one matrix product ($A_k = G_\mathbb{I}^\dagger E_{\{k\}} G_\mathbb{I}$, Definition 2.2) and, at a setting of a family a submodule constructs from the Grams, by the further operations of that construction as well. Reading item (v) as admitting anything under item (iv) is reading an exact implication as a bound on an evaluation, which is the one substitution Theorem 10.10 (ii) exists to forbid. This step is also why item (iv) is not redundant on route (i), where step (1) holds just as well from $\sum_k A_k \preceq \mathbb{1}$: redundancy of the *exact* condition is not admission of the *computed* one. And $0 \leq c$ is tested where the rest of (6.1) is not because the error term of Remark 10.8, on which step (3) turns, is stated for a negative computed c and bounded through this test alone.

(3) *The band is not empty of consequence, and the consequence is quantitative and unsafe.* Remark 10.8 bounds the even-n overshoot of the unclamped evaluation by $(K-1)|c|^n \leq (K-1)\varepsilon_2^2$, and that bound is stated in ε_2 *because item (iv) is what bounds* $|c|$. Admitting (10.9) would make $|c|$ as large as ε_1 and the true bound $(K-1)\varepsilon_1^2$, larger by the factor $(\varepsilon_1/\varepsilon_2)^2$, which at an ordinary pair of declared numbers is many orders of magnitude. A stated error term of this contract would then be false on values this contract admits, in the direction where the evaluated $\beta(n)$ *exceeds* the exact one, which is the direction that reaches a key length. Remark 10.6 says what a declared domain buys, "that every error term this contract states holds on every value the contract admits", and that is the property item (v) winning would destroy.

(4) *Nothing is lost by item (iv) deciding.* The case the widening exists for, a receiver with $\lambda_{\max}(B) = 1$ exactly whose computed $\lambda_{\max}(B)$ sits a rounding unit above 1, is admitted by item (iv)'s own $-\varepsilon_2$; and item (iv) tests the very quantity that enters (6.2), where item (v) tests one two computations upstream. A declarer who wants the wider band declares a wider ε_2, which is what a receiver tolerance is for and which carries the stated error term with it, rather than leaving it behind.

Item (v) is one-sided as a test and two-sided in its consequences. $G^\dagger G \preceq \mathbb{1}$ is an upper bound on a spectrum, so the test relaxes in one direction only: it admits $\lambda_{\max}(G^\dagger G) \in (1, 1+\varepsilon_1]$, which the exact condition excludes, and it refuses nothing that the exact condition, applied to the same computed number, would admit. The *errors* are two-sided in the sense of Remark 10.7, and deliberately so. Too tight refuses a legitimate device: a receiver with a unitary G, such as the worked instrument (17.1) of Example 17.1, has $\lambda_{\max}(G^\dagger G) = 1$ *exactly*, so a computed value one rounding unit above 1 would refuse it, which is the argument

for widening item (iv) verbatim. Too loose admits a genuinely non-contractive G, which violates Assumption 11.3. Route (ii) carries no separate Gram admissibility check, because Theorem 4.1 (i) *derives* $\sum_k A_k \preceq G_{\mathbb{I}}^{\dagger} G_{\mathbb{I}} \preceq \mathbb{1}$ from contractivity, so the *exact* condition item (i) tests at route (i) is redundant here rather than dropped.

What that derivation transports, and what it does not. It is an implication between *exact* quantities. It therefore makes item (i)'s exact condition redundant on route (ii), and, through $B \preceq \mathbb{1}$ in (6.3), item (iv)'s exact condition $c \geq 0$ redundant too. It transports *no admission*: the computed $\lambda_{\max}(A_\Sigma)$ and the computed c are separate evaluations, related to the exact ones by nothing this contract asserts (Theorem 10.10 (ii)), so an ε_1-admitted G does not thereby satisfy any test stated at ε_2. Item (iv) accordingly decides the bound data on route (ii) as on every other, and the band (10.9) that opens when $\varepsilon_1 > \varepsilon_2$ is refused. *Each route therefore tests its own declared datum under its own declared number*, ε_1 on the supplied G at route (ii) and ε_2 on the supplied Grams at route (i), because each is the tolerance attached to *that* datum's residual, and neither is thereby a tolerance on the derived c. A declarer who wants one slack on $\lambda_{\max}(A_\Sigma)$ regardless of route declares $\varepsilon_1 = \varepsilon_2$, which also empties (10.9).

Uses Def. 2.2, Thm. 4.1, Def. 6.1, Thm. 6.2, Cor. 8.2, Def. 10.4, Rem. 10.6, Rem. 10.8, Thm. 10.10 · newly derived · accepted

Every quantity a receiver tolerance bounds, decides or reduces lies in [0, 1], so at a tolerance of one or more its test is vacuous or degenerate. The declared domain [0, 1) makes every error term the contract states hold on every value it admits.

Remark 10.6 (The domain of the receiver tolerances). Each of $\varepsilon_1, \ldots, \varepsilon_6$, ε_{gap} and ε_{eig} is declared in $[0, 1)$ and validated at input (Requirement 14.4). A declared number needs a declared domain for the same reason a declared parameter does, and here the domain is not a convention: *a receiver tolerance is a slack on a normalized quantity, never a scale.* Every quantity the eight bound, decide or reduce already lies in $[0, 1]$: eigenvalues of operators between 0 and $\mathbb{1}$ ((10.2), (10.8), Definition 10.4 (i), (v)), the spectral-norm defects (10.1), (10.3), (10.4) and (10.7) of operators bounded by $\mathbb{1}$, entries and column defects of a stochastic matrix (Definition 10.4 (iii)), and $c = 1 - \lambda_{\max}(B) \in [0, 1]$ (Definition 10.4 (iv)). At $\varepsilon \geq 1$ the corresponding test is not merely loose, it is *vacuous or degenerate*, and namably so: Definition 10.4 (i) would admit every Hermitian family with $\lambda_{\min} \geq -1$; (10.8) would return $\text{rank}_{\varepsilon_2}(A_k) = 0$ for *every* port Gram, so the rank check could never refuse (the under-reporting direction of Remark 10.7 taken to its limit); (iii) would call every declared matrix monotone; (iv) would admit every computed c; and (v) would admit a G of operator norm $\sqrt{2}$. ε_{gap} is bounded for the same reason at one remove: it bounds a slack in $p_{\text{mc}} \in [0, 1]$ (Theorem 8.14), so a value ≥ 1 permits the emitted bound to sit below the attainable one by the whole range of the quantity, which is no constraint. The two members the exact window adds are bounded on the same argument, up to a computed $\widehat{\beta}^*$ that rounding puts outside $[0, 1]$, which is not clamped into that interval and is not claimed to lie in it (Remark 8.11). ε_6 bounds the positive part of $\beta(n) - \widehat{\beta}^*(n)$, where $\beta(n) \leq 1$ (Theorem 6.2) and the exact value that $\widehat{\beta}^*(n)$ evaluates lies in $[0, 1]$ (Lemma 7.2 with Definition 8.1); so at $\varepsilon_6 \geq 1$ the test $r_6 \leq \varepsilon_6$ can refuse only an evaluation that puts $\widehat{\beta}^*(n)$ below the evaluated $\beta(n)$ less 1, which is at most 0 up to rounding, outside the interval where the exact value lies. ε_{eig} is subtracted from a quantity whose exact value β^* lies in $[0, 1]$ at every receiver (Lemma 7.2 with (7.3), which Definition 8.1 gives). So

at $\varepsilon_{\text{eig}} \geq 1$ the second entry of (8.6) is positive only where a computed $\widehat{\beta}^*$ exceeds 1; elsewhere it is never positive and the exact window is never the binding term: the declaration pays the full cost of the computation and buys nothing, which is degenerate in the same namable sense as the rest of this list.

Two arguments use the upper bound. Remark 10.8's even-n estimate uses $\varepsilon_2^n \leq \varepsilon_2^2$ for every even $n \geq 2$, which is exactly $\varepsilon_2 \leq 1$; and the justification of (10.8) in Remark 10.7, that the spectrum it thresholds is normalized to $[0, 1]$, is an argument about a threshold *inside* that interval.

The domain is a floor, not sizing advice. Nothing here says a tolerance near 1 is sensible; Remark 10.7 records that the choice of ε_2 decides which declarations are accepted, and Proposition 19.4 gives the inequality a declarer sizes ε_2 and ε_4 against. What the declared domain buys is that every error term this contract states holds on every value the contract admits.

Uses Thm. 6.2, Lem. 7.2, Def. 8.7, Rem. 8.11, Thm. 8.14, Def. 10.1, Def. 10.4 · *Used by* Rem. 10.5, Rem. 10.8, Req. 14.4 · newly derived · accepted

Under any positive threshold a counted rank can exceed or fall short of the exact one, so the rank check is not refuse-only. The contract therefore fixes the threshold at the declared ε_2, absolute rather than relative, instead of an undeclared library default.

Remark 10.7 (Two-sided errors of a rank cutoff, and the choice of threshold)*.* Under any positive threshold τ, the count (10.8) can differ from the exact rank in *both* directions.

- *Over-reported.* An exactly singular declared Gram whose zero eigenvalue evaluates at $+2\tau$ (rounding noise in the declared entries, or in the eigenvalue routine) is counted as one rank higher, and a declaration consistent with its port map is *refused*.
- *Under-reported.* A Gram whose exact rank is 2 but whose second eigenvalue is genuinely small, below τ, is counted as rank 1, and a declaration *inconsistent* with its port map is *admitted* on a single-mode port.

So the check is not refuse-only, and the choice of τ is a choice about which declarations are accepted, not a numerical detail. That is why this contract fixes $\tau = \varepsilon_2$, a number the declarer supplies and can reason about, rather than inheriting whatever cutoff a linear-algebra library applies by default; the usual library default is $\lambda_{\max} \cdot \max(D_1, D_2) \cdot \epsilon_{\text{mach}}$, which is relative, undeclared, and invisible in the artifact. Under (10.8) both directions remain possible (no threshold can remove them), but both are now consequences of a declared number, and both are recorded: the rank the check used is a function of $\{A_k\}$ and ε_2 alone.

The threshold of (10.8) is *absolute*, because a port Gram is already normalized by construction: $0 \preceq A_k \preceq A_\Sigma \preceq \mathbb{1}$ puts every eigenvalue in $[0, 1]$, and $\boldsymbol{\alpha}^\dagger A_k \boldsymbol{\alpha}$ is a mean photon number per unit input intensity, so there is no scale to normalize against. A threshold taken relative to $\lambda_{\max}(A_k)$ would make the decision at a nearly dark port (all eigenvalues $\sim 10^{-6}$) a million times tighter than at a lossless one, which is a property of the port's throughput and not of the declarer's numerics. And the threshold is ε_2, because (10.8) and Definition 10.4 (i) are the same test on the same spectrum from the two sides: item (i) admits eigenvalues down to $-\varepsilon_2$ as "numerically zero", and (10.8) counts as nonzero exactly those above $+\varepsilon_2$, so a

declared Gram admitted by item (i) has its numerically-zero eigenvalues in $[-\varepsilon_2, \varepsilon_2]$ and the two thresholds cannot disagree about which they are.

Uses Def. 10.4 · *Used by* Rem. 19.6 · newly derived · accepted

An admitted negative c lowers the analytic bound at odd photon numbers and can raise it at even ones, by at most $(K-1)\varepsilon_2{}^2$, second order on the declared domain. Neither comparison is an enclosure of the exact bound.

Remark 10.8 (The direction of the unclamped c at odd and even photon numbers). Definition 10.4 (iv) admits $c \in [-\varepsilon_2, 0)$ and feeds that value to (6.2) unchanged. Since $\beta(n) = 1 + (K-1)\gamma c^n - \sum_k \beta_k d_k^n$ is increasing in c^n, and since the exact c satisfies $c_{\mathrm{ex}} \geq 0$, the effect splits by the parity of n, and both sides are stated.

- *Odd n.* $c^n < 0 \leq c_{\mathrm{ex}}^n$, so the evaluated $\beta(n)$ is *below* the exact one. Conservative, and by a first-order amount: measured against the value at $c = 0$ the deficit is $(K-1)\gamma|c|^n \leq (K-1)\varepsilon_2$ at $n = 1$, which is a lower bound on the deficit against $\beta(n)$ at c_{ex}, that one being $(K-1)\gamma\big(c_{\mathrm{ex}}^n + |c|^n\big)$.
- *Even $n \geq 2$.* $c^n = |c|^n > 0$, so the evaluated $\beta(n)$ can *exceed* the exact one, by at most $(K-1)\gamma\big(|c|^n - c_{\mathrm{ex}}^n\big) \leq (K-1)|c|^n \leq (K-1)\varepsilon_2^2$, using $\gamma \leq 1$, $|c| \leq \varepsilon_2$ and $\varepsilon_2 \leq 1$. The bound $|c| \leq \varepsilon_2$ is Definition 10.4 (iv) and no other rule, which is why that item decides the admission of c on *every* route, item (v)'s route-(ii) admission at ε_1 not being a bound on $|c|$ at all. The last step is where the declared upper bound of Remark 10.6 is used, since $\varepsilon_2^n \leq \varepsilon_2^2$ for even $n \geq 2$ holds exactly on $\varepsilon_2 \leq 1$ and fails above it. The overshoot is second order in ε_2 on the declared domain, and this is the one direction in which clamping c to 0 would have been the safer choice.

The contract fixes the unclamped rule because a clamped and an unclamped evaluation are both defensible and they differ, so leaving the choice to an implementation would leave the emitted number undetermined by the contract. Neither statement above is an enclosure: they compare two evaluations of (6.2), and Theorem 10.10 (ii) still governs the distance from either to the exact quantity. The case arises in the floating-point evaluation only: the rigorous evaluation takes its lower end of c in $[0, 1]$ (Remark 10.12, Definition 8.12), so no negative c enters a weight it forms.

Uses Def. 6.1, Def. 10.4, Rem. 10.6 · *Used by* Rem. 10.5, Rem. 10.12 · newly derived · accepted

The claims are exact statements about exactly declared data, evaluated in binary64 with diagnostic residuals, while the blocks are enclosed and, where the port Grams are enclosed, the weights evaluated rigorously, in exact rational arithmetic. The restriction asserts nothing a run could violate.

Definition 10.9 (The evaluation model: floating point, enclosures and rigour)**.** The claims of this contract are exact statements about exactly declared data (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)). This contract *evaluates* them in binary64 floating-point arithmetic (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64)), reports those of the six residuals of Definition 10.1 that exist on the input route taken and for the evaluation requested, and asserts no rigorous enclosure of any quantity it

computes in floating point (Assumption 11.11 is the one exception): this is its *floating-point evaluation*. For a declaration whose model is an admissible receiver exactly it moreover computes, in exact rational arithmetic, complex enclosures (Definition 9.1) of the photon-number blocks of the emitted family: those of Theorem 9.5, and in the submodules those of the setting families a submodule constructs, each enclosed by that submodule, and of their composition with the dark counts (Corollary 20.2). They are exact where the declaration is on the exact route (Definition 9.8) and the entry is known exactly (Theorem 9.9). From the same enclosures of the port Grams it computes, in exact rational arithmetic, at every setting at which the rules of Lemma 9.2 give them, complex enclosures of the setting's operators $B^{(t)}$, $B_k^{(t)}$ and prefactors $\gamma^{(t)}$, $\beta_k^{(t)}$ (Definition 8.1) and, at a setting carrying an exact-window depth, of its blocks $M_{\mathrm{mc}|t}^{(=n)}$ in the window; from these, an upper bound u on $\lambda_{\max}(B^{(t)})$ and lower bounds ℓ_k on $\lambda_{\min}(B_k^{(t)})$ and λ_n on the least eigenvalue of each window block, by the least-eigenvalue bounds over an enclosure (Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound), Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure)), and a lower bound g on $\gamma^{(t)}$ and upper bounds b_k on $\beta_k^{(t)}$, each the real part of a centre less or plus its radius; from these, the reals $g, b_k, 1-u, 1-\ell_k$ each intersected with $[0,1]$ and $\max\{0,\lambda_n\}$; and from those, by the formulas of Definition 8.12 (iii), a weight: this is its *rigorous evaluation* of the weights. A weight emitted at a setting is the weight so formed where the rigorous evaluation is formed there, and otherwise the floating-point evaluation of Definition 8.3 or 8.7. These enclosures and bounds are not evaluations, and a decision taken on them is exact; their interval arithmetic is proved in this contract (Lemma 9.2) and their eigenvalue bounds in the foundations contract, and no interval-arithmetic assumption is imported. This *evaluation model* is a *restriction* this contract places on itself and asserts nothing a run could violate.

Uses Def. 8.1, Def. 8.12, Def. 9.1, Def. 9.8 · *Used by* Thm. 10.10, Ass. 11.11, Req. 13.5, Req. 14.6 · newly derived · accepted

The receiver's identities and inequalities are exact unless a claim names a receiver tolerance or A-EIG; its floating-point numbers carry no error bound except through A-EIG, while rational enclosures contain the model's blocks and rigorously evaluated weights are bounds outright.

Remark 10.11 says what item (i) does not assert, and Remark 8.11 says why the window residual r_6 of item (v) is blind to the direction Assumption 11.11 constrains. Requirement 13.5 is where the engine forms the rigorous evaluation of item (vii), and Requirement 14.5 what its label means for the tier of a run. Items (vi) and (vii) speak of the core's own family; the submodules state the counterparts for theirs, such as the enclosures of the dark-composed

blocks (Corollary 20.2), and a submodule that constructs a setting family states the enclosures of its blocks and the rigorously evaluated weight at its settings.

Theorem 10.10 (Exact statements, floating-point and rigorous evaluation)**.** For the evaluation model of Definition 10.9:

(i) *The asserted mathematics is exact, except where a claim says so in its own statement. Every asserted identity and inequality* of this module and its submodules is between exactly specified quantities and contains no error term, no tolerance and no numerical hypothesis (Theorem 6.2, Lemma 6.5, Lemma 6.8, Lemma 7.2 and Lemma 7.4 included, the last two being stated on the exact β^* and carrying no tolerance), *except where a claim's own statement names a receiver tolerance or Assumption 11.11.*

(ii) *The floating-point evaluation is not.* The emitted $\{A_k\}$, block matrices and those bound values that the floating-point evaluation produces are floating-point evaluations of those quantities; the enclosures of item (vi) and the rigorous evaluation of item (vii) are not. This contract asserts no bound on the difference, and the residuals of Definition 10.1 are *diagnostics*: passing them is necessary for the artifact to be self-consistent and is not sufficient for any enclosure statement.

(iii) *The evaluation label.* An artifact carries the rigorous evaluation label exactly where every per-setting weight it emits is formed by the rigorous evaluation of Definition 10.9, as the enclosed weight of Definition 8.12 or a non-negative real below it; every other artifact, on every input route, the route-(iii) compiler included, carries the floating-point evaluation label. The rigorous label concerns the weights alone: the blocks are evaluated in floating point under (ii) whatever the label. No path in this module produces a label stronger than these two.

(iv) *Where route (i) stands.* If the Grams are *declared*, as binary64 values, which are dyadic rationals (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)), the compilation error of (ii) does not exist: there is no compilation. The eigenvalue quantities of Theorem 6.2 are still evaluated in floating point by the floating-point evaluation, so (ii) and (iii) apply to a weight it forms; what changes is that the rigorous evaluation of (vii) then encloses no matrix product, the enclosures it starts from being the declared Grams themselves. This is also why r_1 and r_3 do not exist on this route: they are written on a G that is not supplied (Definition 10.1). Nothing further is claimed.

(v) *The one place a floating-point number is asserted to bound an exact one, and it is opt-in.* Where a setting t carries an exact-window depth and its emitted weight $p_{\mathrm{mc}}^{(t)}$ is that of Definition 8.7, the weight depends on the computed $\widehat{\beta}^*$ through (8.6), and through the minimum so may the uniform p_{mc}; the inequality $\widehat{\beta}^{*(t)}(n) - \varepsilon_{\mathrm{eig}} \leq \beta^{*(t)}(n)$ that makes each such emitted value a lower bound is Assumption 11.11, not a theorem of this contract: (ii) governs $\widehat{\beta}^*$ exactly as it governs every other computed quantity, and r_6 is a diagnostic in exactly the sense (ii) fixes and is blind to the direction the assumption constrains. Where a setting's emitted weight is an enclosed weight, no computed $\widehat{\beta}^*$ and no $\varepsilon_{\mathrm{eig}}$ enters it and Assumption 11.11 is not invoked there, whatever depth is declared. Where no depth is declared, no β^* is computed, Assumption 11.11 is not invoked, and the emitted weights are the values (8.5) defines, expression for expression, evaluated in floating point or on enclosed bound data. Item (iii) covers all of these: an artifact with a weight resting on Assumption 11.11 carries the floating-point label, so no artifact resting on Assumption 11.11 can claim more.

(vi) *The model's blocks are enclosed, and the evaluation approximates them.* For a declaration whose model is an admissible receiver exactly, the photon-number blocks of the click-pattern and emitted POVMs of that model are quantities the identities and inequalities of (i) are about. Their complex enclosures (Definition 9.1) of Theorem 9.5, where they are formed, are computed in exact rational arithmetic and contain them: every entry of a block lies within its radius of its centre, an entry whose radius is zero is its centre, and on the exact route the entries that Theorem 9.9 names have radius zero when the port Grams are enclosed exactly. A positive semidefiniteness or a positive definiteness established through Lemma "Positive semidefiniteness from an entrywise enclosure" (foundations/lem:enclosure-psd) (i) on such an enclosure with a Hermitian centre holds of the block itself. The floating-point block matrices are evaluations of the same blocks and fall under (ii): this contract asserts no bound on their difference from the blocks, so no decision about the blocks is taken on them.

(vii) *The rigorously evaluated weights are bounds.* Let t be a setting of a setting family (Definition 8.1) at which the rigorous evaluation is formed from complex enclosures of $B^{(t)}$, of every $B_k^{(t)}$, of $\gamma^{(t)}$, of every $\beta_k^{(t)}$ and, at a setting carrying a depth, of the window blocks $M_{\mathrm{mc}|t}^{(=n)}$; for a receiver without a probe, whose setting family is that of Corollary 8.2, the enclosures it forms from the enclosures of the port Grams are such. Then the four clipped reals it forms are enclosed bound data of the setting and, at a setting carrying a depth, the win-

dow values it forms are rigorous window bounds; the weight it forms from them is their enclosed weight, a rational computed exactly; and it and every real between 0 and it bound the multiclick element of the setting from below on every photon-number block above the cutoff and lie in $[0, 1]$. Its validity rests on no number computed in floating point: such a number enters it, if at all, only as the exact rational it is, of which nothing is assumed, as the approximate eigenbasis of Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) does. So (ii) does not govern it, and it rests on no assumption about arithmetic.

Uses Def. 6.1, Thm. 6.2, Lem. 6.5, Lem. 6.8, Lem. 7.2, Lem. 7.4, Def. 8.1, Cor. 8.2, Def. 8.3, Def. 8.7, Def. 8.12, Thm. 8.13, Thm. 8.14, Def. 9.1, Lem. 9.2, Thm. 9.5, Def. 9.8, Thm. 9.9, Def. 10.1, Def. 10.4, Def. 10.9, foundations/cor:least-eigenvalue-over-enclosure, foundations/def:binary64, foundations/def:exact-reference, foundations/lem:enclosure-psd, foundations/lem:least-eigenvalue-bound · *Assumes* A-EIG · *Used by* Rem. 10.5, Req. 14.5 · newly derived · accepted · *Oracles* registry_receiver_povm.py

Proof. (i) By inspection of the statements: each is a Löwner-order or scalar inequality between quantities defined from $\{A_k\}$, G, $\mathcal{T}_{\mathrm{set}}$, $\{\rho^{(t)}\}$, N, $\{m_t\}$ and $\{m_t^*\}$, with no epsilon anywhere. A claim falling under the exception is recognised from its own statement: (8.6) introduces $\varepsilon_{\mathrm{eig}}$ into an emitted value; Theorem 8.14 and the two corollaries that instantiate it name Assumption 11.11; and the submodules' tolerance claims carry their declared slacks in the inequalities they assert. The receiver tolerances occurring elsewhere in the claims this item quantifies over occur in cross-references to Definition 10.4, in remarks quantifying the effect of a rule stated here, and in Example 17.3's evaluation of (8.6), none of them an asserted identity or inequality of the kind this item quantifies over. Theorem 8.13 and Definition 8.12 state inequalities between reals and the exact quantities of the setting and name neither.

(ii) is the content of Definition 10.9. That the residuals are not sufficient is immediate: $r_1 = \cdots = r_6 = 0$ is a statement about the *computed* matrices, which are consistent with each other but are not shown to be close to the exact ones; a compilation that lost precision uniformly would satisfy every residual.

(iii) is a stipulation of the emitted behaviour, together with the refusal convention of the module's interface (`spec/modules/receiver/interface.yaml`); it proves nothing, and Requirement 14.5 is its engine form. It is consistent because the rigorous label is given only where every emitted weight is a bound by (vii) at every setting whose enclosures meet its hypothesis, which (vii) shows for the unconditioned family and each submodule shows for its own, and that is the only thing the label says; the floating-point label claims nothing; and it is sound because a label withheld is never unsafe.

(iv) The compiled matrix is absent by hypothesis, so no product is formed; the remaining floating-point steps of the floating-point evaluation

are the eigenvalue extremes in (6.5) and the arithmetic of (6.2). On this route the enclosures of the port Grams that the rigorous evaluation reads are the declared Grams, exactly (Definition 9.8).

(v) The first sentence is Definition 8.7 together with Assumption 11.11, which is the hypothesis Theorem 8.14's proof discharges (7.5) with on that route. The last conjunct of the first sentence, that r_6 is blind to the direction the assumption constrains, is read off (10.6). Assumption 11.11 bounds each computed $\widehat{\beta}^{*(t)}(n)$ from above at the settings carrying a depth whose emitted weight is that of Definition 8.7 in floating point, which are among those over which the maximum in (10.6) runs, and at the indices $N < n \le m_t^*$; at the other settings of that maximum it constrains nothing, and the argument below is term by term. The direction it constrains is a computed value above $\beta^{*(t)}(n) + \varepsilon_{\mathrm{eig}}$. At such a setting the hypotheses of Theorem 6.2 hold (Definition 8.1 (b)), so (7.2) of Lemma 7.2 gives $\beta^{(t)}(n) \le \beta^{*(t)}(n)$, and $\varepsilon_{\mathrm{eig}} \ge 0$ (Definition 10.4). A term $\big(\beta^{(t)}(n) - \widehat{\beta}^{*(t)}(n)\big)_+$ is therefore positive only where $\widehat{\beta}^{*(t)}(n) < \beta^{(t)}(n) \le \beta^{*(t)}(n)$, an undershoot. A computed value with $\widehat{\beta}^{*(t)}(n) > \beta^{*(t)}(n) + \varepsilon_{\mathrm{eig}} \ge \beta^{(t)}(n)$ makes its term 0, which is also the term the exact value $\beta^{*(t)}(n)$ gives in its place. Each term depends on its own computed value alone, so r_6 takes the same value whether the computed values that violate Assumption 11.11 stand or are replaced by the exact ones: no value of r_6, and so no comparison with ε_6, tells the two apart. This is a property of r_6 as (10.6) defines it on the computed values; the number reported is its floating-point evaluation, which (ii) governs as it governs every residual. The second sentence is Definition 8.12 (iii): the window entry (8.9) is formed from $\check{\beta}(n)$ and the rigorous window bound $w(n)$, and neither is a computed eigenvalue or contains $\varepsilon_{\mathrm{eig}}$; Theorem 8.13 (iii), which assumes nothing, is then what makes the weight a bound. The third sentence is a property of the definitions rather than an argument: β^* occurs in (8.5) nowhere, r_6 exists only where a depth is declared, and ε_6 and $\varepsilon_{\mathrm{eig}}$ are required only there (Definition 10.1, Requirements 14.1 and 13.2; the inputs table of `spec/modules/receiver/interface.yaml`); and an enclosed weight at a setting without a depth is (8.5) read on enclosed bound data (Definition 8.12 (iii)). The fourth is (iii), since a weight of Definition 8.7 is not formed by the rigorous evaluation.

(vi) Theorem 9.5 (ii) and (iii) state that the enclosures it forms are complex enclosures of the blocks of the click-pattern and emitted POVMs; Definition 10.9 computes them in exact rational arithmetic, every rule of Lemma 9.2 being a finite computation on Gaussian rationals and rationals. By Definition 9.1, an entry x with centre x_c and radius r satisfies $|x - x_c| \le r$, so $x = x_c$ when $r = 0$; the entries named in Theorem 9.9 have radius zero under its hypothesis by its item (ii). A block is Hermitian, and an enclosure with a Hermitian centre supplies the hypotheses of Lemma "Positive semidefiniteness from an entrywise enclosure"

(foundations/lem:enclosure-psd) (i) with X the block, so the conclusion of that lemma is a property of the block itself. The last two sentences are (ii) applied to the floating-point block matrices.

(vii) *The enclosures.* For a receiver without a probe the setting family is that of Corollary 8.2: $B = A_\Sigma$ and $B_k = A_{[K]\setminus\{k\}}$ are sums of port Grams, enclosed from the enclosures of the port Grams by Lemma 9.2 (i); $\gamma = \beta_k = 1$ exactly; and each window block $M_{\mathrm{mc}}^{(=n)} = \mathbb{1} + (K-1)\Pi_{\mathrm{vac}}([K])^{(n)} - \sum_k \Pi_{\mathrm{vac}}([K]\setminus\{k\})^{(n)}$ ((8.3)) is enclosed by Theorem 9.5 (ii), taken with the depth as its cutoff, and the same rules. Each is a complex enclosure of the quantity it names, which is the hypothesis of (vii) for such a receiver. At every setting the hypothesis covers, an enclosure of a Hermitian matrix made Hermitian-centred by Lemma 9.2 (v) is an entrywise enclosure.

The four bounds. For an entrywise enclosure (C, r) of a Hermitian X and a rational $\rho \geq 0$ with $\rho^2 \geq \sum_{i,l} r_{il}^2$, Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (i) and (ii) give $\lambda_{\min}(X) \geq \lambda_{\min}(C) - \rho$ and $\lambda_{\max}(X) \leq \lambda_{\max}(C) + \rho$, and Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) (ii) and (iv) bound $\lambda_{\min}(C)$ from below and $\lambda_{\max}(C)$ from above at every real V that admits a rational θ with $0 \leq \theta < 1$ and $\|V^{\mathsf{T}}V - \mathbb{1}\|_F \leq \theta$, the hypothesis of that lemma, the identity with $\theta = 0$ among them, however V was obtained, and as rationals by its item (iii). For a complex enclosure (x_c, r) of a real x, $|x - \operatorname{Re} x_c| = |\operatorname{Re}(x - x_c)| \leq |x - x_c| \leq r$, so $\operatorname{Re} x_c - r \leq x \leq \operatorname{Re} x_c + r$. Hence the reals of Definition 10.9 satisfy $u \geq \lambda_{\max}(B)$, $\ell_k \leq \lambda_{\min}(B_k)$, $g \leq \gamma$, $b_k \geq \beta_k$ and $\lambda_n \leq \lambda_{\min}(M_{\mathrm{mc}|t}^{(=n)})$.

The weight. Theorem 8.13 (i) makes the clipped reals enclosed bound data and the window values rigorous window bounds, so the weight formed from them by Definition 8.12 (iii) is their enclosed weight; it is a finite expression in rationals built from sums, products, minima and maxima by (8.5) and (8.7), so a rational computed exactly. Theorem 8.13 (iii) gives the bound and the range, for the enclosed weight and for every real between 0 and it, under the hypotheses of Definition 8.1 alone. A floating-point number enters, if at all, only as the basis V read as the rational matrix it is, which decides tightness and never validity, so the difference (ii) speaks of does not arise. □

The eigenvalue-slack assumption is the one relation between a floating-point and an exact number the receiver asserts; enclosures and rigorous weights are proved. A claim outside the exception of Theorem 10.10 may still mention a receiver tolerance, in a cross-reference, a remark or an example.

Remark 10.11 (Scope of the exactness rule). The one relation between a floating-point number and an exact one that this contract does assert is Assumption 11.11; it is a separate assumption because it is invoked only on the opt-in route of Definition 8.7, evaluated in floating point. The enclosures of the photon-number blocks (Theorem 9.5) relate rationals computed in exact arithmetic to the model, and so

do the rigorously evaluated weights (Theorem 8.13); they are proved, not assumed. An artifact carries the rigorous evaluation label exactly where every weight it emits was evaluated rigorously, and the floating-point label otherwise; no path in this module produces another label (Theorem 10.10 (iii), Requirement 14.5).

Item (i) of Theorem 10.10 does not say that a claim outside its exception never *mentions* a receiver tolerance: such claims do, in cross-references to Definition 10.4 recording which declared number decides an exact condition in the evaluation model of Definition 10.9 (Definition 6.1, Definition 19.1, Remark 19.7), and in remarks and an example that quantify the effect of a rule stated in that theorem (Remark 8.6, Remark 8.11, Example 17.3); a cross-reference asserts nothing. Nor does the item *enumerate* the exceptions: the rule decides each claim by reading that claim's own statement.

newly derived · accepted

A rigorous evaluation of the bound needs lower enclosures of c and γ and upper ones of the per-port d and β, each intersected with [0, 1], which excludes no exact value. In the window a rigorous lower bound on the block eigenvalue replaces the computed one, and A-EIG is not invoked.

The floating-point evaluation of Theorem 10.10 leaves a gap between the emitted and the exact bound. This remark records in which direction each input has to be enclosed to close it; Requirement 13.5 closes it in these directions, and Theorem 8.13 proves that the result is a bound.

Remark 10.12 (Directions of rigorous enclosures of the bound inputs). Closing the gap means replacing the four inputs of (6.2) by rigorous *outer* enclosures taken in the conservative direction, formed from enclosures of B, the B_k and the prefactors rather than from their floating-point evaluation. This engine forms them in exact rational arithmetic, from the enclosures of the port Grams behind the reference of the blocks (Theorem 9.5), which are proved and need no interval-arithmetic assumption, and from the least-eigenvalue bounds of the foundations contract (Requirement 13.5). A weight so evaluated can carry a run to the certified tier (Definition "The tiers of a reported key rate" (foundations/def:tiers)) where the run's own certified evaluation succeeds; a weight evaluated in floating point cannot. There are four directions, not two. Reading the signs off $\beta(n) = 1 + (K-1)\gamma c^{\,n} - \sum_k \beta_k d_k^n$ with $n \geq 1$ and $c, d_k \in [0,1]$:

- c: a *lower* enclosure, i.e. an *upper* enclosure of $\lambda_{\max}(B)$ in (6.5);
- d_k: an *upper* enclosure, i.e. a *lower* enclosure of $\lambda_{\min}(B_k)$, since the coefficient $-\beta_k$ is non-positive;
- γ: a *lower* enclosure, since the coefficient $(K-1)c^{\,n}$ is non-negative;
- each β_k: an *upper* enclosure, since the coefficient $-d_k^n$ is non-positive.

The last two also preserve the hypothesis $\gamma \leq \beta_k$ of Theorem 6.2, which an enclosure in the opposite direction could break.

Every one of the four is taken in $[0,1]$, and that is not optional. Each of $c, d_k, \gamma, \beta_k \in [0,1]$ is a *theorem*: the admissibility paragraph of Theorem 6.2 derives (6.1) from $0 \preceq B_k \preceq B \preceq \mathbb{1}$ and $\gamma \leq \beta_k \leq 1$. So intersecting each rigorous enclosure with $[0,1]$ excludes no exact value, keeps it a valid enclosure, and keeps it on the conservative side of (6.2). Without the intersection a rigorous outer

enclosure can leave the interval in each of the four cases, and it can do so at ordinary devices, not only at contrived ones: $\overline{\lambda_{\max}(B)} > 1$ where $\lambda_{\max}(B) = 1$ exactly puts $\underline{c} < 0$; $\underline{\lambda_{\min}(B_k)} < 0$ at a singular B_k (at a receiver without a probe, one with an input polarization that reaches no port other than k, the configuration Remark 6.7 discusses) puts $\overline{d_k} > 1$; and $\overline{\beta_k} > 1$ is possible wherever $\beta_k = 1$ exactly, as at every probe-off setting. Data outside (6.1) is not admissible bound data, and *both* lemmas that read the bound data use the excluded part: Lemma 6.8 (i) obtains its tail step from $d_k \le 1$, and Lemma 6.5's chord argument is taken on $u_k \in [\beta_k c^n, \beta_k]$, which is $d_k \in [c, 1]$. Taken in $[0, 1]$, the enclosed tuple is admissible bound data, both lemmas apply to it unchanged, and β evaluated on it is a lower bound on β evaluated on the exact tuple; these are the enclosed bound data of Definition 8.12, and Theorem 8.13 states both facts.

The domain on which the enclosure of c is taken, and why this is not the clamping Definition 10.4 (iv) forbids. "β is non-decreasing in c" holds on $c \ge 0$ and *fails* below zero at even n, which is exactly what Remark 10.8 states for the floating-point evaluation; and a rigorous outer enclosure at a receiver where $\lambda_{\max}(B) = 1$ exactly can put its lower end below zero. The enclosure of c is therefore taken *in* $[0, 1]$: the evaluation uses $\underline{c} := \max\{0,\, 1 - \overline{\lambda_{\max}(B)}\}$. That is sound, and for a reason available inside this contract: $c_{\text{ex}} \in [0, 1]$ is a *theorem* (the admissibility paragraph of Theorem 6.2, from $0 \preceq B \preceq \mathbb{1}$), so intersecting a rigorous enclosure with $[0, 1]$ loses no exact value, $\underline{c} \le c_{\text{ex}}$ still holds, and on $[0, 1]$ the monotonicity does hold, so $\beta(\underline{c}) \le \beta(c_{\text{ex}})$ at every n. This is the opposite of the situation of Definition 10.4 (iv): there, no enclosure exists, the computed c is not known to bound c_{ex} on either side, and clamping it to 0 would assert a relation to the exact value that nothing supports, so nothing is clamped and Remark 10.8 states the resulting two-sided direction. Here, the enclosure *is* a rigorous bound and the intersection is with a domain the contract proves; nothing is asserted that is not already a theorem.

A fifth direction, where an exact window is declared. The four above are the four inputs of (6.2). Definition 8.7 adds one more computed quantity to the emitted bound, $\beta^*(n)$, and it takes a *lower* enclosure: it enters (8.6) as a lower bound on itself. As with the other four the enclosure is taken in $[0, 1]$, which excludes no exact value because $\beta^*(n) \in [0, 1]$ is a theorem (Lemma 7.2 with Definition 8.1), and as with c this is an intersection with a domain the contract proves rather than a clamp. A rigorous lower enclosure of $\beta^*(n)$ is a rigorous window bound of Definition 8.12: it takes the place of the computed eigenvalue less the slack, so ε_{eig} does not enter the weight and Assumption 11.11 is not invoked at that setting, the weight being a bound by Theorem 8.13. This is the only route in this remark that removes an assumption from the ledger rather than tightening a number; the declared ε_{eig} stays for a setting the engine evaluates in floating point.

Uses Def. 6.1, Thm. 6.2, Lem. 6.5, Rem. 6.7, Lem. 6.8, Lem. 7.2, Def. 8.7, Def. 10.4, Rem. 10.8 · *Used by* Req. 13.5 · newly derived · accepted

11 Assumptions

This section states every assumption the derivations use, the implicit ones included. Most device assumptions spell out the trusted, exactly characterised receiver (Assumption 11.1): the declared instrument is exactly the

physical receiver, with every optical path declared, passive, fed by vacuum on every undeclared mode, read by threshold detectors that respond to any photon in their ports, and fixed independently of the adversary. Round homogeneity of the receiver (Assumption 11.10) refines the protocol compiler's round-homogeneity premise instead. The eigenvalue-slack assumption is the one premise on the arithmetic, and only a setting with an exact window whose weight is evaluated in floating point invokes it. The engine verifies none of them; conditions decided on the declared data alone are definitions with requirements, not assumptions.

Imports foundations/def:round-homogeneity

The declared instrument describes the physical receiver exactly, and the receiver is under the legitimate party's control. It is the parent of most of the receiver's device assumptions, which spell out its parts.

The receiver is trusted and known exactly. No uncertainty set, no outer Gram family, no device family and no characterisation protocol is represented anywhere in this contract; a receiver known only up to a stated uncertainty is future scope (`spec/future-scope.md`).

Assumption 11.1 [A-TRUST] (Trusted and exactly characterised receiver)**.** (i) The declared instrument describes the physical receiver *exactly*: its mode space, its declared element graph or the supplied instrument matrix or port Gram family, the port map, the efficiencies, the dark-count probabilities and the probe states; (ii) the receiver is under the legitimate party's control: it is not selected, tuned or otherwise influenced by the adversary.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: The receiver you declared is the receiver you run. Its mode space, its optical layout (or the instrument or Gram matrices you supplied), the assignment of outputs to detectors, the efficiencies, the dark-count probabilities and the probe states are those of the physical device, with no uncertainty to account for. The device is under your control, and nobody else can select, tune or influence it. A characterisation uncertainty is not modelled: the result applies to the device exactly as declared.

Category receiver · *Affects* secrecy · *Assumed by* Lem. 4.2, Cor. 8.2, Cor. 8.15, Rem. 18.4, Prop. 21.4 · *Machine key* `receiver_trusted_exact` · newly derived · accepted

The assumptions on the physical receiver and the run, and A-EIG on the arithmetic, are never verified. The conditions decided on the declared data are definitions with requirements, and the floating-point evaluation is a restriction the contract places on itself.

Remark 11.2 (Assumptions this module cannot check)*.* Every assumption of this contract that is a statement about the physical receiver or the run is one the engine never verifies, as the contract says of each: no function of the declared data decides it, and validation decides only those of its clauses that constrain the declared data. Assumption 11.11 is a statement about the *arithmetic*; it is likewise never verified, and it is invoked only at a setting where a caller declares an exact-window depth and the weight is evaluated in floating point (Requirement 15.3), so a declaration

that declares none, or whose weights are all evaluated rigorously, is unaffected by it. The conditions that constrain the declared data alone are not assumptions of this contract: the well-formedness of a declared element graph (Definition 3.2) and the occupation-diagonality of the declared probe states (Definition 2.3) are decided on the declared data (Requirements 12.1 and 12.2). The floating-point evaluation of Definition 10.9 is a restriction this contract places on itself and asserts nothing that a run could violate.

newly derived · accepted

Every optical element is passive linear optics, with no gain and no mixing of creation and annihilation operators. Total photon number is then a conserved grading, on which every block statement of the module rests.

Under this assumption the total photon number is a conserved grading of the receiver's modes.

Assumption 11.3 [A-PASSIVE] (Passive linear optics)**.** (i) Every optical element is passive linear optics: it maps annihilation operators to combinations of annihilation operators, $b_i = \sum_j G_{ij} a_j$, with $G^\dagger G \preceq \mathbb{1}$; (ii) no element mixes a with $a^\dagger$, and none provides gain.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Every optical component of your receiver is passive and linear: beam splitters, wave plates, polarisers, attenuators and the like. No component amplifies light, and none is nonlinear (no frequency conversion, no squeezing). The total photon number is then never increased inside the receiver. If a component can add photons, the photon-number blocks the analysis rests on do not exist.

Category receiver · *Affects* secrecy · *Refines* A-TRUST · *Assumed by* Lem. 4.2, Cor. 8.2, Cor. 8.15, Rem. 16.10, Prop. 21.4 · *Machine key* `passive_optics` · newly derived · accepted

Undeclared input modes and dilation modes start in vacuum. A mode that can carry light to a detector is declared, as a channel mode if the adversary can populate it and as a source mode if the legitimate party does.

Vacuum on every undeclared input is what lets the receiver be described by the columns of its declared inputs alone.

Assumption 11.4 [A-VAC] (Vacuum on undeclared input modes)**.** (i) Every mode of $\mathbb{M} \setminus \mathbb{I}$, and every mode of the dilation $\mathbb{E}$, carries the vacuum state at the input; (ii) every mode that can carry a non-vacuum state and can couple to a port is declared in $\mathbb{I}$: in $\mathbb{I}_C$ if the adversary can populate it, in $\mathbb{I}_P$ if the legitimate party does.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Every optical input of your receiver that you did not declare as an input, and every internal loss or ancilla mode, receives no light. A mode that can carry light to a detector is declared: as a channel mode if an attacker could put light into it, and as a source mode if your own equipment, for example a probe laser, does.

Category receiver · *Affects* secrecy · *Refines* A-TRUST · *Assumed by* Lem. 2.6, Cor. 3.5, Lem. 4.2, Cor. 8.2, Cor. 8.15, Rem. 16.8, Rem. 16.9, Prop. 21.4 · *Machine key* `undeclared_inputs_vacuum` · newly derived · accepted

Every optical path from outside the receiver to a detector is a declared mode. A bright-illumination or wavelength-shifted attack targets exactly this premise.

A photon arriving in a spatial, spectral, temporal or polarization mode that the model does not carry does not exist for this module. The assumption is therefore not free: it is exactly the one a bright-illumination or wavelength-shifted attack violates, and violating it makes the emitted POVM a description of a device the adversary is not attacking.

Assumption 11.5 [A-MODES] (Completeness of the declared modes)**.** (i) Every optical path from outside the receiver to a detector is represented by a declared mode of $\mathbb{M}$.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Every way light from outside can reach one of your detectors (every spatial, spectral, temporal and polarisation mode it can arrive in) is represented by one of the declared modes. Light the model does not carry, for example at another wavelength or along a path straight to the detectors as in bright-illumination attacks, violates this. Declare the mode, or declare a model of the attack. This premise is about the completeness of the receiver's optical description; that nothing reaches the station except through its declared inputs, and that nothing leaks out of it, is the laboratory premise of this station's party, A-LAB.

Category receiver · *Affects* secrecy · *Refines* A-TRUST · *Assumed by* Lem. 4.2, Cor. 8.2, Cor. 8.15, Rem. 16.10, Prop. 21.4 · *Machine key* `no_undeclared_paths` · newly derived · accepted

Each detector reports one bit per gate, with no photon-number resolution, no memory or timing effects, and no correlations beyond the declared dark-count channel. It fixes the click-or-no-click outcome alphabet the receiver's measurement is built on.

Assumption 11.6 [A-THRESH] (Threshold detectors without memory)**.** (i) Each detector reports one bit per detection gate: click or no click; (ii) there is no photon-number resolution, no dead time, no afterpulsing and no timing jitter; (iii) there is no correlation of any kind between gates or between detectors beyond the declared dark-count channel; (iv) dark counts are independent across rounds, and across ports unless a correlated channel is declared explicitly (Definition 19.1).

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Your detectors are threshold (click or no-click) detectors read once per gate. They have no photon-number resolution, no dead time, no afterpulsing and no timing jitter. Gates and detectors are correlated only through the dark-count model you declared. Dark counts are independent from round to round, and between detectors unless you declared a correlated dark-count channel. Detectors with dead time or afterpulsing at your gate rate violate this.

Uses Def. 1.1 · *Category* receiver · *Affects* secrecy · *Refines* A-TRUST · *Assumed by* Cor. 8.2, Cor. 8.15, Rem. 16.10, Rem. 19.7, Prop. 21.4 · *Machine key* `threshold_memoryless_detectors` · newly derived · accepted

A port clicks exactly when the modes it collects carry at least one photon, whatever their distribution over those modes. It is the multimode form of A-THRESH, and it gives a detector that collects several modes a meaning.

This is the multimode form of Assumption 11.6. It is easily left implicit, but without it "a detector that collects several modes" is undefined. Under it the click decision is invariant under any passive unitary acting inside the port, since such a unitary preserves the total photon number of the port's modes.

Assumption 11.7 [A-PORT] (Mode-insensitive port response)**.** (i) Port k clicks if and only if the total photon number in the modes $\mathbb{P}_k$ is nonzero; equivalently, its no-click element is the joint vacuum projector $\Pi_k^0 = \mathfrak{F}(\mathbb{1} - E_{\{k\}})$ on those modes.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Each detector port clicks exactly when at least one photon arrives in any of the modes it collects, whatever their distribution over those modes. A detector whose click probability depends on the spatial, spectral or temporal mode of the arriving photon (efficiency mismatch) violates this. Such a mismatch has to be declared as mode-dependent loss in front of the port.

Category receiver · *Affects* secrecy · *Refines* A-THRESH · *Assumed by* Lem. 2.6, Cor. 8.2, Cor. 8.15, Prop. 21.4 · *Machine key* `mode_insensitive_ports` · newly derived · accepted

Declared component parameters, efficiencies, dark-count probabilities and probe mean photon numbers are exact. A non-zero parameter tolerance would be an uncertainty set, which is out of scope and is refused rather than ignored.

This is the parameter part of Assumption 11.1. A parameter tolerance is an uncertainty set and is out of scope: uncertainty sets on the device parameters are future scope (`spec/future-scope.md` §11), and the reserved tolerance field that would carry one is refused unless absent or exactly zero.

Assumption 11.8 [A-EXACT] (Exact declared parameters)**.** (i) Every declared component parameter, efficiency, dark-count probability and probe mean photon number is exact.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Every number you declared for a receiver component (each efficiency, dark-count probability, probe mean photon number and other component parameter) is the device's true value, not an estimate with an error bar. The engine refuses a non-zero tolerance field, and an uncertainty set is not modelled. A result computed from a measured value therefore applies to a device with exactly that value.

Category receiver · *Affects* secrecy · *Refines* A-TRUST · *Assumed by* Lem. 4.2, Cor. 8.2, Cor. 8.15, Prop. 21.4 · *Machine key* `exact_parameters` · newly derived · accepted

The receiver configuration and the probe declaration are fixed before the run and cannot be chosen or influenced by the adversary. With A-TRUST this makes one declared POVM, rather than a family, the right object.

Together with Assumption 11.1, this assumption is what makes a single POVM, rather than a family, the right object.

Assumption 11.9 [A-NOSEL] (Device fixed independently of the adversary)**.** (i) The component list, its parameters, the port map and the probe declaration are fixed before the protocol run; (ii) none of them is chosen by the adversary, and none can be influenced by her, in particular not through a degree of freedom the model does not carry, such as wavelength.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Your receiver's configuration (its components and their settings, which output goes to which detector, and the probe settings) is fixed before the run. An attacker cannot choose or change it, including through properties the model does not describe, such as the wavelength of injected light shifting a beam splitter's ratio. If an attacker can steer the receiver this way, the declared measurement is the wrong one and the result does not apply. This premise is about the configuration of the receiver itself; what may leave or enter the station by other routes is the laboratory premise of this station's party, A-LAB.

Category receiver · *Affects* secrecy · *Refines* A-TRUST · *Assumed by* Lem. 4.2, Cor. 8.2, Cor. 8.15, Prop. 21.4 · *Machine key* `device_not_adversarial` · newly derived · accepted

The same instrument, probe declaration and dark-count channel describe every round, and the receiver has no memory across rounds. One receiver artifact then describes the whole run.

Each round is described by the same three objects, independently of every other round. Rounds are not averaged, and per-round variation is not modelled.

Assumption 11.10 [A-ROUND] (Round homogeneity of the instrument, probe and dark counts)**.** The family of the instrument, the probe declaration and the dark-count channel is round-homogeneous (Definition "Round homogeneity of a per-round family" (foundations/def:round-homogeneity)) in both clauses of that definition: (i) the same instrument, the same probe declaration and the same dark-count channel describe every round of the protocol; (ii) the receiver acts in every round as they describe, whatever the inputs and outputs of the other rounds, with no memory across rounds.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Your receiver behaves the same way in every round: the same optics, the same probe settings and the same dark-count behaviour. Nothing it does in one round depends on the inputs or outputs of other rounds: no detector memory, and no drift that the declaration averages over. A receiver whose efficiency drifts during the run violates this. Only parts of a run fixed before it starts, each processed as its own execution and composed under A-SEPARATE, can be analysed separately; a split chosen after seeing the data is not covered. A receiver whose detectors remember earlier clicks, through afterpulsing, a dead time or any other dependence on earlier rounds, violates it too, and nothing in this engine repairs that: no input describes more than one round.

Uses foundations/def:round-homogeneity · *Category* receiver · *Affects* secrecy · *Refines* A-RHOM · *Assumed by* Lem. 4.2, Cor. 8.2, Cor. 8.15, Prop. 21.4 · *Machine key* `round_homogeneity` · newly derived · accepted

Where an exact window is evaluated in floating point, the declared slack bounds any overestimate of the exact block minimum eigenvalue by the computed one. This one-sided premise on the arithmetic is the price of that window; a setting without one, or evaluated rigorously, does not invoke it.

On finitely many blocks the exact window takes the larger of the analytic bound and a computed smallest eigenvalue less a declared slack. That the computed extreme eigenvalue of a matrix of dimension $\binom{D+n-1}{n}$, D being the dimension of the channel input space, does not overshoot the exact one by more than the slack is a premise about the arithmetic, not a theorem. It is one-sided: since the analytic bound stays as a floor, an undershoot costs tightness only. A declaration that requests no exact window does not invoke it, and neither does a setting whose weight is the enclosed weight of Definition 8.12, where a proved lower bound on the block eigenvalue stands in place of the computed one.

Assumption 11.11 [A-EIG] (Declared slack on the computed block eigenvalues)**.** (i) At every setting t carrying a declared exact-window depth m_t^* whose emitted weight is that of Definition 8.7 formed by the floating-point evaluation of Definition 10.9, and every index $N < n \le m_t^*$, the value $\widehat{\beta}^{*(t)}(n)$ an implementation computes for the exact block minimum eigenvalue $\beta^{*(t)}(n)$ of (7.1) satisfies

$$\widehat{\beta}^{*(t)}(n) - \varepsilon_{\mathrm{eig}} \;\le\; \beta^{*(t)}(n). \tag{11.1}$$

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Applies only if you request an exact-window depth at a setting whose weight the receiver evaluates in floating point. Wherever your declaration describes a valid receiver exactly, the receiver evaluates the weight rigorously instead, from a proved lower bound on each block eigenvalue, and does not use this premise; the record says which evaluation each setting got. Where it falls back to floating point, the eigensolver that computes the smallest eigenvalue of each requested high-photon block may overestimate it by at most the slack ε_eig you declared. The engine cannot check this and trusts its floating-point eigensolver up to ε_eig. Without an exact-window request this premise is not used, and the bound comes from a closed form that rests on no premise about the eigensolver.

Uses Def. 7.1, Def. 8.7, Def. 8.16, Def. 10.9 · *Category* numerical trust · *Affects* secrecy · *Assumed by* Rem. 8.11, Thm. 8.14, Cor. 8.15, Thm. 10.10 · *Machine key* `exact_window_eigenvalue_slack` · newly derived · accepted

Part II
Engine requirements

What an implementation of this module must do, record or refuse; each requirement names the result that motivates it.

12 Requirements: compilation, representation and outcomes

These requirements bind the engine where the mathematics leaves a choice to the implementation. A component graph that is not well formed, and a probe state outside the Gaussian tier, are refused rather than repaired, and the port map, the one-mode-per-port reading and the emitted alphabet are declared or decided, never inferred.

A declared element graph failing any of (E1) to (E3) is refused with the failing condition named, and the offending element or edge for (E1) or element pair for (E3). No matrix is emitted and no ordering is invented.

Theorem 3.4 (iv) makes each condition of a well-formed graph decidable by finite inspection; this requirement says what the engine does when one fails.

REQUIREMENT 12.1 (Refusal of a graph that is not well formed). The engine refuses a declared element graph (input `element_graph`) that fails any of conditions (E1)–(E3) of Definition 3.2, with the failing condition named, together with the offending element or edge for (E1) and the offending element pair for (E3); no matrix is emitted, and no ordering is invented to repair it. The refusal follows the `fail_closed` convention.

Motivated by Def. 3.2, Thm. 3.4 · implemented (effective: accepted) · *Code* certiqs_key.receiver.graph:check_element_graph

The engine admits declared probe states of the Gaussian tier only, which are occupation-diagonal by construction, and refuses any other. Conditioning on an admitted probe therefore keeps the channel photon-number blocks.

The channel grading of the blocks is recovered from the total one only under a condition on the probe state (Remark 4.9); occupation-diagonality is the stronger form of that condition which validation can check.

REQUIREMENT 12.2 (Admission of Gaussian-tier probe states only). The engine admits a probe declaration (input `probe_declaration`) only in the Gaussian tier of Definition 2.3, whose states are occupation-diagonal by construction, and refuses a declared probe state outside it under the `fail_closed` convention.

Motivated by Def. 2.3, Rem. 4.9 · implemented (effective: accepted) · *Code* certiqs_key.receiver.settings:check_probe_declaration

The port map is an input of the declaration and is never inferred from the instrument matrix. Which output modes lie in a port has no observable content beyond the port Grams it produces.

REQUIREMENT 12.3 (Declared port map). The engine takes the detector ports as a declared input (`port_map`) and never infers them from the instrument matrix: by Remark 4.3, the partition of the output modes into "in some port" and "in no port" has no observable content beyond the port Grams it produces.

Motivated by Rem. 4.3 · accepted

The engine uses a one-mode-per-port representation of a receiver only where every emitted port Gram has rank at most one, the criterion of Theorem 5.1 (iv), and never by default. Ranks are decided on the Grams after every detector efficiency is applied.

A receiver architecture must not assume one optical mode per detector unless rank one is established (`spec/architecture.md#32-the-seven-cores`, the receiver core). Remark 5.4 explains why nothing else in this contract needs the assumption.

REQUIREMENT 12.4 (One mode per port only where the rank criterion holds). The engine uses a one-mode-per-port representation of a receiver only where the criterion of Theorem 5.1 (iv), rank $A_k \leq 1$ for every k, holds, and never assumes it by default. The criterion is decided on the emitted port Grams, that is after every detector efficiency is applied (Remark 18.2), with each rank decided as Requirement 14.3 fixes.

Motivated by Thm. 5.1, Rem. 5.4, Req. 14.3, Rem. 18.2 · implemented (effective: accepted) · *Code* certiqs_key.receiver.grams:one_mode_per_port

The engine takes the emitted outcome alphabet and the coarse-graining onto it as declared inputs and never fixes an alphabet internally. The optical profile's six outcomes at four detectors are one declared instance, not the definition.

The optical profile's six-outcome alphabet $\mathcal{Y} = \{\mathrm{nc}, \mathrm{sc}_H, \mathrm{sc}_V, \mathrm{sc}_+, \mathrm{sc}_-, \mathrm{mc}\}$ (`spec/notation.md#37-optical-profile-symbols`) is the instance $K = 4$, $\varpi(\varnothing) = \mathrm{nc}$, $\varpi(\{k\}) = \mathrm{sc}_k$, and $\varpi(S) = \mathrm{mc}$ for $S \in \mathsf{MC}$: one declared instance among many.

REQUIREMENT 12.5 (Declared emitted alphabet and coarse-graining). The engine takes the emitted alphabet $\mathcal{V}$ as an input (`emitted_outcome_alphabet`) and the declared coarse-graining ϖ of Definition 5.5 with it (`outcome_coarse_graining`); it never fixes an outcome alphabet internally. A receiver that fixed six outcomes over four detectors internally would make the outcome alphabet an implementation detail of the receiver rather than a declared property of the protocol, and every downstream index would then depend on an undeclared convention.

Motivated by Def. 5.5 · accepted

13 Requirements: the emitted weight

These requirements fix what a declaration must supply before any weight is computed and which declarations are refused. They also fix that the weight is evaluated rigorously wherever the port Grams are enclosed, with no eigenvalue slack, and what the artifact records beside each emitted weight, so that a reader can recompute it.

Imports foundations/cor:least-eigenvalue-over-enclosure, foundations/def:exact-reference, foundations/lem:least-eigenvalue-bound, foundations/req:rounding-direction

A declaration must supply a window at every setting where predicate (X) fails, and one missing there is refused rather than defaulted. Which settings need one is read off the declaration; a window at or below the cutoff is refused everywhere, and one above it at an exact setting is unused.

Branch 2 of (8.5) is evaluated on a declared window m_t, and the contract fixes no default for it. Whether a setting takes that branch is a property of the declaration: by Remark 8.5 it is read off the declaration alone, before any arithmetic.

Requirement 13.1 (Declared window at every setting whose exact flag is general). A declaration supplying no window m_t (`tail_window`) at a setting where predicate (X) of Definition 8.3 fails is refused rather than defaulted; by Remark 8.5 that condition is decidable at input validation from declared data alone. The engine therefore requires m_t exactly at the settings whose exact flag is general, and at no others. A declared window satisfies $m_t > N$ (Definition 8.3); one that does not is refused at every setting, whatever its flag. A window $m_t > N$ supplied at a setting whose flag is exact is unused, not an error.

Motivated by Def. 8.3, Rem. 8.5 · implemented (effective: accepted) · *Code* certiqs_key.receiver.validation:check_tail_window

A declaration carrying an exact-window depth must also declare the eigenvalue slack and the window-consistency threshold, and a missing one is refused, never defaulted. A depth at or below the cutoff, or above the declared window where predicate (X) fails, is refused.

Requirement 13.2 (Declaration of an exact-window depth and its tolerances). Declaring an exact-window depth (Definition 8.7) at any setting requires ε_{eig} and ε_6 (the bound on r_6 of Definition 10.1); the engine refuses a declaration missing either, rather than defaulting it, exactly as a missing window m_t is refused (Requirement 13.1), with `exact_window_declared_without_required_tolerance`. The engine refuses a depth violating $m_t^* > N$, or violating $m_t^* \le m_t$ at a setting where predicate (X) of Definition 8.3 fails, with `exact_window_depth_outside_declared_range`.

Motivated by Def. 8.3, Def. 8.7, Def. 10.1, Req. 13.1 · implemented (effective: accepted) · *Code* certiqs_key.receiver.validation:check_exact_window

The artifact carries every computed block eigenvalue of an exact window exactly as computed, and the eigenvalue slack enters only as the slack the window value subtracts. Nothing is clamped, so a reader can recompute each window value and see what the slack cost.

REQUIREMENT 13.3 (Computed block eigenvalues carried as computed). Nothing is clamped, and the computed value is emitted: at every setting carrying an exact-window depth, the engine carries each $\widehat{\beta}^{*(t)}(n)$, $N < n \leq m_t^*$, in the artifact exactly as computed, in `bound_quantifier_domain`, and $\varepsilon_{\mathrm{eig}}$ enters only as the declared slack subtracted in (8.6) (Definition 8.7), so that a reader can recompute (8.6) and see what the slack cost.

Motivated by Def. 8.7, Rem. 8.11 · accepted

An exact window is evaluated only on request, never by default. The slot carries the per-setting weights, and the artifact says for each which evaluation and which definition produced it and at what depth; block eigenvalues may also be reported as diagnostics.

REQUIREMENT 13.4 (Provenance of each emitted multiclick weight). The engine evaluates an exact window, that of Definition 8.7 or the rigorous window bounds of Definition 8.12, only where a declaration requests it by an exact-window depth, and never by default. Where it is not requested, the engine may also report block eigenvalues as a diagnostic; the values entering the strengthening slot (Definition 8.16), `high_photon_bounds`, are the per-setting weights $p_{\mathrm{mc}}^{(t)}$ of Theorem 8.14 — at a receiver without a probe the single value p_{mc} — each the enclosed weight of Definition 8.12, rounded down, where Requirement 13.5 forms it, and otherwise the floating-point evaluation of Definition 8.3 or 8.7; and the artifact says, for each, which evaluation produced it, which of Definitions 8.3 and 8.7 gives its formula and at what depth, in `bound_quantifier_domain`.

Motivated by Def. 8.3, Def. 8.7, Def. 8.12, Thm. 8.14, Def. 8.16, Rem. 8.18, Req. 13.5 · accepted

Wherever the port Grams are enclosed, the engine bounds the four inputs and the window blocks from rational enclosures, forms the enclosed weight exactly with no eigenvalue slack and emits it rounded down. Elsewhere it emits the floating-point weight and records why.

Remark 10.12 reads off the bound function the direction in which each input of the weight must be bounded, and Theorem 8.13 shows that a weight formed from bounds taken in those directions is a bound outright. The receiver holds what that needs: rational enclosures of its port Grams for every declaration with a model admissible exactly (Requirement 14.6), from which the operators of the bound, the prefactors and the window blocks are enclosed by the rules that enclose its photon-number blocks, and the least-eigenvalue bounds of Lemma "Least-eigenvalue lower bound from an

approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) and Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure), which turn an enclosure of a matrix into bounds on its extreme eigenvalues.

REQUIREMENT 13.5 (Rigorous evaluation of the multiclick weight). (i) *Where it applies.* On every declaration whose port Grams the engine encloses by Requirement 14.6 (iii), the engine evaluates the weight of every setting by (ii) to (iv), in exact rational arithmetic: the rigorous evaluation of Definition 10.9. At a setting where it cannot, it emits the weight of Definition 8.3 or 8.7 evaluated in floating point, with the reason `port_grams_not_enclosed` where the declaration has no model admissible exactly and `bound_inputs_not_enclosed` where a reciprocal that (ii) forms receives no enclosure.

(ii) *The enclosures.* At each setting t the engine forms, from the enclosures of the port Grams by the rules of Lemma 9.2, complex enclosures of $B^{(t)}$ and of every $B_k^{(t)}$, of $\gamma^{(t)}$ and of every $\beta_k^{(t)}$, and, where t carries an exact-window depth m_t^*, of $M_{\mathrm{mc}|t}^{(=n)}$ for $N < n \le m_t^*$, each enclosure of one of these Hermitian matrices made Hermitian-centred by part (v) of that lemma. For a receiver without a probe these are, by Corollary 8.2, $B = A_\Sigma$ and $B_k = A_{[K]\setminus\{k\}}$, sums of enclosed Grams, $\gamma = \beta_k = 1$ exactly, and the blocks of (8.3), formed from the enclosures of Theorem 9.5 (ii) taken with m_t^* as its cutoff. For a setting family a submodule constructs they are the enclosures of its prefactors, its operators and its window blocks that the submodule states, formed by the same rules.

(iii) *The four inputs and the window bounds.* For an entrywise enclosure (C, r) of a Hermitian matrix X and a rational $\rho \ge 0$ with $\rho^2 \ge \sum_{i,l} r_{il}^2$, an upper bound on $\lambda_{\max}(C)$ from Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) (iv) plus ρ bounds $\lambda_{\max}(X)$ from above, and a lower bound on $\lambda_{\min}(C)$ from part (ii) of that lemma less ρ bounds $\lambda_{\min}(X)$ from below (Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (i), (ii)); for a complex enclosure (x_c, r) of a real x, $\operatorname{Re} x_c - r \le x \le \operatorname{Re} x_c + r$. By these rules the engine forms an upper bound u on $\lambda_{\max}(B^{(t)})$, lower bounds ℓ_k on $\lambda_{\min}(B_k^{(t)})$, a lower bound g on $\gamma^{(t)}$, upper bounds b_k on $\beta_k^{(t)}$ and lower bounds λ_n on $\lambda_{\min}(M_{\mathrm{mc}|t}^{(=n)})$. Its enclosed bound data are (8.10) of Theorem 8.13 (i), each of the four intersected with $[0, 1]$, and its rigorous window bounds are $\max\{0, \lambda_n\}$. The output of a floating-point eigensolver may enter only as the basis V of Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound), read as

the exact rational matrix it is; a basis for which the engine forms no rational $\theta < 1$ satisfying the hypothesis of that lemma is replaced by the identity, which admits $\theta = 0$. No computed eigenvalue is used as a bound, and $\varepsilon_{\mathrm{eig}}$ is not subtracted.

(iv) *The weight.* The engine computes the enclosed weight $\check{p}_{\mathrm{mc}}^{(t)}$ (Definition 8.12) as an exact rational and emits, as the setting's weight in the strengthening slot `high_photon_bounds` (Definition 8.16), the largest binary64 number at or below it, rounded down as Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) rounds an emitted lower bound; by Theorem 8.13 (iii) the emitted number bounds the multiclick element of its setting on every block above the cutoff. The uniform weight is the minimum of the emitted per-setting weights.

(v) *What stays.* The floating-point evaluation of the blocks, the residuals and their tests, and, at a setting carrying a depth, the computed values of Requirement 13.3, are formed as they are without this requirement. At a setting evaluated by (ii) to (iv), the floating-point weight of Definition 8.3 or 8.7 enters no emitted number, and it may be recorded beside the emitted one as a diagnostic.

(vi) *The record.* Per setting, in `bound_quantifier_domain`: the evaluation that produced the emitted weight, `rigorous` or `floating_point`, with the reason of (i) where it is `floating_point`; and, where it is `rigorous`, the enclosed bound data, the rigorous window bounds and the enclosed weight as exact rationals, so that a reader can recompute the weight. A setting whose weight is `rigorous` is not one at which Assumption 11.11 is invoked (Requirement 15.3), and the artifact's evaluation label follows Requirement 14.5.

Reading: Wherever your receiver declaration describes a valid receiver exactly, the engine computes each high-photon multiclick weight it can from rational enclosures of the receiver's model, every bound taken on the safe side, and emits that weight rounded down: no floating-point eigenvalue is trusted and no eigenvalue slack is subtracted. The artifact records, setting by setting, whether it did. Where the declaration is valid only within the receiver tolerances, or a needed enclosure cannot be formed at a setting, the engine emits the floating-point weight there instead and records why; a key rate that reads such a weight is not certified.

Motivated by Thm. 6.2, Def. 7.1, Def. 8.1, Cor. 8.2, Def. 8.3, Def. 8.7, Def. 8.12, Thm. 8.13, Def. 8.16, Def. 9.1, Lem. 9.2, Thm. 9.5, Def. 10.9, Rem. 10.12, Ass. 11.11, Req. 13.3, Req. 14.5, Req. 14.6, Req. 15.3, foundations/cor:least-eigenvalue-over-enclosure, foundations/def:exact-reference, foundations/lem:least-eigenvalue-bound, foundations/req:rounding-direction · *Used by* Req. 13.4, Req. 14.5, Req. 15.3 · implemented (effective: accepted) · *Code* certiqs_key.receiver.rigorous:rigorous_weights

14 Requirements: residuals, tolerances and the evaluation label

These requirements say which residuals the engine emits, and that a residual or an admission test that fails refuses while nothing is repaired or clamped. The receiver tolerances have a declared domain, and every artifact carries an evaluation label, the rigorous one only where every weight was evaluated rigorously. Whether a declaration defines an admissible receiver exactly, and whether it is on the exact route, is decided in exact arithmetic and recorded with its cause, and the port Grams of every such declaration are enclosed.

Imports foundations/def:binary64

Every artifact carries exactly the residuals that exist on its input route and for its request; the key of any other residual is absent, never zero. The tolerances of the first and third residual are declared on every route, that of the sixth is required only with an exact-window depth.

The artifact already follows this rule for the record of the receiver construction, which is present exactly on route (iii); a residual that does not exist on the route or for the request taken is likewise absent, because emitting 0 would report a check that was not made.

Requirement 14.1 (Emission of the residuals that exist on the route and request). The engine emits with every artifact, in `validation_results`, those residuals of Definition 10.1 that exist on the input route taken and for the evaluation requested, and no other: the keys of r_1 and r_3 are absent, not zero, on route (i), and the key of r_6 is absent, not zero, at a declaration that requests no exact window. The blocks over which (10.5) maximises are the floating-point blocks $M_{v|t}^{(=n)}$ of the coarse-grained elements that Definition 10.1 names, before any dark-count channel, which the `photon_blocks` entry carries beside its dark-composed blocks (Requirement 22.9); r_5 does not run over the dark-composed blocks, on which the exact value 0 of Proposition 10.2 (iv) is not asserted. ε_1 and ε_3 are declared on every route, route (i) included, where they are unused (`numerical_tolerances`); ε_6 is required exactly where an exact-window depth is declared, on the same condition as r_6, and where it is declared elsewhere it is unused and not an error.

Motivated by Def. 10.1, Prop. 10.2, Req. 22.9 · implemented (effective: accepted) · *Code* certiqs_key.receiver.construct:emitted_residuals

Each emitted residual, and the residual gap of a declared window, is compared with its receiver tolerance, and one above it refuses the request. No block is Hermitized, symmetrized, projected, clamped or otherwise repaired: the artifact carries the computed block and its own defect.

A repaired block is a different operator, and no claim of this contract holds for it (Remark 10.3); so a residual only ever refuses.

REQUIREMENT 14.2 (Refusal on a residual above its tolerance, and no repair). The engine compares each residual that exists (Definition 10.1) with its receiver tolerance ε_i, and a residual above its tolerance is a *refusal*, not a warning; in particular $r_5 > \varepsilon_5$ refuses with the reason `photon_block_positivity_residual_above_tolerance`, and $r_6 > \varepsilon_6$ with the reason `window_eigenvalue_consistency_residual_above_tolerance`. In the same way, at every setting where predicate (X) of Definition 8.3 fails, the engine compares the residual gap g_{N,m_t} of (6.19) on the declared window m_t with $\varepsilon_{\mathrm{gap}}$, and a gap above it refuses. The direction is refuse-only: no block is Hermitized, symmetrized, projected onto the positive cone, clamped or repaired in any way; the artifact carries the computed block itself, and r_5 reports that block's own defect. The same holds of r_6: it refuses and repairs nothing (Remark 8.11 states the direction of failure it is blind to).

Motivated by Lem. 6.8, Def. 8.3, Rem. 8.11, Def. 10.1, Rem. 10.3 · implemented (effective: accepted) · *Code* certiqs_key.receiver.residuals:check_residuals

The engine decides each exact condition of Definition 10.4 by the test given there, every rank by the ε_2-rank and by no other rule. Bound data failing their admission test are refused and no artifact is emitted.

The refusal rules of the module's interface (`spec/modules/receiver/interface.yaml`) tabulate no reason string for a bound-data failure, and this contract adds none: fixing a contract-verbatim key here would settle an interface detail that no claim of this contract consumes.

REQUIREMENT 14.3 (Decisions at the receiver tolerances, and refusal of failed bound data). The engine decides each exact condition that Definition 10.4 lists by the test given there and by no other threshold, and every rank statement this module decides on floating-point data by (10.8); no other rule is licensed, there or at Theorem 5.1 (iv). Bound data failing item (iv) of Definition 10.4 are a failed validation, so `fail_closed` applies at this site: no artifact is emitted. The reason string is the implementer's to name, under the rule the refusal rules of the module's interface state for the validations they do not tabulate, none of the tabulated strings covering this check.

Motivated by Thm. 5.1, Def. 10.4 · implemented (effective: accepted) · *Code* certiqs_key.receiver.construct:construct_receiver

Each receiver tolerance is validated at input against its declared domain [0, 1), and a value outside it is refused. At one or more every test the tolerance decides would be vacuous or degenerate (Remark 10.6).

REQUIREMENT 14.4 (Validation of the receiver tolerances against their domain). The engine validates each of the receiver tolerances $\varepsilon_1, \ldots, \varepsilon_6$, ε_{gap} and ε_{eig} of Definition 10.4 at input against the declared domain $[0, 1)$, and refuses a value outside that interval with the reason `tolerance_outside_declared_domain`.

Motivated by Def. 10.4, Rem. 10.6 · accepted

Every receiver artifact says whether its weights were all evaluated rigorously or not, and no path produces another label. A run whose declaration reads a per-setting weight evaluated in floating point, directly or through a uniform weight, reports at most a first-order tier.

This contract claims no enclosure of any quantity it computes in floating point (Definition 10.9), while the weights it evaluates rigorously are bounds by theorem (Theorem 8.13). The label says which of the two an artifact's weights are (Theorem 10.10 (iii)), and the requirement's last sentence states the consequence for the tier of a run.

REQUIREMENT 14.5 (The evaluation label on every artifact). The engine writes `bound_evaluation` on every artifact of this module, on every input route: `rigorous` where the weight of every setting was formed by the rigorous evaluation of Requirement 13.5, and `floating_point` otherwise, an artifact with a weight resting on Assumption 11.11 among them; which evaluation formed each setting's weight is recorded as Requirement 13.5 (vi) says. No path in this module produces another label (Theorem 10.10 (iii)): the label `certified`, which would name a tier and not an evaluation, and the retired fallback reason `receiver_grams_not_enclosed` are never produced. A run whose declaration carries a weight into which a per-setting weight enters that the record of Requirement 13.5 (vi) names as evaluated in floating point, the uniform weight of a slot entry counting the weight of every setting as entering it, is not reported at the certified tier (Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (vii)); in every other respect the tier a run reports from these artifacts follows Requirement "Reporting of the tiers" (foundations/req:tiers).

Motivated by Thm. 8.13, Def. 10.9, Thm. 10.10, Ass. 11.11, Req. 13.5, foundations/req:tiers, meat-core/frank-wolfe/req:certified-evaluation · accepted

The engine decides exactly whether a declaration has a model admissible exactly and, where its arithmetic represents the parameters, whether it is on the exact route. For every such model it encloses the port Grams in rational arithmetic, exactly on that route, and records which.

The engine encloses the port Grams of every declaration that defines an admissible receiver exactly, so the receiver's blocks have a reference whatever the parameters are (Theorem 9.5); the reference the receiver offers is formed from these enclosures (Requirement 22.9). The exact route is decided in addition, because only exact entries let a snap keep a block at weight zero (Theorem 9.9). The entries of the catalogue (Definition 16.2) carry the square roots of their transmittances, reflectivities, efficiencies and pass and leak parameters, and of the complements $1-x$, $1-t_H$, $1-t_V$ and $1-y$ of the splitting ratios and the losses; at axis angles that are multiples of $\pi/8$ they involve otherwise only cos and sin of multiples of $\pi/4$ and phases that are multiples of $\pi/4$, which lie in $\mathbb{Q}(i,\sqrt{2})$. That is why the floor of item (ii) names those square roots and those angles. A compiled graph whose generic blocks are contractions exactly is admissible exactly at every other parameter in its domain (Theorem 16.3), so on that route the model needs no decision beyond those blocks and the domain comparisons.

REQUIREMENT 14.6 (Deciding the exact route and enclosing the port Grams). (i) *Declared Grams or matrix.* On a route that declares the port Grams or the instrument matrix, the engine computes the port Grams in exact rational arithmetic from the declared values, by (2.2) where the instrument matrix is declared, and decides in exact rational arithmetic whether every A_k is Hermitian and whether the family satisfies (9.7), that is, whether it is an admissible Gram family exactly. The declaration has a model admissible exactly, and is then on the exact route (Definition 9.8), exactly when both hold.

(ii) *Component graph.* On the route that declares a component graph, the engine decides in exact rational arithmetic whether the matrix of every generic block, whose entries are declared values, is a contraction, $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}$. Where every one is, the declaration has a model admissible exactly: the compiled instrument matrix is a contraction (Theorem 3.4 (ii)) and its port Grams are an admissible Gram family (Theorem 4.1 (i)), at the exact values of the declared parameters. For a declaration that has one and whose parameters its exact arithmetic represents, the engine moreover compiles the instrument matrix of Definition 3.3 and the port Grams exactly and decides whether every port Gram has entries in $\mathbb{Q}(i)$, which places the declaration on the exact route. Every transmittance, loss, reflectivity, extinction and efficiency is the rational number the artifact carries for it: the declared value, or, for a loss declared in decibels, the converted efficiency that `presentation_rule` makes the declared datum, a binary64 num-

ber and so a dyadic rational (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)). An angle parameter given in the presentation `angle_presentation`, as a rational coefficient c of π, is $c\pi$ exactly; an angle given in radians is a declared value, the dyadic rational it denotes. The arithmetic represents at least the field generated over $\mathbb{Q}(i)$ by $\sqrt{2}$ and by $\sqrt{p}$ and $\sqrt{1-p}$ for every declared rational parameter p in $[0,1]$, which contains every entry of every catalogue entry whose angles are given as coefficients of π, whose axis angles are multiples of $\pi/8$, whose retardances are multiples of $\pi/2$ and whose rotations and phases are multiples of $\pi/4$. A declaration whose parameters the arithmetic does not represent, a non-zero angle given in radians among them, is treated as off the exact route, with its cause recorded by (iv), even where its port Grams happen to have entries in $\mathbb{Q}(i)$ (two half-wave plates in sequence at one such angle, for instance, compose to $-\mathbb{1}$).

(iii) *Enclosure of the port Grams.* For every declaration with a model admissible exactly, the engine computes, in exact rational arithmetic, complex enclosures (Definition 9.1) of its port Grams at the exact values of the declared parameters. On the exact route as decided by (i) or (ii) each port Gram is enclosed exactly, by itself. Otherwise, on the route of (ii), the enclosures are those of Theorem 9.5 (i), with the matrix of every element enclosed by the rules of Lemma 9.2 applied to the formula of its catalogue entry from square-root enclosures and angle enclosures of its parameters, and $\sqrt{\eta_k}$ by a square-root enclosure; where the exact arithmetic of (ii) has computed a port Gram, its entries, which are expressions in square roots of rationals, may be enclosed from those expressions instead. A phase $e^{i\varphi}$ is $\cos\varphi + i\sin\varphi$, and the angle $\delta/2$ of a retarder is again a rational or a rational multiple of π. No floating-point value enters an enclosure.

(iv) *Record.* The `photon_blocks` entry of the artifact records whether (i) or (ii) places the declaration on the exact route and, where neither does, why: a parameter or an angle that the exact arithmetic of (ii) does not represent (a non-zero angle given in radians among them), port Grams that are not Gaussian-rational, or no model admissible exactly, which is declared port Grams that are not Hermitian, a declared family that is not an admissible Gram family exactly (a declaration admitted only at the receiver tolerances), or a generic block that is not a contraction exactly. Being off the exact route is not a refusal of the receiver and takes nothing away from the floating-point blocks (Definition 10.9); where the declaration has a model admissible exactly, its port Grams are enclosed by (iii).

Motivated by Def. 2.2, Def. 3.3, Thm. 3.4, Thm. 4.1, Def. 9.1, Lem. 9.2, Thm. 9.5, Def. 9.8, Thm. 9.9, Def. 10.4, Def. 10.9, Def. 16.2, Thm. 16.3, Req. 22.9, foundations/def:binary64 · *Used by* Req. 13.5, Req. 22.9, Req. 22.10 · implemented (effective: accepted) · *Code* certiqs_key.receiver.reference:enclose_port_grams

15 Requirements: the assumptions

The engine validates, where it is decidable, every clause of an assumption that constrains the declared data, and it refuses a non-zero parameter tolerance. It carries the eigenvalue-slack assumption beside the settings at which it was invoked, those whose weight uses a computed block eigenvalue.

Every clause of an assumption that constrains the declared data is enforced by validation wherever it is decidable. The assumptions the engine never verifies form the never-verified set, and no code path decides more of them than their clauses on the declared data.

Which assumptions the engine never verifies is recorded in each one's verification field (`spec/claim-format.md#32-whether-the-engine-verifies-an-assumption`), from which the contract generates the sentence that says so.

Requirement 15.1 (Validation of the assumptions on the declared data). The engine enforces by validation, wherever it is decidable, every clause of an assumption of this contract that constrains the declared data. The assumptions that the engine never verifies, as the contract says of each (Remark 11.2), form `never_verified_set`, and no code path decides what they say beyond those clauses.

Motivated by Rem. 11.2 · implemented · *Code* certiqs_key.receiver.construct:construct_receiver

The reserved tolerance field of a declared parameter is admitted only when absent or exactly zero and is refused otherwise. An ignored tolerance would be a silent claim that the device is exact (A-EXACT).

Assumption 11.8 makes every declared parameter exact, and a tolerance silently ignored would be a claim of exactness that was not made; the field is therefore validated and never ignored. Its place in the declaration is the parameters table of the interface (`spec/modules/receiver/interface.yaml`).

Requirement 15.2 (Refusal of a non-zero parameter tolerance). The engine validates the reserved `tolerance` field of every declared parameter as absent or exactly zero, and refuses it otherwise.

Motivated by Ass. 11.8 · implemented · *Code* certiqs_key.receiver.validation: check_parameter_tolerances

The assumption ledger carries A-EIG beside the settings at which it was invoked: those whose emitted weight uses a computed block eigenvalue. A reader of the artifact sees which emitted weights rest on the declared eigenvalue slack.

Requirement 15.3 (Invoking settings on the eigenvalue-slack row of the ledger). The engine carries Assumption 11.11 in the artifact's assumption ledger `assumption_ledger` beside the settings at which it was invoked. A setting is one at which it was invoked exactly when its emitted weight is that of Definition 8.7, evaluated in floating point; a setting whose weight is an enclosed weight of Definition 8.12, formed by Requirement 13.5, is not one, whatever depth it declares.

Motivated by Def. 8.7, Def. 8.12, Ass. 11.11, Req. 13.5 · accepted

Part III
Component catalogue

The component catalogue lets a receiver be declared as an optical layout instead of a single matrix. Each of its ten entries fixes a support, a parameter domain and a passive matrix with every phase and every loss the part has written out (Definition 16.2). The main result is that every entry maps its whole declared domain into local elements, lossless exactly on a named subfamily (Theorem 16.3); one attenuation lemma carries most of the proof. The amplitude each entry removes has rows of its own, none on the lossless subfamily, and along a compiled graph they make a port dilation of the receiver (Lemma 16.4). Detector efficiency and the probe source are declared layers outside the catalogue because they act at different points of the measurement. A worked BB84 receiver follows one declaration through compilation to its port Grams, its analytic high-photon weight and the improvement an exact window buys at a smaller cutoff.

16 Catalogue entries and their admissibility

The main result, Theorem 16.3, is that every entry maps its whole declared domain into local elements, lossless exactly on the subfamily its table row names. Every entry but the generic block is written as a unitary core with a positive contractive attenuation on one or both sides, so one lemma gives this for each of them. The generic block is the one entry whose admission is a spectral test, decided at the route-(ii) receiver tolerance (Definition 16.5). The remarks place the rotator outside the retarder family, give the physical roles of the entries, separate the detector and source layers from the chain, and say which convention factors are free. The rows that carry what each entry removes, stacked along a compiled graph with those of the detector efficiencies, make a port dilation of the receiver (Lemma 16.4).

A unitary with a positive semidefinite contraction on either side is a passive local element, lossless exactly when both contractions are the identity. Every catalogue entry has this form, so the lemma carries most of the catalogue's admissibility proof.

The catalogue has ten entry types, each a short matrix with a short verification. Every one of them is a unitary core with a positive semidefinite attenuation on one or both sides, so the following lemma does almost all of the work.

Lemma 16.1 (Admissibility of an attenuated unitary). Let V be unitary on $\mathbb{C}^{\Sigma}$ and let D_1, D_2 be positive semidefinite contractions on $\mathbb{C}^{\Sigma}$. Then $G_{\Sigma} \coloneqq D_2 V D_1$ satisfies $G_{\Sigma}^{\dagger} G_{\Sigma} \preceq \mathbb{1}_{\Sigma}$, and $G_{\Sigma}^{\dagger} G_{\Sigma} = \mathbb{1}_{\Sigma}$ if and only if $D_1 = D_2 = \mathbb{1}_{\Sigma}$.

Used by Thm. 16.3 · newly derived · accepted

Proof. The bound. Conjugation by a unitary preserves the Löwner order, so $V^\dagger D_2^2 V \preceq V^\dagger \mathbb{1} V = \mathbb{1}$. Since $X \preceq Y$ implies $Z^\dagger X Z \preceq Z^\dagger Y Z$ for any Z, $G_\Sigma^\dagger G_\Sigma = D_1 (V^\dagger D_2^2 V) D_1 \preceq D_1 \mathbb{1} D_1 = D_1^2 \preceq \mathbb{1}$.

The equality case. If the composite equals $\mathbb{1}$, the chain forces $D_1^2 = \mathbb{1}$. A positive semidefinite contraction with $D_1^2 = \mathbb{1}$ has all its eigenvalues in $[0,1]$ squaring to 1, hence equal to 1, so $D_1 = \mathbb{1}$. Then $V^\dagger D_2^2 V = \mathbb{1}$ gives $D_2^2 = \mathbb{1}$, and the same argument gives $D_2 = \mathbb{1}$. Conversely, $D_1 = D_2 = \mathbb{1}$ gives $G_\Sigma = V$, which is unitary. □

Ten entry types, from the beamsplitter C1 to the generic block C10, each fix a support, a parameter domain and a passive matrix with every phase and every loss the part has written out. A table collects the supports, the domains and the lossless subfamilies.

A receiver declared as an optical layout is built from the entries below, which form the component catalogue of Definition 3.1. Each entry is written the way the part is specified: every phase and every loss the part has appears as a parameter, and none is dropped because it could have been absorbed elsewhere.

Definition 16.2 (Entries of the component catalogue)**.** The *catalogue entries* are the entry types C1–C10 below. Throughout, a *path* is a spatial index p and carries the two polarization modes (p,H) and (p,V); a matrix written on two paths acts as $\cdot \otimes \mathbb{1}_2$ on polarization unless stated otherwise, and a matrix written on polarization acts on one path.

C1 — Beamsplitter. Support: two paths (p_1, p_2), four modes. Parameters: transmittance $x \in [0,1]$, reflection phase $\varphi \in [0, 2\pi)$, per-output losses $y_1, y_2 \in [0,1]$. Core, on the ordered path pair with rows = outputs:

$$B(x,\varphi) = \begin{pmatrix} \sqrt{x} & \sqrt{1-x}\, e^{i\varphi} \\ \sqrt{1-x}\, e^{-i\varphi} & -\sqrt{x} \end{pmatrix}, \qquad G_{\mathrm{C1}} = \Big[\operatorname{diag}\big(\sqrt{1-y_1}, \sqrt{1-y_2}\big)\, B(x,\varphi) \Big] \otimes \mathbb{1}_2. \tag{16.1}$$

A *polarization-dependent beamsplitter* declares (x_H, φ_H) and (x_V, φ_V) separately, with the per-output losses y_1, y_2 of C1, and is diag $\big(\sqrt{1-y_1}, \sqrt{1-y_2}, \sqrt{1-y_1}, \sqrt{1-y_2}\big) \big[B(x_H, \varphi_H) \oplus B(x_V, \varphi_V)\big]$ in the reordered basis $(p_1 H, p_2 H, p_1 V, p_2 V)$; this is the matrix of entry C2 at $(t_H, \varphi_H, t_V, \varphi_V) = (x_H, \varphi_H, x_V, \varphi_V)$ written in that basis.

C2 — Polarizing beamsplitter. Support: two paths, four modes. Parameters: $t_H, t_V \in [0,1]$ (ideal $t_H = 1$, $t_V = 0$), phases $\varphi_H, \varphi_V \in [0, 2\pi)$, losses $y_1, y_2 \in [0,1]$. In the ordering $(p_1 H, p_1 V, p_2 H, p_2 V)$,

$$G_{\mathrm{C2}} = \operatorname{diag}\big(\sqrt{1-y_1}, \sqrt{1-y_1}, \sqrt{1-y_2}, \sqrt{1-y_2}\big)\, \Xi^\dagger \big[B(t_H, \varphi_H) \oplus B(t_V, \varphi_V)\big]\, \Xi, \tag{16.2}$$

where Ξ is the permutation from $(p_1 H, p_1 V, p_2 H, p_2 V)$ to $(p_1 H, p_2 H, p_1 V, p_2 V)$.

C3 — Retarder*, with half- and quarter-wave presets.* Support: the two polarization modes of one path. Parameters: retardance $\delta \in [0, 2\pi)$, fast-

axis angle $\theta \in [0, \pi)$, scalar loss $y \in [0, 1]$. With $R(\theta) = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$,

$$W(\delta, \theta) \coloneqq e^{-i\delta/2}\, R(\theta)\, \mathrm{diag}(1, e^{i\delta}) R(\theta)^{\mathsf{T}} = e^{-i\delta/2} \begin{pmatrix} \cos^2\theta + e^{i\delta}\sin^2\theta & (1 - e^{i\delta})\sin\theta\cos\theta \\ (1 - e^{i\delta})\sin\theta\cos\theta & \sin^2\theta + e^{i\delta}\cos^2\theta \end{pmatrix}, \tag{16.3}$$

and $G_{\mathrm{C3}} = \sqrt{1 - y}\, W(\delta, \theta)$. The presets are

$$\mathrm{HWP}(\theta) \coloneqq W(\pi, \theta) = -i \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}, \qquad \mathrm{QWP}(\theta) \coloneqq W(\tfrac{\pi}{2}, \theta), \tag{16.4}$$

so that $\mathrm{HWP}(\pi/8) = -\frac{i}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$.

C4 — Circular retarder, or polarization rotator. Support: the two polarization modes of one path. Parameter: rotation $\rho \in [0, 2\pi)$. $G_{\mathrm{C4}} = R(\rho)$.

C5 — Mirror or fold. Support: one path. Parameters: reflectivities $R_H, R_V \in [0, 1]$, phases $\varphi_H, \varphi_V \in [0, 2\pi)$, handedness flip $\mathrm{fl} \in \{\mathrm{false}, \mathrm{true}\}$ (default false).

$$G_{\mathrm{C5}} = \mathrm{diag}\left(\sqrt{R_H}, \sqrt{R_V}\right) \mathrm{diag}\left(e^{i\varphi_H}, e^{i\varphi_V}\right) F, \qquad F = \begin{cases} \mathrm{diag}(1, -1) & \mathrm{fl} = \mathrm{true} \\ \mathbb{1}_2 & \text{otherwise.} \end{cases} \tag{16.5}$$

C6 — Phase shifter. Support: any declared mode subset Σ. Parameters: $\varphi_m \in [0, 2\pi)$ per $m \in \Sigma$. $G_{\mathrm{C6}} = \mathrm{diag}(e^{i\varphi_m})_{m \in \Sigma}$.

C7 — Loss. Support: any declared mode subset Σ. Parameters: $\eta_m \in [0, 1]$ per $m \in \Sigma$, optionally declared as $L_{\mathrm{dB}} \geq 0$ with $\eta = 10^{-L_{\mathrm{dB}}/10}$. $G_{\mathrm{C7}} = \mathrm{diag}(\sqrt{\eta_m})_{m \in \Sigma}$.

C8 — Polarizer. Support: the two polarization modes of one path. Parameters: transmission axis $\theta \in [0, \pi)$, pass amplitude $\sqrt{t}$ and leak amplitude $\sqrt{t_\perp}$ with $t, t_\perp \in [0, 1]$. With $|e_\theta\rangle = \cos\theta\,|H\rangle + \sin\theta\,|V\rangle$,

$$G_{\mathrm{C8}} = \sqrt{t}\; |e_\theta\rangle\langle e_\theta| + \sqrt{t_\perp}\; \left|e_\theta^\perp\right\rangle\!\left\langle e_\theta^\perp\right| = R(\theta)\, \mathrm{diag}\left(\sqrt{t}, \sqrt{t_\perp}\right) R(\theta)^{\mathsf{T}}. \tag{16.6}$$

C9 — Mode routing. Support: any declared mode subset. Parameter: a permutation π of the modes of Σ; $(G_{\mathrm{C9}})_{ij} = \delta_{i, \pi(j)}$.

C10 — Generic block. Support: any declared mode subset Σ. Parameter: a matrix $G_\Sigma \in \mathbb{C}^{\Sigma \times \Sigma}$, admitted after the explicit check $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}_\Sigma$.

Table 4 collects the supports and the parameter domains.

Used by Thm. 16.3, Lem. 16.4 · newly derived · implemented · *Code* certiqs_key.receiver.catalogue:Component

ID	Support	Parameters (domain)	Lossless exactly when
C1	2 paths (4 modes)	$x \in [0,1]$, $\varphi \in [0,2\pi)$, $y_1, y_2 \in [0,1]$	$y_1 = y_2 = 0$
C2	2 paths (4 modes)	$t_H, t_V \in [0,1]$, $\varphi_H, \varphi_V \in [0,2\pi)$, $y_1, y_2 \in [0,1]$	$y_1 = y_2 = 0$
C3	1 path, polarization	$\delta \in [0,2\pi)$, $\theta \in [0,\pi)$, $y \in [0,1]$	$y = 0$
C4	1 path, polarization	$\rho \in [0,2\pi)$	always
C5	1 path, polarization	$R_H, R_V \in [0,1]$, $\varphi_H, \varphi_V \in [0,2\pi)$, fl	$R_H = R_V = 1$
C6	any Σ	$\varphi_m \in [0,2\pi)$	always
C7	any Σ	$\eta_m \in [0,1]$	every $\eta_m = 1$
C8	1 path, polarization	$\theta \in [0,\pi)$, $t, t_\perp \in [0,1]$	$t = t_\perp = 1$ (then $G_\Sigma = \mathbb{1}$)
C9	any Σ	a permutation of Σ	always
C10	any Σ	G_Σ with $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}$ (checked)	$G_\Sigma^\dagger G_\Sigma = \mathbb{1}$ (checked)

Table 4: The catalogue entries. Support shapes are stated in modes; the parameter domains are the declared domains of Definition 3.2 (E1); the last column names the subfamily on which Theorem 16.3 proves the entry lossless.

Every catalogue entry maps its whole declared parameter domain to contractions on its modes, lossless exactly on the subfamily the catalogue's table lists for it. Only the generic block, an arbitrary matrix on declared modes, needs its contractivity checked at admission.

A declared element graph is well formed only if every element is a contraction on its support. For entries C1–C9 this holds by theorem on the whole declared domain, so for each of them condition (E1) of Definition 3.2 reduces to checking that its parameters lie in that domain.

Theorem 16.3 (Admissibility of the catalogue entries). Let G_Σ be the matrix of an entry type C1–C10 of Definition 16.2 at a parameter tuple in its declared domain. Then:

(i) $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}_\Sigma$, so G_Σ is a local element in the sense of Definition 1.3;

(ii) G_Σ is lossless, $G_\Sigma^\dagger G_\Sigma = \mathbb{1}_\Sigma$, exactly on the subfamily named in Table 4;

(iii) for C1–C9, a matrix whose parameter tuple passes the domain comparison of (E1) (Definition 3.2) is contractive by (i); C10 is the only entry whose declared domain is the contractivity condition itself, so that its check of Theorem 3.4 (iv)(E1) is a spectral test on the supplied matrix.

Uses Def. 1.3, Def. 3.2, Def. 3.3, Thm. 3.4, Lem. 16.1, Def. 16.2 · *Used by* Def. 16.5, Rem. 16.6, Rem. 16.7, Ex. 17.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.catalogue: Component · *Oracles* registry_receiver_povm.py

Proof. Idea. Each of the entries C1–C9 is exhibited in the form $D_2 V D_1$ of Lemma 16.1, after which only the unitarity of the core has to be checked, and the lemma then gives (i) and the lossless case (ii) together. Entry C10 is decided by its hypothesis, checked on the supplied matrix.

C1. $B(x,\varphi)$ is unitary: its first column has squared norm $x+(1-x)=1$, its second $(1-x)+x=1$, and their inner product is $\sqrt{x}\sqrt{1-x}e^{i\varphi} + \overline{\sqrt{1-x}}e^{-i\varphi}(-\sqrt{x}) = \sqrt{x(1-x)}e^{i\varphi} - \sqrt{x(1-x)}e^{i\varphi} = 0$. Tensoring with $\mathbb{1}_2$ preserves unitarity, and $\mathrm{diag}(\sqrt{1-y_1},\sqrt{1-y_2})\otimes\mathbb{1}_2$ is a positive semidefinite contraction because $1-y_i\in[0,1]$. Lemma 16.1 with $D_1=\mathbb{1}$ gives admissibility and losslessness exactly at $y_1=y_2=0$. (At $\varphi=0$ the core is real with $\det B=-1$; that sign is an artefact of the convention, see Remark 16.9.)

C2. $B(t_H,\varphi_H)\oplus B(t_V,\varphi_V)$ is a direct sum of unitaries, hence unitary; conjugating by the permutation Ξ preserves unitarity; the prefactor is again a positive semidefinite diagonal contraction. The polarization-dependent beamsplitter of entry C1 is this matrix written in the reordered basis (p_1H,p_2H,p_1V,p_2V), so the same argument covers it. At $t_H=1$, $t_V=0$, $\varphi_H=\varphi_V=0$, $y_1=y_2=0$ the element is the signed permutation $p_1H\mapsto p_1H$, $p_1V\mapsto p_2V$, $p_2H\mapsto -p_2H$, $p_2V\mapsto p_1V$, which is the intended routing up to per-mode phases.

C3. $R(\theta)$ is real orthogonal ($R(\theta)^{\mathsf{T}}R(\theta)=\mathbb{1}$ by $\cos^2+\sin^2=1$) and $\mathrm{diag}(1,e^{i\delta})$ is unitary, so $W(\delta,\theta)$ is a product of unitaries times the unimodular scalar $e^{-i\delta/2}$, hence unitary; the expanded form in (16.3) is the product carried out, and (16.4) is $\delta=\pi$, using $e^{i\pi}=-1$, $\cos^2\theta-\sin^2\theta=\cos 2\theta$ and $2\sin\theta\cos\theta=\sin 2\theta$, with the scalar $e^{-i\pi/2}=-i$. The prefactor $\sqrt{1-y}$ is a scalar contraction.

C4. $R(\rho)$ is real orthogonal, hence unitary, with no attenuation factor.

C5. $\mathrm{diag}(e^{i\varphi_H},e^{i\varphi_V})F$ is a product of unitaries (F is diagonal with entries of modulus one in both cases), and $\mathrm{diag}(\sqrt{R_H},\sqrt{R_V})$ is a positive semidefinite contraction, equal to $\mathbb{1}$ exactly when $R_H=R_V=1$.

C6, C9. Diagonal with unimodular entries, respectively a permutation matrix: both unitary, both with $D_1=D_2=\mathbb{1}$.

C7. $V=\mathbb{1}$, $D_2=\mathrm{diag}(\sqrt{\eta_m})$, a positive semidefinite contraction equal to $\mathbb{1}$ exactly when every $\eta_m=1$.

C8. The two displayed forms agree because $R(\theta)\,|H\rangle=|e_\theta\rangle$ and $R(\theta)\,|V\rangle=|e_\theta^\perp\rangle$, so conjugating $\mathrm{diag}(\sqrt{t},\sqrt{t_\perp})$ by $R(\theta)$ is the spectral decomposition written in the rotated basis. The result is Hermitian positive semidefinite with eigenvalues $\sqrt{t},\sqrt{t_\perp}\in[0,1]$, so it is itself a positive semidefinite contraction: Lemma 16.1 applies with $V=\mathbb{1}$, $D_1=\mathbb{1}$, and equality holds

exactly at $t = t_\perp = 1$.

C10. The condition is the hypothesis, checked on the supplied matrix; losslessness is the same check with equality.

Item (iii). For C1–C9 it is (i), which holds on the whole declared domain; for C10 the declared domain is, by Definition 16.2, the set of blocks G_Σ with $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}_\Sigma$. □

Every catalogue entry has a loss factor, one row per amplitude it can remove and no row on its lossless subfamily; a Gaussian-rational generic block has one from an exact decomposition, which exists. Stacked along a compiled graph with the efficiency layer's loss rows, they make a port dilation.

Theorem 16.3 writes every entry but the generic block as a unitary core with positive semidefinite attenuations on its sides (Lemma 16.1). The amplitude an attenuation removes can be sent to new modes, one per attenuated mode: the core and the attenuation together with those rows are then an isometry, and the rows vanish exactly where nothing is attenuated. Done element by element along a compiled graph, and once more for the detector efficiencies, this gives a port dilation of the receiver whose loss modes are only those its declared losses need, which Theorem 9.7 reads the photon-number blocks through.

Lemma 16.4 (Loss factors of the catalogue entries and the dilation of a graph)**.** For a diagonal matrix D write $D^\flat$ for the matrix of the rows of D whose diagonal entry is not zero, so that $(D^\flat)^\dagger D^\flat = D^\dagger D$.

(i) *Loss factors.* Let G_Σ be the matrix of an entry type of Definition 16.2 at a parameter tuple in its declared domain, in the notation of that definition, and let Θ be the matrix given for its type below, each of its rows indexed by a mode of its own:

- C1: $\Theta = \big(\operatorname{diag}(\sqrt{y_1}, \sqrt{y_2})^\flat\, B(x,\varphi)\big) \otimes \mathbb{1}_2$;
- C2: $\Theta = \operatorname{diag}(\sqrt{y_1}, \sqrt{y_1}, \sqrt{y_2}, \sqrt{y_2})^\flat\, \Xi^\dagger \big[B(t_H,\varphi_H) \oplus B(t_V,\varphi_V)\big] \Xi$;
- C3: $\Theta = \operatorname{diag}(\sqrt{y}, \sqrt{y})^\flat$;
- C4, C6 and C9: Θ with no row;
- C5: $\Theta = \operatorname{diag}\big(\sqrt{1-R_H}, \sqrt{1-R_V}\big)^\flat$;
- C7: $\Theta = \operatorname{diag}\big(\sqrt{1-\eta_m}\big)^\flat_{m\in\Sigma}$;
- C8: $\Theta = \operatorname{diag}\big(\sqrt{1-t}, \sqrt{1-t_\perp}\big)^\flat\, R(\theta)^\mathsf{T}$;
- C10, with the entries of G_Σ in $\mathbb{Q}(i)$: $\Theta = D^{1/2}E$, where $\mathbb{1}_\Sigma - G_\Sigma^\dagger G_\Sigma = E^\dagger D E$ with D diagonal with positive rational diagonal entries and E with entries in $\mathbb{Q}(i)$; such a decomposition exists, with at most $|\Sigma|$ rows.

Then $G_\Sigma^\dagger G_\Sigma + \Theta^\dagger\Theta = \mathbb{1}_\Sigma$, and Θ is a *loss factor* of the entry. For C1–C9, Θ has no row exactly on the lossless subfamily named in Table 4.

(ii) *The dilation of a compiled graph.* Let $\mathfrak{R}$ be a receiver instrument whose instrument matrix is the compiled instrument matrix $G = D_\eta G^\sigma_{\mathfrak{E}}$ of a well-formed graph $\mathfrak{E}$ of catalogue elements for a linear extension $\sigma = (e_1, \dots, e_L)$ (Definition 3.3), with ports $\mathbb{P}_1, \dots, \mathbb{P}_K$ and efficiencies η_k. For each j let Θ_j be a loss factor of (i) of the element e_j, its rows indexed by a mode set $\mathbb{E}_j$, and let $\hat{\Theta}_j : \mathbb{C}^{\mathbb{M}} \to \mathbb{C}^{\mathbb{E}_j}$ act as Θ_j on the modes of the support Σ_{e_j} and as zero on the others. Let $\Theta_\eta := \big(\sum_k \sqrt{1-\eta_k}\, E_{\{k\}}\big)^\flat$, its rows indexed by a mode set $\mathbb{E}_\eta$, one for each mode of a port with $\eta_k < 1$, the sets $\mathbb{M}$, $\mathbb{E}_\eta$ and the $\mathbb{E}_j$ being pairwise disjoint. With $\iota : \mathbb{C}^{\mathbb{I}} \to \mathbb{C}^{\mathbb{M}}$ the inclusion of the declared input modes, put $X_0 := \iota$ and $X_j := \hat{G}_{e_j} X_{j-1}$ for $j = 1, \dots, L$, and let

$$\Omega : \mathbb{C}^{\mathbb{I}} \to \mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}_\eta \sqcup \mathbb{E}_L \sqcup \cdots \sqcup \mathbb{E}_1} \tag{16.7}$$

have the rows $D_\eta X_L$ on $\mathbb{M}$, $\Theta_\eta X_L$ on $\mathbb{E}_\eta$ and $\hat{\Theta}_j X_{j-1}$ on $\mathbb{E}_j$. Then Ω, with the ports $\mathbb{P}_1, \dots, \mathbb{P}_K$ of $\mathfrak{R}$, is a port dilation of $\mathfrak{R}$ (Definition 9.6), and its rows on $\mathbb{M}$ are those of $G_{\mathbb{I}}$.

Uses Def. 1.3, Def. 2.1, Def. 2.2, Def. 3.2, Def. 3.3, Def. 9.6, Def. 16.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.factor:port_dilation

Proof. The flat matrix. For $D = \mathrm{diag}(a_1, \dots, a_r)$, $(D^\flat)^\dagger D^\flat = \sum_{i:\, a_i \neq 0} |a_i|^2 e_i e_i^\dagger = D^\dagger D$, the rows omitted being zero. So in every case of (i) but C10, $\Theta^\dagger\Theta$ is the same product with the full diagonal matrix, and below $\Theta^\dagger\Theta$ is computed with it.

(i). C1. $B(x,\varphi)$ is unitary: its columns have squared norms $x + (1-x) = 1$ and $(1-x) + x = 1$, and their inner product is $\sqrt{x}\sqrt{1-x}\, e^{i\varphi} - \sqrt{1-x}\, e^{i\varphi}\sqrt{x} = 0$. With $D_2 := \mathrm{diag}(\sqrt{1-y_1}, \sqrt{1-y_2})$ and $(Z \otimes \mathbb{1}_2)^\dagger (Z \otimes \mathbb{1}_2) = Z^\dagger Z \otimes \mathbb{1}_2$, $G_\Sigma^\dagger G_\Sigma + \Theta^\dagger\Theta = \big(B^\dagger(D_2^2 + \mathrm{diag}(y_1, y_2))B\big) \otimes \mathbb{1}_2 = B^\dagger B \otimes \mathbb{1}_2 = \mathbb{1}$. *C2.* $V := \Xi^\dagger[B(t_H, \varphi_H) \oplus B(t_V, \varphi_V)]\Xi$ is unitary, a direct sum of unitaries conjugated by a permutation; with D_2 the diagonal prefactor of (16.2), $G_\Sigma^\dagger G_\Sigma + \Theta^\dagger\Theta = V^\dagger\big(D_2^2 + \mathrm{diag}(y_1, y_1, y_2, y_2)\big)V = V^\dagger V = \mathbb{1}$. *C3.* $W(\delta, \theta)$ is unitary, a product of the real orthogonal $R(\theta)$, the unitary $\mathrm{diag}(1, e^{i\delta})$, $R(\theta)^\mathsf{T}$ and the unimodular $e^{-i\delta/2}$; so $G_\Sigma^\dagger G_\Sigma = (1-y)\mathbb{1}_2$ and $\Theta^\dagger\Theta = y\,\mathbb{1}_2$. *C4, C6, C9.* $R(\rho)$ is real orthogonal, a diagonal matrix with unimodular entries is unitary, and so is a permutation matrix; so $G_\Sigma^\dagger G_\Sigma = \mathbb{1}$, and $\Theta^\dagger\Theta = 0$, an empty sum. *C5.* The three factors of (16.5) are diagonal, so they commute, $F^\dagger F = \mathbb{1}$ and $|e^{i\varphi_H}| = |e^{i\varphi_V}| = 1$; so $G_\Sigma^\dagger G_\Sigma = \mathrm{diag}(R_H, R_V)$ and $\Theta^\dagger\Theta = \mathrm{diag}(1 - R_H, 1 - R_V)$. *C7.* $G_\Sigma^\dagger G_\Sigma = \mathrm{diag}(\eta_m)$ and $\Theta^\dagger\Theta = \mathrm{diag}(1 - \eta_m)$. *C8.* With $R := R(\theta)$ real orthogonal, the second form of (16.6) gives $G_\Sigma^\dagger G_\Sigma = R\,\mathrm{diag}(t, t_\perp)R^\mathsf{T}$, and $\Theta^\dagger\Theta = R\,\mathrm{diag}(1-t, 1-t_\perp)R^\mathsf{T}$; their sum is $RR^\mathsf{T} = \mathbb{1}$. *C10.* $\Theta^\dagger\Theta = E^\dagger(D^{1/2})^2 E = E^\dagger D E = \mathbb{1}_\Sigma - G_\Sigma^\dagger G_\Sigma$, $D^{1/2}$ being the real diagonal matrix of the positive square roots. Such a

decomposition exists. Put $H \coloneqq \mathbb{1}_\Sigma - G_\Sigma^\dagger G_\Sigma$: it is Hermitian, positive semidefinite, since $G_\Sigma^\dagger G_\Sigma \preceq \mathbb{1}$ on the declared domain (Definition 16.2), and has entries in $\mathbb{Q}(i)$. A diagonal entry $H_{jj} = 0$ forces $H_{jl} = 0$ for every l, since $|H_{jl}|^2 \le H_{jj} H_{ll}$ for a positive semidefinite H. If $H = 0$, take E and D with no row. Otherwise some $h \coloneqq H_{pp} \neq 0$; it is real and lies in $\mathbb{Q}(i)$, so it is a positive rational. Let e be the row with entries $e_l \coloneqq H_{pl}/h$, in $\mathbb{Q}(i)$, and put $H' \coloneqq H - h\, e^\dagger e$, whose entries $H_{il} - H_{ip}H_{pl}/h$ lie in $\mathbb{Q}(i)$. For a vector x put $c \coloneqq ex$ and $y \coloneqq x - c\,\delta_p$, δ_p the p-th coordinate vector; as $(Hx)_p = hc$, $y^\dagger H y = x^\dagger H x - 2h|c|^2 + h|c|^2 = x^\dagger H' x$, so H' is positive semidefinite. Row p of H' is zero, and so is every row of H' whose row in H is zero, its entry H_{jp} being zero; so H' has fewer non-zero diagonal entries than H. Repeating the step on H' ends after at most $|\Sigma|$ steps at the zero matrix, and H is the sum of the terms $h\, e^\dagger e$ formed, which is $E^\dagger D E$ with the rows e stacked into E and the pivots h on the diagonal of D. In every case of C1–C9 the sum is $\mathbb{1}_\Sigma$. A matrix $D^\flat$ has no row exactly when every diagonal entry of D is zero, and a product with it on the left then has none either; so Θ has no row exactly when $y_1 = y_2 = 0$ for C1 and C2, $y = 0$ for C3, $R_H = R_V = 1$ for C5, every $\eta_m = 1$ for C7 and $t = t_\perp = 1$ for C8, and always for C4, C6 and C9: the subfamilies of Table 4.

(ii). Each layer keeps the norm. Fix j and split $\mathbb{C}^{\mathbb{M}} = \mathbb{C}^{\Sigma} \oplus \mathbb{C}^{\mathbb{M}\setminus\Sigma}$ with $\Sigma \coloneqq \Sigma_{e_j}$. The global embedding is $\hat{G}_{e_j} = G_\Sigma \oplus \mathbb{1}$ (Definition 1.3) and $\hat{\Theta}_j = \begin{pmatrix}\Theta_j & 0\end{pmatrix}$, so $\hat{G}_{e_j}^\dagger \hat{G}_{e_j} + \hat{\Theta}_j^\dagger \hat{\Theta}_j = \big(G_\Sigma^\dagger G_\Sigma + \Theta_j^\dagger \Theta_j\big) \oplus \mathbb{1} = \mathbb{1}_{\mathbb{M}}$ by (i). The terminal layer $D_\eta = \sum_k \sqrt{\eta_k}\, E_{\{k\}} + E_{\mathbb{M}\setminus\bigcup_k \mathbb{P}_k}$ (Definition 3.3) is real and diagonal, the ports being pairwise disjoint (Definition 2.1 (iii)), and $D_\eta^2 + \Theta_\eta^\dagger \Theta_\eta = \sum_k \big(\eta_k + (1-\eta_k)\big) E_{\{k\}} + E_{\mathbb{M}\setminus\bigcup_k \mathbb{P}_k} = \mathbb{1}_{\mathbb{M}}$. *Telescoping.* The first identity gives $X_j^\dagger X_j = X_{j-1}^\dagger \hat{G}_{e_j}^\dagger \hat{G}_{e_j} X_{j-1} = X_{j-1}^\dagger X_{j-1} - (\hat{\Theta}_j X_{j-1})^\dagger (\hat{\Theta}_j X_{j-1})$, and the blocks of rows of Ω lying on pairwise disjoint mode sets,

$$\Omega^\dagger \Omega = X_L^\dagger \big(D_\eta^2 + \Theta_\eta^\dagger \Theta_\eta\big) X_L + \sum_{j=1}^{L} (\hat{\Theta}_j X_{j-1})^\dagger (\hat{\Theta}_j X_{j-1}) = X_L^\dagger X_L + \sum_{j=1}^{L} \big(X_{j-1}^\dagger X_{j-1} - X_j^\dagger X_j\big) = X_0^\dagger X_0 = \iota^\dagger \iota$$

The ports. $X_L = \hat{G}_{e_L} \cdots \hat{G}_{e_1} \iota = G_{\mathfrak{E}}^\sigma \iota$ by (3.1), so the rows of Ω on $\mathbb{M}$ are $D_\eta G_{\mathfrak{E}}^\sigma \iota = G\iota = G_{\mathbb{I}}$, the columns $\mathbb{I}$ of G (Definition 2.1). Every port $\mathbb{P}_k$ lies in $\mathbb{M}$, so the rows of Ω on it are those of $E_{\{k\}} G_{\mathbb{I}}$, and $\Omega^\dagger E_{\mathbb{P}_k} \Omega = G_{\mathbb{I}}^\dagger E_{\{k\}} G_{\mathbb{I}} = A_k$ by (2.2). The ports are pairwise disjoint subsets of the mode set of Ω, so (9.3) holds. □

The contractivity of the generic block C10 is decided by the route-(ii) test of Definition 10.4 (v), at the same widened threshold, and its lossless case, an equality, is decided two-sidedly within ε_1. Without the widening, rounding one unit above one would refuse a unitary block.

The generic block is the one catalogue entry whose admission is a spectral test on a supplied matrix, and that test is evaluated in floating point. It is decided at ε_1, one of the receiver tolerances, for the same reason as at route (ii): a lossless block has $\lambda_{\max}(G_\Sigma^\dagger G_\Sigma) = 1$ *exactly* (the worked instrument (17.1) of Example 17.1 is unitary), so a computed value one rounding unit above 1 would refuse it.

Definition 16.5 (Admission test of the generic block)**.** The (E1) predicate of catalogue entry C10 of Definition 16.2 is decided by the route-(ii) contractivity test of Definition 10.4 (v) applied to the supplied block G_Σ: the block is admitted when $\lambda_{\max}(G_\Sigma^\dagger G_\Sigma) \le 1 + \varepsilon_1$. C10 is the one entry whose (E1) is a spectral condition rather than a parameter-domain comparison (Theorem 16.3 (iii)). Its lossless sub-case $G_\Sigma^\dagger G_\Sigma = \mathbb{1}$ is an *equality* and is decided two-sidedly, as $\|G_\Sigma^\dagger G_\Sigma - \mathbb{1}\| \le \varepsilon_1$.

Uses Def. 10.4, Thm. 16.3 · newly derived · accepted

Every retarder of the catalogue is generated in the σz–σx plane, while the rotator is generated by σy, so no single retarder is a non-trivial rotation. A chain of two half-wave plates is one, and the rotator is the one-element primitive.

Remark 16.6 (The rotator outside the retarder family). The rotator C4 of Definition 16.2 is *not* a special case of the retarder C3. Write $\sigma_x, \sigma_y, \sigma_z$ for the Pauli matrices in the (H, V) basis. The scalar of (16.3) cancels exactly, $e^{-i\delta/2}\,\mathrm{diag}(1, e^{i\delta}) = e^{-i(\delta/2)\sigma_z}$, so that, using $R(\theta)\sigma_z R(\theta)^{\mathsf{T}} = \cos 2\theta\,\sigma_z + \sin 2\theta\,\sigma_x$,

$$W(\delta,\theta) = R(\theta)\, e^{-i(\delta/2)\sigma_z}\, R(\theta)^{\mathsf{T}} = \exp\Big(-\tfrac{i\delta}{2}\big(\cos 2\theta\,\sigma_z + \sin 2\theta\,\sigma_x\big)\Big), \tag{16.8}$$

an identity, not an equality up to a phase. Its generator lies in the σ_z–σ_x plane, while $R(\rho) = \exp(-i\rho\sigma_y)$ generates about the missing axis. So no single C3 element equals $R(\rho)$ for $\rho \in (0, 2\pi)$: a lossy one is not unitary, and $\exp\big(-\tfrac{i\delta}{2}\boldsymbol{n}\cdot\boldsymbol{\sigma}\big)$ with $\boldsymbol{n}$ in the z–x plane equals $\exp(-i\rho\sigma_y)$ only where both are real multiples of $\mathbb{1}$, that is $\sin\rho = 0$ and $\cos\tfrac{\delta}{2} = \cos\rho$, which leaves $\rho = \pi$ with $\delta = 2\pi$, outside the domain of C3. A chain of two half-wave plates does equal a rotation: with $M(a) := \left(\begin{smallmatrix} \cos a & \sin a \\ \sin a & -\cos a \end{smallmatrix}\right)$, (16.4) reads $\mathrm{HWP}(\theta) = -i\,M(2\theta)$, and $M(a)M(b) = R(a-b)$, so $\mathrm{HWP}(t_2)\,\mathrm{HWP}(t_1) = -R\big(2(t_2 - t_1)\big) = R\big(2(t_2 - t_1) + \pi\big)$, which reaches every $R(\rho)$. C4 is the one-element primitive.

Uses Thm. 16.3 · newly derived · accepted

The mirror's handedness flip is a choice of frame, the phase shifter is the interferometer's control knob, and fibre, coupling, filter and connector losses are loss entries at different sites. Route (ii) is the special case of route (iii) with one generic block spanning the receiver.

Remark 16.7 (Physical roles of the catalogue entries). Read physically, the entries of Definition 16.2 play the following roles.

- *C2.* Entry C2 is the polarization-dependent beamsplitter of entry C1 with its per-output losses, written in the mode order (p_1H, p_1V, p_2H, p_2V); its ideal preset $t_H = 1$, $t_V = 0$ is the extreme case, the polarizing beamsplitter.
- *C3.* The preset HWP$(\pi/8)$ is the 22.5° plate that maps the rectilinear basis to the diagonal one.
- *C5.* The handedness flip is a declared choice of coordinate frame, not a physical claim.
- *C6.* This is the interferometric receiver's control knob.
- *C7.* This is the loss primitive: fibre attenuation, coupling efficiency, spectral-filter insertion loss and connector loss are all C7 at different *sites*, and the sites are what Theorem 18.1 is about.
- *C8.* It is contractive and not unitary unless $t = t_\perp = 1$: the blocked polarization is loss.
- *C10.* This entry makes route (ii) a special case of route (iii). A user with a measured instrument matrix, or with a component the catalogue does not have, declares one block, and a route-(ii) declaration is recovered as the route-(iii) declaration of exactly one C10 block spanning the whole receiver.

Uses Thm. 16.3 · newly derived · accepted

Detector efficiency with dark counts, and the probe source, are declared layers rather than catalogue entries. Efficiency enters as the terminal factor, dark counts act on the outcome label, and source modes, unlike vacuum inputs, are never eliminated.

Remark 16.8 (Detector and source layers outside the catalogue). Two further declared objects are *layers*, not chain elements, and are therefore not rows of Table 4 of Definition 16.2.

L1 — Detector layer. A port map $\{\mathbb{P}_k\}_{k=1}^{K}$ with per-port efficiency $\eta_k \in [0, 1]$ and dark-count probability $p_{\mathrm{dc}|k} \in [0, 1]$. The efficiency enters as the terminal factor D_η of Definition 3.3; the dark counts never enter G at all and act on the outcome label (Theorem 19.2). *Detector* efficiency is *not* C7: it is the terminal layer, and the schema keeps the two apart (Theorem 18.1 (iii) and Requirement 22.1).

L2 — Source (probe) layer. The input split $\mathbb{I} = \mathbb{I}_C \sqcup \mathbb{I}_P$ and the probe declaration of Definition 2.3. Undeclared input modes carry vacuum and may be eliminated (Theorem 4.1 (iv)); declared source modes carry a declared state and *may not*. That difference is the entire structural distinction between "loss" and "probe", and it is a definition rather than a convention (Remark 2.7).

Uses Def. 2.3, Def. 3.3 · *Assumes* A-VAC · newly derived · accepted

The symmetric beamsplitter differs from the catalogue's real one by a phase on each side. The two conventions give the same port Grams whenever the element is terminal with each output path inside one port and the input-side phase is a scalar on the declared input space.

A reader who imports a beamsplitter matrix written in another convention needs to know which of its phase factors is free. The catalogue fixes one convention, the core $B(x,\varphi)$ of (16.1), and the interface transcribes it (`spec/modules/receiver/interface.yaml`).

Remark 16.9 (Beamsplitter conventions and the port Grams). The "symmetric" beamsplitter convention $B_{\mathrm{sym}}(x) \coloneqq \left(\begin{smallmatrix} \sqrt{x} & i\sqrt{1-x} \\ i\sqrt{1-x} & \sqrt{x} \end{smallmatrix}\right)$ differs from $B(x,0)$ by phases on *both* sides, and it matters which side each one is on. Writing $\Upsilon \coloneqq \mathrm{diag}(1,i)$ on the ordered path pair, multiplication gives

$$\Upsilon\, B(x,0)\, \Upsilon = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} \sqrt{x} & \sqrt{1-x} \\ \sqrt{1-x} & -\sqrt{x} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} \sqrt{x} & i\sqrt{1-x} \\ i\sqrt{1-x} & \sqrt{x} \end{pmatrix} = B_{\mathrm{sym}}(x), \tag{16.9}$$

the sign of the $(2,2)$ entry being restored by $i \cdot i = -1$. The two factors have different status.

- The *left* factor is an output-side phase. If the element is terminal and each of its two output paths lies inside one detector port, Corollary 5.2 applies and it changes no port Gram.
- The *right* factor is an *input*-side phase, which Corollary 5.2 does not cover. Consider the case in which the element acts directly on the inputs, each of its two input paths being either declared input modes or an undeclared vacuum port (Assumption 11.4), so that the factor acts on $\mathbb{C}^{\mathbb{I}}$ as a fixed diagonal unitary $\widetilde{\Upsilon}$, the vacuum columns being eliminated exactly (Theorem 4.1 (iv)). (With optics upstream of the element the pull-back of a diagonal phase need be neither diagonal nor unitary on $\mathbb{C}^{\mathbb{I}}$, and the comparison that follows is not the right one to make; the conclusion, to convert before declaring, still holds, and more strongly.) The two conventions then give $A_k^{\mathrm{sym}} = \widetilde{\Upsilon}^{\dagger} A_k \widetilde{\Upsilon}$ for every k: the *same* family only up to that conjugation. The conjugation is the identity map exactly when $\widetilde{\Upsilon}$ is a scalar on $\mathbb{C}^{\mathbb{I}}$, that is, when every declared input mode receives the same phase from the factor, as when the element's one declared input path is the whole declared input space and the other input path is the vacuum port: a global phase on all of $\mathbb{C}^{\mathbb{I}}$ cancels in $A_k = G_{\mathbb{I}}^{\dagger} E_{\{k\}} G_{\mathbb{I}}$. When declared input modes receive different phases, as when both input paths are declared, $\widetilde{\Upsilon}$ is a genuine relative phase between them, and the two conventions can describe *different* devices, whose Grams differ by that conjugation.

So the two conventions agree on the port Grams, and hence on the POVM, whenever both conditions hold: the element is terminal with each of its output paths inside one detector port, and the input-side factor is a scalar on the declared input space. Neither is automatic, which is why the catalogue fixes the one convention $B(x,\varphi)$ of (16.1), and why no catalogue entry drops a phase it could have absorbed.

Assumes A-VAC · newly derived · accepted

Active elements, modulators and switches driven by a party's random choice, cavities and feedback paths, and mode-shape-dependent effects are outside the catalogue, each for a stated reason. A spectral filter reduced to a scalar transmittance is a loss entry and is in scope.

Each exclusion from the catalogue follows from an assumption of the receiver or from where the protocol declaration puts classical data. The interface records that placement in its test-statistics model (`spec/modules/receiver/interface.yaml`).

Remark 16.10 (Elements outside the catalogue). Squeezers, parametric amplifiers and any active element are excluded by Assumption 11.3: their Bogoliubov map mixes a with $a^\dagger$, so $G^\dagger G \preceq \mathbb{1}$ fails, total photon number is not conserved, and the photon-number block structure on which the receiver's analysis rests does not exist. Modulators and optical switches driven by a party's random choice are excluded because a classical choice belongs in the protocol declaration as a weight, not in the mode algebra: the receiver states its measurement per setting (Definition 8.1), and the weights of the settings enter only downstream, as the class weights of Definition "Joint alphabet, classes and the joint family of a private setting" (reductions/def:joint-family). Cavities and feedback paths are excluded by condition (E2) of Definition 3.2. Anything mode-shape-dependent is outside the declared-mode model, in which each label of Definition 1.1 is one mode: dispersion and frequency conversion would act on the reserved temporal and spectral index, which Definition 1.1 fixes to a single value that no element reads, and a mode-profile mismatch would carry light into modes the declaration does not contain, which Assumption 11.5 excludes; a spectral filter reduced to a scalar transmittance is C7 and is in scope.

Uses Def. 3.2 · *Assumes* A-PASSIVE, A-THRESH, A-MODES · newly derived · accepted

17 A worked BB84 receiver

A four-detector BB84 receiver with a probe path is declared from three catalogue entries, compiled, and reduced to its channel port Grams (Example 17.1). Its analytic multiclick weight is computed in closed form, and an exact window of depth two, evaluated in floating point, reaches at cutoff one, less the declared eigenvalue slack, the weight the analytic bound reaches only at cutoff two.

A four-detector BB84 receiver with a probe path, declared from three catalogue entries, compiles to a unitary instrument matrix with rank-one channel port Grams. Its dropped wave-plate phase is invisible at the ports, and eliminating the vacuum probe inputs is exact.

A four-detector BB84 receiver with a probe path can be declared from three catalogue entries. The example shows that the catalogue and its conventions express such a receiver, exhibits a case where a dropped wave-plate phase is harmless *for the reason Corollary 5.2 gives*, and supplies the numbers used

in Example 17.2.

Example 17.1 (A BB84 receiver compiled from the catalogue). Declare eight modes: a channel path C with (C_H, C_V), a probe path P with (P_H, P_V), a Z arm with (Z_H, Z_V) and an X arm with (X_H, X_V). Declare $\mathbb{I}_C = \{C_H, C_V\}$, $\mathbb{I}_P = \{P_H, P_V\}$, and four single-mode ports $\mathbb{P}_H = \{Z_H\}$, $\mathbb{P}_V = \{Z_V\}$, $\mathbb{P}_+ = \{X_H\}$, $\mathbb{P}_- = \{X_V\}$, with unit efficiencies $\eta_H = \eta_V = \eta_+ = \eta_- = 1$ on the four ports, so that the terminal detector-efficiency layer D_η of Definition 3.3 is the identity and the compiled instrument matrix $G = D_\eta G_{\mathfrak{C}}$ is $G_{\mathfrak{C}}$ itself. Declare the chain

(1) C1 with $x = p_Z$, $\varphi = 0$, $y_1 = y_2 = 0$, on the ordered path pair (C, P), so that (per polarization) the first output, on the C path, carries $\sqrt{p_Z}\,C + \sqrt{p_X}\,P$ and the second, on the P path, carries $\sqrt{p_X}\,C - \sqrt{p_Z}\,P$, with $p_X = 1 - p_Z$;

(2) C9 on all eight modes, the permutation exchanging C_H, C_V with Z_H, Z_V and P_H, P_V with X_H, X_V, so that $Z \leftarrow \sqrt{p_Z}\,C + \sqrt{p_X}\,P$ and $X \leftarrow \sqrt{p_X}\,C - \sqrt{p_Z}\,P$;

(3) C3 preset HWP$(\pi/8)$ on the X arm only.

Every two of these elements whose supports intersect are ordered by the declared chain ((1) and (2) share the C and P modes, (2) and (3) the X arm), so (E3) holds; (E1) holds because the three declared entries carry in-domain parameters, and (E2) because the declaration is a chain and a chain has no cycle. With (E1), (E2) and (E3) the declaration is well formed, so by Corollary 3.5 this declaration is a receiver instrument, and the results stated of a receiver instrument apply to it. Compiling by (3.1) and reading off $G_{\mathbb{I}}$ with rows ordered $(H, V, +, -)$ and columns ordered (C_H, C_V, P_H, P_V):

$$G_{\mathbb{I}} = \begin{pmatrix} \sqrt{p_Z} & 0 & \sqrt{p_X} & 0 \\ 0 & \sqrt{p_Z} & 0 & \sqrt{p_X} \\ -i\sqrt{\frac{p_X}{2}} & -i\sqrt{\frac{p_X}{2}} & i\sqrt{\frac{p_Z}{2}} & i\sqrt{\frac{p_Z}{2}} \\ -i\sqrt{\frac{p_X}{2}} & i\sqrt{\frac{p_X}{2}} & i\sqrt{\frac{p_Z}{2}} & -i\sqrt{\frac{p_Z}{2}} \end{pmatrix}, \tag{17.1}$$

which is unitary: each column has squared norm $p_Z + 2 \cdot \frac{p_X}{2} = 1$ or $p_X + 2 \cdot \frac{p_Z}{2} = 1$, and the six pairwise inner products vanish (for instance $\langle C_H, C_V \rangle = \frac{p_X}{2} - \frac{p_X}{2} = 0$ and $\langle C_H, P_H \rangle = \sqrt{p_Z p_X} - 2 \cdot \frac{\sqrt{p_Z p_X}}{2} = 0$).

The phase. Rows 3 and 4 of (17.1) carry the overall factor $-i$: it is the physical retarder phase $e^{-i\delta/2}$ of (16.3) at $\delta = \pi$, which an implementation that writes HWP as the real matrix $\left(\begin{smallmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{smallmatrix}\right)$ silently drops. Here that factor is a *per-port* phase — both affected rows belong to ports that are never recombined — so by Theorem 5.1 (iii) the port Grams, and hence the POVM, are identical either way. That is a statement about this receiver and this factor, not about phases in general.

The channel-only receiver. With the probe off (vacuum on $\mathbb{I}_P$), Lemma 2.6 lets the P columns be dropped: the channel-only instrument is the left 4×2 block of (17.1), and the channel port Grams are

$$A_H = p_Z\,|H\rangle\langle H|\,, \quad A_V = p_Z\,|V\rangle\langle V|\,, \quad A_+ = p_X\,|+\rangle\langle +|\,, \quad A_- = p_X\,|-\rangle\langle -|\,, \tag{17.2}$$

each of rank one — so by Theorem 5.1 (iv) this receiver *is* exactly a one-mode-per-port device, and $A_\Sigma = p_Z(|H\rangle\langle H| + |V\rangle\langle V|) + p_X(|+\rangle\langle +| + |-\rangle\langle -|) = \mathbb{1}_2$: lossless, as the construction promises. Dropping the P columns is the vacuum elimination of Theorem 4.1 (iv), not a second model.

Uses Lem. 2.6, Def. 3.3, Cor. 3.5, Thm. 4.1, Thm. 5.1, Cor. 5.2, Thm. 16.3 · newly derived · accepted

The channel-only BB84 receiver has a closed-form analytic multiclick bound, and its one setting is exact, so the emitted weight is the positive part of the bound at the first block above the cutoff, with no tail window. At balanced bases the cutoffs $N = 2$ and $N = 3$ give 1/2 and 3/4.

The rank-one Grams of the BB84 receiver make every quantity of the analytic multiclick bound explicit, and its single setting shows the collapse branch at work.

Example 17.2 (Analytic multiclick weight of the BB84 receiver). Take the *channel-only* receiver of the last paragraph of Example 17.1 (the probe off and its P columns eliminated, Lemma 2.6), which declares $\mathbb{I}_P = \varnothing$ and therefore has $\mathcal{T}_{\text{set}} = \{\text{off}\}$ by Definition 2.3. Then $\gamma = \beta_k = 1$ and the Grams are (17.2) with $A_\Sigma = \mathbb{1}_2$, so $c = 1 - \lambda_{\max}(A_\Sigma) = 0$, and

$$A_\Sigma - A_H = \mathbb{1} - p_Z\,|H\rangle\langle H| \;\Rightarrow\; \lambda_{\min} = 1 - p_Z = p_X \;\Rightarrow\; d_H = p_Z, \tag{17.3}$$

and likewise $d_V = p_Z$, $d_+ = d_- = p_X$. Hence, for $n \geq 1$,

$$\beta(n) = 1 - 2p_Z^n - 2p_X^n. \tag{17.4}$$

This receiver declares no probe modes, so its one setting is flagged exact (Corollary 8.2) and predicate (X) of Definition 8.3 holds, (6.8) holds with equality ($\gamma = \beta_k = 1$), branch 1 is taken, and no tail window is declared or required (Remark 8.5, Requirement 13.1). Since $\mathcal{T}_{\text{set}} = \{\text{off}\}$, the minimum over $\mathcal{T}_{\text{set}}$ in Definition 8.3 is over that one setting, so what is computed here is the emitted p_{mc} and not one of several per-setting values. Hence $p_{\text{mc}} = \max\{0, \beta(N+1)\}$ exactly. At $p_Z = p_X = \frac{1}{2}$ this is $\beta(n) = 1 - 2^{2-n}$: $\beta(1) = -1$, $\beta(2) = 0$, $\beta(3) = \frac{1}{2}$, $\beta(4) = \frac{3}{4}$, non-decreasing from the first non-negative index, as Lemma 6.5 requires. A cutoff $N = 2$ therefore yields $p_{\text{mc}} = \frac{1}{2}$, and $N = 3$ yields $p_{\text{mc}} = \frac{3}{4}$. Note that $p_{\text{mc}} \to 1$ exponentially exactly when $\max(p_Z, p_X) < 1$, i.e. when no single basis absorbs the whole input.

Uses Def. 2.3, Lem. 2.6, Def. 3.3, Lem. 6.5, Def. 8.3 · newly derived · accepted

The exact two-photon multiclick block of the channel-only BB84 receiver shows the slack in the analytic bound. An exact window of depth two reaches at cutoff one, less the declared eigenvalue slack, the weight the analytic bound reaches only at cutoff two.

An exact window takes, on the first blocks above the cutoff, the larger of the analytic bound and a computed block eigenvalue less the eigenvalue slack. On the BB84 receiver the two-photon block can be diagonalised by hand, which shows what the window buys.

Example 17.3 (Exact window on the BB84 receiver). Continue with the channel-only receiver of Example 17.2: $K = 4$, $\gamma = \beta_k = 1$, $\mathbb{I}_C = \{C_H, C_V\}$ so $D = |\mathbb{I}_C| = 2$, and the Grams (17.2) with $A_\Sigma = \mathbb{1}_2$. Here $B = A_\Sigma = \mathbb{1}_2$ and $B_k = A_{[K]\setminus\{k\}} = \mathbb{1} - A_k$, so $\mathbb{1} - B = 0$ and $\mathbb{1} - B_k = A_k$, and (7.1) reads, for $n \geq 1$,

$$M_{\text{mc}}^{(=n)} = \mathbb{1}^{(=n)} - \sum_k \mathfrak{F}(A_k)^{(n)} = \mathbb{1}^{(=n)} - p_Z^n\big(\,|n_H\rangle\langle n_H| + |n_V\rangle\langle n_V|\,\big) - p_X^n\big(\,|n_+\rangle\langle n_+| + |n_-\rangle\langle n_-|\,\big). \tag{17.5}$$

Two steps: $\mathfrak{F}(0)^{(n)} = 0$ for $n \geq 1$ kills the γ term; and for a rank-one $A_k = p\,|\psi\rangle\langle\psi|$, $(p\,|\psi\rangle\langle\psi|)^{\otimes n}$ is p^n times the orthogonal projector onto $(\mathrm{span}\,|\psi\rangle)^{\otimes n}$, whose restriction to Sym^n is $|n_\psi\rangle\langle n_\psi|$ with $|n_\psi\rangle = |\psi\rangle^{\otimes n}$ the occupation vector carrying all n photons in $|\psi\rangle$ (Lemma 1.6 (iv) with $P = |\psi\rangle\langle\psi|$).

The two-photon block. The space $\mathrm{Sym}^2(\mathbb{C}^2)$ has dimension $\binom{2+2-1}{2} = 3$ and occupation basis $\{|2_H\rangle, |1_H 1_V\rangle, |2_V\rangle\}$, so $|2_H\rangle\langle 2_H| + |2_V\rangle\langle 2_V| = \mathbb{1} - |1_H 1_V\rangle\langle 1_H 1_V|$; likewise $|2_+\rangle\langle 2_+| + |2_-\rangle\langle 2_-| = \mathbb{1} - |1_+ 1_-\rangle\langle 1_+ 1_-|$. Expanding $|\pm\rangle = (|H\rangle \pm |V\rangle)/\sqrt{2}$,

$$|1_+ 1_-\rangle = \tfrac{1}{\sqrt{2}}\big(\,|+\rangle\otimes|-\rangle + |-\rangle\otimes|+\rangle\,\big) = \tfrac{1}{\sqrt{2}}\big(\,|2_H\rangle - |2_V\rangle\,\big), \tag{17.6}$$

which is orthogonal to $|1_H 1_V\rangle$. Substituting into (17.5) at $n = 2$,

$$M_{\mathrm{mc}}^{((=2))} = \big(1 - p_Z^2 - p_X^2\big)\,\mathbb{1} + p_Z^2\;|1_H 1_V\rangle\langle 1_H 1_V| + p_X^2\;|1_+ 1_-\rangle\langle 1_+ 1_-|\,, \tag{17.7}$$

whose spectrum is therefore $\{1 - p_X^2,\ 1 - p_Z^2,\ 1 - p_Z^2 - p_X^2\}$, the first two on the two displayed orthonormal vectors, the third on $(|2_H\rangle + |2_V\rangle)/\sqrt{2}$. So

$$\beta^*(2) = 1 - p_Z^2 - p_X^2 \qquad \text{against} \qquad \beta(2) = 1 - 2p_Z^2 - 2p_X^2, \tag{17.8}$$

a gap of $p_Z^2 + p_X^2 > 0$ at every (p_Z, p_X).

At the cutoff that matters. Take $p_Z = p_X = \frac{1}{2}$, where $\beta(n) = 1 - 2^{2-n}$ and $\beta^*(2) = \frac{1}{2}$. At the cutoff $N = 1$, Definition 8.3 emits $\max\{0, \beta(2)\} = 0$: the two-photon block is discarded entirely, although half of its weight is guaranteed. Declare the minimal depth $m^* = N + 1 = 2$. Predicate (X) holds (this receiver declares $\mathbb{I}_P = \varnothing$), so (8.7) takes branch 1 with the remainder $\beta(m^* + 1) = \beta(3) = \frac{1}{2}$, and, at an evaluation $\widehat{\beta}^*(2) = \beta^*(2)$,

$$p_{\mathrm{mc}} = \max\big\{0,\ \min\big\{\ell(2),\ \beta(3)\big\}\big\} = \max\big\{0,\ \tfrac{1}{2} - \varepsilon_{\mathrm{eig}}\big\}, \tag{17.9}$$

since $\ell(2) = \max\{\beta(2), \widehat{\beta}^*(2) - \varepsilon_{\mathrm{eig}}\} = \max\{0, \frac{1}{2} - \varepsilon_{\mathrm{eig}}\}$, which is $\leq \frac{1}{2} = \beta(3)$ on the whole declared domain $\varepsilon_{\mathrm{eig}} \in [0, 1)$, so the minimum is $\ell(2)$ throughout. The value is $\frac{1}{2} - \varepsilon_{\mathrm{eig}}$ at $\varepsilon_{\mathrm{eig}} \leq \frac{1}{2}$ and 0 above it; the outer $\max\{0, \cdot\}$ is carried through the chain rather than dropped, because the declared domain reaches past $\frac{1}{2}$. The declared slack is subtracted once, from the computed block eigenvalue $\widehat{\beta}^*(2) = \frac{1}{2}$.

What that buys, and what it does not change. Example 17.2 reaches $p_{\mathrm{mc}} = \frac{1}{2}$ only at the cutoff $N = 2$; the exact window reaches $\max\{0, \frac{1}{2} - \varepsilon_{\mathrm{eig}}\}$ at $N = 1$, from a single 3×3 eigenvalue problem, and N is the parameter every downstream target dimension grows with. Theorem 6.2 is untouched: $\beta(3) = \frac{1}{2}$ is still the remainder covering every block above the second, and the analytic bound is still what makes the quantifier over $n > N$ finite. Note also that the declaration must supply $\varepsilon_{\mathrm{eig}}$ and ε_6 to declare the depth at all, and that this receiver still declares no tail window, predicate (X) holding (Remark 8.5).

Uses Lem. 1.6, Def. 3.3, Thm. 6.2, Def. 7.1, Def. 8.3, Def. 8.7 · newly derived · accepted

Part IV
Detector imperfections

This submodule adds detector efficiency and dark counts to the passive receiver, and its two main results say how each acts on the measurement. Efficiency uniform within a port scales that port's Gram and enters at the single terminal layer of the compiled matrix (Theorem 18.1); efficiency that varies within a port is not a scalar and can lower a Gram's rank. Dark counts are a classical channel on the recorded click pattern that adds clicks and never removes them. Such a channel preserves the POVM structure and improves the multiclick bound p_mc to $(1 - \mathrm{r_min})\mathrm{p_mc} + \mathrm{r_min}$, r_min being the least favourable transition from every non-multiclick outcome, the no-click outcome included (Theorem 19.2). A declared channel is admitted at two receiver tolerances, and the bound a downstream module composes with carries their cost explicitly. The emitted blocks are those of the dark-composed family, enclosed rigorously for any declared parameters and exact where the enclosures have radius zero; loss and a dark count in every port make each of its elements positive definite on every block, and neither can be dropped (Proposition 20.3), which is the floor a model needs if no snap of its blocks is to revive a zero row. A receiver without a probe is a uniform loss of transmittance t followed by a normalized station whose efficiencies are divided by t, for any t from its largest efficiency η_max to one, so that loss can be handed to the adversary with the observed statistics unchanged (Proposition 21.4); by default the engine compiles the normalized station at $\mathrm{t} = \eta_\mathrm{max}/(99/100)$, and the declared receiver (t = 1) where $\eta_\mathrm{max} \geq 99/100$.

18 Detector efficiency

The main result, Theorem 18.1, is that detector efficiency is a parameter of the terminal layer, applied after every chain element, that multiplies each port Gram by its efficiency when uniform within the port. The remarks show a within-port attenuation that lowers a rank, why a product of transmittances is the wrong loss audit in an interferometer, and why power data and click data differ by exactly the efficiencies.

Efficiency η_k uniform on detector port k scales that port's Gram by η_k, keeping its rank if η_k > 0; varying within a port, it is no scalar and can lower the rank. It enters only as the compiled instrument matrix's last layer, and the loss decomposition Lemma 4.2 checks that loss is counted once.

"Apply the efficiency once" is easy to state for a fixed device and does not survive verbatim once a receiver is compiled from components: a component graph has many legitimate loss sites, among them every beamsplitter's loss,

every C7 element, every polarizer's extinction, every mirror's reflectivity and the detector efficiency itself. The theorem fixes where the detector efficiency enters, what it does to the port Grams and what the emitted record of the loss sites lists. What the engine keeps apart in its input and records in its output is stated in Requirements 22.1 and 22.2.

Theorem 18.1 (Detector efficiency in the port Grams). Let $\mathfrak{R}$ be a receiver instrument with port Grams $\{A_k\}$ and an instrument matrix G that does not yet contain a detector-efficiency layer (for a declared element graph, the product $G^{\sigma}_{\mathfrak{E}}$ of Definition 3.3), let $\eta_1, \dots, \eta_K \in [0,1]$, and let $D_\eta = \sum_k \sqrt{\eta_k}\, E_{\{k\}} + E_{\mathbb{M}\setminus\bigcup_k \mathbb{P}_k}$ be the terminal detector-efficiency layer of Definition 3.3.

(i) *Uniform within a port: a scalar on the Gram.* $D_\eta G$ is a contraction, and its port Grams are

$$A_k \;\longmapsto\; \eta_k\, A_k. \tag{18.1}$$

For $\eta_k > 0$ this preserves $\operatorname{rank} A_k$, so it leaves the one-mode-per-port criterion of Theorem 5.1 (iv) untouched. Every formula of Theorems 4.5, 5.6, 6.2 and 8.14 therefore covers inefficient detectors under the single substitution (18.1); in particular $A_\Sigma \mapsto \sum_k \eta_k A_k$, which is in general not a multiple of A_Σ unless the η_k agree.

(ii) *Non-uniform within a port: not a scalar, and the rank can drop.* If attenuation varies *within* a port, that is, for a diagonal contraction D that is not constant on $\mathbb{P}_k$, the effective Gram is

$$A_k^{\mathrm{eff}} = G_{\mathbb{I}}^{\dagger} D^{\dagger} E_{\{k\}} D\, G_{\mathbb{I}}, \tag{18.2}$$

which is in general *not* a multiple of A_k and satisfies only $\operatorname{rank} A_k^{\mathrm{eff}} \le \operatorname{rank} A_k$, with strict inequality possible. All formulas of the theorems named in (i) still apply, to A_k^{eff}.

(iii) *One structural site.* Detector efficiency is a parameter of the terminal layer L1 and enters the compiled instrument matrix $D_\eta G^{\sigma}_{\mathfrak{E}}$ only through the single left factor D_η of Definition 3.3, which is applied after every chain element. It is therefore not an audit but an invariant of the compiler: there is exactly one site at which it can enter.

(iv) *The composition record.* The emitted efficiency record lists every declared loss parameter q_i, under its name, with the element that declares it and its position in the compiled product: the transmittance x of a beamsplitter (C1) and t_H, t_V of a polarizing beamsplitter (C2), which is also the entry of a polarization-dependent beamsplitter, its x_H, x_V recorded as t_H, t_V; the output losses y_1, y_2 of C1 and C2 and the loss y of a retarder (C3), whose transmittances are $1-y_1$, $1-y_2$

and $1-y$; the reflectivities R_H, R_V of a mirror (C5); the transmittances η_m of a loss element (C7); the pass and leak transmittances $t, t_\perp$ of a polarizer (C8); and the detector efficiencies η_k. Two elements may carry equal numerical values; they are still two losses and appear as two rows.

(v) *The deficiency identity.* The identity (4.3) of Lemma 4.2 holds for every receiver instrument, before and after the layer D_η, and is the correct check that loss was accounted for exactly once: its two summands are the amplitude that left through the dilation and the amplitude that reached an unmonitored declared mode, and neither carries a separate efficiency term; after the layer, the efficiency loss is part of the dilation summand.

Uses Prop. 1.4, Def. 3.3, Lem. 4.2, Thm. 5.1 · *Used by* Rem. 18.4, Rem. 18.5, Prop. 20.3, Def. 21.2, Rem. 21.6, Req. 22.1, Req. 22.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.efficiency: efficiency_record · *Oracles* registry_receiver_povm.py

Proof. (i) D_η is diagonal with entries in $[0,1]$, hence a positive semidefinite contraction, and a product of contractions is a contraction (Proposition 1.4 (ii)). The ports are pairwise disjoint, so $E_{\{k\}} D_\eta = \sqrt{\eta_k}\, E_{\{k\}}$, whence

$$(D_\eta G)_{\mathbb{I}}^\dagger E_{\{k\}} (D_\eta G)_{\mathbb{I}} = G_{\mathbb{I}}^\dagger D_\eta E_{\{k\}} D_\eta G_{\mathbb{I}} = \eta_k\, G_{\mathbb{I}}^\dagger E_{\{k\}} G_{\mathbb{I}} = \eta_k A_k. \tag{18.3}$$

A strictly positive scalar multiple of a matrix has the same rank.

(ii) The displayed formula is the definition of the port Gram for the instrument DG. For the rank, $E_{\{k\}} DG_{\mathbb{I}} = D_k\big(E_{\{k\}} G_{\mathbb{I}}\big)$ where D_k is the diagonal restriction of D to $\mathbb{P}_k$, and $\operatorname{rank}(D_k Y) \le \operatorname{rank} Y$ for any Y; the Gram of a matrix has the rank of that matrix. Strictness is exhibited in Remark 18.2.

(iii) is read off Definition 3.3, in which D_η is the one factor that carries the detector efficiencies.

(iv) is a statement about the emitted artifact.

(v) is Lemma 4.2, which uses only that the instrument matrix is a contraction; $D_\eta G$ is one by (i). □

A port of two modes with Gram diag(1, ε^2) has rank two; attenuating one of its modes to zero leaves rank one. Apart from a zero efficiency, within-port attenuation is the one operation in this contract that can change the answer to the one-mode-per-port rank test.

Remark 18.2 (A port Gram whose rank drops under within-port attenuation). Take $|\mathbb{I}| = 2$ and a port collecting two modes, with the two rows of $E_{\{k\}} G_{\mathbb{I}}$ equal to $(1,0)$ and $(0,\varepsilon)$ for some $\varepsilon \in (0,1]$. Then $A_k = \operatorname{diag}(1, \varepsilon^2)$ has rank 2. Attenuating the second of those modes to zero, by a diagonal D with entries 1 and 0 inside the same port, gives $A_k^{\text{eff}} = \operatorname{diag}(1,0)$, of rank 1. So a device that does *not* satisfy the one-mode-per-port criterion of Theorem 5.1 (iv) satisfies it after a within-port

attenuation, and the rank test gives its answer only on the Grams after every efficiency is applied, never before (Requirement 12.4). For single-mode ports the situation never arises, which is why the layer L1 declares one η_k per port (the uniform case of Theorem 18.1 (i)) by construction. Apart from declaring some $\eta_k = 0$, which sends A_k to 0 by Theorem 18.1 (i), this is the one operation in this contract that can change the answer to the rank test (Remark 5.4).

Uses Thm. 5.1, Rem. 5.4 · newly derived · accepted

In a lossless balanced Mach–Zehnder interferometer at zero phase, a product of transmittances along paths reports a spurious loss of one half at the first port, as it cannot see the interference cross term. The deficiency identity is the audit that holds, and the artifact reports it as a residual.

Remark 18.3 (Failure of the path-product loss audit under interference). The obvious audit, multiplying the declared transmittances along each path and comparing with the port throughput, is *wrong* in any interferometric topology, and the failure is not small. Consider a balanced Mach–Zehnder interferometer built from the component catalogue: C1 with $x = \frac{1}{2}, \varphi = 0$ and no loss, then C6 with phase ϕ on the second path, then C1 with $x = \frac{1}{2}, \varphi = 0$ again, with the two output paths monitored by two ports, one per path. Every declared loss parameter is 0 ($y_1 = y_2 = 0$ at both beamsplitters, and the phase shifter is unitary), and (4.3) correctly returns $\mathbb{L} = 0$. The compiled matrix is, on the two paths (each entry acting as that multiple of $\mathbb{1}_2$ on polarization),

$$G = B(\tfrac{1}{2}, 0)\,\mathrm{diag}(1, e^{i\phi})B(\tfrac{1}{2}, 0) = \tfrac{1}{2}\begin{pmatrix} 1 + e^{i\phi} & 1 - e^{i\phi} \\ 1 - e^{i\phi} & 1 + e^{i\phi} \end{pmatrix}, \tag{18.4}$$

so at $\phi = 0$ it is the identity and *all* the light entering path 1 reaches port 1. A path-product audit sees two paths from input 1 to output 1, transmit–transmit and reflect–reflect, each of power weight $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, and concludes a throughput of $\frac{1}{2}$, reporting a spurious 50% loss that does not exist. The missing $\frac{1}{2}$ is the interference cross term, which no product of transmittances can see. The checkable invariant is (4.3), and it is what the artifact reports (residual r_3 of Definition 10.1).

Uses Lem. 4.2, Def. 10.1 · newly derived · accepted

A power characterisation of the optics returns the port Grams A_k, click statistics return η_k A_k. Double counting is avoided exactly when the efficiencies enter once, and which convention the declared numbers follow cannot be read off the numbers.

Remark 18.4 (Power data and click data measure different objects). A classical power characterisation of the optics measures intensities independently of the single-photon detectors and returns A_k; detector click statistics return $\eta_k A_k$. Double counting is avoided exactly when the efficiencies enter through (18.1) once: either the Grams are declared from power data and then rescaled by η_k, or they are declared from click data and the η_k are already inside them and must be declared as 1. Which of the two a declaration means is not inferable from the numbers (Requirement 22.3 says where it is recorded). *Characterising* the device is out of scope (Assumption 11.1); recording which convention the declared numbers follow is not.

Uses Thm. 18.1 · *Assumes* A-TRUST · *Used by* Req. 22.3 · newly derived · accepted

For one mode the efficiency substitution gives an on/off detector of efficiency η the normally ordered no-click element $\exp(-\eta n)$, the zero-count, noise-free case of a published photo-counting formula. The agreement corroborates and backs no step.

Remark 18.5 (The single-mode efficiency substitution in the photo-counting formula). For a single mode, (18.1) together with Proposition 1.11 says that an on/off detector of efficiency η has no-click element $:e^{-\eta\hat{n}}:$ rather than $:e^{-\hat{n}}:$. The same substitution appears in [4, Eq. (3)], whose photo-counting formula with finite quantum efficiency η and noise ν reads $p_k = \langle :(\eta\hat{n}+\nu)^k e^{-(\eta\hat{n}+\nu)}:\rangle/k!$ and whose $k=0$, $\nu=0$ case is $\langle :e^{-\eta\hat{n}}:\rangle$. This is a corroboration only; no proof relies on it.

Uses Prop. 1.11, Thm. 18.1 · newly derived · accepted

19 Dark counts

A dark-count channel is column-stochastic and monotone. The main result, Theorem 19.2, is that at every setting it keeps the composed family a photon-number block-diagonal POVM, never moves weight out of the multiclick set, and improves the multiclick bound; the minimum defining the improvement must run over the no-click column too. A declared channel is admitted at two receiver tolerances, and the composed inequality that holds for every admitted channel carries both slacks (Proposition 19.4).

A dark-count channel is a column-stochastic matrix on click patterns that can add clicks and never removes one. It is generated from per-port dark-count probabilities in product form, or declared as a matrix and then checked at the receiver tolerances.

With the efficiencies inside the Grams, the remaining detector imperfection is a click produced with no incident light. Dark counts change the recorded outcome, not the optical modes, so they are a classical channel on the outcome label, and a channel satisfying the definition's conditions (D1) and (D2) exactly can only *help* the multiclick bound. The care is in *how much* it helps. A declared channel is checked in the floating-point evaluation model of Definition 10.9.

Definition 19.1 (Dark-count channel and its independent product form)**.** A *dark-count channel* is a matrix $R = (R_{S|S'})_{S,S'\subseteq[K]}$ that is

(D1) *column-stochastic*: $R_{S|S'} \geq 0$ and $\sum_S R_{S|S'} = 1$ for every S';

(D2) *monotone*: $R_{S|S'} > 0$ implies $S' \subseteq S$; a dark count adds a click and never removes one.

The *independent product form* generated by per-port dark-count probabilities $p_{\mathrm{dc}|k} \in [0,1]$ is

$$R_{S|S'} := \mathbb{1}\left[S' \subseteq S\right] \prod_{k\in S\setminus S'} p_{\mathrm{dc}|k} \prod_{k\notin S} \left(1 - p_{\mathrm{dc}|k}\right). \tag{19.1}$$

For $\mathcal{O} \subseteq 2^{[K]}$ we write $R_{\mathcal{O}|S'} \coloneqq \sum_{S\in\mathcal{O}} R_{S|S'}$. A dark-count channel is supplied on one of two routes: on the *generated dark-count route* it is (19.1), and on the *declared dark-count route* it is a declared matrix R. (D1) and (D2) hold by construction for (19.1) (Theorem 19.2 (i)); for a *declared* R they are conditions to be checked, and in floating point they are decided at the receiver tolerances ε_4 and ε_2 respectively, by item (iii) of Definition 10.4, whose content is Definition 19.3.

Uses Def. 10.4 · *Used by* Thm. 19.2, Def. 19.3, Prop. 19.4, Def. 20.1, Cor. 20.2, Prop. 20.3, Prop. 21.4, Req. 22.7, Req. 22.9, Req. 22.10, reductions/wpfss/prop:route, reductions/wpfss/rem:darkcompose, reductions/wpfss/rem:receiver-imports · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.dark:check_dark_channel

A dark-count channel keeps the photon-number block-diagonal POVM structure and never moves weight out of the multiclick set. It raises the multiclick bound p_mc to (1 − r_min)p_mc + r_min, where r_min is the least multiclick mass over every non-multiclick column, the no-click column included.

The theorem is stated for a channel that satisfies (D1) and (D2) exactly. For a channel admitted at the receiver tolerances, Proposition 19.4 states what holds in place of items (iii) and (iv), and Remark 19.6 (f) states what remains of item (ii). Of the POVM it uses only that it is a POVM whose elements commute with the photon number, so it holds at every setting of a setting family, the click-pattern POVM being the one setting of a receiver without a probe.

Theorem 19.2 (Dark counts on a photon-number block-diagonal POVM). Let R be a dark-count channel, let $\{M_S\}_{S\subseteq[K]}$ be the click-pattern POVM of Theorem 4.5 or, at a setting t of a setting family of K ports, its POVM $\{M_{S|t}\}_{S\subseteq[K]}$ (Definition 8.1 (a)), written $\{M_S\}$, and put $\widetilde{M}_S \coloneqq \sum_{S'\subseteq[K]} R_{S|S'} M_{S'}$.

(i) *The product form is admissible.* (19.1) satisfies (D1) and (D2).

(ii) *Structure is preserved.* $\{\widetilde{M}_S\}_{S\subseteq[K]}$ is a POVM, every $\widetilde{M}_S$ is block diagonal in the photon number with $\widetilde{M}_S^{(=n)} = \sum_{S'} R_{S|S'} M_{S'}^{(=n)}$, and R acts on the outcome label alone: it commutes with every operation on the optical modes.

(iii) *Multiclick absorption.* (D2) implies $R_{\mathsf{MC}|S'} = 1$ for every $S' \in \mathsf{MC}$: dark counts never move weight out of the multiclick set.

(iv) *The bound improves.* With

$$r_{\min} \coloneqq \min_{S'\in 2^{[K]}\setminus \mathsf{MC}} R_{\mathsf{MC}|S'}, \tag{19.2}$$

the minimum running over *every* non-multiclick column, the no-click

column $S' = \varnothing$ included, one has, writing $\widetilde{M}_{\mathsf{MC}} = \sum_{S \in \mathsf{MC}} \widetilde{M}_S$,

$$\widetilde{M}_{\mathsf{MC}} \;\succeq\; (1 - r_{\min})\, M_{\mathsf{MC}} + r_{\min}\, \mathbb{1}, \tag{19.3}$$

and consequently, if $M_{\mathsf{MC}}^{(>N)} \succeq p_{\mathrm{mc}}\, \mathbb{1}^{(>N)}$, then

$$\widetilde{M}_{\mathsf{MC}}^{(>N)} \;\succeq\; \big((1 - r_{\min})\, p_{\mathrm{mc}} + r_{\min}\big)\, \mathbb{1}^{(>N)}, \tag{19.4}$$

which is $\geq p_{\mathrm{mc}}$ since $p_{\mathrm{mc}} \leq 1$.

Uses Thm. 4.5, Def. 8.1, Def. 19.1 · *Used by* Prop. 19.4, Rem. 19.5, reductions/wpfss/prop:route, reductions/wpfss/rem:darkcompose, reductions/wpfss/rem:receiver-imports · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.dark: compose_blocks · *Oracles* registry_receiver_povm.py

Proof. (i) Non-negativity is clear. Fix S' and write $S = S' \sqcup W$ with $W \subseteq [K] \setminus S'$; then $\sum_S R_{S|S'} = \sum_{W \subseteq [K]\setminus S'} \prod_{k \in W} p_{\mathrm{dc}|k} \prod_{k \in [K]\setminus S' \setminus W} (1 - p_{\mathrm{dc}|k}) = \prod_{k \notin S'} \big(p_{\mathrm{dc}|k} + (1 - p_{\mathrm{dc}|k})\big) = 1$, the middle equality being the expansion of the product. (D2) is the indicator in (19.1).

Of $\{M_S\}$ the proof uses only that it is a POVM and that every element commutes with the photon number: for the click-pattern POVM these are Theorem 4.5 (i) and (iii), and at a setting of a setting family they are Definition 8.1 (a).

(ii) $\widetilde{M}_S \succeq 0$ as a non-negative combination of positive operators, and, by (D1) and the completeness of $\{M_S\}$,

$$\sum_S \widetilde{M}_S = \sum_{S'} \Big(\sum_S R_{S|S'} \Big) M_{S'} = \sum_{S'} M_{S'} = \mathbb{1}. \tag{19.5}$$

Block diagonality is inherited from each $M_{S'}$; the coefficients are scalars, so they commute with everything.

(iii) If $S' \in \mathsf{MC}$ and $R_{S|S'} > 0$ then $S \supseteq S'$ by (D2), so $|S| \geq |S'| \geq 2$ and $S \in \mathsf{MC}$. Hence every non-zero entry of the column S' lies in MC, and column-stochasticity gives $R_{\mathsf{MC}|S'} = 1$.

(iv) Split the sum over columns:

$$\begin{aligned} \widetilde{M}_{\mathsf{MC}} &= \sum_{S' \in \mathsf{MC}} R_{\mathsf{MC}|S'} M_{S'} + \sum_{S' \notin \mathsf{MC}} R_{\mathsf{MC}|S'} M_{S'} \overset{\text{(iii)}}{=} M_{\mathsf{MC}} + \sum_{S' \notin \mathsf{MC}} R_{\mathsf{MC}|S'} M_{S'} \\ &\succeq M_{\mathsf{MC}} + r_{\min} \sum_{S' \notin \mathsf{MC}} M_{S'} = M_{\mathsf{MC}} + r_{\min}\big(\mathbb{1} - M_{\mathsf{MC}}\big) = (1 - r_{\min}) M_{\mathsf{MC}} + r_{\min}\mathbb{1}, \end{aligned} \tag{19.6}$$

using $M_{S'} \succeq 0$ and $R_{\mathsf{MC}|S'} \geq r_{\min}$ in the inequality and the completeness of $\{M_S\}$ in the last line. All operators are block diagonal in the photon number, so the inequality restricts to the $> N$ blocks; since $1 - r_{\min} \geq 0$, substituting the assumed bound gives (19.4). The final comparison is $(1 - r_{\min})p_{\mathrm{mc}} + r_{\min} - p_{\mathrm{mc}} = r_{\min}(1 - p_{\mathrm{mc}}) \geq 0$. □

For a declared dark-count matrix, column-stochasticity is decided at ε_4 and monotonicity at ε_2. This is item (iii) of Definition 10.4; the slack the two tests admit is carried into the bound by Proposition 19.4.

Proposition 19.4 states what both slacks cost and which side of each test each step uses: the lower side of (D1)'s column sum together with the upper side of (D2)'s entry bound gives (19.9), the upper side of (D1)'s column sum permits $r_{\min} > 1$ (19.10), and what a downstream module composes with is (19.13) rather than (19.4). The floating-point evaluation model in which the two tests decide is Definition 10.9.

Definition 19.3 (Receiver tolerances deciding dark-channel admissibility)**.** This definition extends the list of Definition 10.4 by its item (iii), under the same rule: an exact condition about declared data is decided in floating point by a receiver tolerance, fixed here, and by no other threshold.

Dark-channel admissibility (Definition 19.1) for a *declared* R: (D1) column-stochasticity is decided at ε_4, admitted when $R_{S|S'} \geq -\varepsilon_4$ for every entry and $\left|\sum_S R_{S|S'} - 1\right| \leq \varepsilon_4$ for every column; and (D2) monotonicity is decided at ε_2: an entry counts as positive when $R_{S|S'} > \varepsilon_2$, so the channel is refused exactly when some entry with $S' \not\subseteq S$ exceeds ε_2. ε_4 bounds the completeness residual (10.4), which is the same kind of quantity as a column sum's defect from one; ε_2 is the threshold at which this contract already calls a spectral quantity zero.

Uses Def. 10.1, Def. 10.4, Def. 19.1 · *Used by* Prop. 19.4, Req. 22.9, reductions/wpfss/rem:darkcompose · newly derived · accepted

For a declared dark-count matrix admitted at ε_2 and ε_4, each multiclick column keeps mass at least $1 - \delta_dc$, r_min can exceed one by at most $(K+2)\varepsilon_4$, and the dark-count improvement holds with rbar = min{r_min, 1} and δ_dc subtracted. This is the inequality a downstream module composes with.

Theorem 19.2 needs (D1) and (D2) exactly, and the tests of Definition 19.3 admit a declared channel that satisfies them only up to ε_4 and ε_2. The proposition carries both slacks into the inequality that is composed with the multiclick bound: the lower side of (D1)'s column sum and the upper side of (D2)'s entry bound cost a constant δ_{dc}. The upper side of (D1), column sums above one and entries down to $-\varepsilon_4$, lets $r_{\min}$ exceed one; the coefficient is capped at one, at a cost bounded in (ii).

Proposition 19.4 (Multiclick bound under a dark-count channel admitted with slack)**.** Let $\{M_S\}_{S\subseteq[K]}$ be the click-pattern POVM of Theorem 4.5 or, at a setting t of a setting family of K ports, its POVM $\{M_{S|t}\}_{S\subseteq[K]}$ (Definition 8.1 (a)), written $\{M_S\}$, and for a matrix $R = (R_{S|S'})_{S,S'\subseteq[K]}$ let $\widetilde{M}_S$, $\widetilde{M}_{\mathsf{MC}}$ and $r_{\min}$ be formed as in Theorem 19.2 and (19.2). Put

$$\delta_{\mathrm{dc}} := \begin{cases} 0 & \text{generated route: } R \text{ is the product form (19.1),} \\ \varepsilon_4 + (K+1)\,\varepsilon_2 & \text{declared route: } R \text{ admitted under Def. 10.4 (iii),} \end{cases} \tag{19.7}$$

the two cases being the two routes of Definition 19.1, and

$$\bar{r} \;\coloneqq\; \min\{\, r_{\min},\, 1 \,\}. \tag{19.8}$$

Let R be declared and admitted under item (iii) of Definition 10.4 (Definition 19.3), on either side of either test and whatever the sign of $r_{\min}$. Then:

(i) *(the multiclick columns)* for every $S' \in \mathsf{MC}$,

$$R_{\mathsf{MC}|S'} \;\geq\; (1-\varepsilon_4) - (K+1)\varepsilon_2 \;=\; 1-\delta_{\mathrm{dc}}; \tag{19.9}$$

(ii) *(the upper side)* for every column S',

$$R_{\mathsf{MC}|S'} \;\leq\; (1+\varepsilon_4) + (K+1)\varepsilon_4 \;=\; 1+(K+2)\varepsilon_4, \tag{19.10}$$

and consequently

$$0 \;\leq\; r_{\min} - \bar{r} \;\leq\; (K+2)\,\varepsilon_4; \tag{19.11}$$

(iii) *(the composed inequality)*

$$\widetilde{M}_{\mathsf{MC}} \;\succeq\; (1-\delta_{\mathrm{dc}})\,M_{\mathsf{MC}} + \bar{r}\,\big(\mathbb{1} - M_{\mathsf{MC}}\big) \;\succeq\; \Big[(1-\bar{r})\,M_{\mathsf{MC}} + \bar{r}\,\mathbb{1}\Big] - \delta_{\mathrm{dc}}\,\mathbb{1}, \tag{19.12}$$

and consequently, whenever $M_{\mathsf{MC}}^{(>N)} \succeq p_{\mathrm{mc}}\,\mathbb{1}^{(>N)}$,

$$\widetilde{M}_{\mathsf{MC}}^{(>N)} \;\succeq\; \big((1-\bar{r})\,p_{\mathrm{mc}} + \bar{r} - \delta_{\mathrm{dc}}\big)\,\mathbb{1}^{(>N)}. \tag{19.13}$$

Uses Thm. 4.5, Def. 8.1, Def. 10.4, Def. 19.1, Thm. 19.2, Def. 19.3 · *Used by* Rem. 19.5, Rem. 19.6, Rem. 19.7, Req. 22.8 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.dark:composed_slack

Proof. Idea. A multiclick column mass is the column sum minus the entries in the $K+1$ rows with at most one click; the two tests bound each of these from both sides, which gives (i) and (ii). The composed operator is a combination of the positive operators $M_{S'}$ with the column masses as coefficients, and lowering every coefficient to its bound gives (iii).

(i) Let $S' \in \mathsf{MC}$. The rows $S \notin \mathsf{MC}$ are exactly those with $|S| \leq 1$, of which there are $K+1$, and none of them contains S' (as $|S| \leq 1 < 2 \leq |S'|$), so every entry $R_{S|S'}$ with $S \notin \mathsf{MC}$ sits in a (D2)-forbidden position and is at most ε_2. The (D1) test bounds the column sum below by $1-\varepsilon_4$. Hence

$$R_{\mathsf{MC}|S'} \;=\; \sum_{S} R_{S|S'} - \sum_{S\notin\mathsf{MC}} R_{S|S'} \;\geq\; (1-\varepsilon_4) - (K+1)\varepsilon_2,$$

which is $1-\delta_{\mathrm{dc}}$ by the declared case of (19.7).

(ii) The widened (D1) admits column sums up to $1+\varepsilon_4$ and entries down to $-\varepsilon_4$, so for every column

$$R_{\mathsf{MC}|S'} \;=\; \sum_{S} R_{S|S'} - \sum_{S\notin\mathsf{MC}} R_{S|S'} \;\le\; (1+\varepsilon_4) + (K+1)\varepsilon_4,$$

which is (19.10). The cost (19.11) is bounded by (19.10): $\bar{r} \le r_{\min}$ by (19.8), and $r_{\min} - \bar{r}$ is 0 when $r_{\min} \le 1$ and $r_{\min} - 1$ otherwise, where $r_{\min}$, a minimum of column masses, is at most $1 + (K+2)\varepsilon_4$.

(iii) $\widetilde{M}_{\mathsf{MC}} = \sum_{S'} R_{\mathsf{MC}|S'} M_{S'}$ with every $M_{S'} \succeq 0$, $\{M_S\}$ being a POVM (Theorem 4.5 (i), or Definition 8.1 (a) at a setting), so replacing each coefficient by a smaller real gives a Löwner-smaller operator. On the columns $S' \in \mathsf{MC}$ use $R_{\mathsf{MC}|S'} \ge 1 - \delta_{\mathrm{dc}}$, which is (19.9); on the columns $S' \notin \mathsf{MC}$ use $R_{\mathsf{MC}|S'} \ge r_{\min} \ge \bar{r}$, the first by the definition (19.2) of $r_{\min}$ as their minimum and the second by (19.8). Summing, with $\sum_{S'\in\mathsf{MC}} M_{S'} = M_{\mathsf{MC}}$ and $\sum_{S'\notin\mathsf{MC}} M_{S'} = \mathbb{1} - M_{\mathsf{MC}}$ by completeness, gives the first inequality of (19.12); regrouping and applying $0 \preceq M_{\mathsf{MC}} \preceq \mathbb{1}$ with $\delta_{\mathrm{dc}} \ge 0$ gives the second. Every operator here is block diagonal in the photon number (Theorem 4.5 (iii), or Definition 8.1 (a) at a setting), so the inequality restricts to the blocks $n > N$. There $1 - \bar{r} \ge 0$ by construction of (19.8), so $(1-\bar{r})M_{\mathsf{MC}}^{(>N)} \succeq (1-\bar{r})p_{\mathrm{mc}}\,\mathbb{1}^{(>N)}$, and adding $(\bar{r} - \delta_{\mathrm{dc}})\,\mathbb{1}^{(>N)}$ gives (19.13). No step uses a bound on $r_{\min}$ from either side, and no step uses the sign of $r_{\min}$ or of $\bar{r}$. □

Taking r_min over the single-click columns only gives an invalid bound whenever the no-click column has less multiclick mass, the generic case: at K = 4 and p = 0.00001 the two differ by four orders of magnitude. A lossy receiver keeps no-click weight in every high-photon block.

Remark 19.5 (The no-click column in the minimum defining $r_{\min}$). The minimum in (19.2) must run over *all* non-multiclick columns, the no-click column $S' = \varnothing$ included. Minimising only over the single-click columns gives an *invalid* bound whenever $R_{\mathsf{MC}|\varnothing} < R_{\mathsf{MC}|\{k\}}$, and that is the generic case rather than a corner one. With equal per-port probabilities $p_{\mathrm{dc}|k} = p$ and $K = 4$,

$$R_{\mathsf{MC}|\varnothing} = 1-(1-p)^4-4p(1-p)^3 = 6p^2+O(p^3), \qquad R_{\mathsf{MC}|\{k\}} = 1-(1-p)^3 = 3p+O(p^2), \tag{19.14}$$

so at a realistic $p = 10^{-5}$ the two differ by more than four orders of magnitude: $6\cdot10^{-10}$ against $3\cdot10^{-5}$. Using the larger number would claim a multiclick weight the device does not have.

The no-click column cannot be excluded because the no-click element does not vanish in the high-photon sectors of a lossy receiver: by Lemma 1.9, $\lambda_{\min}\big(M_{\varnothing}^{(=n)}\big) = \big(1-\lambda_{\max}(A_\Sigma)\big)^n > 0$ whenever $\lambda_{\max}(A_\Sigma) < 1$. So in exactly the regime where the high-photon bound matters, the column that the shortcut discards carries weight. The correct statement is Theorem 19.2 (iv) as written, and (19.13) of Proposition 19.4 for a declared R admitted under item (iii) of Definition 10.4 (Definition 19.3). Which columns the minimum runs over is a question about the index

set and is untouched by either widened test; the numbers above are those of the generated dark-count route, where both conditions are theorems.

Uses Lem. 1.9, Def. 10.4, Thm. 19.2, Prop. 19.4 · newly derived · accepted

Admitting a declared channel at ε_2 and ε_4 relaxes a hypothesis, not a precision; r_min above one is reachable, and ignoring it or δ_dc overstates the multiclick bound in the unsafe direction. The composed family is a POVM only up to ε_4, a defect left visible and not repaired.

Remark 19.6 (Cost of admitting a dark-count channel with slack). *(a) What is relaxed.* Definition 10.4 (iii) admits a declared R that violates (D2) by up to ε_2 in every forbidden position and (D1) by up to ε_4 in every column. Theorem 19.2 is stated and proved for an R satisfying (D1) and (D2) *exactly*, and admitting one that satisfies them only up to $(\varepsilon_2, \varepsilon_4)$ does not weaken the theorem; but the module then emits an object the theorem's hypothesis does not cover, and what that costs has to be stated at the inequality a downstream module actually uses, not only at the identity it is proved through. Proposition 19.4 states it. This is not a matter of the evaluation model of Definition 10.9: the datum is exactly as declared, and what is relaxed is a hypothesis, not the precision of an evaluation, so Theorem 10.10 (ii) does not reach it.

(b) The two values of δ_{dc}. On the generated dark-count route (D1) and (D2) are *theorems* (Theorem 19.2 (i)) and not tests, so there is no slack to carry and the correction in (19.7) is exactly 0; on the declared dark-count route they are tested, and Proposition 19.4 is what the test admits. In (19.9), against the exact value 1 of Theorem 19.2 (iii), *both* slacks appear: reading (19.9) off an exact column sum would drop the ε_4 and overstate the bound for every channel this contract admits.

(c) The upper side of the same tests, and where it bites. (19.2) defines $r_{\min}$ as a minimum of multiclick column masses, and for an R satisfying (D1) and (D2) exactly it satisfies $r_{\min} \le 1$ for free: entries are non-negative and columns sum to 1, so $R_{\mathsf{MC}|S'} \le \sum_S R_{S|S'} = 1$. The widened (D1) does not supply that, and $r_{\min} > 1$ is reachable within (19.10): a declared R placing mass $1+\varepsilon_4$ on the row $S = [K]$ in every column $S' \notin \mathsf{MC}$ and $1-\varepsilon_4$ on that same row in every column $S' \in \mathsf{MC}$ occupies no (D2)-forbidden position at all, is admitted at $(\varepsilon_2, \varepsilon_4) = (0, \varepsilon_4)$, and has $r_{\min} = 1 + \varepsilon_4$. Using the lower side of (D1) in (19.9) and passing over the upper side here would be exactly the one-sided reading that Remarks 10.7 and 10.8 guard against, and it would not be harmless: the coefficient $1 - r_{\min}$ that Theorem 19.2 (iv)'s substitution step requires to be non-negative would not be, and the composed bound would fail, in the unsafe direction, by $(r_{\min} - 1)(1 - p_{\mathrm{mc}})$.

(d) $\bar{r}$ is not a repair, and it costs a bounded amount. Nothing is clamped (Requirement 22.8): $\bar{r}$ is a separate named scalar appearing as a *coefficient in a stated inequality*. Lowering that coefficient *weakens* the bound, and needs nothing beyond $\mathbb{1} - M_{\mathsf{MC}} \succeq 0$; no claim is made that $\bar{r}$ approximates $r_{\min}$ or any exact quantity. That is the distinction Definition 10.4 (iv) draws in the other direction, where clamping a computed c before it enters (6.2) would assert a relation to an exact value that nothing supports. The cost (19.11) is first order in ε_4 and no larger.

(e) Relation to the exact composed inequality. On the generated route $\delta_{\mathrm{dc}} = 0$ and $r_{\min} \le 1$, so $\bar{r} = r_{\min}$ and (19.13) *is* (19.4). On the declared route a downstream

module that used (19.4) instead would overstate by

$$\big(r_{\min} - \bar{r}\big)\big(1 - p_{\mathrm{mc}}\big) + \delta_{\mathrm{dc}}, \tag{19.15}$$

first order in the receiver tolerances and in the *unsafe* direction: a (D2) violation is precisely weight moved out of MC, and $r_{\min} > 1$ is multiclick mass the column arithmetic does not have. And (19.13) improves on p_{mc}, which is the point of Theorem 19.2 (iv), only where $\bar{r}\,(1 - p_{\mathrm{mc}}) \geq \delta_{\mathrm{dc}}$; below that threshold the declared channel's admitted slack costs more than its dark counts gain, and the two emitted numbers say so.

(f) Theorem 19.2 (ii) is outside the widened hypotheses too, and is not repaired either. For an R admitted at $(\varepsilon_2, \varepsilon_4)$ the composed family $\{\widetilde{M}_S\}$ need not be a POVM. Writing $\sigma_{S'} \coloneqq \sum_S R_{S|S'}$, one has $\sum_S \widetilde{M}_S = \sum_{S'} \sigma_{S'} M_{S'}$ with $\sigma_{S'} \in [1 - \varepsilon_4, 1 + \varepsilon_4]$, so $(1 - \varepsilon_4)\mathbb{1} \preceq \sum_S \widetilde{M}_S \preceq (1 + \varepsilon_4)\mathbb{1}$ in place of $= \mathbb{1}$; and a single entry at $-\varepsilon_4$ gives only $\widetilde{M}_S \succeq -\varepsilon_4\,\mathbb{1}$ in place of $\succeq 0$. Both defects are bounded by ε_4, both are properties of the declared datum, and neither is repaired. They are not silent: every consumer decides the admission predicates it states of the data it reads exactly and refuses on a failure (Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly)), so a consumer that requires the composed family to be a POVM, or the channel to satisfy (D1) and (D2), refuses it; and on this side a positivity defect of an emitted block is what r_5 makes visible (Remark 10.3).

Uses Thm. 4.5, Def. 6.1, Rem. 10.3, Def. 10.4, Rem. 10.7, Prop. 19.4 · *Used by* Rem. 19.7 · newly derived · accepted

A declared dark-count channel need not have product form: if it satisfies (D1) and (D2) exactly, the dark-count theorem holds for it verbatim, and if it passes the widened tests it holds in the weakened forms of Proposition 19.4 and Remark 19.6. Correlations across rounds are excluded.

Remark 19.7 (Correlated dark counts within a round). Nothing in Theorem 19.2 uses the product form beyond part (i): a declared, experimentally certified R satisfying (D1) and (D2) *exactly* is admissible, and (ii)–(iv) hold for it verbatim. What must then be *checked* rather than derived is (D1) and (D2), and the artifact records which of the two routes the emitted R came from (Requirement 22.8). Where the check is the one this contract performs, item (iii) of Definition 10.4 (Definition 19.3) at ε_4 and ε_2, the declared R satisfies the two conditions only up to those tolerances, and then (ii), (iii) and (iv) hold only as Proposition 19.4 and Remark 19.6 restate them: the composed inequality is (19.13) in $\bar{r}$ and δ_{dc}, and the composed family is a POVM only up to ε_4. Correlations *across rounds* are a different matter and are excluded by Assumption 11.6: they would violate Assumption 11.10 downstream, and no channel on a single round's label can repair that.

Uses Def. 10.4, Prop. 19.4, Rem. 19.6 · *Assumes* A-THRESH · newly derived · accepted

20 The dark-composed family and the noise floor

The dark-composed family applies the dark-count channel to every setting's click patterns and then the coarse-graining (Definition 20.1). A dark-count

channel has rational entries and carries enclosures of the click-pattern blocks to enclosures of the dark-composed ones, which form the reference handed beside the snap (Corollary 20.2). The main point of the section is Proposition 20.3: loss and a dark-count probability in (0, 1) in every port make every dark-composed element positive definite on every photon-number block, with an explicit floor, and without either some element is singular. The closing remark says what a declaration should therefore model, why an ideal device measuring in a rotated basis dominates no lattice family beyond one photon at a small weight (for the two-photon block of the passive BB84 receiver at a balanced basis choice, 1/4), and why a run at a cutoff of two or more photons that uses no enclosed reference typically rests on its snap.

Imports foundations/def:binary64, foundations/def:evaluated-off-snap, foundations/def:exact-reference, foundations/def:sector-map, foundations/lem:rational-obstruction, foundations/rem:noise-floor

The dark-composed family applies a dark-count channel to the click patterns of each setting's POVM and then the declared coarse-graining. It is the measurement the device performs, and the family a snap is taken from.

Theorem 19.2 composes a dark-count channel with the click-pattern POVM of one receiver. The definition names the composed object for every setting at once, and after the coarse-graining, because that is the measurement a declaration describes when its device has dark counts. The families it is applied to are the unconditioned receiver (Corollary 8.2) and every setting family a submodule constructs.

Definition 20.1 (The dark-composed family)**.** Let $\mathcal{T}_{\mathrm{set}}$ be a finite non-empty set of settings and, for every $t \in \mathcal{T}_{\mathrm{set}}$, let $\{M_{S|t}\}_{S\subseteq[K]}$ be a POVM on a Fock space $\mathcal{F}(\mathbb{C}^I)$ (Definition 1.2) every element of which commutes with the photon number. Let R be a dark-count channel (Definition 19.1) and ϖ a declared coarse-graining onto $\mathcal{V}$ (Definition 5.5). The *dark-composed family* is

$$\widetilde{M}_{S|t} \coloneqq \sum_{S'\subseteq[K]} R_{S|S'}\, M_{S'|t}, \qquad \widetilde{F}_{v|t} \coloneqq \sum_{S\in\varpi^{-1}(v)} \widetilde{M}_{S|t} = \sum_{S'\subseteq[K]} R_{\varpi^{-1}(v)|S'}\, M_{S'|t}, \tag{20.1}$$

for $S \subseteq [K]$, $v \in \mathcal{V}$ and $t \in \mathcal{T}_{\mathrm{set}}$. Each $\{\widetilde{M}_{S|t}\}_S$ and each $\{\widetilde{F}_{v|t}\}_v$ is a POVM every element of which commutes with the photon number, with $\widetilde{F}^{(=n)}_{v|t} = \sum_{S'} R_{\varpi^{-1}(v)|S'} M^{(=n)}_{S'|t}$: the coefficients are non-negative scalars, they sum to one over S in every column by (D1), and the fibres of ϖ partition $2^{[K]}$.

Uses Def. 1.2, Def. 5.5, Def. 19.1 · *Used by* Cor. 20.2, Prop. 20.3, Rem. 20.4, Prop. 21.4, Req. 22.9, reductions/wpfss/cor:normalized-flat · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.generic:dark_composed_blocks

A dark-count channel has rational entries and mixes whole elements with non-negative weights, so enclosures of the click-pattern blocks give enclosures of the dark-composed blocks, exact where theirs are. With Hermitian centres they make the dark-composed blocks up to the cutoff a reference.

A dark count acts on the recorded click pattern, not on the light, so it mixes whole elements with scalar weights. An enclosure therefore passes through it with the same weights, and an exact one stays exact. Item (iii) applies the corollary to the unconditioned receiver; for a setting family a submodule constructs, the hypothesis of item (i) is the enclosure of its blocks that the submodule states. Where the click-pattern blocks are enclosed exactly up to one photon (Theorem 9.9), so are the dark-composed ones.

Corollary 20.2 (The dark-composed reference, enclosed or exact)**.** Let $\{\widetilde{M}_{S|t}\}$ and $\{\widetilde{F}_{v|t}\}$ be the dark-composed family of POVMs $\{M_{S|t}\}$ on $\mathcal{F}(\mathbb{C}^I)$ (Definition 20.1), with a dark-count channel R whose entries are rational.

(i) Suppose that for some $N \geq 0$ every block $M^{(=n)}_{S|t}$ with $n \leq N$, read in the occupation basis of $\mathrm{Sym}^n(\mathbb{C}^I)$ in an order fixed for that block, has a complex enclosure $\big(C^{(n)}_{S|t}, r^{(n)}_{S|t}\big)$ (Definition 9.1). Then

$$\Big(\sum_{S' \subseteq [K]} R_{\varpi^{-1}(v)|S'}\, C^{(n)}_{S'|t},\ \sum_{S' \subseteq [K]} R_{\varpi^{-1}(v)|S'}\, r^{(n)}_{S'|t} \Big) \tag{20.2}$$

is a complex enclosure of $\widetilde{F}^{(=n)}_{v|t}$, exact on every entry on which every $r^{(n)}_{S'|t}$ with $R_{\varpi^{-1}(v)|S'} \neq 0$ is zero; the same holds for $\widetilde{M}^{(=n)}_{S|t}$, with $R_{S|S'}$ in place of $R_{\varpi^{-1}(v)|S'}$.

(ii) Made Hermitian-centred by Lemma 9.2 (v), the enclosures of (i) make $\{\widetilde{F}_{v|t}\}_v$, restricted to the blocks $n \leq N$, a reference with constant 1 (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)) at every setting, against every sector map whose blocks are the photon numbers $n = 0, \ldots, N$ in increasing order, with any block kind and identifiers, each block read in its fixed order; it is an exact reference when every radius is zero. The same holds for $\{\widetilde{M}_{S|t}\}_S$.

(iii) For a receiver instrument with $\mathbb{I}_P = \varnothing$, the hypothesis of (i) holds with $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$, $M_{S|\mathrm{off}} = M_S$ and the enclosures of Theorem 9.5 (ii), at every N.

(iv) R has rational entries on the generated dark-count route, whose $p_{\mathrm{dc}|k}$ are declared values, and on the declared route, whose entries are declared values; a declared value is the dyadic rational it denotes (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)).

Uses Def. 9.1, Lem. 9.2, Thm. 9.5, Def. 19.1, Def. 20.1, foundations/def:binary64, foundations/def:exact-reference, foundations/def:sector-map · *Used by* Rem. 20.4, Req. 22.9, Req. 22.10 · newly derived · accepted

Proof. (i) By Definition 20.1, $\widetilde{F}^{(=n)}_{v|t} = \sum_{S'} R_{\varpi^{-1}(v)|S'} M^{(=n)}_{S'|t}$ with non-negative rational coefficients $R_{\varpi^{-1}(v)|S'}$, a sum of entries of R. Entry by entry,

$$\left| \left(\widetilde{F}^{(=n)}_{v|t}\right)_{il} - \sum_{S'} R_{\varpi^{-1}(v)|S'} \left(C^{(n)}_{S'|t}\right)_{il} \right| \le \sum_{S'} R_{\varpi^{-1}(v)|S'} \left(r^{(n)}_{S'|t}\right)_{il},$$

the centre has entries in $\mathbb{Q}(i)$, and the radius is a non-negative rational, zero exactly where every term with a non-zero coefficient has radius zero. The statement for $\widetilde{M}_{S|t}$ is the same computation with ϖ the identity.

(ii) $\{\widetilde{F}_{v|t}\}_v$ is a POVM every element of which commutes with the photon number (Definition 20.1), so each block is Hermitian and positive semidefinite and the blocks of one n sum to the identity. Lemma 9.2 (v) turns the enclosures of (i) into entrywise enclosures with Hermitian Gaussian-rational centres, exact where both an entry and its transposed entry were. The block of photon number n has the dimension $\binom{|I|+n-1}{n}$ of $\mathrm{Sym}^n(\mathbb{C}^I)$, and the constant 1 is a positive declared value. These are the conditions of Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference); the block kind and identifiers enter nowhere.

(iii) Theorem 9.5 (ii) gives complex enclosures of every $M^{(=n)}_S$, formed in each block's fixed order.

(iv) Each entry of (19.1) is a product of the numbers $p_{\mathrm{dc}|k}$ and $1 - p_{\mathrm{dc}|k}$, rational when the $p_{\mathrm{dc}|k}$ are, and a declared value denotes the dyadic rational it is. □

With loss and a dark-count probability in (0,1) in every port, every dark-composed element is positive definite on every photon-number block, with a least eigenvalue bounded below through the no-click element. Without loss, or without a dark count in some port, some element is singular.

The no-click outcome is the only one that the vacuum and a lossy device both reach on every photon-number block, and a dark count can turn it into any other outcome. That single column of the dark-count channel therefore spreads a positive multiple of the no-click element into every element. Items (i)–(iii) say when this makes every element positive definite; items (iv) and (v) show that neither ingredient can be dropped. Item (iii) applies at every setting whose operator satisfies $\lambda_{\max}(B^{(t)}) \le \lambda_{\max}(A_\Sigma)$; the unconditioned receiver has $B = A_\Sigma$ and $\gamma = 1$ (Corollary 8.2), and a submodule that constructs a setting family states whether its operators do.

Proposition 20.3 (Loss and dark counts make every dark-composed element definite)**.** Let $\{\widetilde{M}_{S|t}\}$ and $\{\widetilde{F}_{v|t}\}$ be the dark-composed family (Definition 20.1) of a setting family $\left(\mathcal{T}_{\mathrm{set}}, \{M_{S|t}\}\right)$ with operators $B^{(t)}$ and pref-

actors $\gamma^{(t)}$ (Definition 8.1), a dark-count channel R and a declared coarse-graining ϖ, and write $r_S \coloneqq R_{S|\varnothing}$.

(i) *The floor.* For every t, S, v and $n \geq 0$,

$$\widetilde{M}^{(=n)}_{S|t} \succeq r_S\, \gamma^{(t)}\big(1-\lambda_{\max}(B^{(t)})\big)^n\, \mathbb{1}, \qquad \widetilde{F}^{(=n)}_{v|t} \succeq R_{\varpi^{-1}(v)|\varnothing}\, \gamma^{(t)}\big(1-\lambda_{\max}(B^{(t)})\big)^n\, \mathbb{1}. \tag{20.3}$$

Hence, if $r_S > 0$ for every $S \subseteq [K]$ and $\lambda_{\max}(B^{(t)}) < 1$, every $\widetilde{M}_{S|t}$ and every $\widetilde{F}_{v|t}$ is positive definite on every photon-number block.

(ii) *Dark counts in product form.* On the generated dark-count route, $r_S = \prod_{k\in S} p_{\mathrm{dc}|k} \prod_{k\notin S}(1-p_{\mathrm{dc}|k})$, and $r_S > 0$ for every S if and only if $0 < p_{\mathrm{dc}|k} < 1$ for every k.

(iii) *Loss through the efficiencies.* If the instrument matrix of a receiver instrument is $D_\eta G'$, with G' a contraction and D_η the terminal detector-efficiency layer, then $\lambda_{\max}(A_\Sigma) \leq \max_k \eta_k$. For the unconditioned receiver, with $\mathbb{I}_P = \varnothing$, $B = A_\Sigma$ and $\gamma = 1$, so $\eta_k < 1$ and $0 < p_{\mathrm{dc}|k} < 1$ for every k give $\lambda_{\min}\big(\widetilde{M}^{(=n)}_S\big) \geq r_S\,(1-\max_k \eta_k)^n > 0$.

(iv) *Loss is necessary.* If $\lambda_{\max}(B^{(t)}) = 1$, then for every dark-count channel and every $n \geq 1$ the element $\widetilde{M}^{(=n)}_{\varnothing|t} = r_\varnothing\, M^{(=n)}_{\varnothing|t}$ is singular; if $B^{(t)} = \mathbb{1}$ it is zero.

(v) *Dark counts in every port are necessary.* For the unconditioned receiver on the generated route, let $Z \coloneqq \{k : p_{\mathrm{dc}|k} = 0\}$. Then $\widetilde{M}^{(=n)}_S = 0$ for every S with $|S \cap Z| > n$; in particular $\widetilde{M}^{((=0))}_{\{k\}} = 0$ for every $k \in Z$. If $p_{\mathrm{dc}|k} = 1$, then $\widetilde{M}_S = 0$ for every $S \not\ni k$.

(vi) *Equivalence.* For the unconditioned receiver on the generated route, every $\widetilde{M}_S$ is positive definite on every photon-number block if and only if $\lambda_{\max}(A_\Sigma) < 1$ and $0 < p_{\mathrm{dc}|k} < 1$ for every k.

Uses Def. 1.2, Lem. 1.6, Lem. 1.9, Def. 2.1, Def. 2.5, Thm. 4.1, Thm. 4.5, Def. 5.5, Thm. 6.2, Def. 8.1, Thm. 18.1, Def. 19.1, Def. 20.1 · *Used by* Rem. 20.4, Rem. 21.5, Rem. 21.6, Req. 22.9, Req. 22.10 · newly derived · accepted

Proof. (i) By (8.1), $\Pi_{\mathrm{vac}}([K])^{(t)} = M_{\varnothing|t}$, so (8.2) gives $M_{\varnothing|t} = \gamma^{(t)}\mathfrak{F}(\mathbb{1} - B^{(t)})$, and $0 \preceq B^{(t)} \preceq \mathbb{1}$ by (6.3). Lemma 1.9 gives $M^{(=n)}_{\varnothing|t} \succeq \gamma^{(t)}(1-\lambda_{\max}(B^{(t)}))^n\mathbb{1}$. Every term of $\widetilde{M}_{S|t} = \sum_{S'} R_{S|S'} M_{S'|t}$ is positive semidefinite, so dropping all but $S' = \varnothing$ gives $\widetilde{M}_{S|t} \succeq r_S M_{\varnothing|t}$, and the blocks inherit the inequality because every term is block diagonal. Summing over the fibre of v gives the second inequality, with $\sum_{S\in\varpi^{-1}(v)} r_S = R_{\varpi^{-1}(v)|\varnothing}$. If every $r_S > 0$ and $\lambda_{\max}(B^{(t)}) < 1$, the right-hand sides are positive multiples of $\mathbb{1}$:

$\gamma^{(t)} > 0$ by Definition 8.1, and every fibre is non-empty because ϖ is onto (Definition 5.5).

(ii) (19.1) at $S' = \varnothing$ is the stated product. It is positive for $S = [K]$ only if every $p_{\mathrm{dc}|k} > 0$, and for $S = \varnothing$ only if every $p_{\mathrm{dc}|k} < 1$; conversely, under both, every factor is positive.

(iii) By Theorem 18.1 (i), $A_k = \eta_k A'_k$ with A'_k the port Grams of G', and $\sum_k A'_k \preceq \mathbb{1}$ by Theorem 4.1 (i) applied to G'. Hence $A_\Sigma = \sum_k \eta_k A'_k \preceq (\max_k \eta_k) \sum_k A'_k \preceq (\max_k \eta_k)\mathbb{1}$, each A'_k being positive semidefinite. For $\mathbb{I}_P = \varnothing$, $M_\varnothing = \mathfrak{F}(\mathbb{1} - A_\Sigma)$ by (4.8) of Theorem 4.5, which is (8.2) with $\gamma = 1$ and $B = A_\Sigma$; the argument of (i), which uses only this form of $M_\varnothing$ and $0 \preceq A_\Sigma \preceq \mathbb{1}$ (Theorem 4.1 (i)), gives the bound, and (ii) makes r_S positive.

(iv) By (D2), $R_{\varnothing|S'} > 0$ forces $S' \subseteq \varnothing$, so $\widetilde{M}_{\varnothing|t} = r_\varnothing M_{\varnothing|t} = r_\varnothing \gamma^{(t)} \mathfrak{F}(\mathbb{1} - B^{(t)})$. By Lemma 1.9 the least eigenvalue of its n-photon block is $r_\varnothing \gamma^{(t)}(1 - \lambda_{\max}(B^{(t)}))^n = 0$ for $n \geq 1$. If $B^{(t)} = \mathbb{1}$, the block is $r_\varnothing \gamma^{(t)}(0^{\otimes n})|_{\mathrm{Sym}^n} = 0$ for $n \geq 1$.

(v) *Step 1 (no more clicks than photons).* For $\mathbb{I}_P = \varnothing$, $M^{(=n)}_{S'} = 0$ whenever $|S'| > n$. Indeed $M_{S'} = V^\dagger X V$ with $X = \prod_{k \in S'} (\mathbb{1} - \Pi^0_k) \prod_{k \notin S'} \Pi^0_k$ (Theorem 4.5, (4.5)). The measurement isometry $V = \mathfrak{F}(U)\mathfrak{F}(\iota)$ of Definition 2.5 maps $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$ into the n-photon sector of $\mathcal{F}(\mathbb{C}^{\mathbb{M} \sqcup \mathbb{E}})$, because both lifts preserve the photon number (Lemma 1.6 (i) and Definition 1.2). Each Π^0_k is the projector onto the span of the occupation vectors with no photon in $\mathbb{P}_k$ (Lemma 1.6 (iv)), so all of them are diagonal in the occupation basis and $\prod_{k \in S'} (\mathbb{1} - \Pi^0_k)$ is the projector onto the span of the occupation vectors with at least one photon in every port of S'. The ports are pairwise disjoint (Definition 2.1 (iii)), so such a vector carries at least $|S'|$ photons, and X vanishes on the n-photon sector when $|S'| > n$. Hence $M_{S'}$ vanishes on $\mathrm{Sym}^n(\mathbb{C}^{\mathbb{I}})$.

Step 2. By (19.1), $R_{S|S'} \neq 0$ requires $S' \subseteq S$ and $p_{\mathrm{dc}|k} \neq 0$ for every $k \in S \setminus S'$, that is, $S \cap Z \subseteq S'$. If $|S \cap Z| > n$, every S' with $R_{S|S'} \neq 0$ has $|S'| > n$, and Step 1 gives $\widetilde{M}^{(=n)}_S = \sum_{S'} R_{S|S'} M^{(=n)}_{S'} = 0$. For $k \in Z$ and $S = \{k\}$, $|S \cap Z| = 1 > 0$. If $p_{\mathrm{dc}|k} = 1$, every entry $R_{S|S'}$ with $k \notin S$ carries the factor $1 - p_{\mathrm{dc}|k} = 0$.

(vi) Sufficiency is (i) with (ii), at $\gamma = 1$ and $B = A_\Sigma$ as in (iii). Necessity: if $\lambda_{\max}(A_\Sigma) = 1$, (iv) gives a singular block of $\widetilde{M}_\varnothing$ at $n = 1$; if some $p_{\mathrm{dc}|k} = 0$, (v) gives $\widetilde{M}^{((=0))}_{\{k\}} = 0$; if some $p_{\mathrm{dc}|k} = 1$, (v) gives $\widetilde{M}_\varnothing = 0$. □

Declare the loss and dark counts the device surely has, in every port; then every element is positive definite and no snap revives a zero row. A run that uses no enclosed reference rests, at a cutoff of two or more, on its snap; an angle decides whether the reference can be exact.

The remark ends with reductions that apply a channel to the model before comparing it with a lattice family; they are planned beyond these contracts

(`spec/future-scope.md`).

Remark 20.4 (What a receiver declaration needs for a rate that rests on its model). The emitted blocks of a receiver are snapped onto a lattice before a key rate is computed, and the dark-composed reference (Corollary 20.2), enclosed for any declared parameters and exact where its radii are zero, is handed beside the snap. Where the run's first-order bound is evaluated on that reference and its key length is formed from a number on it, the blocks carrying it are evaluated off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)) and the key rate rests on the declared model, not on the snap. Where the key length is formed instead from a number certified on the snapped declaration (Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (i)), the moved ones among those blocks rest on the snapped-measurement premise, as every other moved block does (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)). This remark says what that means for a receiver declaration.

What to declare. Model the loss and the dark counts the device has, in every port: an efficiency $\eta_k < 1$ (or any loss that leaves $\lambda_{\max}(A_\Sigma) < 1$) and a dark-count probability $0 < p_{\mathrm{dc}|k} < 1$. By Proposition 20.3 every element of the dark-composed family is then positive definite on every photon-number block. At a setting t whose operator and prefactor in the setting family are $B^{(t)}$ and $\gamma^{(t)}$, the least eigenvalue on block n is at least $r_S \gamma^{(t)} \big(1 - \lambda_{\max}(B^{(t)})\big)^n$ for a click pattern S, and at least the sum of these over a fibre for an emitted outcome (Proposition 20.3 (i)). For the unconditioned receiver, $B = A_\Sigma$ and $\gamma = 1$, so the bound is $r_S \big(1 - \lambda_{\max}(A_\Sigma)\big)^n$, which is at least $r_S (1 - \max_k \eta_k)^n$ when the loss sits in the terminal efficiency layer (Proposition 20.3 (iii)); a setting whose operator has $\lambda_{\max}(B^{(t)}) \le \lambda_{\max}(A_\Sigma)$ and whose prefactor $\gamma^{(t)}$ is below one has at least $\gamma^{(t)}$ times that floor. No row of such a block is structurally zero, so even the last resort of Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) (iv) gives no element of it support that the data deny. The same floor is what makes domination by such a model possible at a small weight (Remark "What domination controls, and why it needs a noise floor" (foundations/rem:noise-floor)); no run relates a snap to its model that way. The rates to declare are the smallest ones the user is sure the device has. A larger rate gives a larger floor, but a model noisier than the device is a wrong model, which nothing in the engine can detect.

The enclosure. An enclosed reference is evaluated at its centres, which displaces the measured family by up to the enclosure's radius (Remark "Exact real arithmetic in the first-order bound" (meat-core/frank-wolfe/rem:fo-arithmetic)). The radii of the square-root enclosures and angle enclosures (Definition 9.1) from which the blocks are computed are amplified on the way to the radius of a block: through the permanents of the Fock blocks, the inclusion–exclusion sums over click patterns and the sums over a fibre of the coarse-graining. The radius of a block bears on a run in two places. The companion's entries are relaxed outward by it, so a larger radius loosens them, which never raises the bound formed on the companion (Lemma "A rounded constraint entry with its constant relaxed" (reductions/lem:row-relaxation)); and the evaluated family lies up to the radius from the model, a perturbation that no claim bounds in its effect on the key rate (Remark "Exact real arithmetic in the first-order bound" (meat-core/frank-wolfe/rem:fo-arithmetic)). No size of the radius is a threshold for either: the enclosures are chosen as tight as

their exact computation allows, and the record shows the largest radius of every handed block and of the companion's datum.

A run that uses no enclosed reference. A run may declare that enclosed references are not used (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i)). Then only the blocks whose reference is exact are handed, and every other moved block rests on the snapped-measurement premise, in the group no usable reference, unless the run declares the strict option (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)), which refuses. The receiver's reference is in general exact only on the exact route, and there with certainty only on the blocks of at most one photon (Theorem 9.9 (iii)): on a block of two or more photons, an entry between occupation vectors $\boldsymbol{m}, \boldsymbol{m}'$ with $\boldsymbol{m}!\,\boldsymbol{m}'!$ not a square is in general a Gaussian rational times an irrational factor such as $1/\sqrt{2}$, and is enclosed with a positive radius. In general, therefore, every run with a cutoff $N \geq 2$ that uses no enclosed reference rests on its snap there, and so does every such run whose declaration is off the exact route; its record and a report on it say so.

An ideal device beyond one photon. Without loss the no-click element vanishes on every block $n \geq 1$, and without dark counts every click pattern vanishes on the vacuum block (Proposition 20.3 (iv), (v)); so an ideal device has rank-deficient elements on every block, and a snap of it may need its last resort, which the record flags. A receiver that measures in a rotated basis can have some of them rank-deficient along irrational directions: on the one-photon block where a direction of the basis has no Gaussian-rational representative, and on blocks of more photons even where every direction has one. In an ideal passive BB84 receiver, the two-photon element in which only the detector of the diagonal polarization clicks, behind a half-wave plate at $\pi/8$, is a multiple of the projector onto $\frac{1}{2}\,|2,0\rangle + \frac{1}{\sqrt{2}}\,|1,1\rangle + \frac{1}{2}\,|0,2\rangle$ in the occupation basis of the channel's two polarization modes, and that ray contains no Gaussian-rational vector, although the one-photon directions of the diagonal basis are rational. By Lemma "Rationality obstruction to the domination of a lattice family" (foundations/lem:rational-obstruction), every weight at which the model dominates a lattice family declared in this basis is then at least the largest eigenvalue of that element, at every lattice exponent, which is $1/4$ for a balanced basis choice; that is why a declared device weight, and not a snap, is where domination could serve. At a cutoff $N \leq 1$ on the exact route the blocks are known exactly (Theorem 9.9 (iii)); where they are moreover dyadic rationals of bounded denominator, as at a balanced basis choice with dyadic efficiencies, they are lattice values and the snap keeps them. The obstruction belongs to domination, which compares the model with a lattice family element by element in the occupation basis. A reduction that first maps the m-photon block of the two polarization modes isometrically into the m-fold tensor power of $\mathbb{C}^2$ carries the ray above to a vector with rational coordinates (Remark "What domination controls, and why it needs a noise floor" (foundations/rem:noise-floor)); for an ideal device on these two modes whose port Grams are Gaussian-rational it would give a target with entries in $\mathbb{Q}(i)$ at every cutoff, at dimension 2^m instead of $m+1$ (with d input modes, d^m instead of $\binom{d+m-1}{m}$). The images of the elements sum to the projector onto the symmetric subspace, not to the identity, so such a target is complete only after an outcome is added on the complement of that subspace. No claim of these contracts makes that reduction.

Angles. A half-wave plate at axis angle θ has entries $\pm\cos 2\theta$ and $\pm\sin 2\theta$ ((16.4)

of Definition 16.2). Every angle is enclosed, so the angle decides nothing about whether a reference exists; it decides whether the reference can be exact. Behind an ideal polarizing beamsplitter the plate's port Gram on the two polarization modes has the diagonal entries $(1 \pm \cos 4\theta)/2$, up to a rational factor, and the exact route needs Gaussian-rational Grams (Definition 9.8), so it needs $\cos 4\theta$ rational. At multiples of $\pi/8$, $\cos 4\theta \in \{0, \pm 1\}$; at $\theta = \pi/24$, $\cos 4\theta = \sqrt{3}/2$ and the reference is enclosed, which the run hands and evaluates as it does an exact one. The exact route reads an angle given as a rational coefficient of π (Requirement 14.6 (ii)); an angle given in radians is a dyadic rational, and a non-zero one is no rational multiple of π, so a plate declared at the binary64 value nearest $\pi/8$ is enclosed, while one declared as $\frac{1}{8}\pi$ can be on the exact route.

Uses Def. 9.1, Def. 9.8, Thm. 9.9, Def. 20.1, Cor. 20.2, Prop. 20.3, foundations/def:evaluated-off-snap, foundations/lem:rational-obstruction, foundations/rem:noise-floor · newly derived · accepted

21 Efficiency normalization

A common factor of the port Grams is a uniform loss that commutes with every passive element, so it can be moved in front of the receiver and counted as part of the channel. The section defines the pure-loss channel and the normalized station, whose efficiencies are the receiver's divided by a factored loss t, and its main point is Proposition 21.4: every click pattern of the receiver, and every dark-count composition of them, is the pull-back of the normalized station's through the pure-loss channel of transmittance t. The remarks show that at low efficiency the normalization removes a squared-efficiency factor from the two-photon multiclick block, and say what it leaves in place, efficiency mismatch included.

The pure-loss channel of transmittance t sends every channel mode through a beamsplitter of transmittance t whose other input is vacuum and discards the reflected light. It is the uniform loss a normalized station hands to the channel.

Uniform loss in front of a receiver acts on the light the adversary sends, before any optical element. Writing it as a channel on the channel system lets Proposition 21.4 hand it to the adversary.

Definition 21.1 (Pure-loss channel on the channel modes)**.** Let $\mathbb{I}_C$ be the channel modes of a receiver instrument, so that the channel system is $\mathcal{H}(C) = \mathcal{F}(\mathbb{C}^{\mathbb{I}_C})$ (Definition 2.1), let $\mathbb{I}'_C$ be a disjoint copy of $\mathbb{I}_C$, with $m \mapsto m'$ the copying bijection, and let $t \in [0,1]$. Put

$$w_t \colon \mathbb{C}^{\mathbb{I}_C} \to \mathbb{C}^{\mathbb{I}_C \sqcup \mathbb{I}'_C}, \qquad w_t\,|m\rangle := \sqrt{t}\,|m\rangle + \sqrt{1-t}\,|m'\rangle\,, \tag{21.1}$$

and identify $\mathcal{F}(\mathbb{C}^{\mathbb{I}_C \sqcup \mathbb{I}'_C})$ with $\mathcal{F}(\mathbb{C}^{\mathbb{I}_C}) \otimes \mathcal{F}(\mathbb{C}^{\mathbb{I}'_C})$ by the occupation identification of Lemma 1.7 (ii), which is unitary. The *pure-loss channel* of transmittance t is

$$\Lambda_t^{\mathrm{loss}}(\rho) \;:=\; \mathrm{tr}_{\mathcal{F}(\mathbb{C}^{\mathbb{I}'_C})}\big[\mathfrak{F}(w_t)\,\rho\,\mathfrak{F}(w_t)^\dagger\big], \qquad \rho \in \mathcal{T}(C), \tag{21.2}$$

and its adjoint is

$$(\Lambda_t^{\mathrm{loss}})^\dagger(Y) \;\coloneqq\; \mathfrak{F}(w_t)^\dagger\big(Y \otimes \mathbb{1}_{\mathcal{F}(\mathbb{C}^{\mathbb{I}'_C})}\big)\mathfrak{F}(w_t), \qquad Y \in \mathcal{L}(C). \tag{21.3}$$

Since $w_t^\dagger w_t = t\,\mathbb{1} + (1-t)\,\mathbb{1} = \mathbb{1}$, the map $\mathfrak{F}(w_t)$ is an isometry (Lemma 1.7 (i)); so $\Lambda_t^{\mathrm{loss}}$ is completely positive and trace preserving, $(\Lambda_t^{\mathrm{loss}})^\dagger$ is completely positive and unital, and $\operatorname{tr}\big[Y\,\Lambda_t^{\mathrm{loss}}(\rho)\big] = \operatorname{tr}\big[(\Lambda_t^{\mathrm{loss}})^\dagger(Y)\,\rho\big]$ for every bounded Y and trace-class ρ, by the defining property of the partial trace and cyclicity of the trace. At $t = 1$ the channel is the identity.

Uses Def. 1.2, Lem. 1.7, Def. 2.1 · *Used by* Lem. 21.3, Prop. 21.4, Rem. 21.5, Req. 22.6, reductions/wpfss/cor:normalized-flat · newly derived · accepted

A normalized station at factor t divides every detector efficiency of a probe-free receiver by t, for t between the largest efficiency and one; t is the factored loss. Its port Grams are those of the receiver divided by t.

The efficiencies enter a receiver's port Grams as one scalar per port (Theorem 18.1 (i)), on each of the three input routes: as the terminal detector-efficiency layer of an instrument matrix, or as the scalars a declared Gram family is multiplied by. The definition divides those scalars by a common factor and keeps everything else.

Definition 21.2 (Normalized station and factored loss)**.** Let $\mathfrak{R}$ be a receiver instrument without declared source modes, $\mathbb{I}_P = \varnothing$, with K ports, whose port Grams are

$$A_k = \eta_k\,A_k^\circ, \qquad k = 1, \dots, K, \tag{21.4}$$

for an admissible Gram family $\{A_k^\circ\}$ on $\mathbb{C}^{\mathbb{I}_C}$ and efficiencies $\eta_k \in [0,1]$ with $\eta_{\max} \coloneqq \max_k \eta_k > 0$. This is the case for an instrument matrix $D_\eta G_0$, G_0 a contraction, with A_k° the port Grams of G_0 (Theorem 18.1 (i)), and for a declared Gram family $\{A_k^\circ\}$ with declared efficiencies. For $t \in [\eta_{\max}, 1]$, a *normalized station* of $\mathfrak{R}$ at the *factored loss* t is a receiver instrument $\mathfrak{R}'$ without declared source modes, with the channel modes and the K ports of $\mathfrak{R}$, whose port Grams are

$$A'_k = \frac{\eta_k}{t}\,A_k^\circ, \qquad \text{so that} \qquad A_k = t\,A'_k. \tag{21.5}$$

One exists: $\eta_k/t \in [0,1]$, so $0 \preceq A'_k \preceq A_k^\circ$ and $\sum_k A'_k \preceq \sum_k A_k^\circ \preceq \mathbb{1}$, and an admissible family is the port-Gram family of some receiver instrument (Theorem 5.1 (ii)); for the instrument matrix $D_\eta G_0$ one is $D_{\eta/t} G_0$, a product of two contractions (Proposition 1.4 (ii)), with $D_{\eta/t}$ the terminal layer of Definition 3.3 at the efficiencies η_k/t. Its click-pattern POVM depends on the port Grams alone (Theorem 5.1 (i)), so it is one POVM for every choice. The choice $t = \eta_{\max}$ is the *full* normalization, at which the largest efficiency becomes one, and $t = 1$ gives $\mathfrak{R}'$ the port Grams of $\mathfrak{R}$.

Uses Prop. 1.4, Def. 2.1, Def. 2.2, Def. 3.3, Thm. 5.1, Thm. 18.1 · *Used by* Prop. 21.4, Rem. 21.5, Rem. 21.6, Req. 22.4, Req. 22.5, Req. 22.6, reductions/wpfss/cor:normalized-flat · newly derived · implemented (effective: accepted) · *Code* certiqs_key.receiver.request:normalized_station

The adjoint of the pure-loss channel of transmittance t takes the lift of a contraction X to the lift of $tX + (1-t)\cdot 1$. Every joint no-click element is such a lift, so loss in front of a receiver acts on its Grams as a scalar.

The joint no-click elements of a receiver are lifts $\mathfrak{F}(\mathbb{1} - A_J)$ of contractions (Theorem 4.5 (ii)), so the following identity is what Proposition 21.4 needs from the pure-loss channel.

Lemma 21.3 (The pure-loss channel on the lift of a contraction). Let $t \in [0,1]$ and let X be an operator on $\mathbb{C}^{\mathbb{I}_C}$ with $X^\dagger X \preceq \mathbb{1}$. Then

$$(\Lambda_t^{\mathrm{loss}})^\dagger\big(\mathfrak{F}(X)\big) \;=\; \mathfrak{F}\big(t\,X + (1-t)\,\mathbb{1}\big). \tag{21.6}$$

Uses Lem. 1.7, Def. 21.1 · *Used by* Prop. 21.4 · newly derived · accepted

Proof. Step 1 (the adjoint). By Lemma 1.7 (iii) with $X_2 = \mathbb{1}$, under the identification of Definition 21.1, $\mathfrak{F}(X) \otimes \mathbb{1}_{\mathcal{F}(\mathbb{C}^{\mathbb{I}'_C})} = \mathfrak{F}(X \oplus \mathbb{1})$, an operator of norm at most one (Lemma 1.7 (i)). So (21.3) and Lemma 1.7 (i) give

$$(\Lambda_t^{\mathrm{loss}})^\dagger\big(\mathfrak{F}(X)\big) = \mathfrak{F}(w_t)^\dagger\,\mathfrak{F}(X \oplus \mathbb{1})\,\mathfrak{F}(w_t) = \mathfrak{F}\big(w_t^\dagger (X \oplus \mathbb{1}) w_t\big).$$

Step 2 (the compression). For $v \in \mathbb{C}^{\mathbb{I}_C}$ write v' for its copy in $\mathbb{C}^{\mathbb{I}'_C}$. Then $w_t v = \sqrt{t}\,v + \sqrt{1-t}\,v'$, $(X \oplus \mathbb{1}) w_t v = \sqrt{t}\,Xv + \sqrt{1-t}\,v'$, and $w_t^\dagger$ maps $a + b'$ to $\sqrt{t}\,a + \sqrt{1-t}\,b$, so $w_t^\dagger (X \oplus \mathbb{1}) w_t = t\,X + (1-t)\,\mathbb{1}$. □

Every click pattern of a probe-free receiver is the pull-back, through the pure-loss channel of the factored loss t, of the same pattern of its normalized station, and so is every dark-count composition. The statistics agree, and the loss can be handed to the adversary.

The idea is that a scalar factor common to all port Grams is a loss that commutes with every passive element, so it can be moved to the front of the receiver, where it acts on the channel modes alone.

Proposition 21.4 (A receiver as uniform loss followed by its normalized station)**.** Let $\mathfrak{R}$ be a receiver instrument without declared source modes, with efficiencies η_k, $\eta_{\max} > 0$, as in Definition 21.2, let $t \in [\eta_{\max}, 1]$ and let $\mathfrak{R}'$ be a normalized station of $\mathfrak{R}$ at the factored loss t. Let $\{M_S\}$ and $\{M'_S\}$, $S \subseteq [K]$, be their click-pattern POVMs on $\mathcal{H}(C) = \mathcal{F}(\mathbb{C}^{\mathbb{I}_C})$ (Theorem 4.5).

(i) *The click patterns.* For every $S \subseteq [K]$,

$$M_S \;=\; (\Lambda_t^{\mathrm{loss}})^\dagger\big(M'_S\big), \qquad \text{equivalently} \qquad \mathrm{tr}[M_S\,\rho] = \mathrm{tr}\left[M'_S\,\Lambda_t^{\mathrm{loss}}(\rho)\right] \text{ for every } \rho \in \mathcal{T}(C). \tag{21.7}$$

(ii) *Classical post-processing, dark counts included.* For every column-stochastic matrix R on $2^{[K]}$, in particular every dark-count channel, and every declared coarse-graining ϖ onto $\mathcal{V}$,

$$\sum_{S'} R_{S|S'} M_{S'} = (\Lambda_t^{\mathrm{loss}})^\dagger\Big(\sum_{S'} R_{S|S'} M'_{S'}\Big), \qquad \sum_{S'} R_{\varpi^{-1}(v)|S'} M_{S'} = (\Lambda_t^{\mathrm{loss}})^\dagger\Big(\sum_{S'} R_{\varpi^{-1}(v)|S'} M'_{S'}\Big) \tag{21.8}$$

so the dark-composed family of $\mathfrak{R}$ with R and ϖ (Definition 20.1) is the pull-back of that of $\mathfrak{R}'$ with the same R and ϖ: the dark counts act after the factored loss.

(iii) *The device.* Under the receiver's device assumptions, Assumptions 11.1, 11.3, 11.4, 11.5, 11.6, 11.7, 11.8, 11.9 and 11.10, $\{M_S\}$ is the measurement the physical receiver makes on the light it receives in its channel modes, and, by Assumptions 11.5 and 11.7, every port responds alike to every mode through which light reaches it; the physical receiver, followed by any dark-count channel, therefore acts on that light exactly as $\Lambda_t^{\mathrm{loss}}$ followed by $\mathfrak{R}'$ and the same dark-count channel.

Reading: If your receiver has no probe source, the largest of its detector efficiencies can be treated as uniform loss in front of the whole receiver, followed by a receiver whose best detector is perfect and whose other detectors keep their efficiencies relative to it. The dark counts stay at the detectors. Both descriptions give the same click probabilities for every input, so the loss in front can be counted as part of the channel the adversary controls. A detector efficiency that depends on time slot, wavelength or spatial mode is covered if declared as mode-dependent loss in front of the detector, which stays in place; a dependence on a degree of freedom the adversary can use but your declaration does not carry is efficiency mismatch, which this does not address.

Uses Def. 2.2, Def. 2.5, Thm. 4.1, Thm. 4.5, Def. 5.5, Def. 19.1, Def. 20.1, Def. 21.1, Def. 21.2, Lem. 21.3 · *Assumes* A-TRUST, A-PASSIVE, A-VAC, A-MODES, A-THRESH, A-PORT, A-EXACT, A-NOSEL, A-ROUND · *Used by* Rem. 21.5, Rem. 21.6, Req. 22.4, Req. 22.5, Req. 22.6, reductions/wpfss/cor:normalized-flat, reductions/wpfss/rem:receiver-imports · newly derived · accepted

Proof. (i) By (21.5), $A_k = t\,A'_k$ for every k, so $A_J = t\,A'_J$ for every $J \subseteq [K]$, in the notation of Definition 2.2. The family $\{A'_k\}$ is the port-Gram family of a receiver instrument, hence admissible (Theorem 4.1 (i)), so $0 \preceq A'_J \preceq \mathbb{1}$ and $X_J := \mathbb{1} - A'_J$ satisfies $0 \preceq X_J \preceq \mathbb{1}$, in particular $X_J^\dagger X_J \preceq \mathbb{1}$. By Theorem 4.5 (ii) and Lemma 21.3,

$$\Pi_{\mathrm{vac}}(J) = \mathfrak{F}(\mathbb{1}-A_J) = \mathfrak{F}\big(t\,X_J+(1-t)\,\mathbb{1}\big) = (\Lambda_t^{\mathrm{loss}})^\dagger\big(\mathfrak{F}(X_J)\big) = (\Lambda_t^{\mathrm{loss}})^\dagger\big(\Pi_{\mathrm{vac}}(J)'\big),$$

with $\Pi_{\mathrm{vac}}(J)' = \mathfrak{F}(\mathbb{1} - A'_J)$ the joint no-click element of $\mathfrak{R}'$. Each M_S is the same signed sum (4.6) of the elements $\Pi_{\mathrm{vac}}(([K] \setminus S) \cup T)$ as M'_S is of the elements $\Pi_{\mathrm{vac}}(([K] \setminus S) \cup T)'$, and $(\Lambda_t^{\mathrm{loss}})^\dagger$ is linear, which gives the first identity of (21.7). The second is the first paired with ρ, by the pairing of Definition 21.1.

(ii) Both sides of each identity are finite linear combinations with the same scalar coefficients, of $\{M_{S'}\}$ on the left and of $\{M'_{S'}\}$ inside the adjoint on the right, so (i) and linearity give them; the fibre sums are the coefficients $R_{\varpi^{-1}(v)|S'} = \sum_{S\in\varpi^{-1}(v)} R_{S|S'}$ of Definition 20.1. Neither column-stochasticity nor any other property of R (Definition 19.1) is used.

(iii) These are the device assumptions under which the click-pattern POVM of a probe-free receiver is the physical receiver's measurement, the set Corollary 8.2 reads its setting family under: Assumption 11.1 (i) makes the declared instrument, efficiencies included, the physical one; Assumption 11.3 makes its optics a passive linear map; Assumption 11.4 puts vacuum on every undeclared input, the probe-free receiver having no other declared input than its channel modes; Assumptions 11.6 and 11.7 make each detector a threshold detector on its port's total photon number; Assumption 11.8 makes the declared parameters exact; and Assumptions 11.9 and 11.10 fix the device independently of the adversary and identically in every round. Under them the measurement on the channel modes is the click-pattern POVM of $\mathfrak{R}$ of Definition 2.5 and Theorem 4.5. Assumption 11.5 (i) says that every optical path from outside to a detector is a declared mode, and Assumption 11.7 (i) that a port clicks exactly when its modes carry a photon, whichever mode carries it; so no mode outside the declaration reaches a detector with an efficiency of its own, and each port's efficiency is the scalar η_k of (21.4). The rest is (i) and (ii). □

At a common efficiency η and no dark counts, the two-photon multiclick block of a receiver is η^2 times that of its normalized station, so every multiclick weight it can emit at a cutoff of one photon is at most η^2. Normalization removes that factor without enlarging the adversary's reach.

Remark 21.5 (The two-photon multiclick block scales with the squared efficiency). Let every port of a receiver without declared source modes have the same efficiency $\eta \in (0,1]$, and let $\mathfrak{R}'$ be its normalized station at the full normalization $t = \eta$, whose efficiencies are all one. Write M_{mc} and M'_{mc} for the multiclick elements (4.10) of the receiver and of $\mathfrak{R}'$. Without dark counts, M'_{mc} vanishes on the photon-number blocks $n = 0$ and $n = 1$, a pattern of two or more clicks needing two photons (Proposition 20.3 (v) with every port dark-free). For $\psi \in \mathrm{Sym}^2(\mathbb{C}^{\mathbb{I}_C})$ the component of $\mathfrak{F}(w_t)\psi = w_t^{\otimes 2}\psi$ with no photon in the copy $\mathbb{I}'_C$ (Definition 21.1) is $t\,\psi \otimes |\mathrm{vac}\rangle$, and its other components carry one or two photons in the copy and so at most one in $\mathbb{I}_C$. The operator $M'_{\mathrm{mc}} \otimes \mathbb{1}$ acts as the identity on the copy, so it does not couple components with different photon numbers there, and it vanishes on the components with at most one photon in $\mathbb{I}_C$. By Proposition 21.4 (i) and (21.3), therefore, $\langle\psi|\, M_{\mathrm{mc}}\,|\psi\rangle = t^2\,\langle\psi|\, M'_{\mathrm{mc}}\,|\psi\rangle$ for every such ψ, and since both blocks are Hermitian,

$$M_{\mathrm{mc}}^{((=2))} \;=\; \eta^2\, M_{\mathrm{mc}}'^{((=2))}. \tag{21.9}$$

A multiclick weight p_{mc} emitted for the receiver at the cutoff $N = 1$ satisfies $M_{\mathrm{mc}}^{((=2))} \succeq p_{\mathrm{mc}}\,\mathbb{1}$, the block $n = 2$ lying above the cutoff ((8.15) of Corollary 8.15), so $p_{\mathrm{mc}} \le \eta^2 \lambda_{\min}\big(M_{\mathrm{mc}}'^{((=2))}\big) \le \eta^2$, whatever window or enclosure produced it; the normalized station's own weight is computed at unit efficiency and carries no such

factor. The normalization does not buy this by enlarging what the adversary can do: by Proposition 21.4 (i), every attack on the receiver followed by $\Lambda_t^{\mathrm{loss}}$ is an attack on $\mathfrak{R}'$ with the same statistics, so an exact analysis of the receiver over all attacks gives a rate at least that of $\mathfrak{R}'$, and what the normalization improves is the multiclick weight a reduction to finitely many photons reads.

Uses Thm. 4.5, Cor. 8.15, Prop. 20.3, Def. 21.1, Def. 21.2, Prop. 21.4 · *Used by* Req. 22.4 · newly derived · accepted

Only the per-port efficiencies are divided: declared mode-dependent loss stays, undeclared efficiency mismatch is outside the assumptions either way, and a probe-fed receiver is not covered. The full normalization can remove the loss the noise floor needs, which a larger factor keeps.

Remark 21.6 (What the normalization factors out and what it leaves in place). *Only the terminal scalars are divided.* The family $\{A_k^\circ\}$ of Definition 21.2 is kept: every loss declared in the element graph, a polarizer's extinction, and an attenuation that varies within a port (Theorem 18.1 (ii)) stay in the normalized station unchanged. A detector whose efficiency depends on a spatial, spectral or temporal mode is declared as such a mode-dependent loss in front of its port, and the normalization leaves it where it is.

Efficiency mismatch is a separate problem. The identity of Proposition 21.4 (i) holds of the declared instrument. That it describes the physical receiver, clause (iii), rests on Assumptions 11.5 and 11.7: an efficiency that differs between degrees of freedom the adversary can address but the declaration does not carry violates one of them, with or without the normalization, which neither creates such a mismatch nor removes one.

A probe-fed receiver is not covered. With declared source modes the probe light passes the same detectors, so a uniform loss moved to the front would act on the probe modes too and replace each probe state by its image under the loss; no claim here states that form.

The choice of the factor. The identity holds at every $t \in [\eta_{\max}, 1]$. At the full normalization $t = \eta_{\max}$ with lossless optics the normalized station can have $\lambda_{\max}(A'_\Sigma) = 1$, and then some element of its dark-composed family is singular on every block with a photon, dark counts or not (Proposition 20.3 (iv)). At $t > \eta_{\max}$ the largest efficiency of the station is $\eta_{\max}/t < 1$, so $\lambda_{\max}(A'_\Sigma) \le \eta_{\max}/t$ and, with a dark count in every port, every such element is positive definite with the floor of Proposition 20.3 (iii) at $\eta_{\max}/t$. Where $\eta_{\max} < 99/100$, the default factor $t = \eta_{\max}/(99/100)$ of Requirement 22.4 is of this kind, with $\lambda_{\max}(A'_\Sigma) \le 99/100$; where $\eta_{\max} \ge 99/100$ the default is $t = 1$, the declared receiver, whose largest efficiency can itself be one. Which factor in $[\eta_{\max}, 1]$ a run is better served by is a matter of the engine's evaluation, not of the identity.

Only declared efficiencies are factored. The proof of Proposition 21.4 (i) uses only $A_k = t\,A'_k$ with $\{A'_k\}$ admissible, which holds for every $t \in [\lambda_{\max}(A_\Sigma), 1]$ with $t > 0$ and $A'_k = A_k/t$; such a factor can be smaller than $\eta_{\max}$, but it is an eigenvalue rather than a declared number. A Gram family declared from click data, with every efficiency declared as one (Remark 18.4), has $\eta_{\max} = 1$, and nothing is factored from it.

Uses Thm. 18.1, Prop. 20.3, Def. 21.2, Prop. 21.4 · *Used by* Req. 22.4 · newly derived · accepted

22 Requirements

The engine keeps detector efficiency and component loss in separate declared fields, and it records every declared transmittance and the efficiency convention of the declared numbers. It compiles a receiver without a probe at its normalized station, by default at the largest declared efficiency divided by 99/100, just short of the full normalization, or at one where that efficiency is at least 99/100, records the factored loss, and has the loss applied to every input on which the emitted measurement stands for the receiver's statistics. It refuses a request that declares a dark-count matrix while some per-port dark-count probability is non-zero, and it emits the dark-count channel with the scalars of the composed bound. The emitted blocks are the dark-composed ones and carry their reference, recorded block by block as exact, enclosed or absent; the snap itself runs only on the exact ones. A moved block of their snap that rests on the snap premise is recorded with its cause, the group of the cause and the fix for it; a block snapped by the last resort is flagged; only a run that declares the strict option refuses.

Imports foundations/cor:rank-over-factor-enclosure, foundations/def:evaluated-off-snap, foundations/def:exact-island, foundations/def:exact-reference, foundations/def:factor-enclosure, foundations/def:sector-map, foundations/def:snap, foundations/def:tiers, foundations/req:snap-reference, foundations/req:support-safe-snap, foundations/req:who-snaps

The input schema keeps a detector efficiency and a C7 element in different fields, so a C7 element is never read as a detector efficiency. The two enter the compiled matrix at different sites (Theorem 18.1 (iii)).

A detector efficiency enters the compiled matrix only through the terminal factor D_η of Definition 3.3 (Theorem 18.1 (iii)); a C7 element is a component of the graph, applied at its own position in the chain.

Requirement 22.1 (Detector efficiency and component loss in separate fields). A C7 element is not a detector efficiency, and the engine's input schema keeps the two in different fields.

Motivated by Def. 3.3, Thm. 18.1 · implemented (effective: accepted) · *Code* certiqs_key.receiver.request:DetectorPort

The emitted efficiency record lists every declared loss parameter under its name, with the element that declares it and its position in the compiled product. Two elements with equal values are two losses and two rows.

Theorem 18.1 (iii) gives the detector efficiency one site in the compiled matrix, and item (iv) of that theorem states the record this requirement obliges the engine to emit: it lists every loss site, the detector efficiency included.

Requirement 22.2 (The efficiency record lists every declared loss parameter). The emitted `efficiency_record` lists every declared loss parameter q_i that Theorem 18.1 (iv) names, by catalogue entry and parameter, with

the element that declares it and its position in the compiled product of Definition 3.3. Every row names its parameter beside its value, which is read as that parameter: a transmittance, a reflectivity, or, for y, y_1 and y_2, a loss whose transmittance is $1 - q_i$. Two elements may carry equal numerical values; they are still two losses and appear as two rows. A polarizer's pass transmittance t is one such row, under its element; the factored loss t that the same record carries by Requirement 22.5 is not a row and is recorded under its own name.

Motivated by Def. 3.3, Thm. 18.1, Req. 22.5 · implemented (effective: accepted) · *Code* certiqs_key.receiver.efficiency:efficiency_record

Whether the declared Grams come from power data or from click data is a declared field, written down in the efficiency record. The numbers alone cannot say which, and the two readings differ by exactly the efficiencies.

Power data return the port Grams A_k and click data return $\eta_k A_k$ (Remark 18.4); the efficiencies must enter through (18.1) exactly once.

Requirement 22.3 (Declared efficiency convention of the port Grams). Which of the two conventions a declaration means, Grams from power data rescaled by η_k or Grams from click data with every η_k declared as 1, is a declared field, and the `efficiency_record` (Requirement 22.2) is where the answer is written down.

Motivated by Thm. 18.1, Rem. 18.4, Req. 22.2 · accepted

By default the engine compiles a probe-free receiver at the factor $t = \eta_max/(99/100)$, or $t = 1$ where $\eta_max \geq 99/100$, and emits that station's blocks, weights and dark-composed family. A declared factor in $[\eta_max, 1]$ replaces it, and one outside is refused.

A receiver's measurement is uniform loss followed by its normalized station (Proposition 21.4), and the weight a consumer reads loses a factor of the squared efficiency without it (Remark 21.5); the factor itself is a choice, of which Remark 21.6 gives the trade-off. The reason the default stops short of the full normalization where the largest efficiency is below 99/100 is the engine's present handling of a singular block, not a property of the identity, and is stated in `spec/modules/receiver/design.md`.

Requirement 22.4 (Compilation of a probe-free receiver at its normalized station). For a request whose receiver declares no source modes and some efficiency $\eta_k > 0$, the engine compiles, in place of the declared receiver, its normalized station at the factored loss t, with the efficiency of port k equal to η_k/t, and emits that station's blocks, multiclick weights and dark-composed family; t is the value of `efficiency_normalization` where the request declares one; otherwise, with $\eta_{\max}$ the largest declared efficiency, $t = \eta_{\max}/(99/100)$, formed exactly, where $\eta_{\max} < 99/100$, and $t = 1$ where $\eta_{\max} \geq 99/100$. On every route the station compiled is the one

at the exact rational efficiencies η_k/t: its exact statements, its reference and its enclosures are formed from those rationals, its floating-point evaluation evaluates them as the evaluation model evaluates every declared number, and no value rounded from η_k/t is declared in their place; the quotient is exactly one at a port with $\eta_k = t$. A declared t outside $[\eta_{\max}, 1]$ is refused with `efficiency_normalization_out_of_range`. A request that declares source modes, or whose efficiencies are all zero, is compiled at $t = 1$, and a declared $t < 1$ in such a request is refused with the same reason.

Reading: Unless you declare otherwise, a receiver without a probe source is analysed with nearly all of its largest detector efficiency moved into the channel as loss: every efficiency is divided by t, the largest one divided by 99/100, so the best detector is analysed at 99/100; at a largest efficiency of 99/100 or more none is divided. This leaves your observed statistics unchanged and typically raises the certified key rate at low efficiencies. You may declare the factor, between the largest efficiency and one; declaring one turns the normalization off. Declaring the largest efficiency itself is allowed and sound, but with dark counts at a photon cutoff of two or more the engine can then report no key where your model gives one, which is why, below 99/100, the default stops short of it; a receiver whose largest efficiency is one or close to it can meet the same limitation.

Motivated by Def. 21.2, Prop. 21.4, Rem. 21.5, Rem. 21.6 · implemented (effective: accepted) · *Code* certiqs_key.receiver.efficiency:normalize_efficiencies

The efficiency record carries the factored loss and whether it was declared, every declared efficiency beside the one compiled, and whether the normalization was applied, exactly where $t < 1$, and why not. Where $t < 1$ the artefact says that its measurement is the normalized station's.

A consumer of the artefact reads a measurement that is the receiver's only after the factored loss is put in front of it (Proposition 21.4), so the factor travels with the blocks. Requirement 22.4 fixes the factor, and Requirement 22.2 the rows of the declared transmittances. For a probe-free receiver at $t = 1$ the normalized station is the declared receiver, however the factor was reached, so whether the normalization was applied is read off t alone, and who chose t is recorded beside it.

Requirement 22.5 (The efficiency record names the normalization). The emitted `efficiency_record` carries the factored loss t, whether t was declared in the request or set by Requirement 22.4, every declared efficiency η_k beside the exact rational η_k/t at which the receiver was compiled, and whether the normalization was applied, which it was exactly where $t < 1$. Where it was not, it carries the reason, the first that holds of: source modes declared, every efficiency zero, or the factor one, declared or by default. Where $t < 1$ the artefact states that its blocks, multiclick weights and dark-composed family are those of the normalized station at t, whose pull-back through the pure-loss channel of transmittance t is the receiver's measurement.

Motivated by Def. 21.2, Prop. 21.4, Req. 22.2, Req. 22.4 · implemented (effective: accepted) · *Code* certiqs_key.receiver.efficiency:EfficiencyNormalization

Where the receiver was compiled at a factored loss t below one, every evaluation of its emitted measurement on a state of the channel system that stands for the receiver's statistics, the honest statistics among them, is made on the state after the pure-loss channel; one that cannot is refused.

The emitted measurement of a normalized receiver is not the receiver's: the receiver's click statistics on an input ρ are those of the emitted measurement on $\Lambda_t^{\mathrm{loss}}(\rho)$ (Proposition 21.4 (i), second form, and (ii)). A consumer that evaluated the emitted measurement on ρ itself would form the statistics of a more efficient receiver. The factor travels in the record of Requirement 22.5. The states the key-rate program optimises over are another matter: the loss is handed to the adversary (Proposition 21.4 (iii)), so it is already part of the channel those states range over, and applying it to them again would be sound but would give up the gain the normalization exists for.

Requirement 22.6 (The factored loss is applied before the artefact is evaluated). Where a receiver artefact's `efficiency_record` carries a factored loss $t < 1$, every evaluation of its emitted blocks or dark-composed family, or of a target formed from them, on a state that stands for the receiver's statistics on an input ρ of the channel system (an honest-statistics provider forming the honest law among them) is made on $\Lambda_t^{\mathrm{loss}}(\rho)$, or on the state the same construction forms from $\Lambda_t^{\mathrm{loss}}(\rho)$, in place of ρ, with the t of that receiver's record. Where the input is a joint state of which that receiver's channel system is one factor, such as a state shared with another party or with a second normalized receiver, $\Lambda_t^{\mathrm{loss}}$ stands here for $\mathrm{id} \otimes \Lambda_t^{\mathrm{loss}}$ acting on that receiver's channel-system factor, each receiver with the t of its own record. The states the key-rate program optimises over are not inputs in this sense, and the loss is not applied to them. A consumer that cannot apply $\Lambda_t^{\mathrm{loss}}$ to the input it is handed refuses with `factored_loss_not_applied`.

Reading: When the receiver is analysed at its normalized station, the expected (honest) statistics the analysis is built around are those of your actual receiver: the efficiency moved into the channel is applied to the expected input before the normalized measurement is evaluated. Without this the acceptance test would be set for a receiver more efficient than yours, and the reported key would be one your device does not reach.

Motivated by Def. 21.1, Def. 21.2, Prop. 21.4, Req. 22.5 · implemented (effective: accepted) · *Code* certiqs_key.runs.pipeline:LossAppliedProvider

A request that declares a dark-count matrix R while some per-port dark-count probability is non-zero is refused. A declared R therefore comes with every per-port probability zero, and exactly one route is taken.

A request carrying both would supply the channel twice with no rule that makes the two agree, and the record of which route the emitted channel came from (Remark 19.7, Requirement 22.8) presupposes that exactly one was taken.

Requirement 22.7 (Refusal of a dark-count channel supplied on both routes). The two routes of Definition 19.1 are exclusive: a request declaring

R while some $p_{\mathrm{dc}|k} \neq 0$ is refused with `dark_count_channel_declared_twice`.

Motivated by Def. 19.1, Rem. 19.7, Req. 22.8 · implemented (effective: accepted) · *Code* certiqs_key.receiver.dark:check_dark_count_source

The artifact carries the dark-count matrix as declared or generated, its route, and r_min, rbar and δ_dc under distinct names, none modified. The inequality a downstream module composes with is that of Proposition 19.4.

Both $r_{\min}$ and $\bar{r}$ are emitted: $\bar{r}$ is the number the composition uses, while $r_{\min}$ as computed is what (19.2) defines and is the only one from which a reader can see that a declared channel produced $r_{\min} > 1$ at all, itself a signal that the declaration is off by more than rounding (Remark 19.6 (c)).

Requirement 22.8 (The emitted dark-count channel and the scalars of its composed bound). The artifact's `dark_count_channel` entry carries R, which of the two routes of Definition 19.1 it came from (Remark 19.7), and $r_{\min}$, $\bar{r}$ and δ_{dc} under distinct names. Nothing is clamped: R is emitted as declared and $r_{\min}$ as (19.2) computes it. (19.13), *not* (19.4), is the inequality a downstream module composes with.

Motivated by Def. 19.1, Thm. 19.2, Prop. 19.4, Rem. 19.6, Rem. 19.7 · implemented (effective: accepted) · *Code* certiqs_key.receiver.artifact:DarkCountRecord

The emitted low-photon blocks are the dark-composed family's, the dark-count channel still emitted. For a model admissible exactly they carry sound enclosures, the handed reference, and a factor enclosure where a port dilation admits one; only exact ones where enclosed references are not used.

The run evaluates its first-order bound on the reference handed beside the snap, and the noise floor of the device (Proposition 20.3) belongs to the model the bound is evaluated on, not to a step after it. The reference offered is therefore the dark-composed family, and the blocks the receiver emits are its blocks. For a setting family a submodule constructs, the enclosures at each setting are those that submodule states. The enclosures cannot recover every zero of the model: an entry that vanishes through cancellation, in an inclusion–exclusion sum or in the composition with the dark-count channel, is enclosed with a positive radius wherever its enclosure is not exact, and a snap then does not treat it as known to be zero. That happens off the exact route, and on it as well, at the entries of a block $n \geq 2$ between occupation vectors $\boldsymbol{m}, \boldsymbol{m}'$ with $\boldsymbol{m}!\,\boldsymbol{m}'!$ not a square, which Theorem 9.9 does not make exact. Item (ii) lists the zeros that are kept exact. An enclosure with positive radii does not fix the rank of a block whose rank is below its dimension and whose support no coordinate vectors span, such as a block of an ideal analyser at an angle whose projector has irrational entries; item (vii) offers such a block besides in factor form, read through a port dilation (Theorem 9.7), wherever its columns are few enough to fix its rank. The instance hands the reference beside its snap, and the run evaluates

its first-order bound on it, so that item (vi) asks for it in every run that uses enclosed references. The emitted blocks remain a decomposition of the physical measurement and are not a squashing argument (Remark 4.9).

REQUIREMENT 22.9 (The emitted blocks are dark-composed and carry their reference). (i) The `photon_blocks` entry carries, for every probe setting t and every $n \le N$, the blocks $\widetilde{F}^{(=n)}_{v|t} = \sum_{S' \subseteq [K]} R_{\varpi^{-1}(v)|S'}\, M^{(=n)}_{S'|t}$ given by the formula (20.1) for the emitted matrix R, in floating point. They are the blocks of the dark-composed family (Definition 20.1) when R is a dark-count channel, satisfying (D1) and (D2) exactly; for a declared R admitted only at the receiver tolerances they are the same formula evaluated at that R.

(ii) For a declaration with a model admissible exactly (Requirement 14.6) whose R satisfies (D1) and (D2) exactly, decided in exact arithmetic on its declared entries on the declared dark-count route, the entry also carries, for every setting and every $n \le N$, a complex enclosure with a Hermitian centre of every block $\widetilde{F}^{(=n)}_{v|t}$ (Definition 9.1), computed in exact rational arithmetic on the exact island (Definition "The exact island" (foundations/def:exact-island) (i)) and with no binary64 value as an input: from the enclosures of the port Grams of Requirement 14.6 (iii), by Theorem 9.5 (ii) for an unconditioned receiver and by the enclosure of its blocks that a submodule constructing the setting family states for that family, then by Corollary 20.2 (i) and Lemma 9.2 (v). The square-root enclosures and angle enclosures used may be any that Definition 9.1 admits. The following entries are enclosed exactly, as zero with radius zero:

(a) every entry of a block $\widetilde{F}^{(=n)}_{v|t}$ such that every $\widetilde{M}^{(=n)}_{S|t}$ with $S \in \varpi^{-1}(v)$, whose sum it is (Definition 20.1), is shown to vanish: for the unconditioned receiver on the generated dark-count route, and for no other receiver or route, by Proposition 20.3 (v), decided on its declared per-port dark-count probabilities; or, for $S = \varnothing$ and $n \ge 1$, by Proposition 20.3 (iv) at a setting whose operator $B^{(t)}$ the engine has computed in exact arithmetic and found to be the identity, which for the unconditioned receiver is A_Σ computed from the port Grams of Requirement 14.6 (i) or (ii);

(b) every entry whose enclosure is exact by Theorem 9.9 (ii) or by the exactness of its blocks that a submodule constructing the setting family states, carried through Corollary 20.2 (i), and whose value is zero.

No other entry is required to be enclosed as zero.

(iii) The enclosures of (ii), with the family they enclose, are the reference

(Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)) that the receiver offers for a snap of its emitted family (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference)), one per setting, with constant 1 (Corollary 20.2 (ii)). The entry records the *offered* reference kind of every block: *exact* where every radius of the block is zero, and *enclosed* otherwise, with the largest radius of the block and the largest radius of the enclosures of irrational inputs (square roots, trigonometric values) from which the centres are computed exactly. The record of every snap of the family carries besides the kind of the reference the snap ran on (Requirement "Snapping by the declarer only" (foundations/req:who-snaps)): *exact* on a block whose offered kind is exact, the snap running on its centres, and *none* on every other block, the snap running on the floating-point blocks of (i). The reference is an exact reference where every block is exact. It also names the claims that establish the enclosure, positive semidefiniteness and completeness of the reference, which Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i) leaves to the constructing module: Corollary 20.2 (ii), with Theorem 9.5 (ii) through Corollary 20.2 (iii) for an unconditioned receiver and the enclosure claim of the constructing submodule for a setting family a submodule constructs. On the exact route, with the port Grams enclosed exactly, every block $n \leq 1$ has radius zero (Theorem 9.9 (iii) for an unconditioned receiver, the exactness claim of the constructing submodule for a setting family a submodule constructs, and Corollary 20.2 (i)). The reference is read against the sector map of the floating-point blocks of (i) in the same entry (Definition "Sector maps and block-list operators" (foundations/def:sector-map)): the same block kind and block identifiers, the photon numbers $n = 0, \ldots, N$ in increasing order, and within each block the occupation vectors in the one order in which the entry lists the floating-point blocks, which is also the order of the rows and columns of every centre and radius matrix. The entries of the reference that a snap treats as known to be exactly zero (Definition "Snapping declared data onto a lattice" (foundations/def:snap)) are therefore exactly those enclosed as zero with radius zero.

(iv) Where the declaration has no model admissible exactly, where a declared R fails (D1) or (D2) exactly, or where a reciprocal that the enclosure of a setting family a submodule constructs forms receives no enclosure, the entry carries no reference, records the offered reference kind of every block as *none*, and records the cause.

(v) R remains emitted in `dark_count_channel` (Requirement 22.8), where

it serves the high-photon weight. It is already composed into the blocks of (i)–(iii), and a consumer does not compose it into them again.

(vi) Items (ii) and (iii) bind every run, save one that declares that enclosed references are not used (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i)). Such a run need form no enclosure of (ii) with a radius that is not zero, beyond what deciding that a block is not enclosed exactly takes, and carries none: it carries the reference of (iii) only on the blocks every entry of which it encloses exactly, with radius zero, from port Grams enclosed exactly, each block so carried being an exact reference of that block, and records their offered kind *exact*; every other block it records as offered with the kind *none*, with the cause that enclosed references are not used (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)). The reference the entry carries, its centres and radii as data, is the handed reference that the instance hands beside its snap of the emitted family (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap) (i)), whatever the snap itself uses of it: the snap uses only the centres of the exact blocks (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (ii)), and the run's first-order bound reads the reference on every block it carries (Requirement "The first-order bound evaluated on the reference companion" (meat-core/frank-wolfe/req:bracket-on-reference)). Item (iv) binds as stated.

(vii) *Factor enclosures.* Where the entry carries the reference of (iii) for a receiver with $\mathbb{I}_P = \varnothing$, whose one setting is off (Corollary 20.2 (iii)), the engine forms besides, on the blocks the count below admits and on no other, a factor enclosure (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) of the block $\widetilde{F}^{(=n)}_{v|\mathrm{off}}$: the one of Theorem 9.7 (iii) at the weights $w_{S'} := R_{\varpi^{-1}(v)|S'}$, rationals by Corollary 20.2 (iv), at which the block of that theorem is $\widetilde{F}^{(=n)}_{v|\mathrm{off}}$ (Definition 20.1), its rows in the order of (iii), for the port dilation (Definition 9.6) formed as follows.

(a) On the route that declares a component graph, the dilation of Lemma 16.4 (ii), with the loss factor of part (i) of that lemma for every element. A row of a loss factor, and of the efficiency layer's, is formed only where the declared value that scales it is decided in exact arithmetic to make it non-zero, so that whether an element is lossless is read off its declared parameters and never off a computed number. The loss factor of a generic block is

formed from a decomposition $\mathbb{1}_\Sigma - G_\Sigma^\dagger G_\Sigma = E^\dagger D E$ formed and decided in exact rational arithmetic; where none is formed, no dilation of the declaration is.

(b) On a route that declares the port Grams or the instrument matrix, the dilation of Theorem 9.7 (iv), where the engine forms decompositions $A_k = E_k^\dagger D_k E_k$ and $\mathbb{L} = E_0^\dagger D_0 E_0$ of the kind named there, formed and decided in exact rational arithmetic.

The entries of the dilation are enclosed by the rules of Lemma 9.2 from square-root enclosures and angle enclosures, as Requirement 14.6 (iii) encloses the elements' matrices and the $\sqrt{\eta_k}$, and the columns as Theorem 9.7 (ii) encloses them, in exact rational arithmetic or by a rigorously enclosed computation (Definition "The tiers of a reported key rate" (foundations/def:tiers)), with no binary64 value as an input; a column enclosed exactly as zero is omitted, and the core is exact. The engine forms the factor enclosure of a block only where the number of occupation vectors $\boldsymbol{m}$ of weight n on the modes of the dilation whose row is not enclosed exactly as zero, with $R_{\varpi^{-1}(v)|S_{\boldsymbol{m}}} > 0$, is at most the block's dimension $\binom{|\mathbb{I}|+n-1}{n}$, a count it decides before it computes any column: a factor enclosure with more columns than that dimension fixes no rank (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (ii)). It forms none for a receiver with $\mathbb{I}_P \neq \varnothing$, whose blocks are not formed through a port dilation; none where no dilation of (a) or (b) is formed; none on a block the count does not admit; and none in a run that declares that enclosed references are not used (vi). A factor enclosure is carried beside the entrywise enclosure of its block, which it leaves unaltered, and is handed with the reference, altered by no step. The entry records per block whether one is carried, with its number of columns and its largest radius, or the cause where none is; and it names Theorem 9.7 and, on the route of (a), Lemma 16.4 as the claims that establish that the block is a member of its factor enclosure, which Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i) leaves to the module that constructs the reference.

Motivated by Rem. 4.9, Def. 9.1, Lem. 9.2, Thm. 9.5, Def. 9.6, Thm. 9.7, Thm. 9.9, Def. 10.4, Req. 14.6, Lem. 16.4, Def. 19.1, Def. 19.3, Def. 20.1, Cor. 20.2, Prop. 20.3, Req. 22.8, foundations/cor:rank-over-factor-enclosure, foundations/def:evaluated-off-snap, foundations/def:exact-island, foundations/def:exact-reference, foundations/def:factor-enclosure, foundations/def:sector-map, foundations/def:snap, foundations/def:tiers, foundations/req:snap-reference, foundations/req:who-snaps, meat-core/frank-wolfe/req:bracket-on-reference · *Used by* Req. 22.10 · implemented (effective: accepted) · *Code* certiqs_key.receiver.reference:dark_composed_reference

A moved block of the receiver's snap that rests on the snap premise records the reference kind, its cause, its group and the fix the cause details: exact data, an enclosed reference used, tighter enclosures, or loss and dark counts. Only a strict run refuses, or one whose last resort fails.

Under Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference), a moved block of the receiver's snap is not refused by default: unless the run evaluates it off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)), the run rests there on the premise that the device measures the snapped family (Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement)), and the record names the block with its cause, its group and a fix. Where the support-safe rules cannot restore the conditions of a snap on a block, a last resort snaps it anyway and the record flags it (Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) (iv)). A user can act on the record only if it says what to change in the receiver declaration, and for a receiver the cause of the group *no usable reference* and the last resort each have a definite remedy, which details the fix of the group; Remark 20.4 explains them. The record also names, block by block, whether the snap ran on the centres of an exact reference or on the floating-point blocks, and which reference the receiver hands, so that a reader sees where an enclosure entered the result. An angle at any value is no cause: every declaration with a model admissible exactly has an enclosed reference (Requirement 22.9).

Requirement 22.10 (Cause and fix recorded for a receiver block resting on its snap). When a moved block of a snap of the receiver's emitted dark-composed blocks (Requirement 22.9) rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) under Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii), its record names, beside the setting, the block with its cause and the group that requirement names for it, the reference kind of the snap, as Requirement "Snapping by the declarer only" (foundations/req:who-snaps) defines it (*exact*, where the snap ran on the centres of an exact reference of the block, or *none*), the kind of the reference handed on the block, if any (*exact*, every radius of the block zero, or *enclosed*, with the largest radius of the block and the largest radius of the enclosures of irrational inputs (square roots, trigonometric values) from which the centres were computed exactly), and the fix below for the cause, which details the fix of its group. A run that declares the strict option of Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv) refuses instead, naming the same; no other run refuses for these causes, save where the last resort of Requirement "Support-safe snaps of a measurement family"

(foundations/req:support-safe-snap) (iv) yields no snap.

(i) *No reference.* The cause recorded by Requirement 22.9 (iv), in the group *no usable reference*, and the fix for it:

  (a) for declared port Grams that are not Hermitian, a declared family that is not an admissible Gram family exactly, or a generic block that is not a contraction exactly (Requirement 14.6 (i), (ii)), declared data that are Hermitian and admissible exactly, in rational arithmetic and not only at the receiver tolerances;

  (b) for a declared dark-count matrix that does not satisfy (D1) and (D2) exactly, one that is exactly column-stochastic and monotone, or the generated dark-count route;

  (c) for a reciprocal that the enclosure of a setting family a submodule constructs forms and that receives no enclosure, square-root and angle enclosures with smaller radii (Definition 9.1).

(ii) *Enclosed references not used.* Where the run declares that enclosed references are not used and the entry offers an enclosed one, the block is handed no reference, in the group *no usable reference*, and the fix names the kind the entry offers for every block (Requirement 22.9 (iii)), with the largest radius of every block offered as enclosed, and the alternative of an exact reference, which the receiver offers for a declaration on the exact route at a cutoff $N \leq 1$ when its port Grams are enclosed exactly (Theorem 9.9 (iii), Corollary 20.2 (ii)), or a run that uses enclosed references.

A block that the snap makes by the last resort of Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) (iv) is flagged `support_not_preserved` in the record, beside its cause and reference kind, and where it rests, in the group *last resort*, the fix names the ports whose declared efficiency is 1 where $\lambda_{\max}(A_\Sigma) = 1$, and the click patterns S whose entry $R_{S|\varnothing}$ of the dark-count channel is 0, since Proposition 20.3 (i) needs $R_{S|\varnothing} > 0$ for every S: on the declared dark-count route it asks for a declared R with $R_{S|\varnothing} > 0$ for every S, and on the generated dark-count route, where every $R_{S|\varnothing}$ is positive exactly when $0 < p_{\mathrm{dc}|k} < 1$ for every k (Proposition 20.3 (ii)), a dark-count probability in $(0, 1)$ in every port; in either case a model with loss, at the smallest loss and dark-count rates the user is sure the device has (Proposition 20.3); and, where the entry offers no reference, the fix of (i) as well. Where every element of a block of the reference is positive definite (Proposition 20.3 (i)), no row of it is structurally zero (Definition "Snapping declared data onto a lattice" (foundations/def:snap)), so every index lies in the structural support of every element and the uniform lift of that item meets the rules (i) to (iii) of Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) on the block. For

a block whose cause is the run's, a handed reference on which the companion's bound is withdrawn or not evaluated, or counts while the key length is formed from a number certified on the snapped declaration in its place, the cause `companion_not_certified` (Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (i)), the record of the run names the fix Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii) names for that cause, and, for such a block snapped by the last resort, which rests in the group *last resort*, that fix beside the one above; the run decides that cause after its search, so the fix is completed there and not at the snap. A handed block that the run evaluates off its snap rests on nothing and is named with no fix, whether or not the last resort snapped it. A reference that fails the decisions of Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i) cannot arise for the receiver: the centres are Hermitian with entries in $\mathbb{Q}(i)$ and the radii symmetric non-negative rationals by construction (Lemma 9.2 (v)), and the enclosure, positive semidefiniteness and completeness that the constructing module establishes are Corollary 20.2 (ii), whose hypothesis Theorem 9.5 (ii), through Corollary 20.2 (iii), and, for a setting family a submodule constructs, that submodule's enclosure of its blocks supply.

Motivated by Def. 9.1, Lem. 9.2, Thm. 9.5, Def. 9.8, Thm. 9.9, Req. 14.6, Def. 19.1, Cor. 20.2, Prop. 20.3, Rem. 20.4, Req. 22.9, foundations/ass:snapped-measurement, foundations/def:evaluated-off-snap, foundations/def:exact-reference, foundations/def:snap, foundations/req:snap-reference, foundations/req:support-safe-snap, foundations/req:who-snaps, meat-core/frank-wolfe/req:certified-evaluation · implemented (effective: accepted) · *Code* certiqs_key.receiver.snapfix:receiver_fixes

Interface

The declared inputs and outputs of the module and its submodules: each key with its symbol, when it is present and the claims that fix it. The domains, types and refusal rules are in the interface files themselves.

Inputs: Receiver POVM (receiver)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`mode_space` M (the declared mode space)		—
`input_modes` Inp = Inp_C (disjoint union) Inp_S		—
`port_map` {P_k}_{k=1..K}		—
`port_grams` {A_k}		—
`instrument_matrix` Gamma		—
`element_graph` G		—
`detector_layer` {(P_k, eta_k, p_dc|k)}		—
`dark_count_channel` R		—
`probe_declaration` (T, {rhoˆ(t)})		—
`photon_cutoff` N		—
`tail_window` {m_t}_{t in T}		—
`exact_window` {m*_t}		—
`numerical_tolerances` {eps_1, ..., eps_6}, eps_gap, eps_eig		—
`outcome_coarse_graining` cg		—
`requested_assurance`		—

Key and symbol	Presence	Fixed by
`efficiency_normalization` t		—

Outputs: Receiver POVM (receiver)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`input_mode_space` Inp = Inp_C (disjoint union) Inp_S	always	Def. 1.1, Def. 2.1
`outcome_alphabet` V with cg	always	Def. 5.5, Thm. 5.6
`port_grams` {A_k}	always	Def. 2.2, Thm. 18.1
`receiver_construction` G, {P_k}, probe declaration	route (iii) only	—
`photon_blocks` blk M^{(=n)}_v, 0 <= n <= N	always	Thm. 4.5, Thm. 5.6, Thm. 19.2, Def. 20.1, Thm. 9.5, Thm. 9.9, Thm. 9.7, Cor. 20.2, Req. 22.9
`block_dimensions` binom(|Inp_C| + n - 1, n)	always	Def. 1.2
`efficiency_record` {(q_i, e_i, s_i)}, t	always	Req. 22.2, Req. 22.5
`dark_count_channel` R, r_min, rbar, delta_dc	always	Thm. 19.2, Prop. 19.4, Req. 22.8
`high_photon_bounds` {(O_i, (p_i^(t))_{t in T})}	always	Def. 8.16, Def. 8.3, Def. 8.7, Def. 8.12, Thm. 8.14, Thm. 8.13, Req. 13.5, Thm. 19.2, Prop. 19.4, Cor. 8.15
`bound_quantifier_domain` N, T, {m_t}, {m*_t}, {beta-hat*^(t)(n)}	always	Thm. 8.14, Def. 8.3, Def. 8.7, Def. 8.12, Req. 13.5, Req. 13.4
`validation_results` r_1..r_6, g_{N,m_t}	always	Def. 10.1, Req. 14.1, Req. 14.2, Prop. 10.2, Lem. 6.8, Thm. 10.10

Key and symbol	Presence	Fixed by
`bound_evaluation`	always	Thm. 10.10, Req. 14.5, Req. 13.5, Def. 10.9
`assumption_ledger` A-...	always	—
`calibration_coverage`	always	Ass. 11.1
`mathematical_spec_version`	always	—

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